



Internal Use Only

MOBILE PHONE SERVICE MANUAL

CAUTION

BEFORE SERVICING THE UNIT, READ THE "SAFETY PRECAUTIONS" IN THIS MANUAL

MODEL : LG-H522y

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1.1 Purpose

This manual provides the information necessary to repair, calibration, description and download the features of this model.

1.2 Regulatory Information

A. Security

This material is prohibited to share and release to unauthorized person, in accordance with the regulations, LG Electronics, Civil / criminal responsibility in accordance with the relevant provisions violate.

B. Precautions for repair

- In case of Disassembly or Assembly to repair product, be careful of a product failure caused by RF signals and Static electricity.
- When using Magnetic tool for the Phone's SVC repair, you should check affect the Electric parts according to effect of Magnet.
- When fastening the screw, be careful not to damage the head of screw and even product.

C. Attention

Boards, which contain Electrostatic Sensitive Device (ESD), are indicated by the  sign.

Following information is ESD handling:

- Service personal should ground themselves by using a wrist, strap when exchange system board.
- When repair are made to a system board, they should spread the floor with anti-static mat which is also grounded.
- Use a suitable, grounded soldering iron.
- Keep sensitive parts in these protective packages until these are used.
- When returning system board or parts like EEPROM to the Factory, use the protective package as described.

2.1 Band Specification

Support Band	TX Freq (MHz)	RX Freq (MHz)
GSM850	824 – 849	869 – 894
GSM900	880 – 915	925 – 960
DCS1800	1710 – 1785	1805 – 1880
PCS1900	1850 – 1910	1930 – 1990
W2100(B1)	1920 – 1980	2110 – 2170
W850(B5)	824 – 849	869 – 894
W900(B8)	880 – 915	925 – 960
LTE B3	1710 – 1785	1805 – 1880
LTE B7	2500 – 2570	2620 – 2690
LTE B28	703 – 748	758 – 803
BT/WiFi	Bluetooth 4.0 + BLE / 802.11 b/g/n	
GPS	1574.42 MHz ~ 1576.42 MHz	

2.2 HW Features

Item	Feature	Comment
Phone type	Touch	
Size	69.8 X 139.7 X 10.2 mm	
Weight	134g(Tocad) / 136g(LGC)	With Battery
Android OS	L-OS	
MTK Chipset	MT6732	
Memory	8GB eMMC/1GB RAM	
Display	5" HD (1280X720)	
Camera	8MP AF / 5MP VT	
Micro SD	Up to 32GB	
Audio	Receiver 1206 Speaker 1511 # of MIC. : 2	
Bluetooth	4.0	
WLAN	802.11 b/g/n	
LTE	B3,B7,B28	
GPS	A-GPS	Glionass

2.3 RSSI Display

Antenna BAR	Specification		
	LTE (RSRP) (B3/B7/B28)	WCDMA (RSCP) (B1/B5/B8)	GSM (RSSI) (850/EGSM/DCS /PCS)
5→4	-85 ± 3	-89 ± 3	-91 ± 3
4→3	-95 ± 3	-94 ± 3	-96 ± 3
3→2	-105 ± 3	-100 ± 3	-99 ± 3
2→1	-115 ± 3	-104 ± 3	-103 ± 3
1→0	-128 ± 3	-110 ± 3	-105 ± 3
0 → No SVC	-131 ↓	-115 ↓	-110 ↓

2.4 Current consumption

	Specification		
	GSM	WCDMA	LTE
Stand by current (Sleep & Idle Avg.)	` BT off : 8 mA (PP = 5) ` BT on : 11 mA (PP = 5)	` BT off : 8 mA (DRX = 7) ` BT on : 11 mA (DRX = 7)	` BT off : 8 mA (DRX = 2.56sec) ` BT on : 11 mA (DRX = 2.56sec)
Talk current	370 mA (TX Level = Max)	250 mA (TX = 10dBm)	370 mA (Tx= 10dBm)
Power off	500uA		

2.5 Battery bar

Battery Bar	Specification	Battery Bar	Specification
Bar 20(Full)	98%이상	Bar 9 -> Bar 8	43% -> 42%
Bar 20 -> Bar 19	98% -> 97%	Bar 8 -> Bar 7	38% -> 37%
Bar 19 -> Bar 18	93% -> 92%	Bar 7 -> Bar 6	33% -> 32%
Bar 18 -> Bar 17	88% -> 87%	Bar 6 -> Bar 5	28% -> 27%
Bar 17 -> Bar 16	83% -> 82%	Bar 5 -> Bar 4	23% -> 22%
Bar 16 -> Bar 15	78% -> 77%	Bar 4 -> Bar 3	16% -> 15%
Bar 15 -> Bar 14	73% -> 72%	Bar 3 -> Bar 2	13% -> 12%
Bar 14 -> Bar 13	68% -> 67%	Bar 2 -> Bar 1	8% -> 7%
Bar 13 -> Bar 12	63% -> 62%	Bar 1 -> Bar 0	3% -> 2%
Bar 12 -> Bar 11	58% -> 57%	Power off	1% -> 0%
Bar 11 -> Bar 10	53% -> 52%	Low battery pop-up	15% , 5%, 1%
Bar 10 -> Bar 9	48% -> 47%		

※Cut Off Voltage : 3.4 ± 0.1 V

2.6 SW Specification

Item	Feature	Comment
RSSI	0 ~ 5 Levels	
Battery Charging	0 ~ 20 Levels	
Key Volume	0 ~ 7 Level	
Audio Volume	0 ~ 15 Level	
Time / Date Display	Yes	
Multi-Language	Yes	depending on build language
PC Sync	Yes	
Speed Dial	Yes	
CLIP / CLIR	Yes	
Phone Book	Name / Number / Email / Groups / Postal addresses / Organizations / IM / Note / Nickname / Website / Event /	There is no limitation on the number of items. It depends on available memory amount.
Last Dial Number	Yes	
Last Received Number	Yes	
Last Missed Number	Yes	
Search by Number/Name	Yes	
Group	Yes	There is no limitation on the number of items. It depends on available memory amount.
Fixed Dial Number	Yes	
Voice Memo	Yes	Support voice recorder
Network Selection	Automatic	
Mute	Yes	
Call Divert	Yes	
Call Barring	Yes	
Call Duration	Yes	
SMS Over GPRS	Yes	
MMS MPEG4 Send / Receive / Save	Yes	ØSend / Receive : Yes ØSave : depends on content type/ Support video content type list 1.video/mp4 2.video/h263 3.video/3gpp2 video/3gpp

2.6 SW Specification

Voice Memo	Yes	Support voice recorder
Network Selection	Automatic	
Mute	Yes	
Call Divert	Yes	
Call Barring	Yes	
Call Charge (AoC)	No	
Call Duration	Yes	
SMS (EMS)	There is no limitation on the number of items It depends on available memory amount.	EMS does not support.
SMS Over GPRS	Yes	
EMS Melody / Picture Send / Receive / Save	No	
MMS MPEG4 Send / Receive / Save	Yes	➤ Send / Receive : Yes ➤ Save : depends on content type Support video content type list 1. video/mp4 2. video/h263 3. video/3gpp2 - video/3gpp
Long Message	MAX 2000 characters	The standard of Open vender
Cell Broadcast	Yes	
Download	Over the Web	
Calendar	Yes	
Memo	Yes	There is no limitation on the number of items. It depends on available memory amount.
World Clock	Yes	

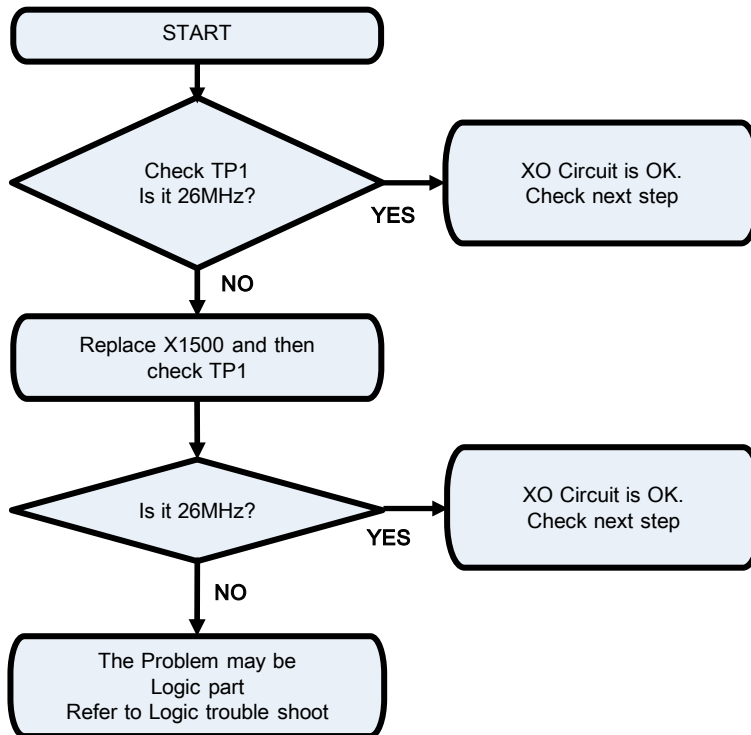
2.6 SW Specification

Item	Feature	Comment
Long Message	It depends on sim inserted	
Cell Broadcast	Yes	
Download	Over the Web	
Calendar	Yes	
Memo	Yes	There is no limitation on the number of items. It depends on available memory amount.
World Clock	Yes	
Stop Watch	Yes	
Wall Paper	Yes	
SIM Toolkit	Yes	
MMS	Yes	LG MMS Client
CPHS	Yes	
Camera	Yes	8M AF / Digital Zoom : x4
Bluetooth	Yes	Ver. 4.1
FM radio	Yes	
GPRS	Yes	class 33
EDGE	Yes	class 33
Hold / Retrieve	Yes	
Conference Call	Yes	Max. 6
DTMF	Yes	
AMR	Yes	
IM	Yes	Google Hangout
Email	Yes	

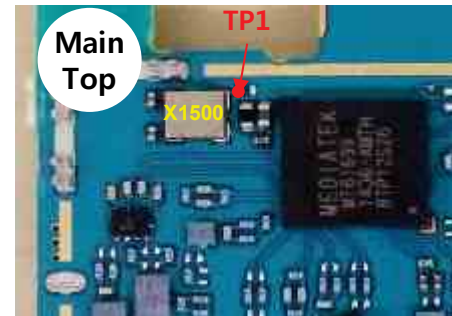
3.1 Checking XO Block

The out put frequency(26MHz) of VTCXO(X1500) is used as the reference one of MT6169

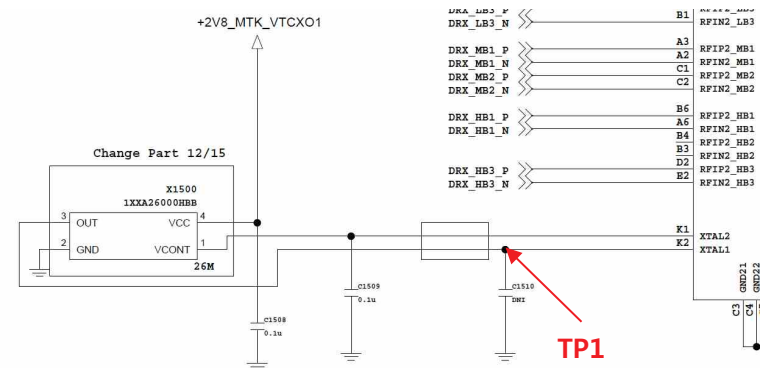
Checking Flow



Image



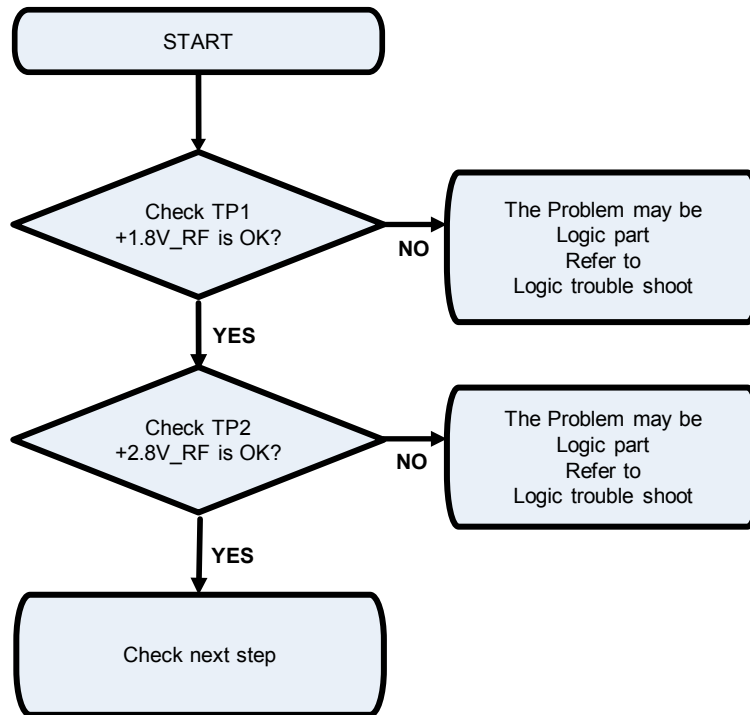
Circuit Diagram



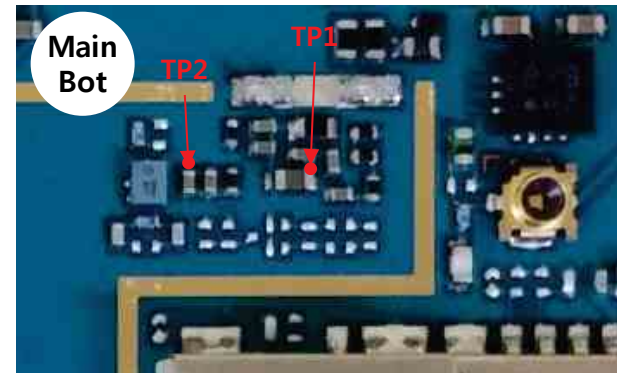
3.2 Checking Transceiver DC Power Supply Circuit Block

The MT6169 operating voltages used two voltage sources 1.8V and 2.8V

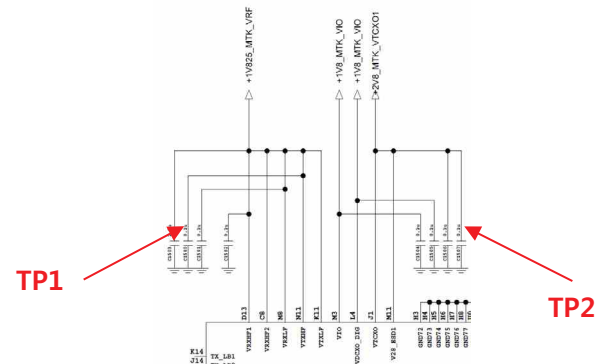
Checking Flow



Image



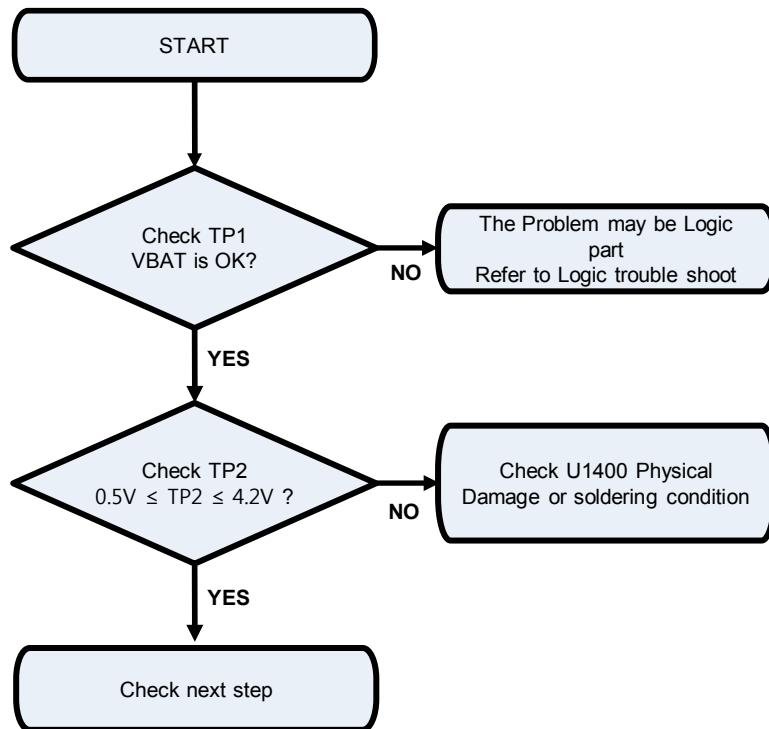
Circuit Diagram



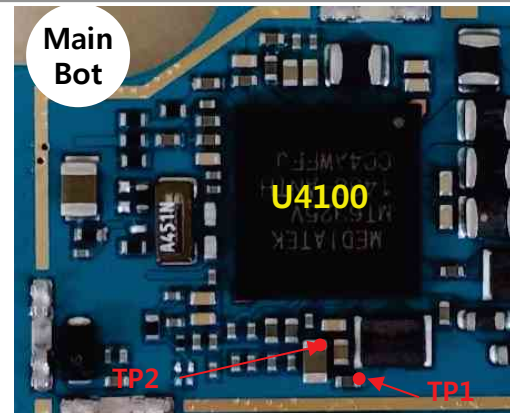
3.3 Checking DC-DC Block

The DC-DC(MT6325, U4100) output voltages is used as the reference one of SKY77621

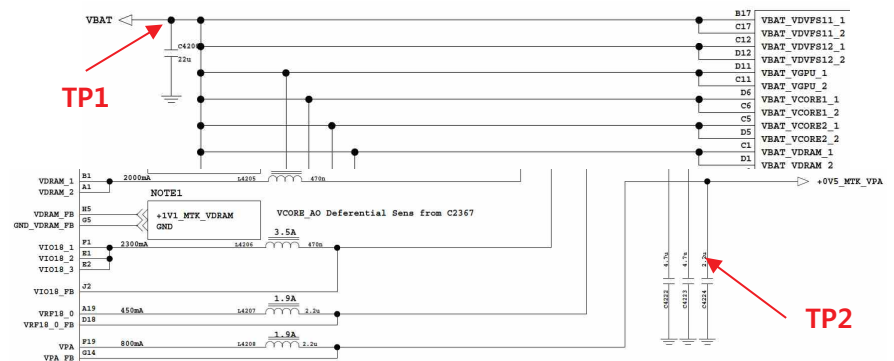
Checking Flow



Image



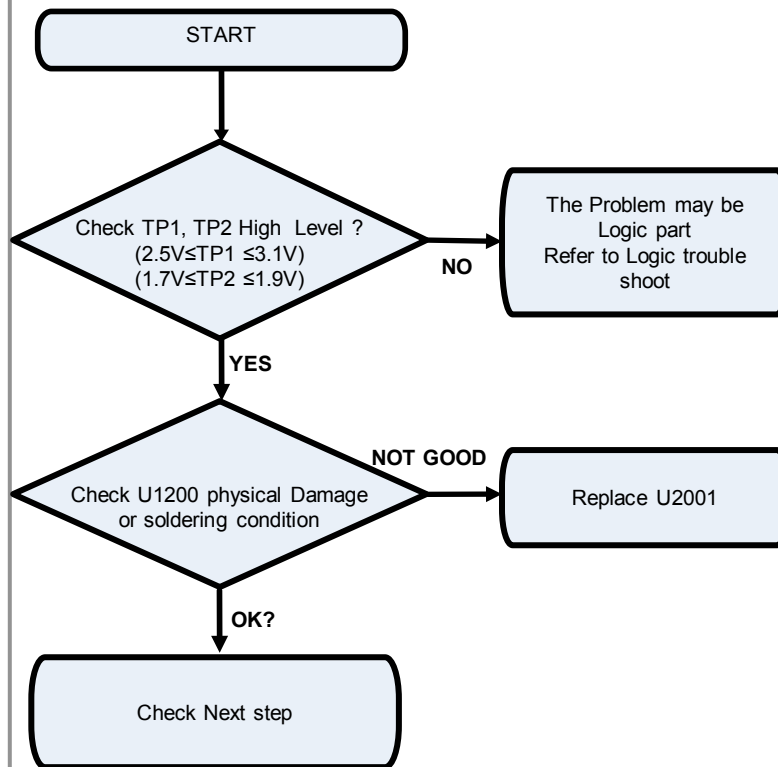
Circuit Diagram



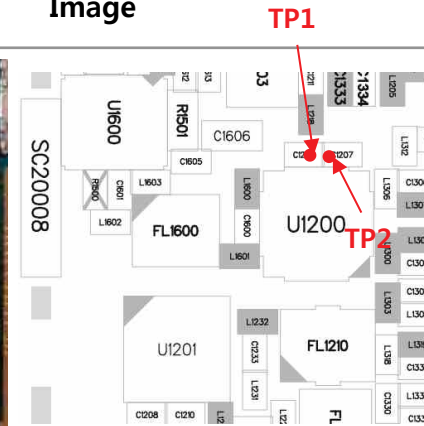
3.4 ASM(Antenna Switch Module) Block

3.4.1 Checking ANT #1 ASM (GSM 850/900, W B5/8, LTE B7/28)

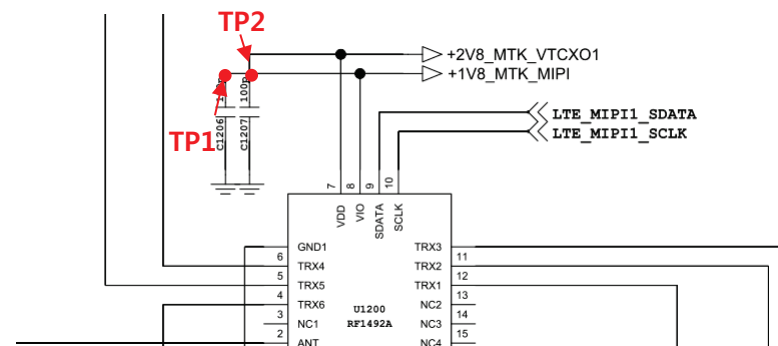
Checking Flow



Image

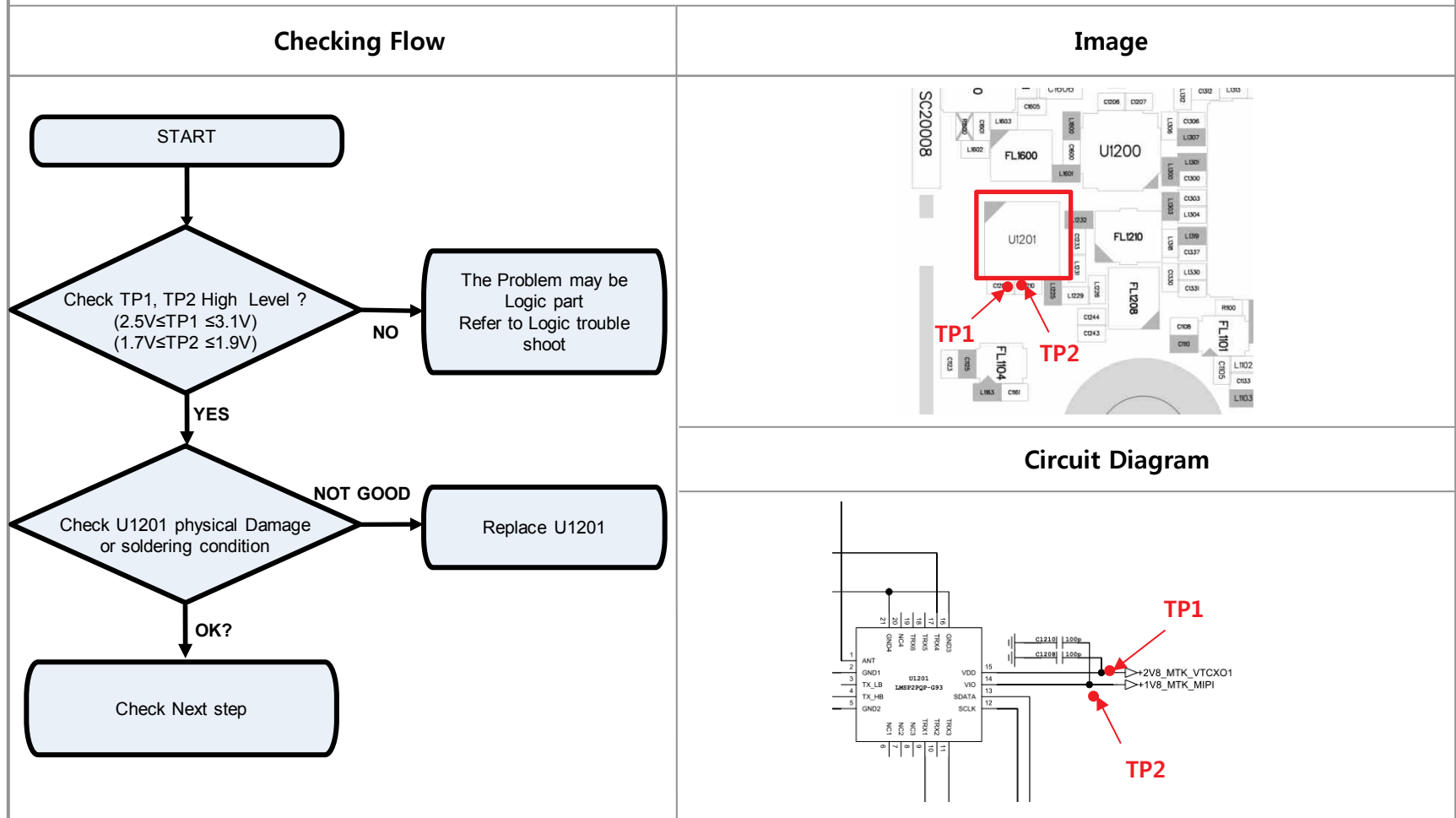


Circuit Diagram



3.4 ASM(Antenna Switch Module) Block

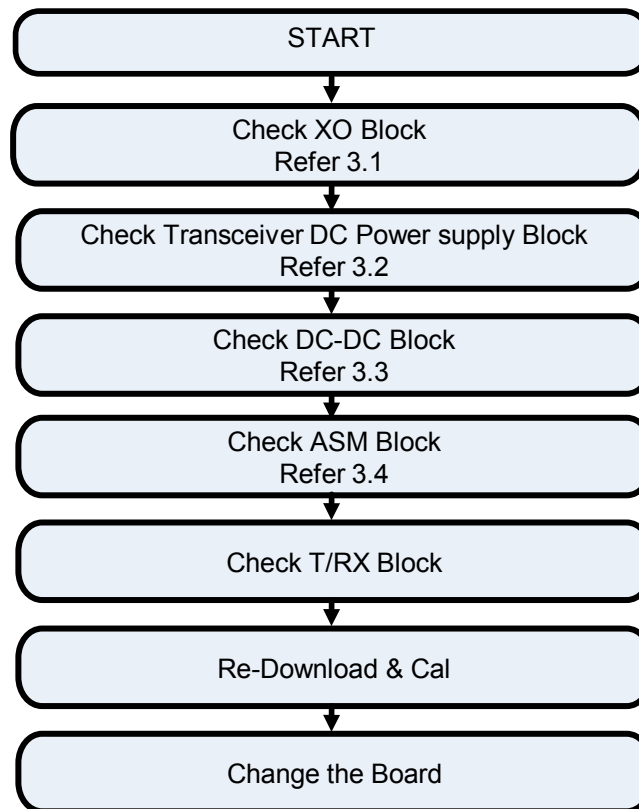
3.4.2 Checking ANT #2 ASM (GSM 1800/1900, W B1, LTE B3)



3.5 GSM RF PART

GSM RF Part support GSM850/900/1800/1900 with ASM, PAM, Transceiver component

Checking Flow



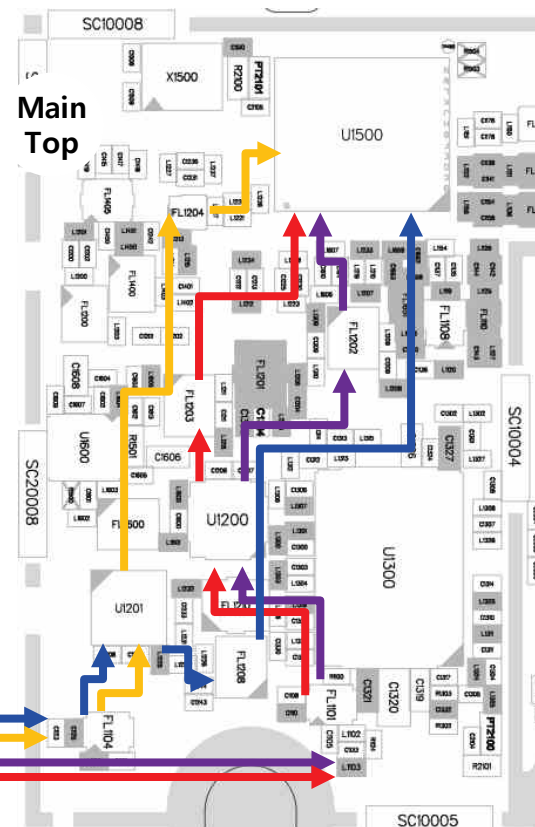
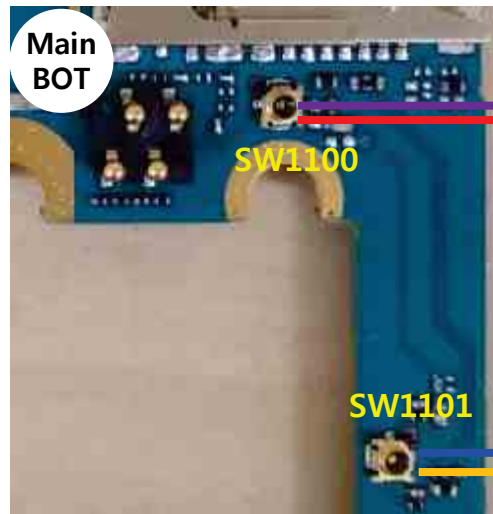
3. TROUBLE SHOOTING

3.5 GSM RF PART

3.5.1 GSM RF Part RX RF PATH

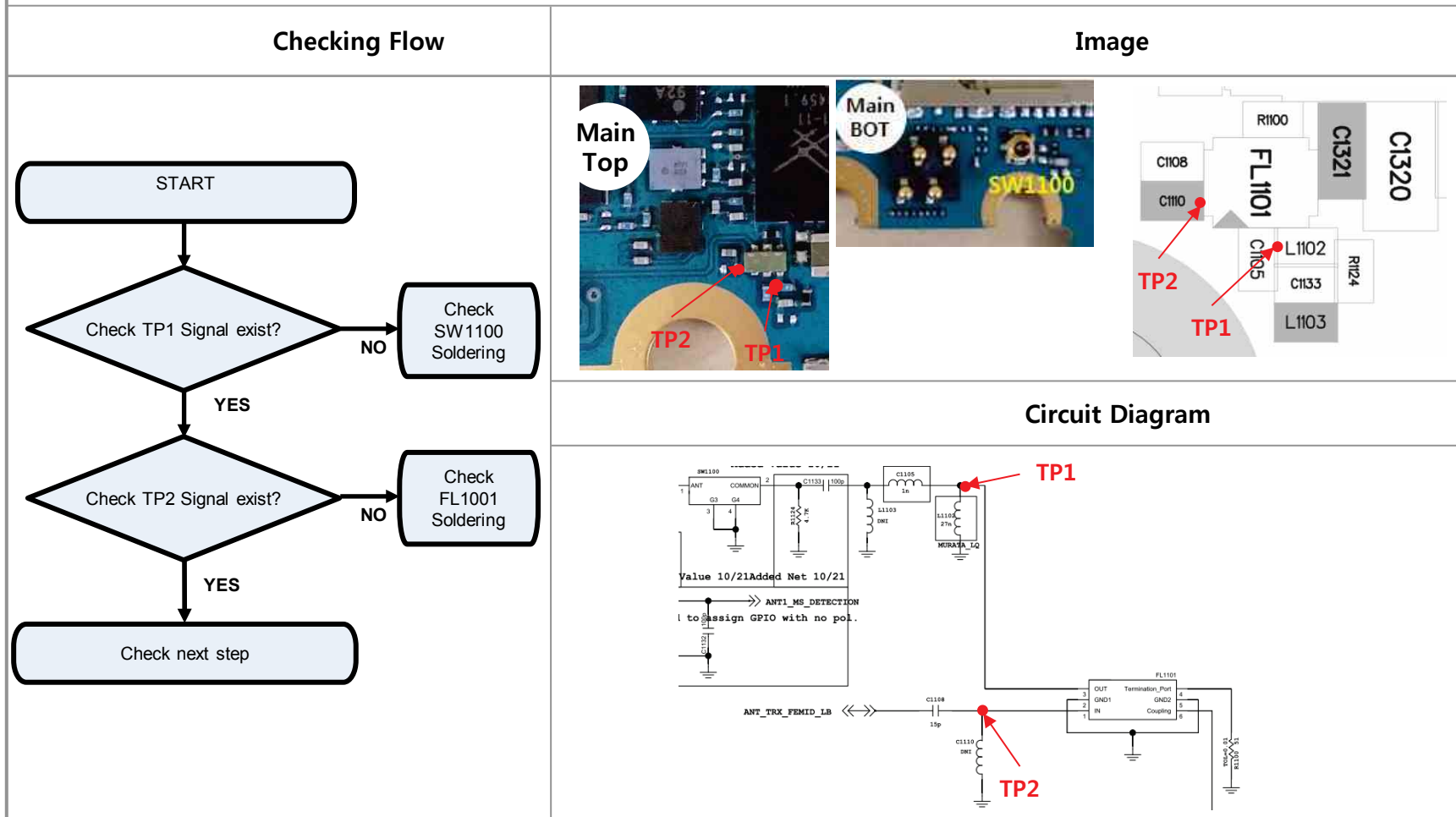
Image

1. GSM850 RX PATH
2. GSM900 RX PATH
3. GSM1800 RX PATH
4. GSM1900 RX PATH



3.5 GSM RF PART

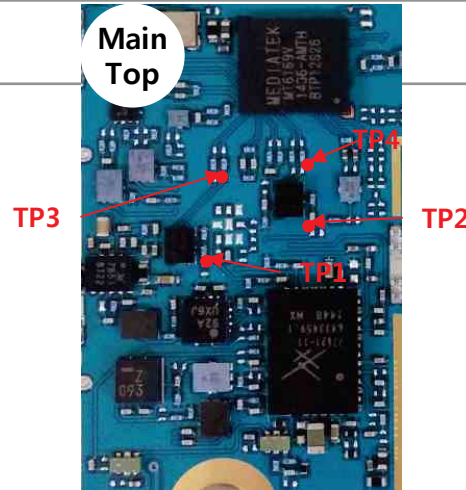
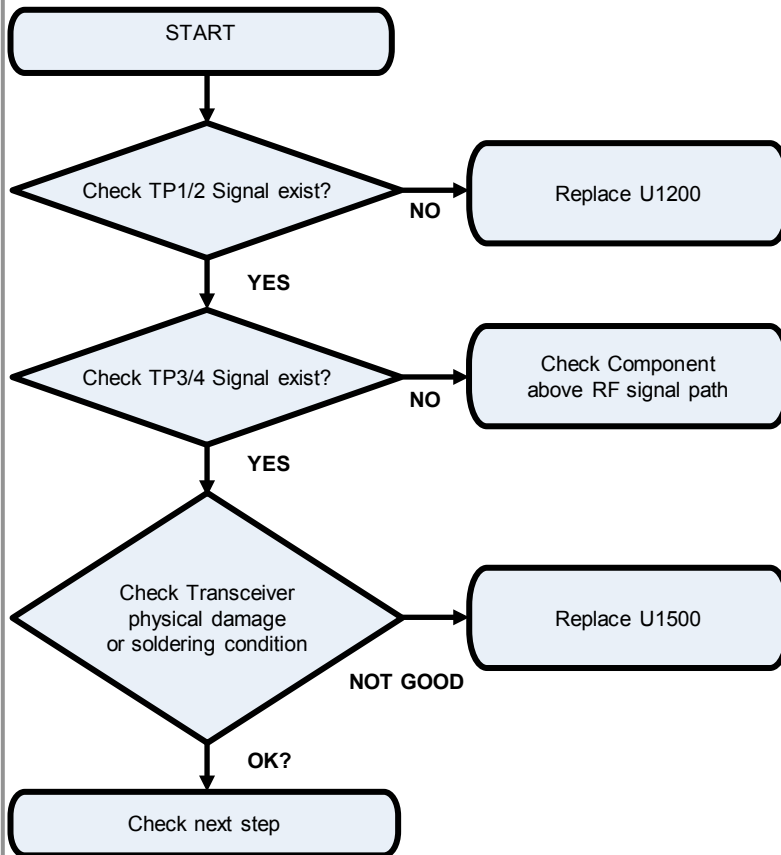
3.5.2 Checking RF Signal RX path(SW, GSM850/900)



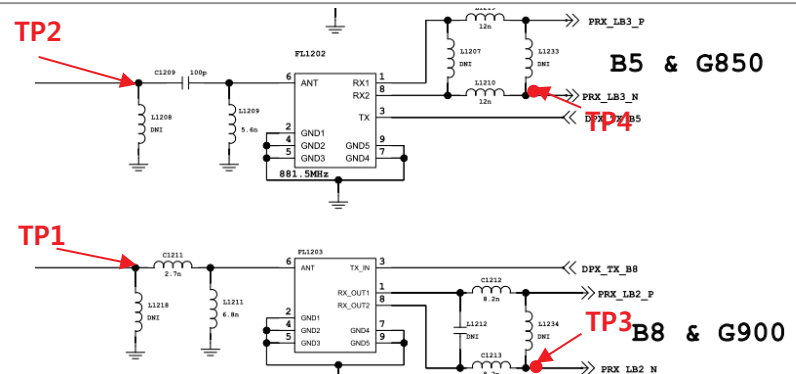
3.5 GSM RF PART

3.5.3 Checking RF Signal RX path(GSM850/900)

Checking Flow



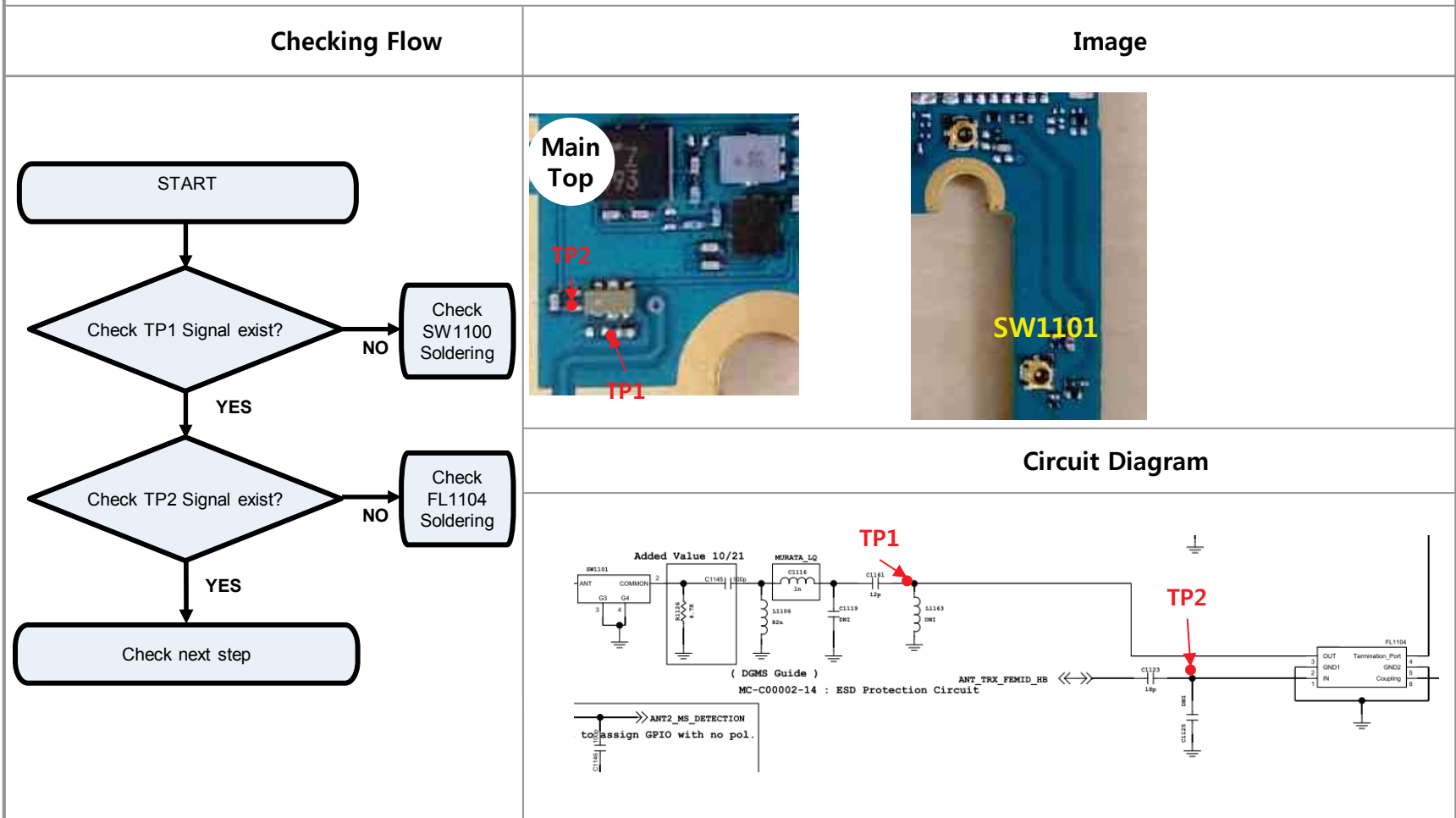
Circuit Diagram



3. TROUBLE SHOOTING

3.5 GSM RF PART

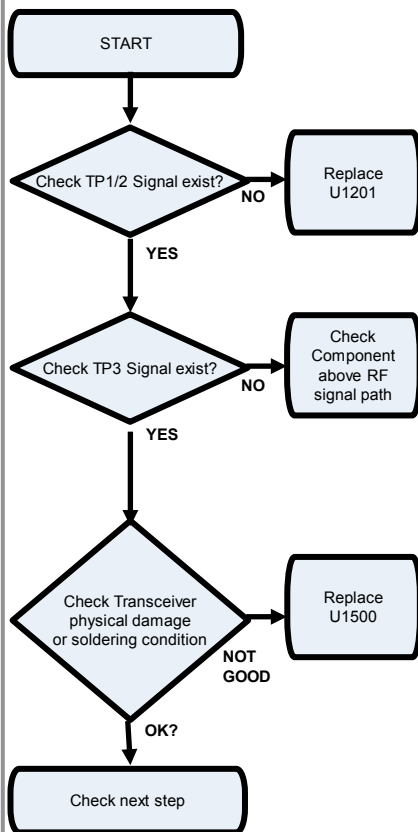
3.5.4 Checking RF Signal RX path(SW, GSM1800/1900)



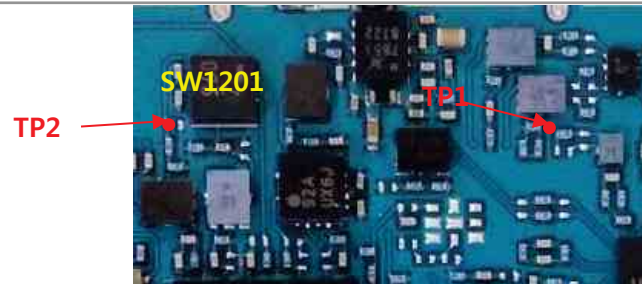
3.5 GSM RF PART

3.5.5 Checking RF Signal RX path(GSM1800/1900)

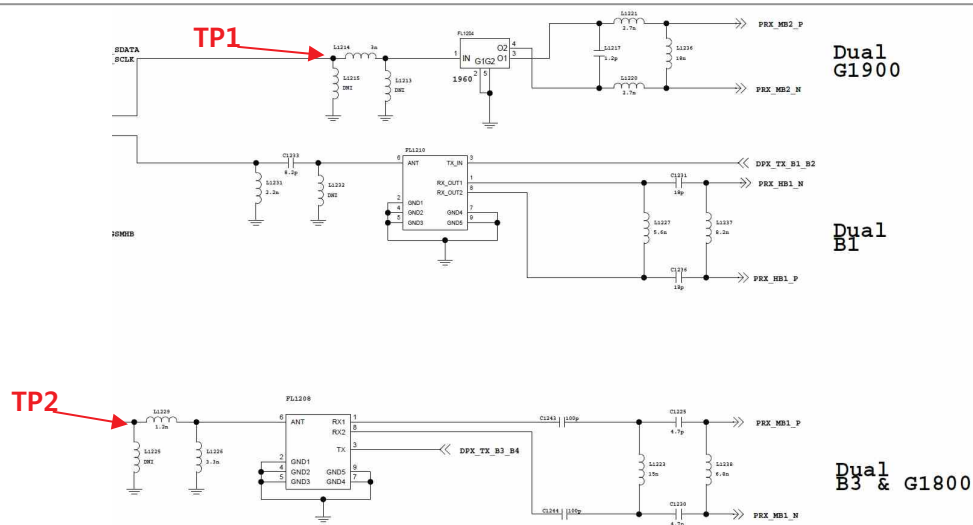
Checking Flow



Image



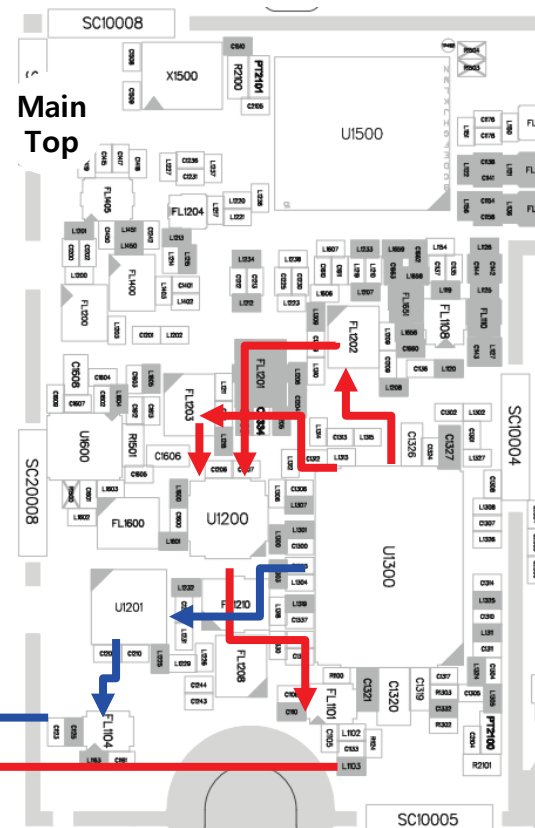
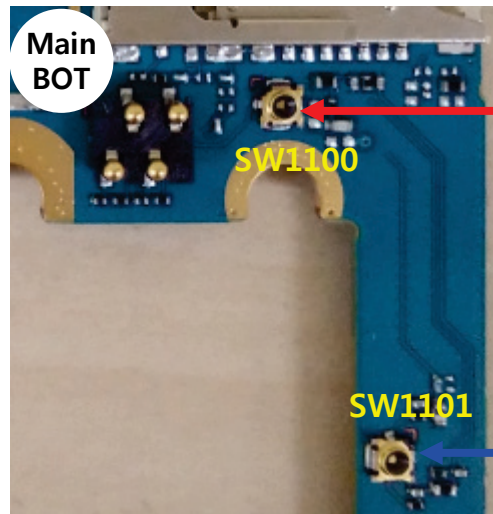
Circuit Diagram



3.5 GSM RF PART

3.5.6 GSM RF Part TX RF PATH

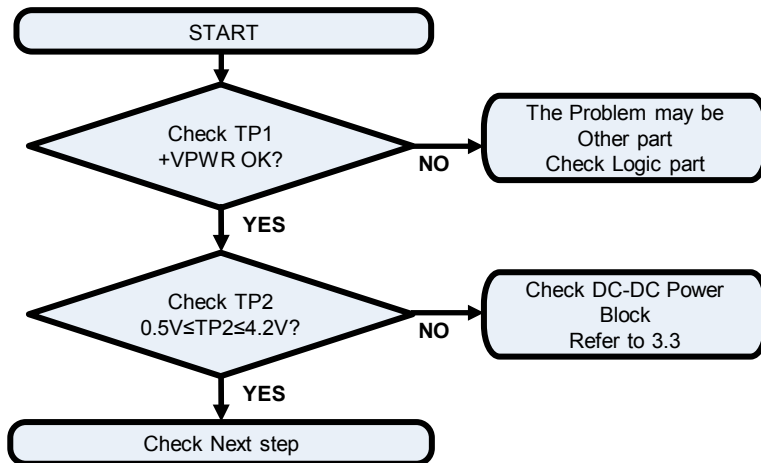
1. GSM850/900 TX PATH
2. GSM1800/1900 TX PATH



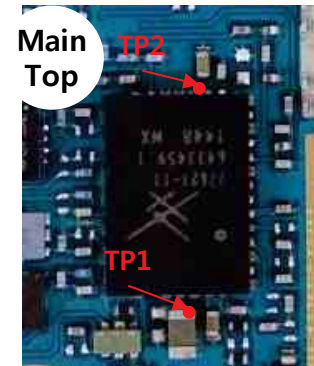
3.5 GSM RF PART

3.5.7 Checking GSM PAM DC Power Circuit

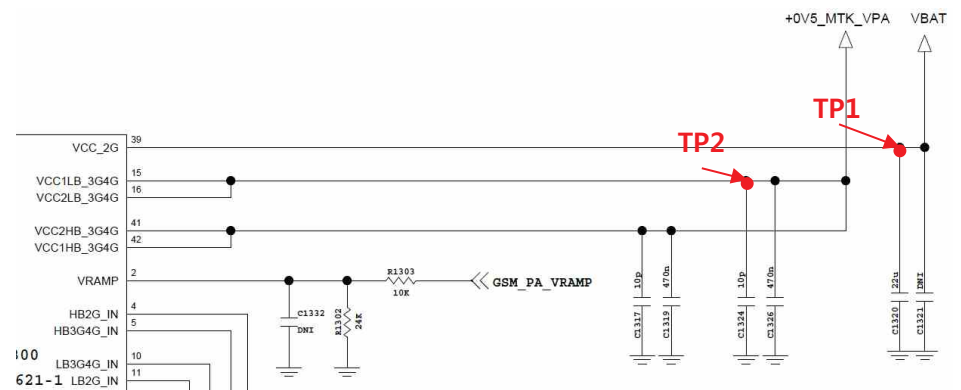
Checking Flow



Image



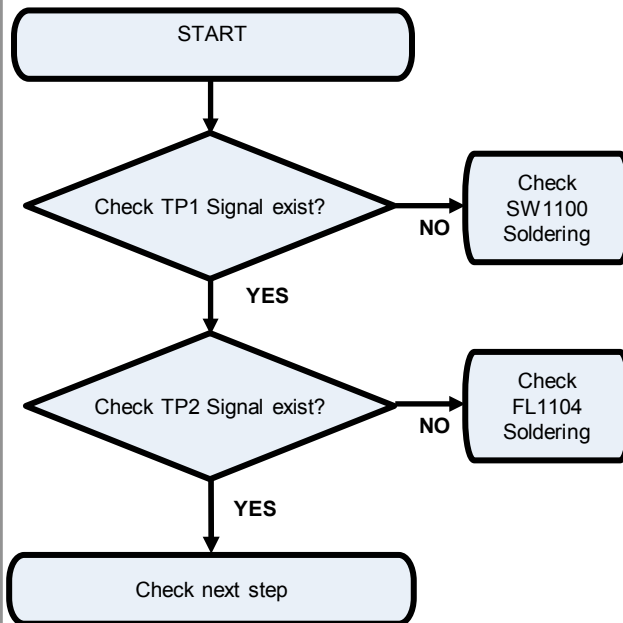
Circuit Diagram



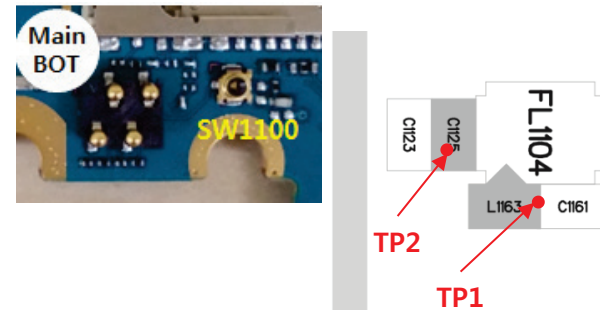
3.5 GSM RF PART

3.5.8 Checking RF Signal TX path(SW, GSM850/900)

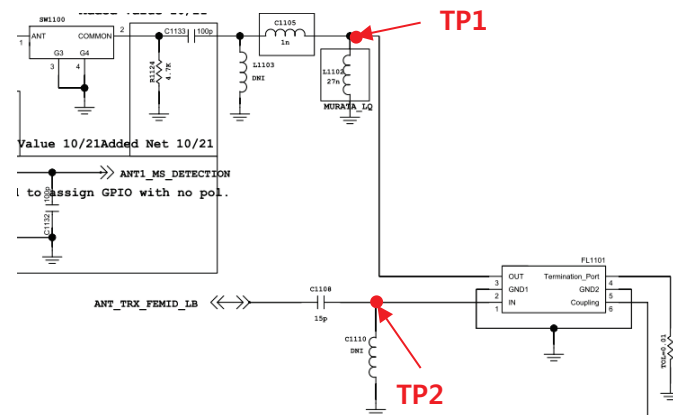
Checking Flow



Image



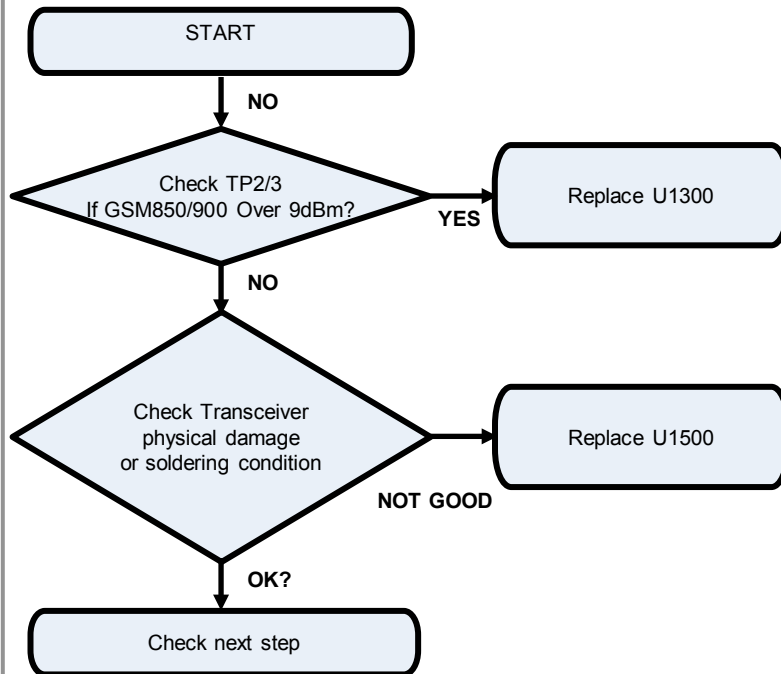
Circuit Diagram



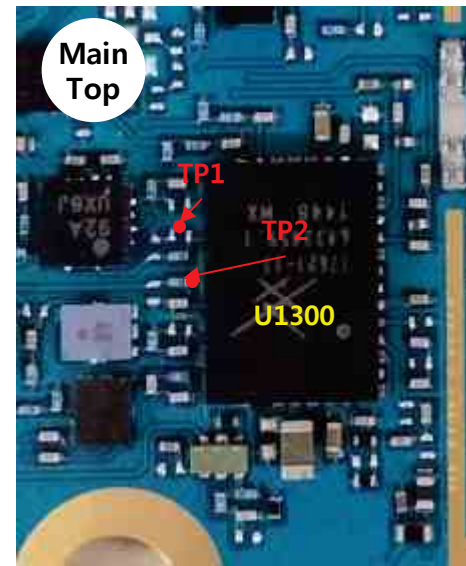
3.5 GSM RF PART

3.5.9 Checking RF Signal TX path(GSM850/900)

Checking Flow



Image



3.5 GSM RF PART

3.5.9 Checking RF Signal TX path(GSM850/900)

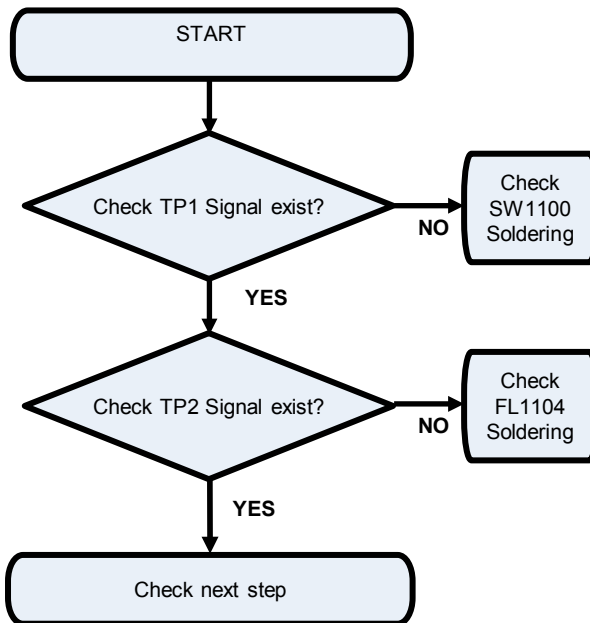
Circuit Diagram



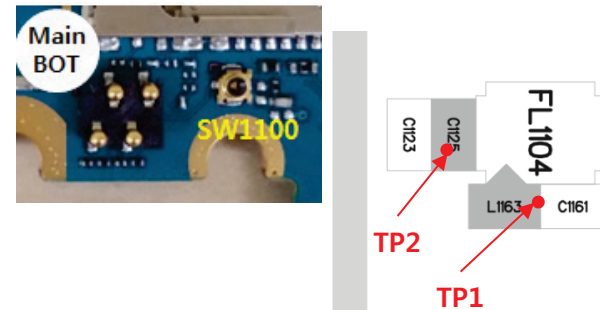
3.5 GSM RF PART

3.5.10 Checking RF Signal TX path(SW, GSM1800/1900)

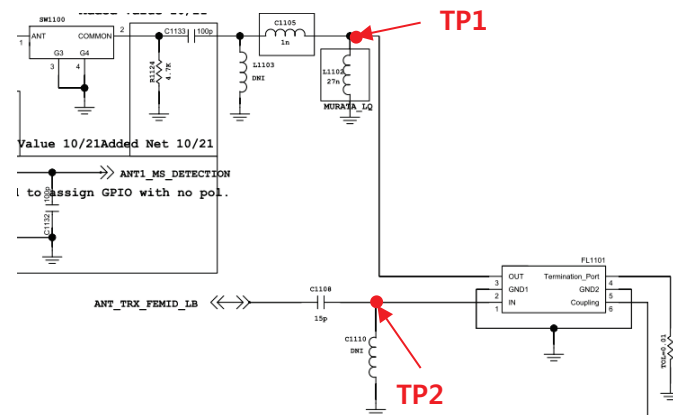
Checking Flow



Image



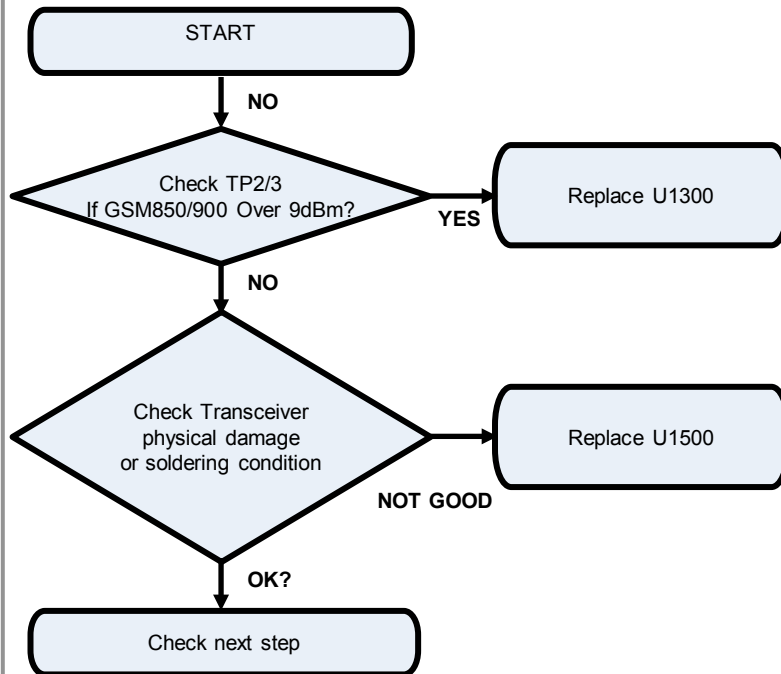
Circuit Diagram



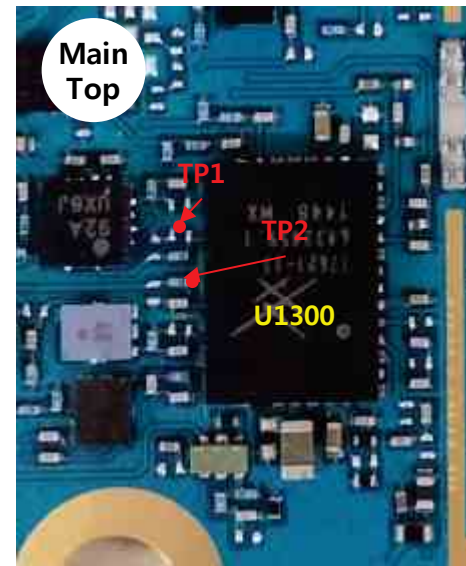
3.5 GSM RF PART

3.5.11 Checking RF Signal TX path(GSM1800/1900)

Checking Flow



Image



3.5 GSM RF PART

3.5.11 Checking RF Signal TX path(GSM1800/1900)

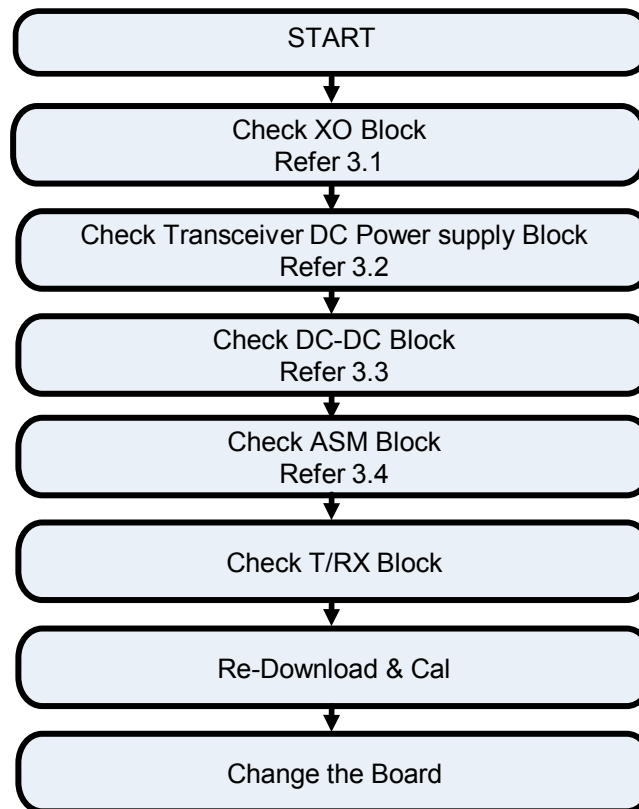
Circuit Diagram



3.6 WCDMA RF PART

WCDMA RF Part support WCDMA B1/5/8 with ASM, PAM, Transceiver component

Checking Flow

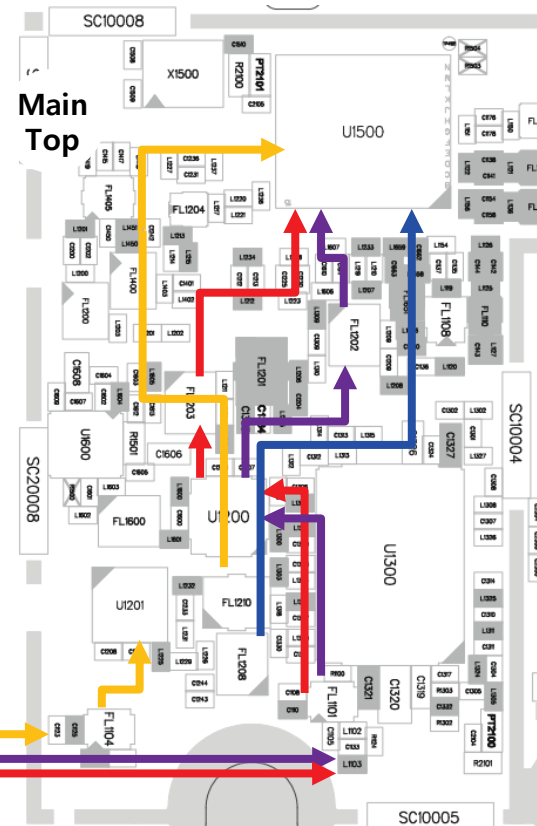
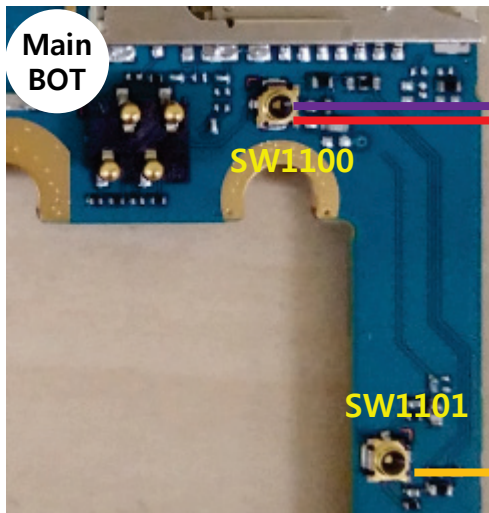


3.6 WCDMA RF PART

3.6.1 WCDMA RF Part RX RF PATH

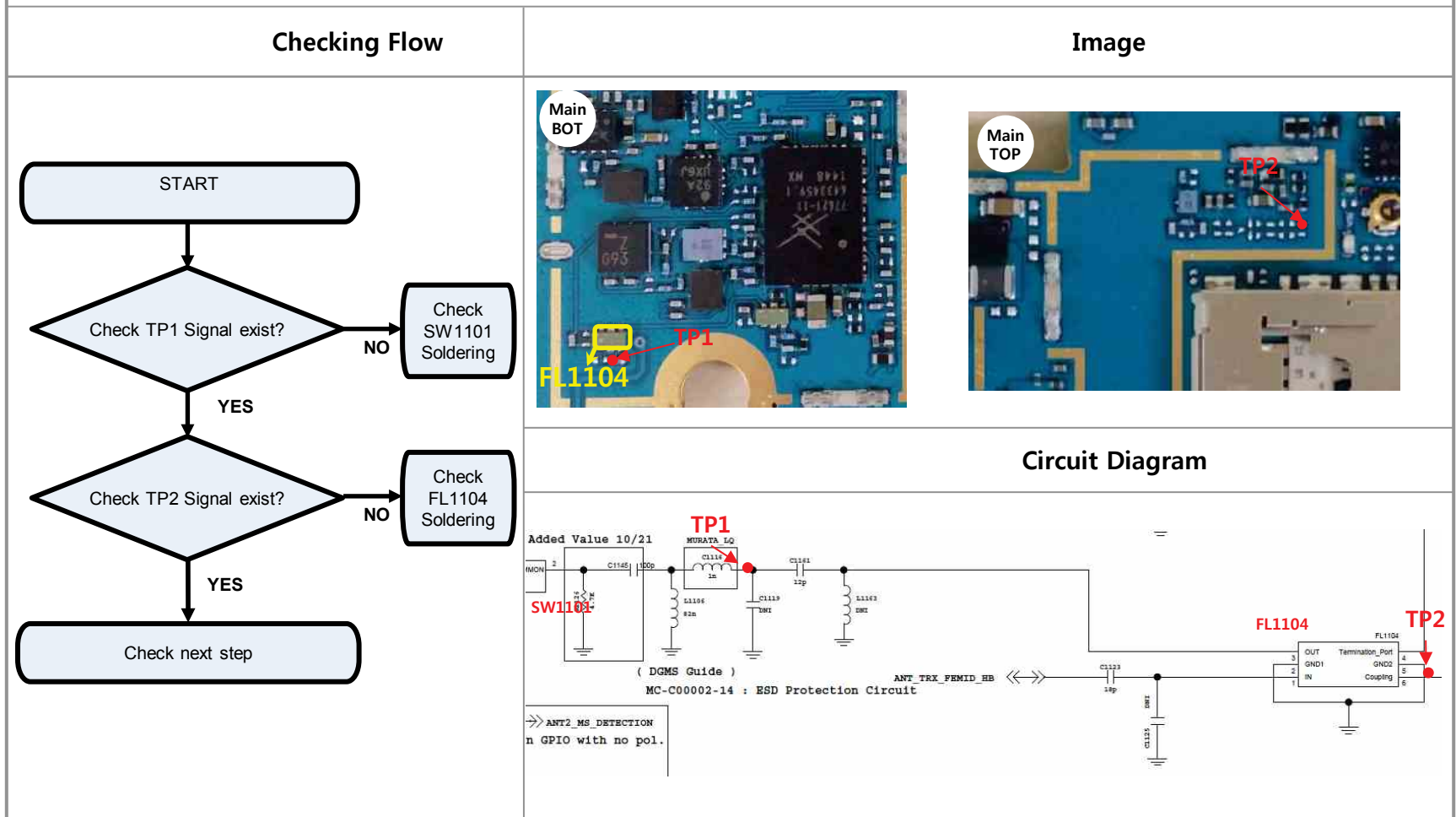
Image

1. WCDMA B1 RX PATH
2. WCDMA B5 RX PATH
3. WCDMA B8 RX PATH



3.6 WCDMA RF PART

3.6.2 Checking RF Signal RX path(SW, WCDMA B1)



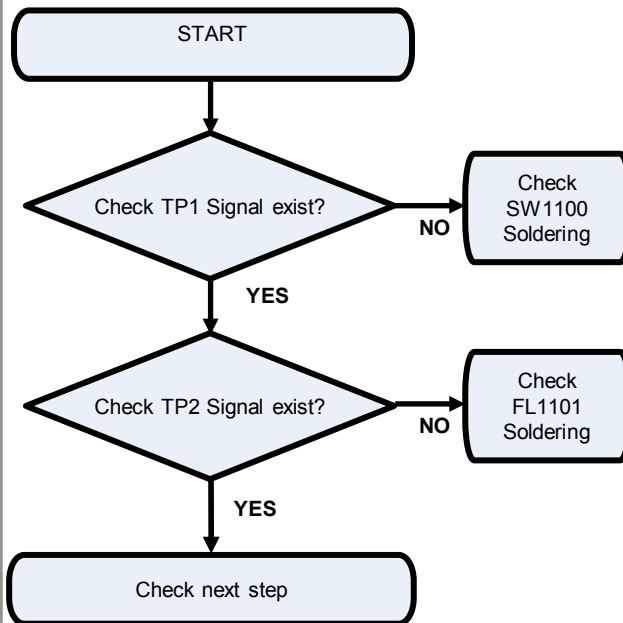
3.6.3 Checking RF Signal RX path(WCDMA B1)



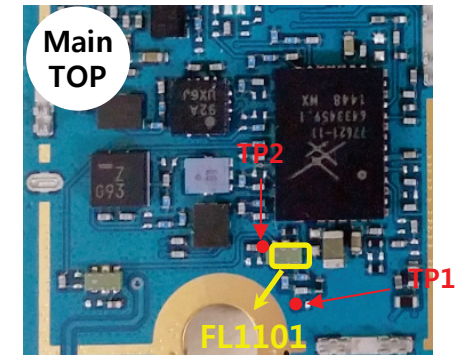
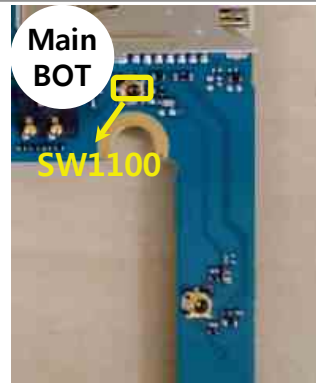
3.6 WCDMA RF PART

3.6.4 Checking RF Signal RX path(SW, WCDMA B5/8)

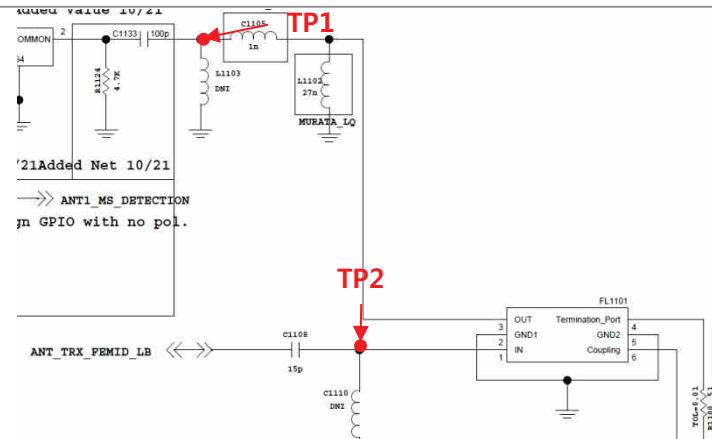
Checking Flow



Image



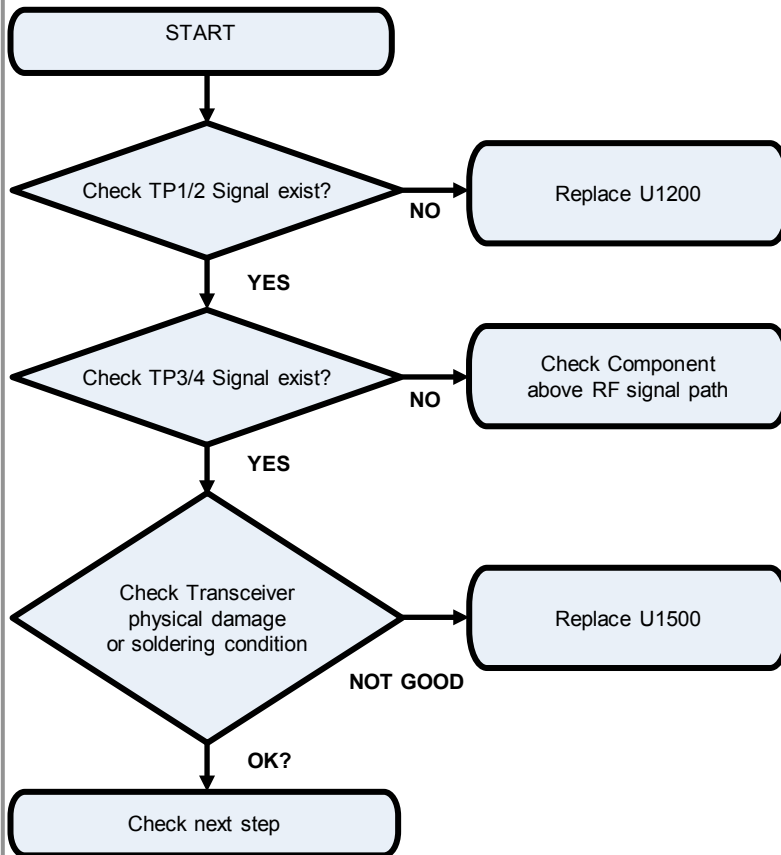
Circuit Diagram



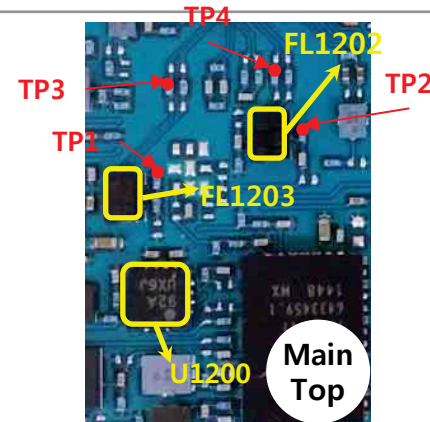
3.6 WCDMA RF PART

3.6.5 Checking RF Signal RX path(WCDMA B5/8)

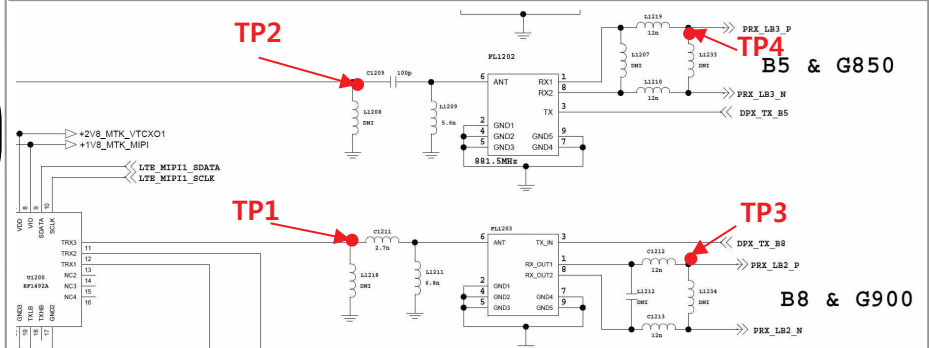
Checking Flow



Image



Circuit Diagram



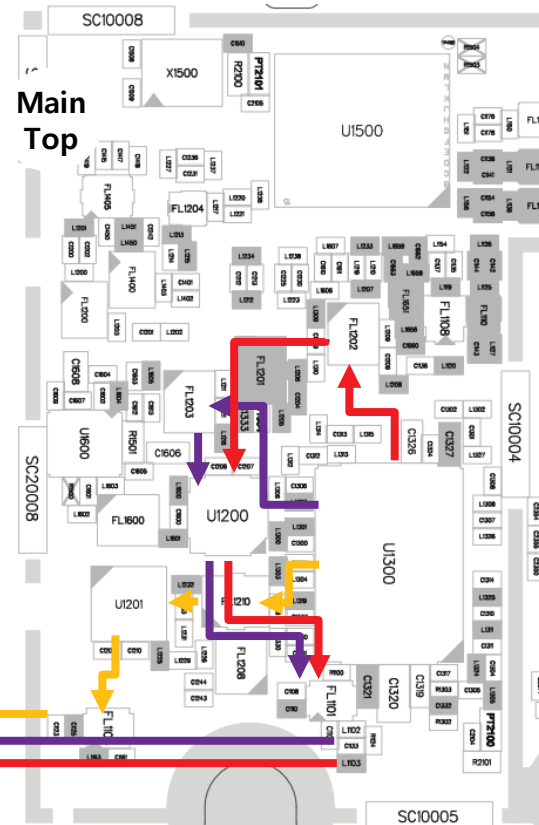
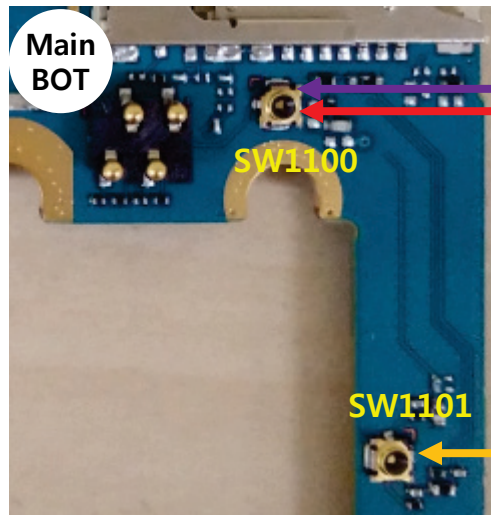
3. TROUBLE SHOOTING

3.6 WCDMA RF PART

3.6.6 WCDMA RF Part TX RF PATH

Image

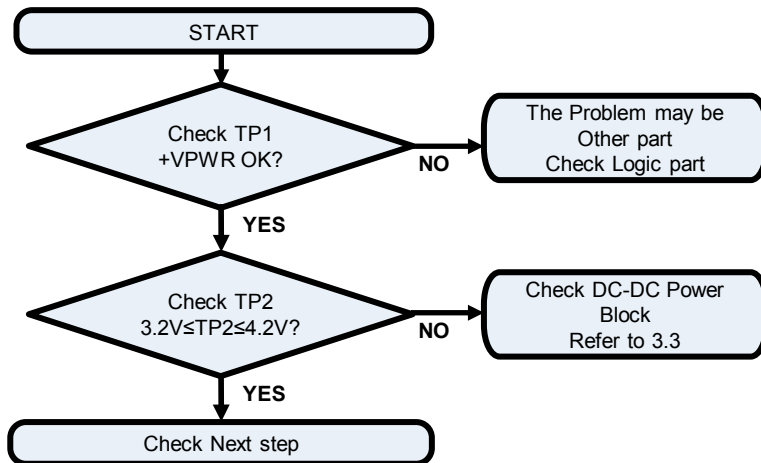
1. WCDMA B1 TX PATH
 2. WCDMA B5 TX PATH
 3. WCDMA B8 TX PATH
-
1. GSM850/900 TX PATH
 2. GSM1800/1900 TX PATH



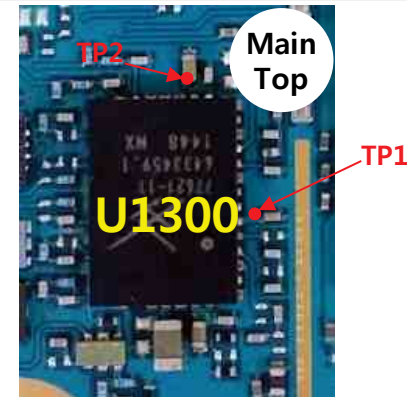
3.6 WCDMA RF PART

3.6.7 Checking WCDMA PAM DC Power Circuit

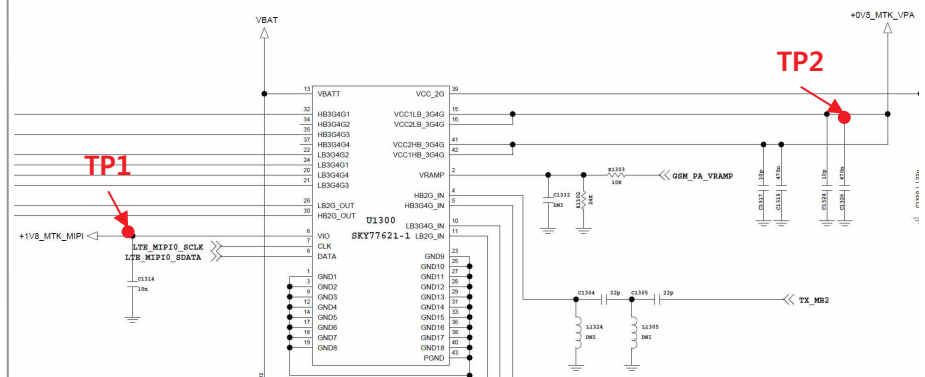
Checking Flow



Image

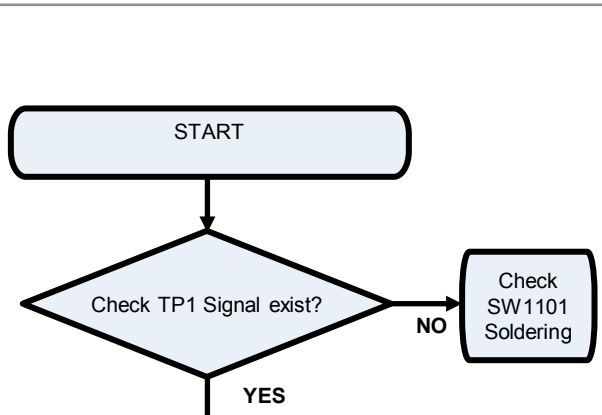
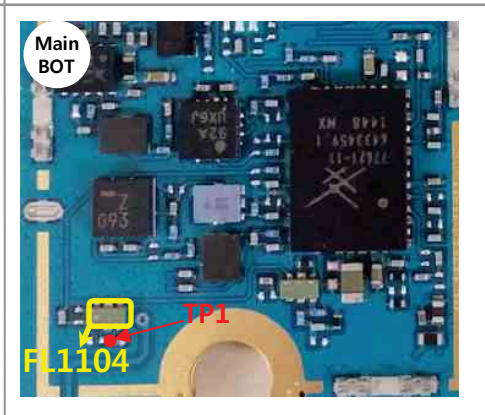
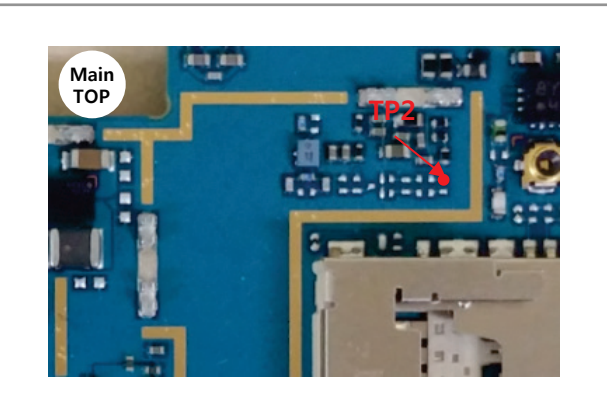


Circuit Diagram



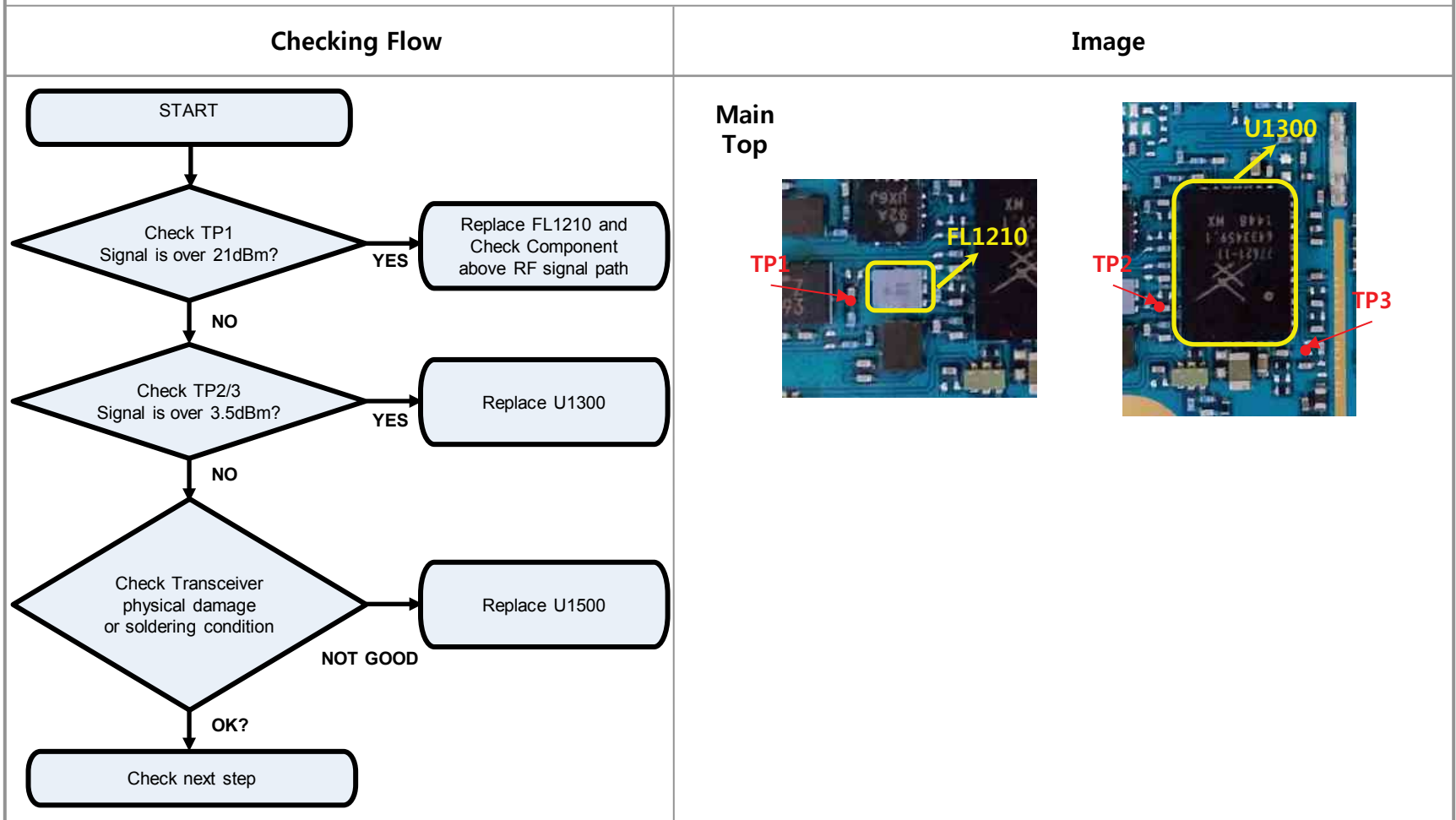
3.6 WCDMA RF PART

3.6.8 Checking RF Signal TX path(SW, WCDMA B1)

Checking Flow	Image	
 <pre> graph TD START([START]) --> TP1{Check TP1 Signal exist?} TP1 -- NO --> SW1101[Check SW1101 Soldering] TP1 -- YES --> TP2{Check TP2 Signal exist?} TP2 -- NO --> FL1104[Check FL1104 Soldering] TP2 -- YES --> NextStep([Check next step]) </pre>		
	Circuit Diagram	

3.6 WCDMA RF PART

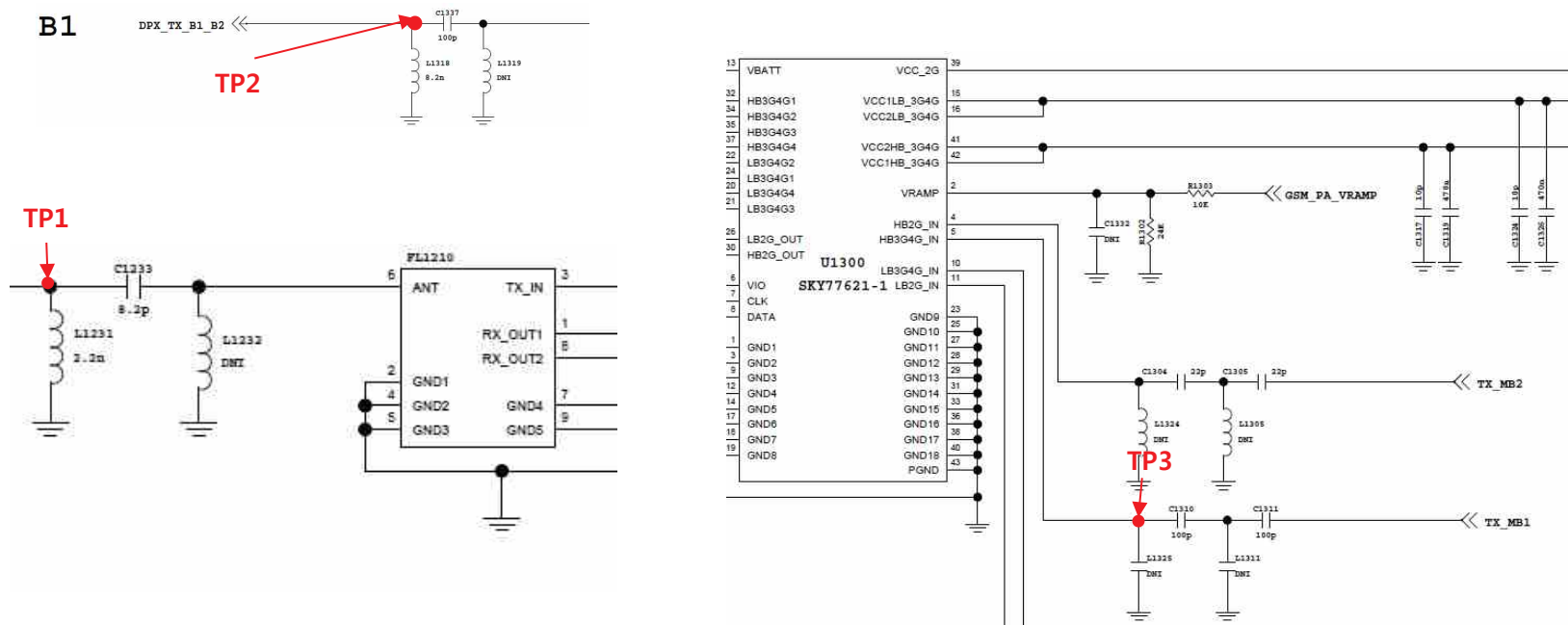
3.6.9 Checking RF Signal TX path(WCDMA B1)



3.6 WCDMA RF PART

3.6.9 Checking RF Signal TX path(WCDMA B1)

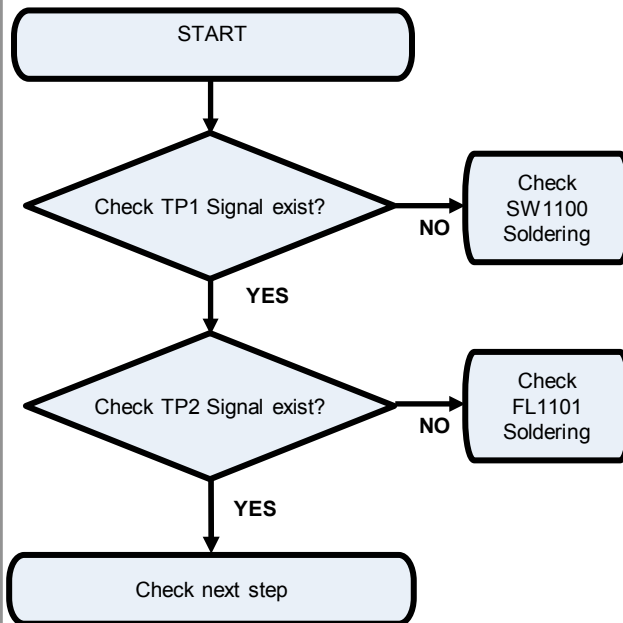
Circuit Diagram



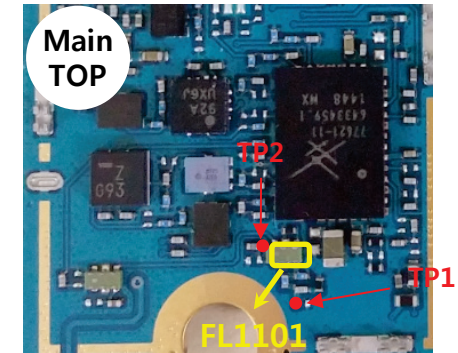
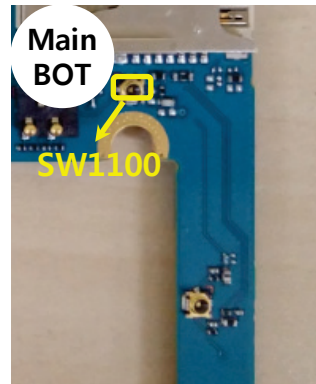
3.6 WCDMA RF PART

3.6.10 Checking RF Signal TX path(SW, WCDMA B5/8)

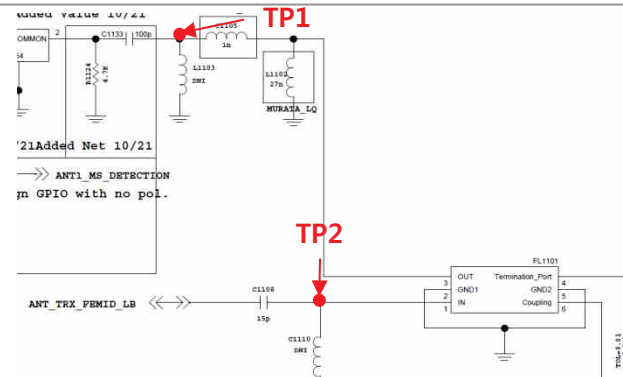
Checking Flow



Image



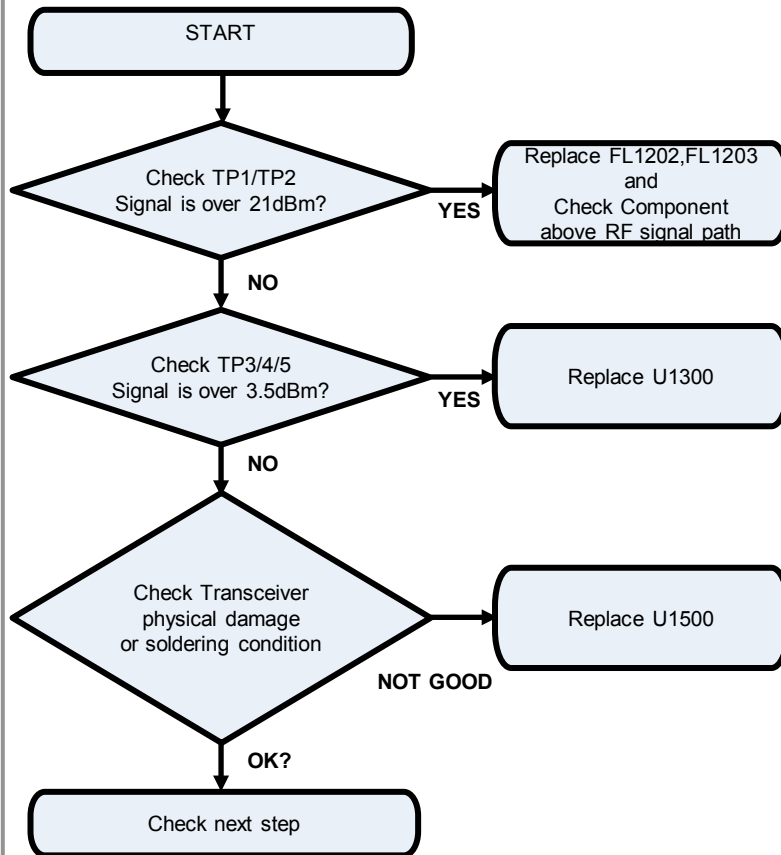
Circuit Diagram



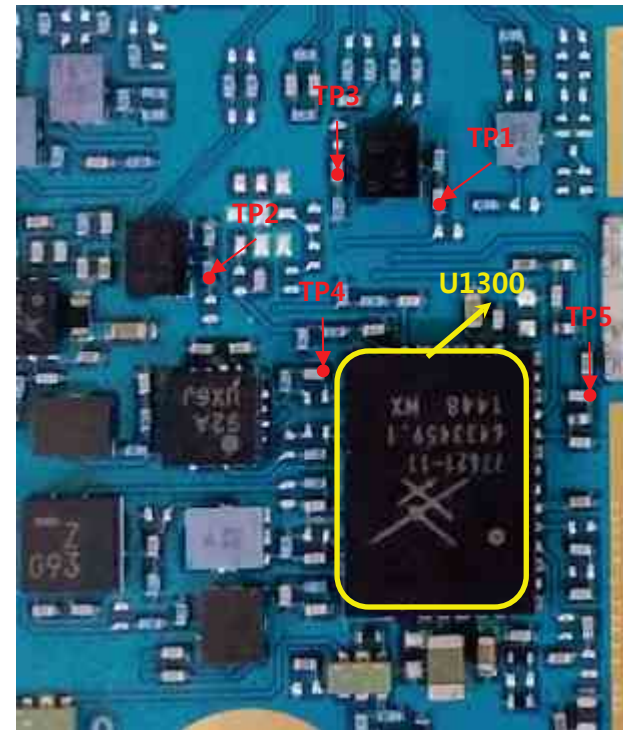
3.6 WCDMA RF PART

3.6.11 Checking RF Signal TX path(WCDMA B5/8)

Checking Flow



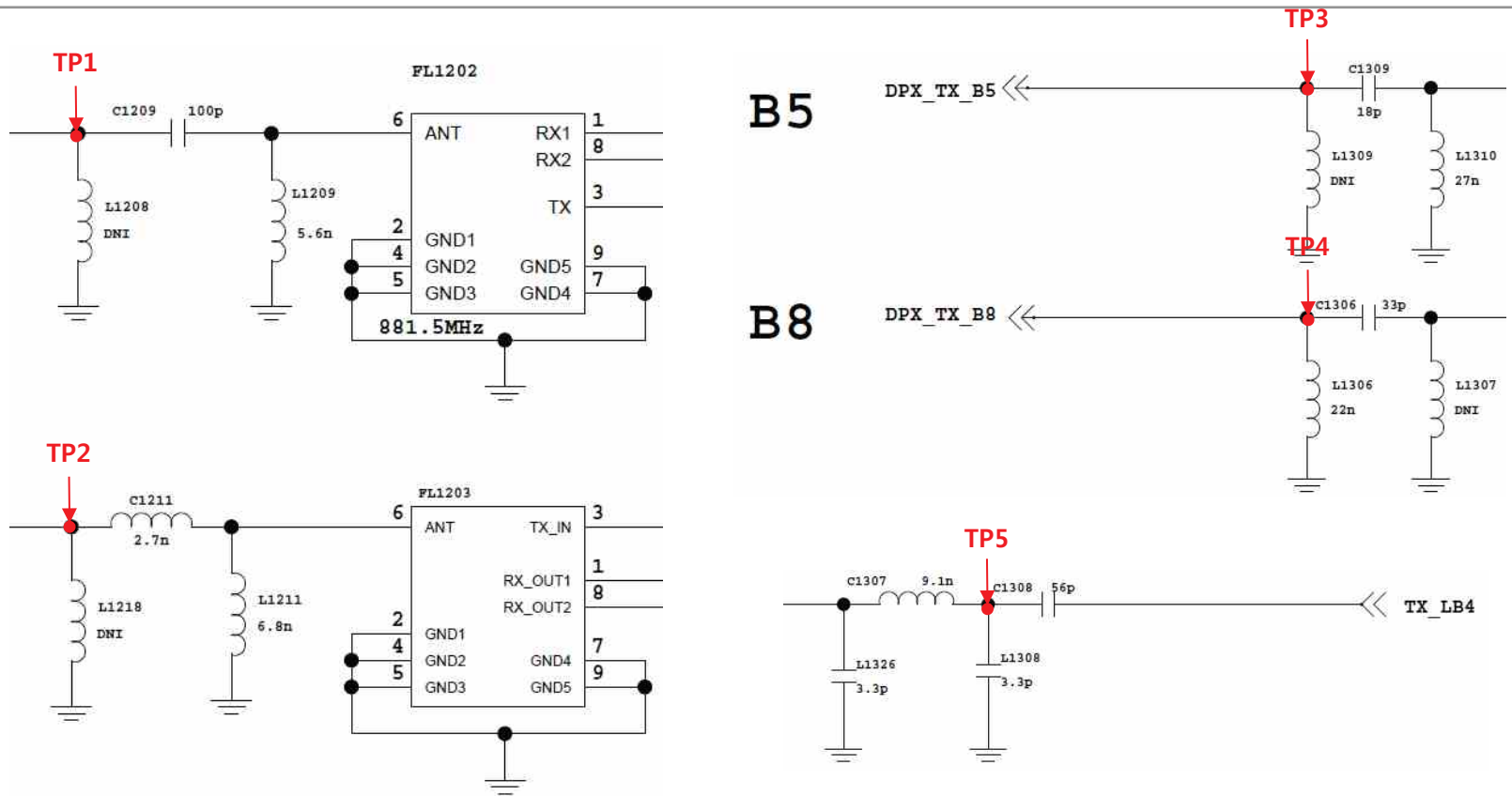
Image



3.6 WCDMA RF PART

3.6.11 Checking RF Signal TX path(WCDMA B5/8)

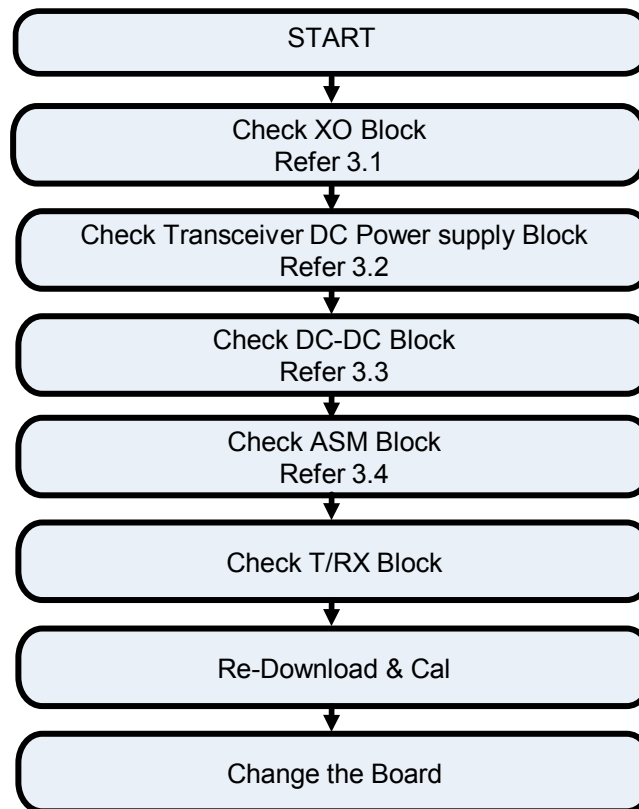
Circuit Diagram



3.7 LTE RF PART

LTE RF Part support LTE B3/7/28 with ASM, PAM, Transceiver component

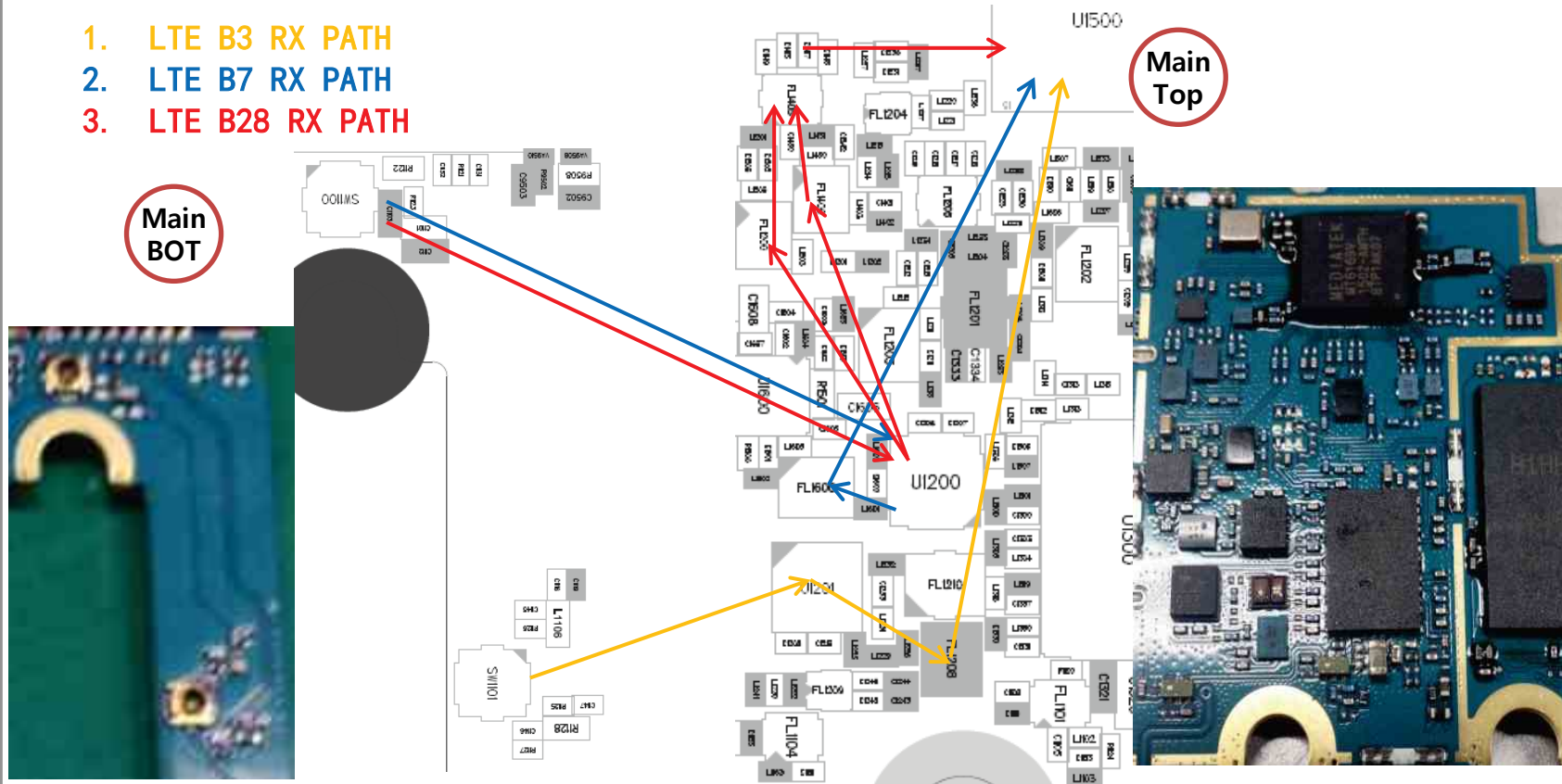
Checking Flow



3.7 LTE RF PART

3.7.1 LTE RF Part RX RF PATH

Image

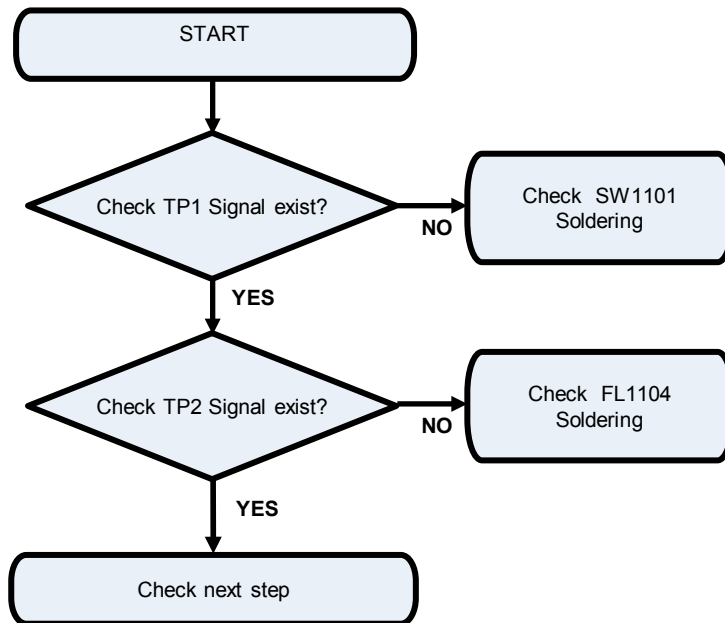


3. TROUBLE SHOOTING

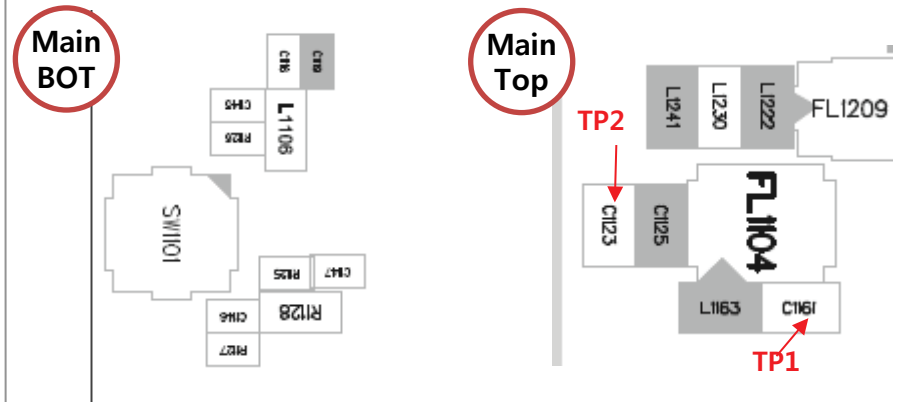
3.7 LTE RF PART

3.7.2 Checking RF Signal RX path(SW, LTE B3)

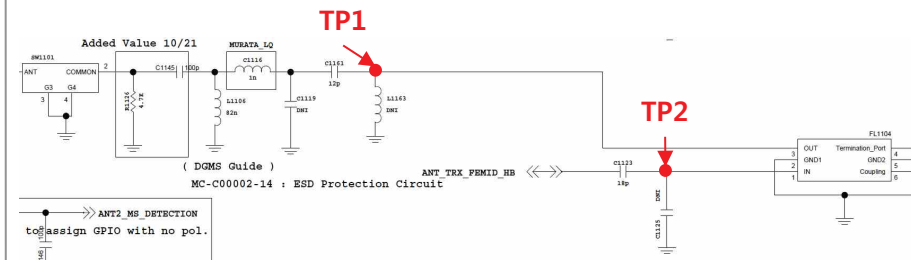
Checking Flow



Image



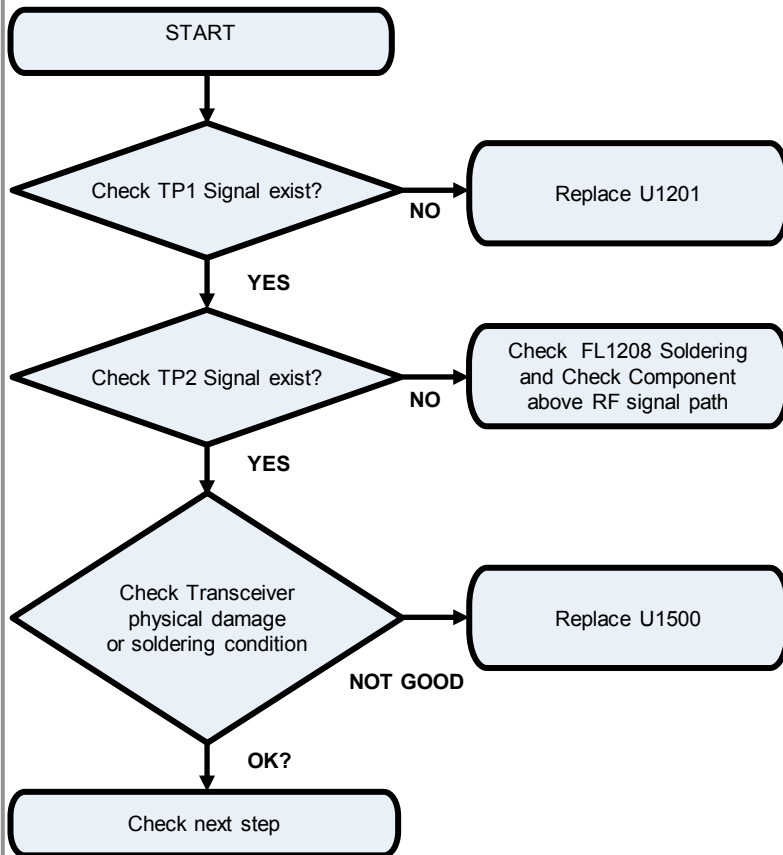
Circuit Diagram



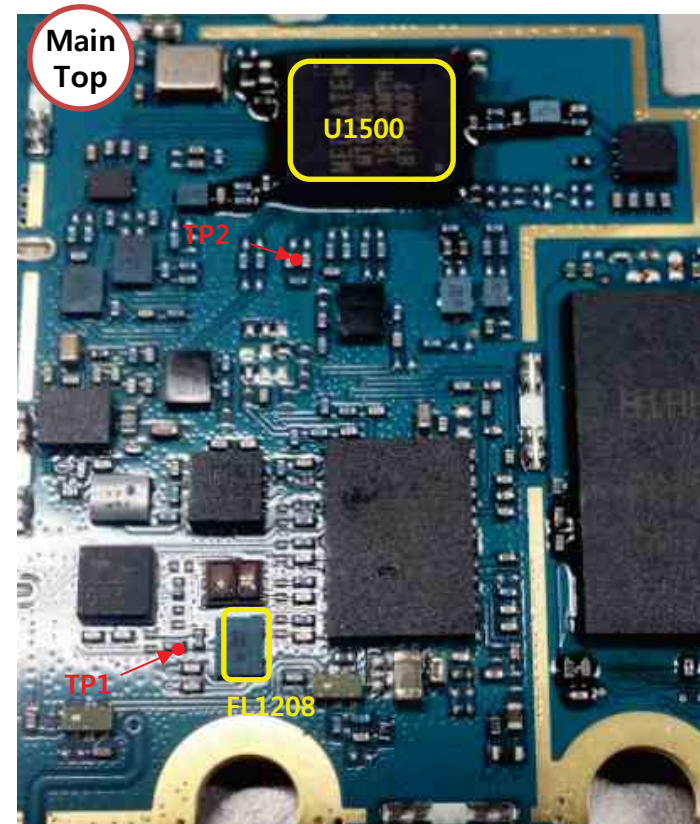
3.7 LTE RF PART

3.7.3 Checking RF Signal RX path(LTE B3)

Checking Flow



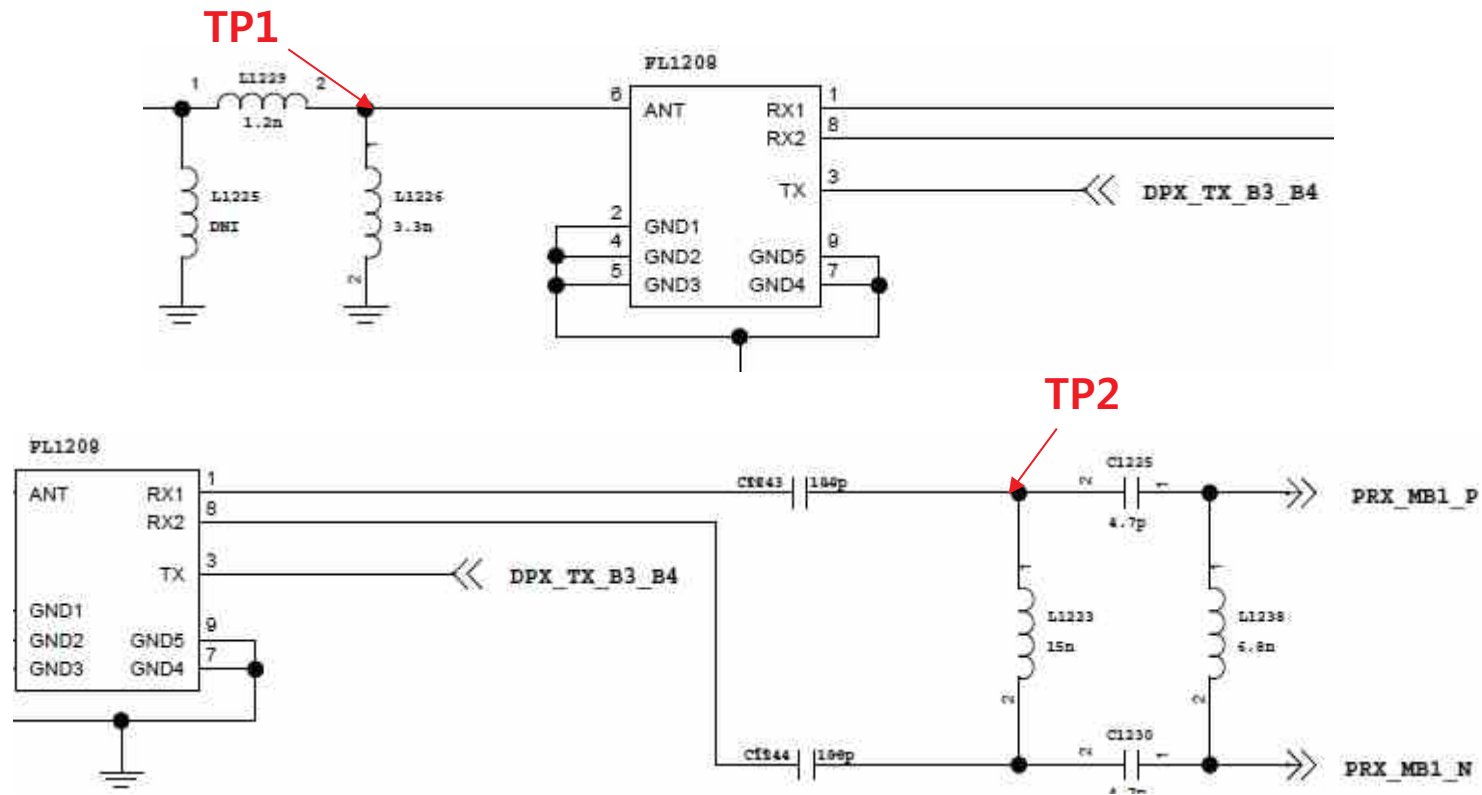
Image



3.7 LTE RF PART

3.7.3 Checking RF Signal RX path(LTE B3)

Circuit Diagram

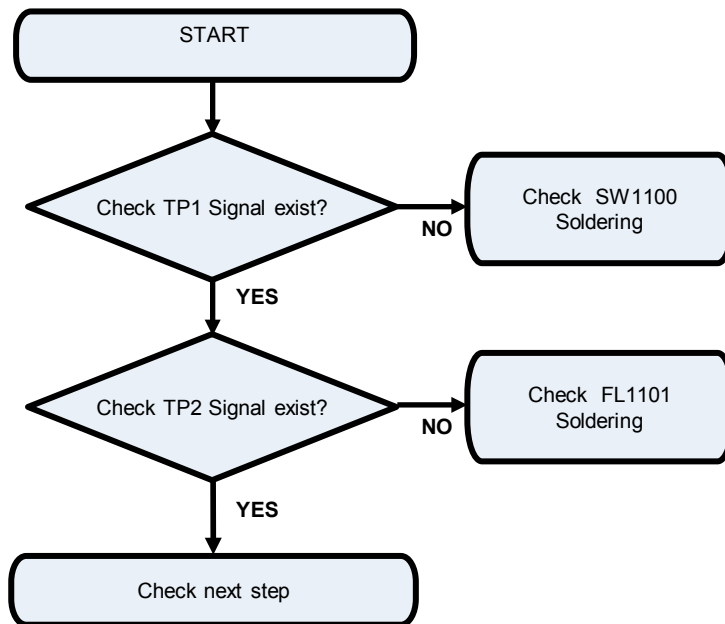


Du
B3

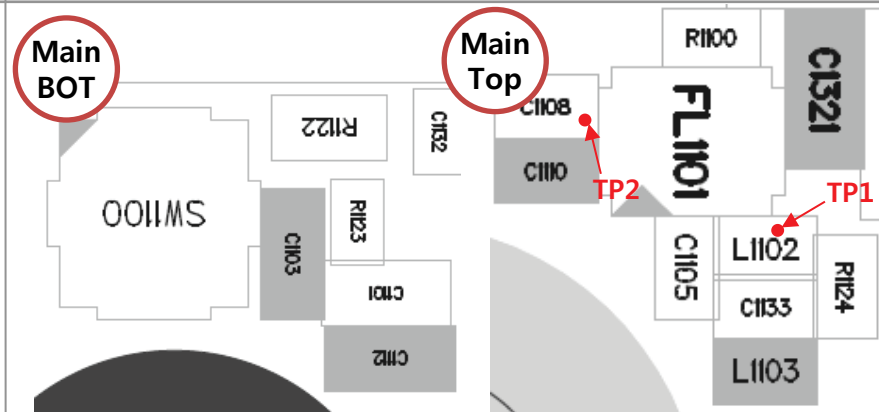
3.7 LTE RF PART

3.7.4 Checking RF Signal RX path(SW, LTEB7/28)

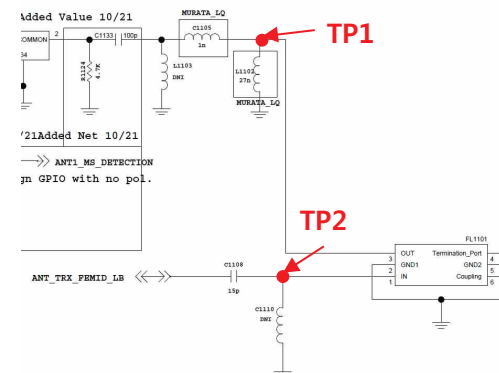
Checking Flow



Image



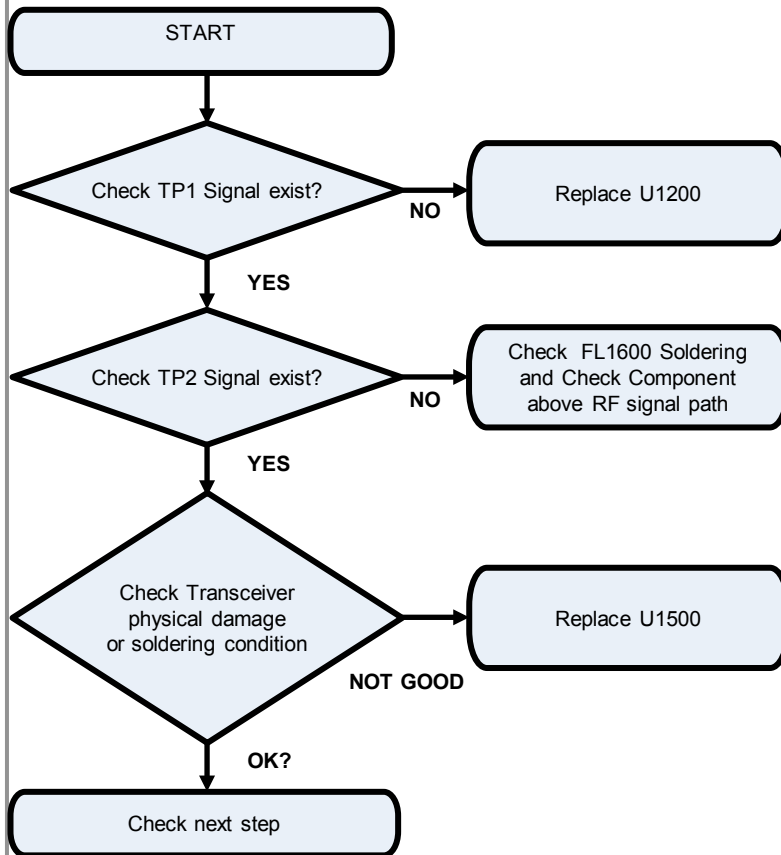
Circuit Diagram



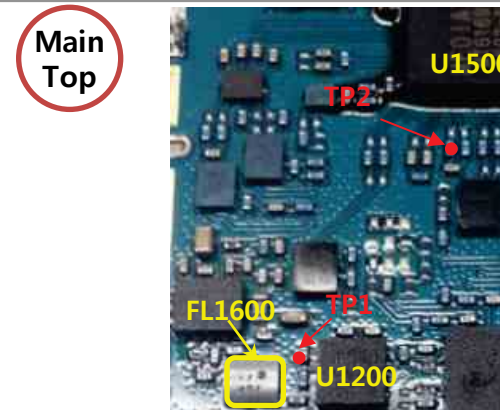
3.7 LTE RF PART

3.7.5 Checking RF Signal RX path(LTE B7)

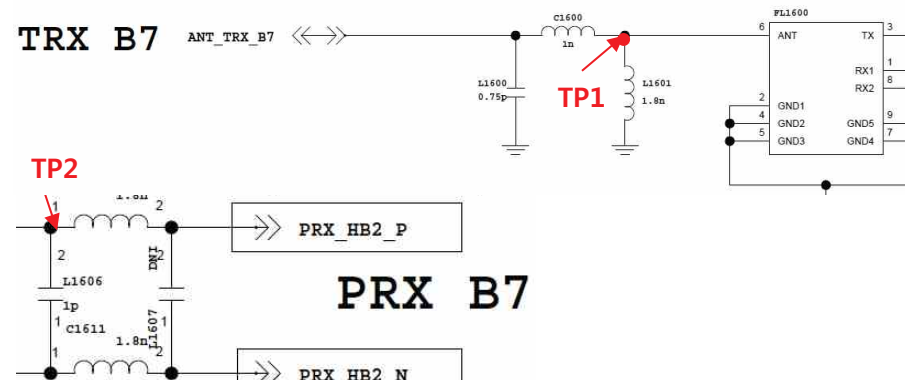
Checking Flow



Image



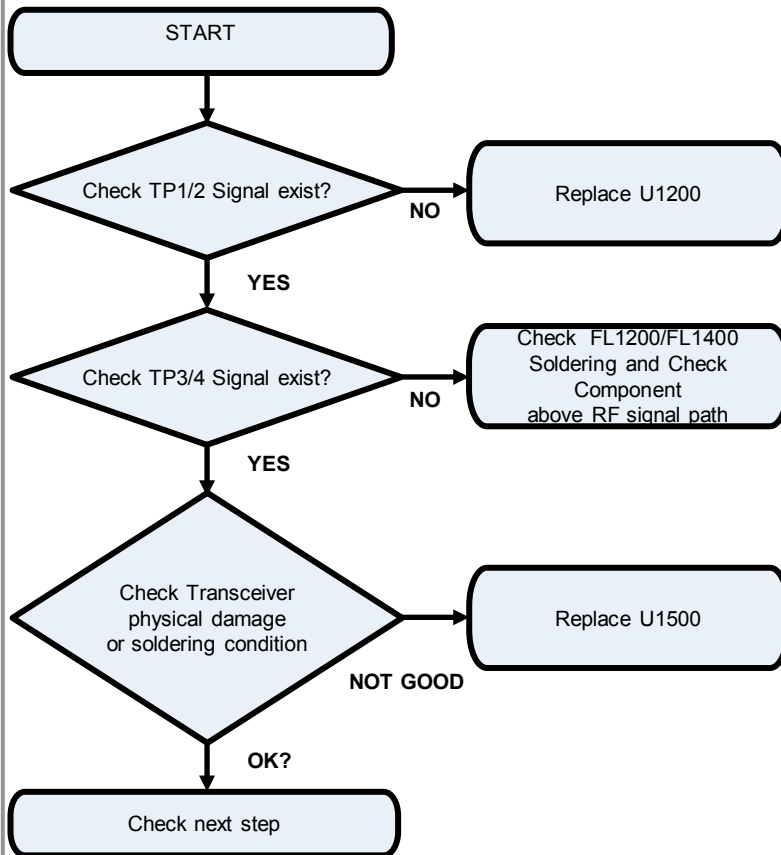
Circuit Diagram



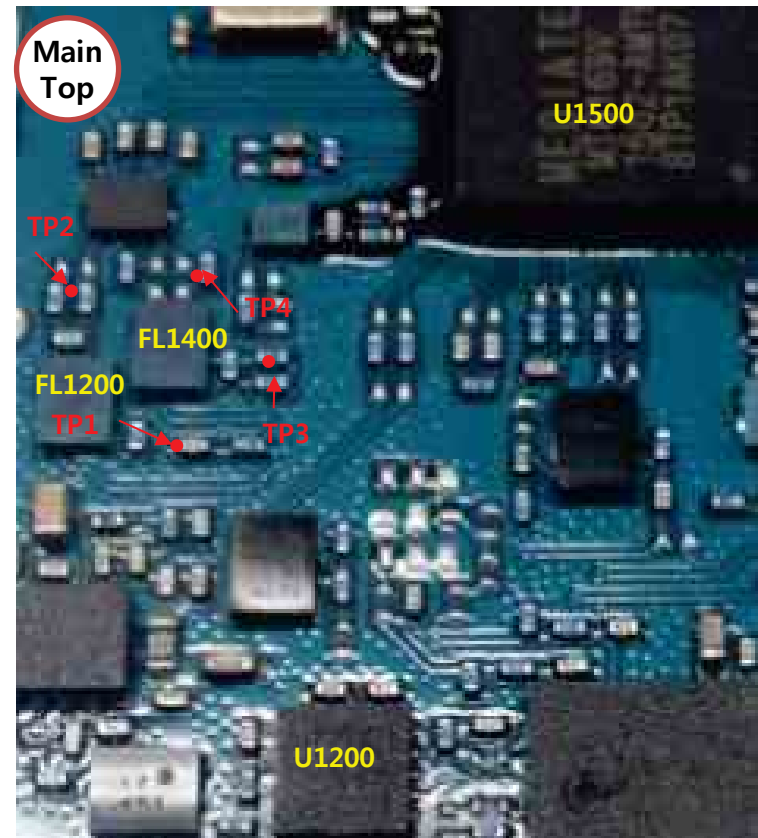
3.7 LTE RF PART

3.7.6 Checking RF Signal RX path(LTE B28)

Checking Flow



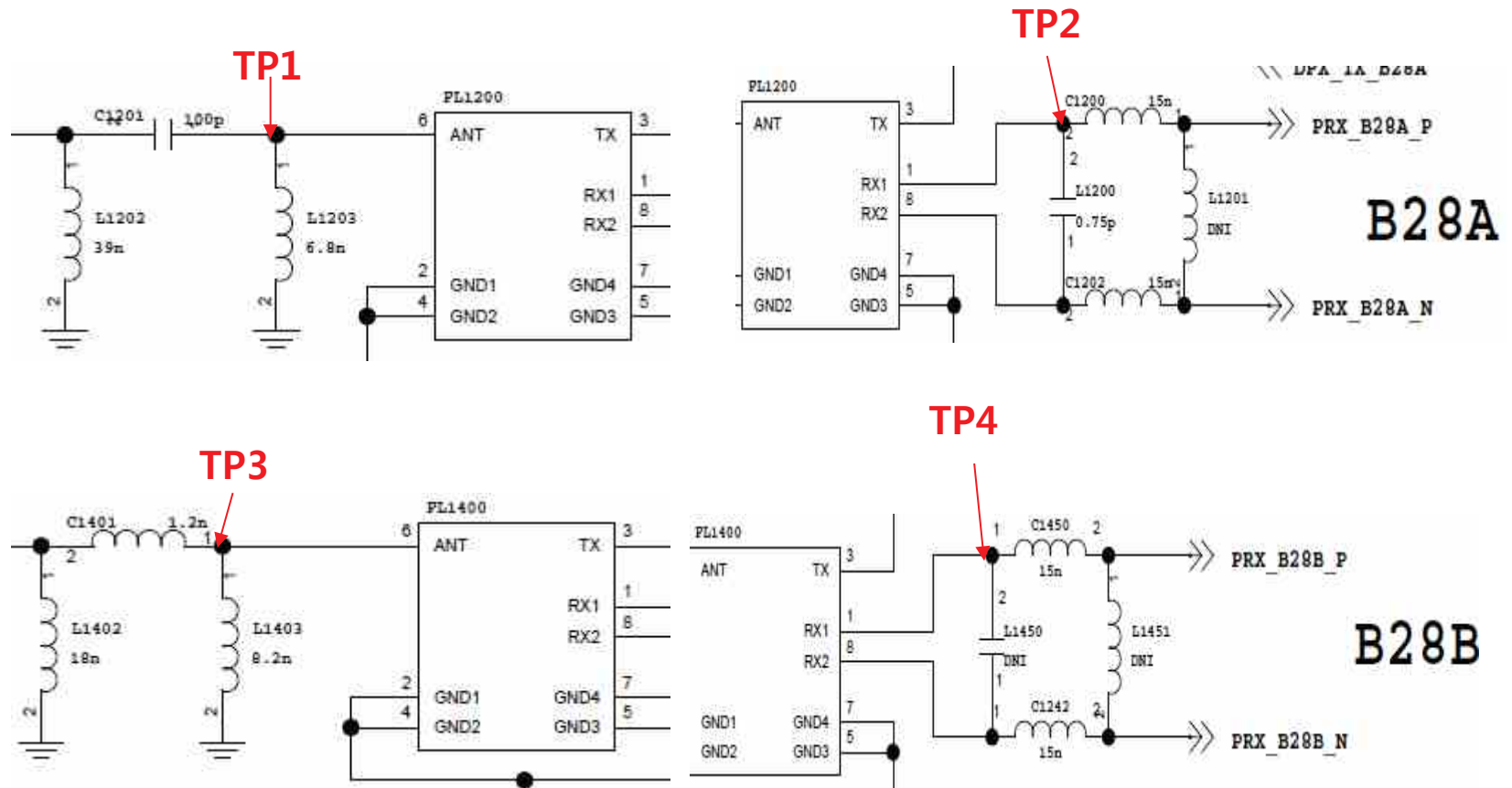
Image



3.7 LTE RF PART

3.7.6 Checking RF Signal RX path(LTE B8/20)

Circuit Diagram

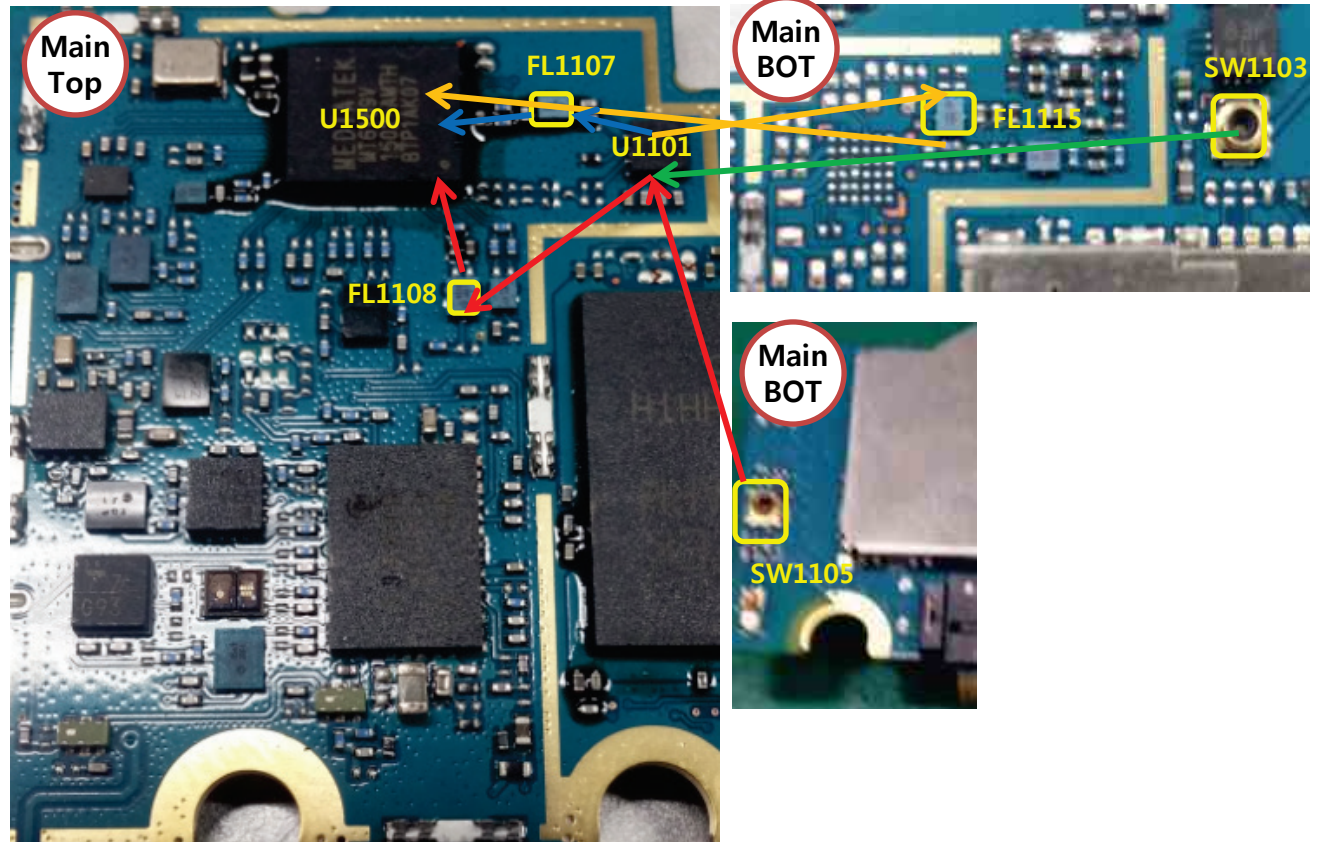


3.7 LTE RF PART

3.7.7 LTE RF Part DRX RF PATH

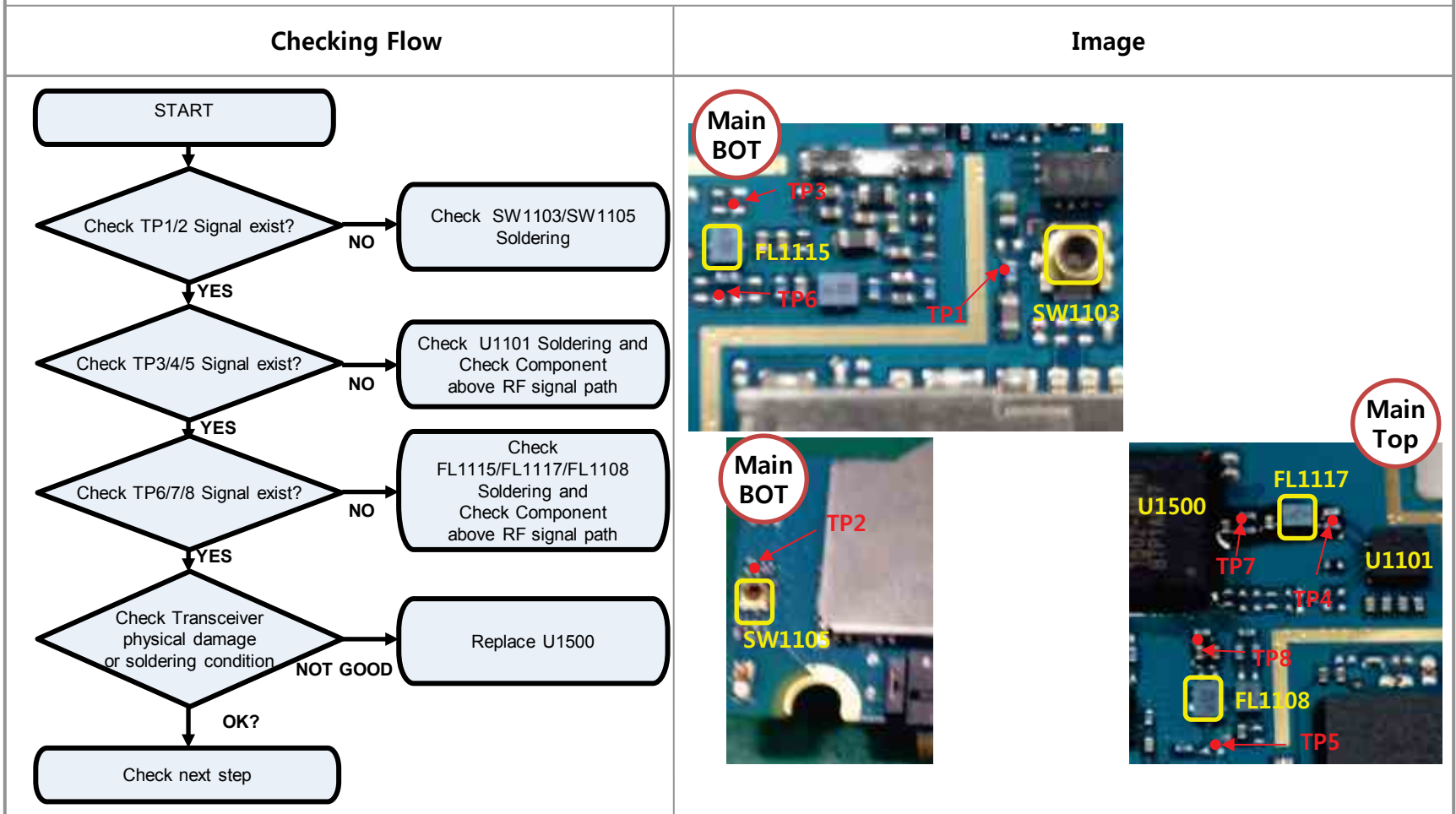
Image

1. LTE B3 DRX PATH
2. LTE B7 DRX PATH
3. LTE B28 DRX PATH
4. LTE B3/28 DRX COMMON PATH



3.7 LTE RF PART

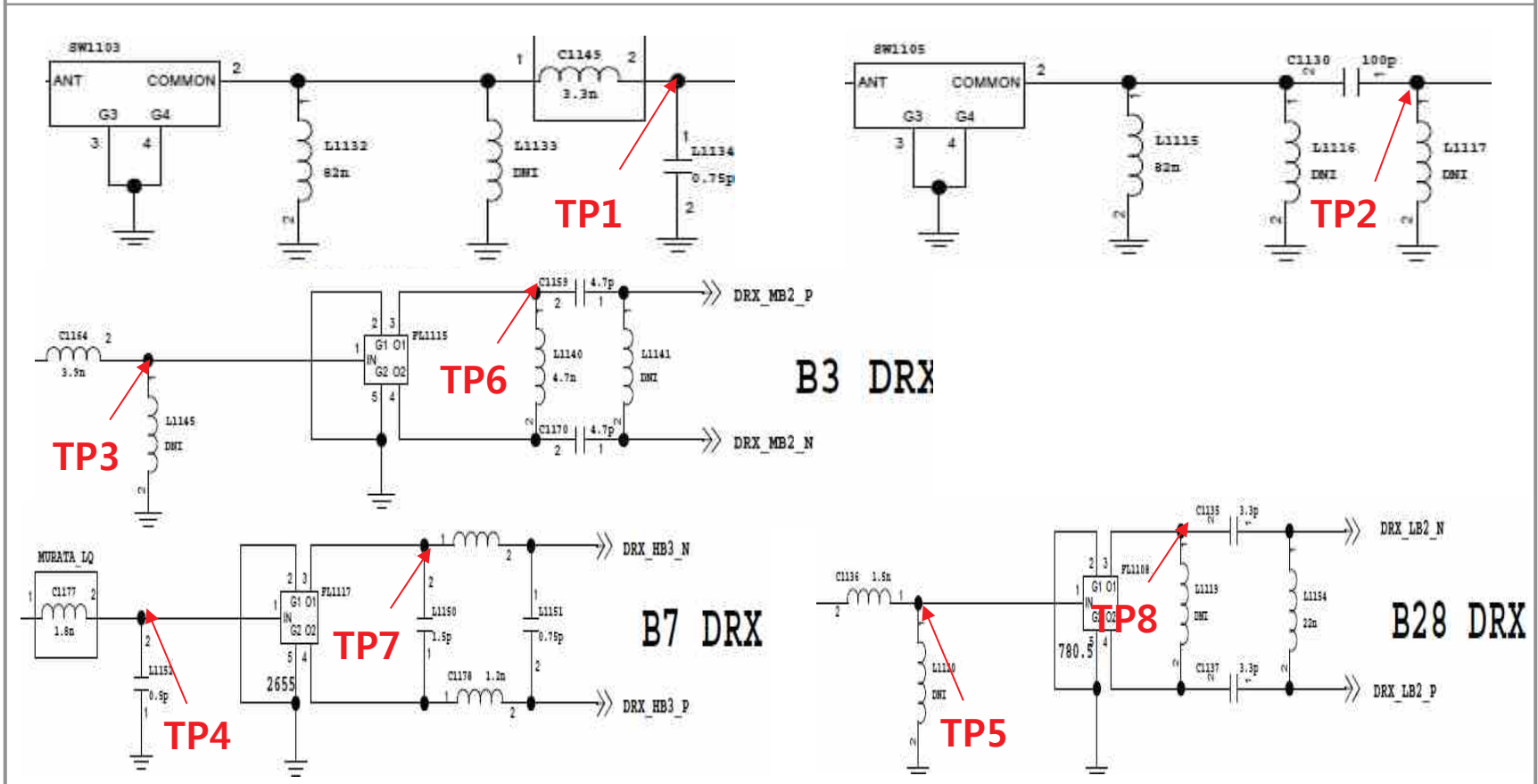
3.7.8 Checking RF Signal DRX path(LTEB3/B7/B28)



3.7 LTE RF PART

3.7.8 Checking RF Signal DRX path(LTEB3/B7/B28)

Circuit Diagram

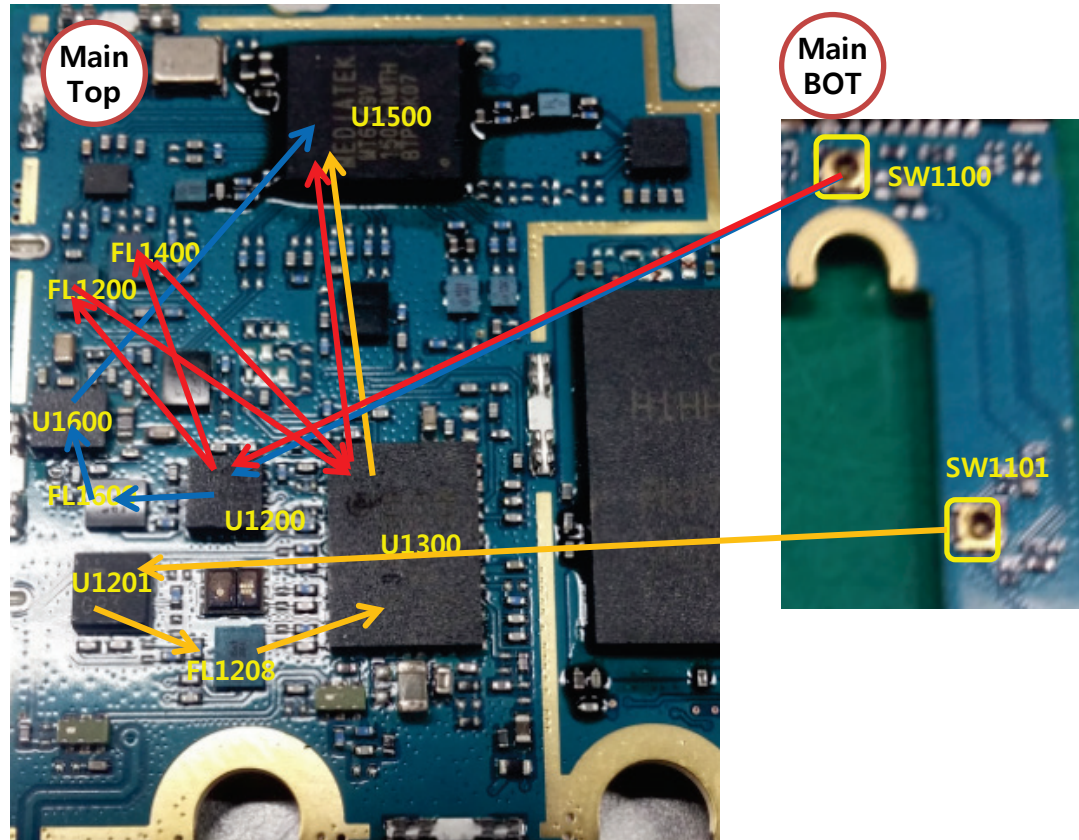


3.7 LTE RF PART

3.7.10 LTE RF Part TX RF PATH

Image

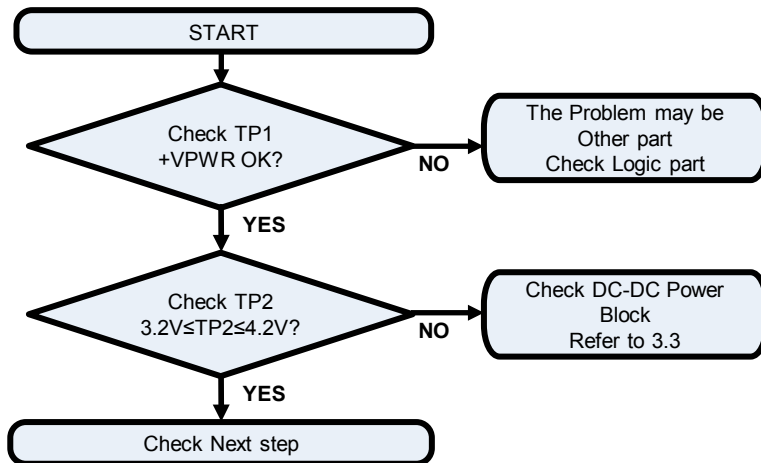
1. LTE B3 TX PATH
2. LTE B7 TX PATH
3. LTE B28 TX PATH



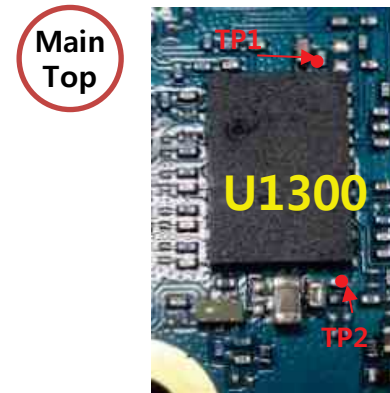
3.7 LTE RF PART

3.7.11 Checking LTE PAM DC Power Circuit(B3/28)

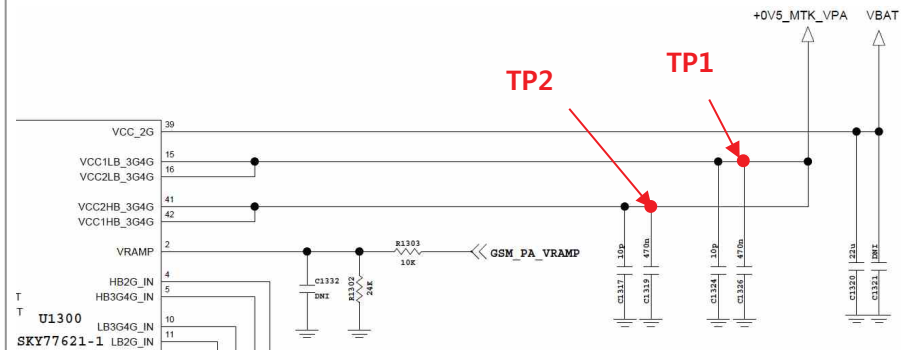
Checking Flow



Image



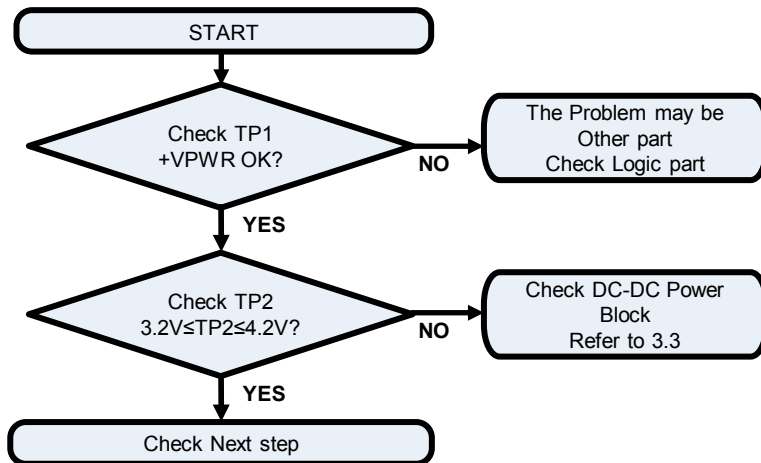
Circuit Diagram



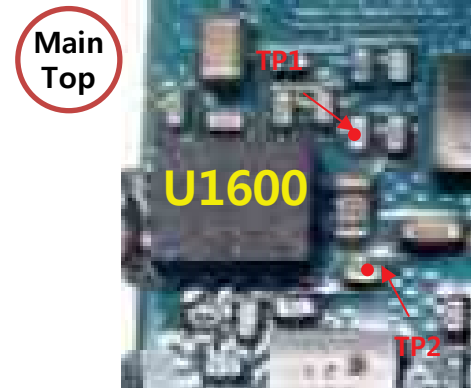
3.7 LTE RF PART

3.7.12 Checking LTE PAM DC Power Circuit(B7)

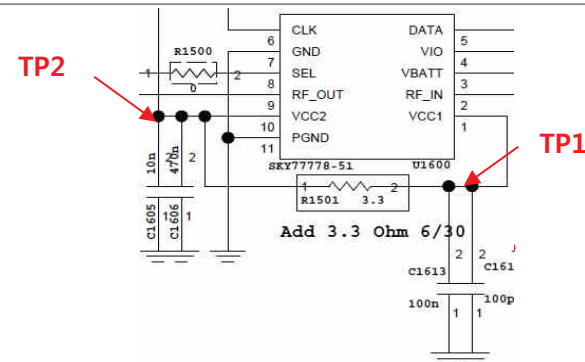
Checking Flow



Image



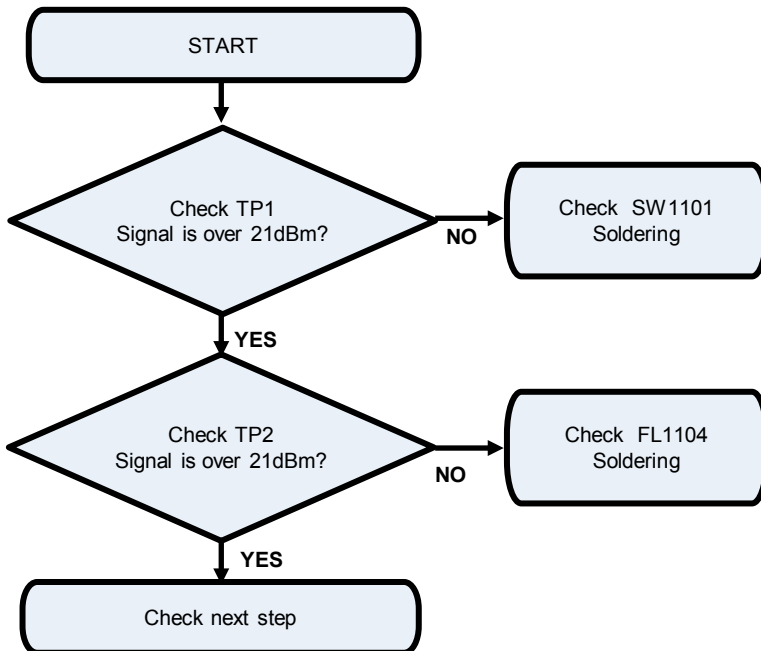
Circuit Diagram



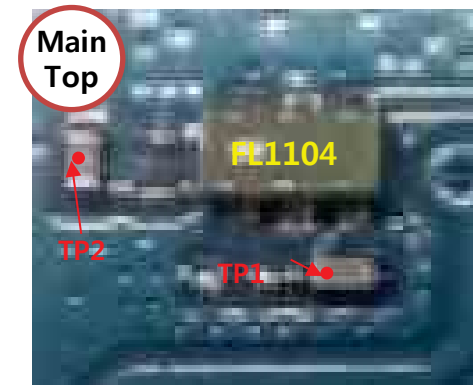
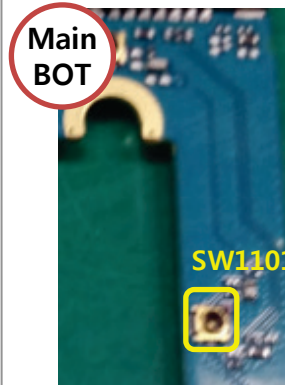
3.7 LTE RF PART

3.7.13 Checking RF Signal TX path(SW, LTE B3)

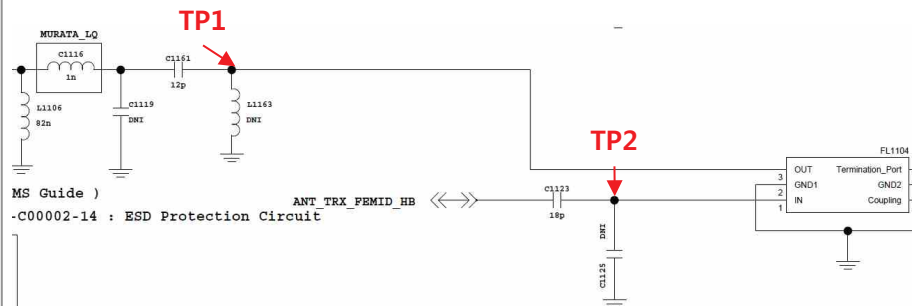
Checking Flow



Image



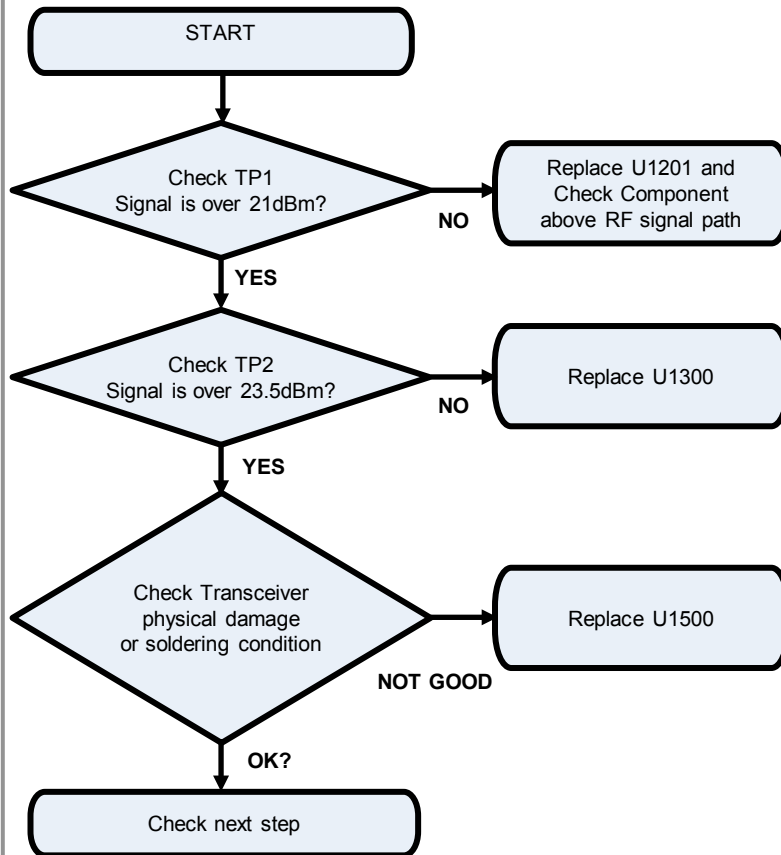
Circuit Diagram



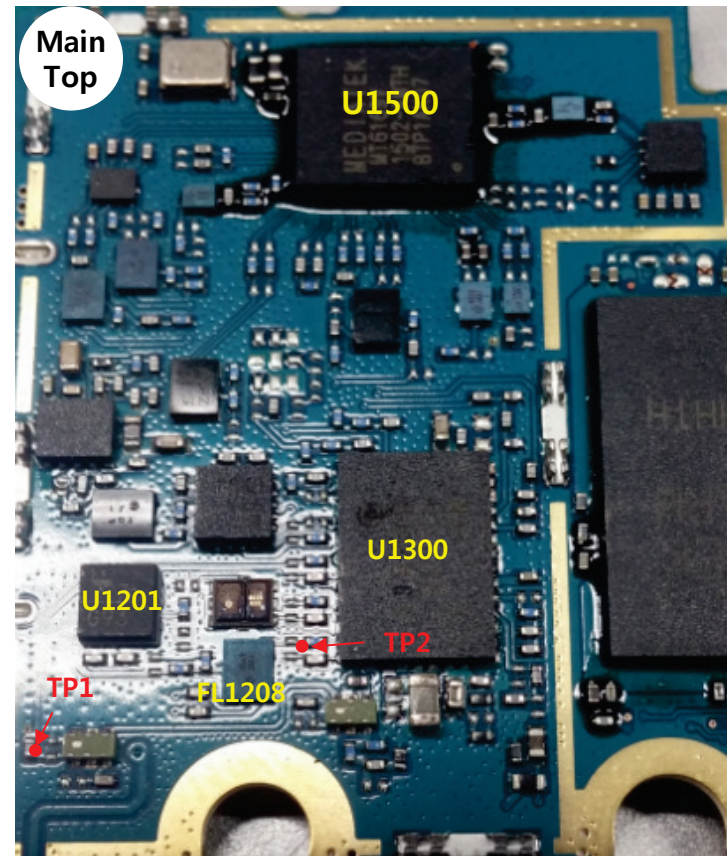
3.7 LTE RF PART

3.7.14 Checking RF Signal TX path(LTE B3)

Checking Flow



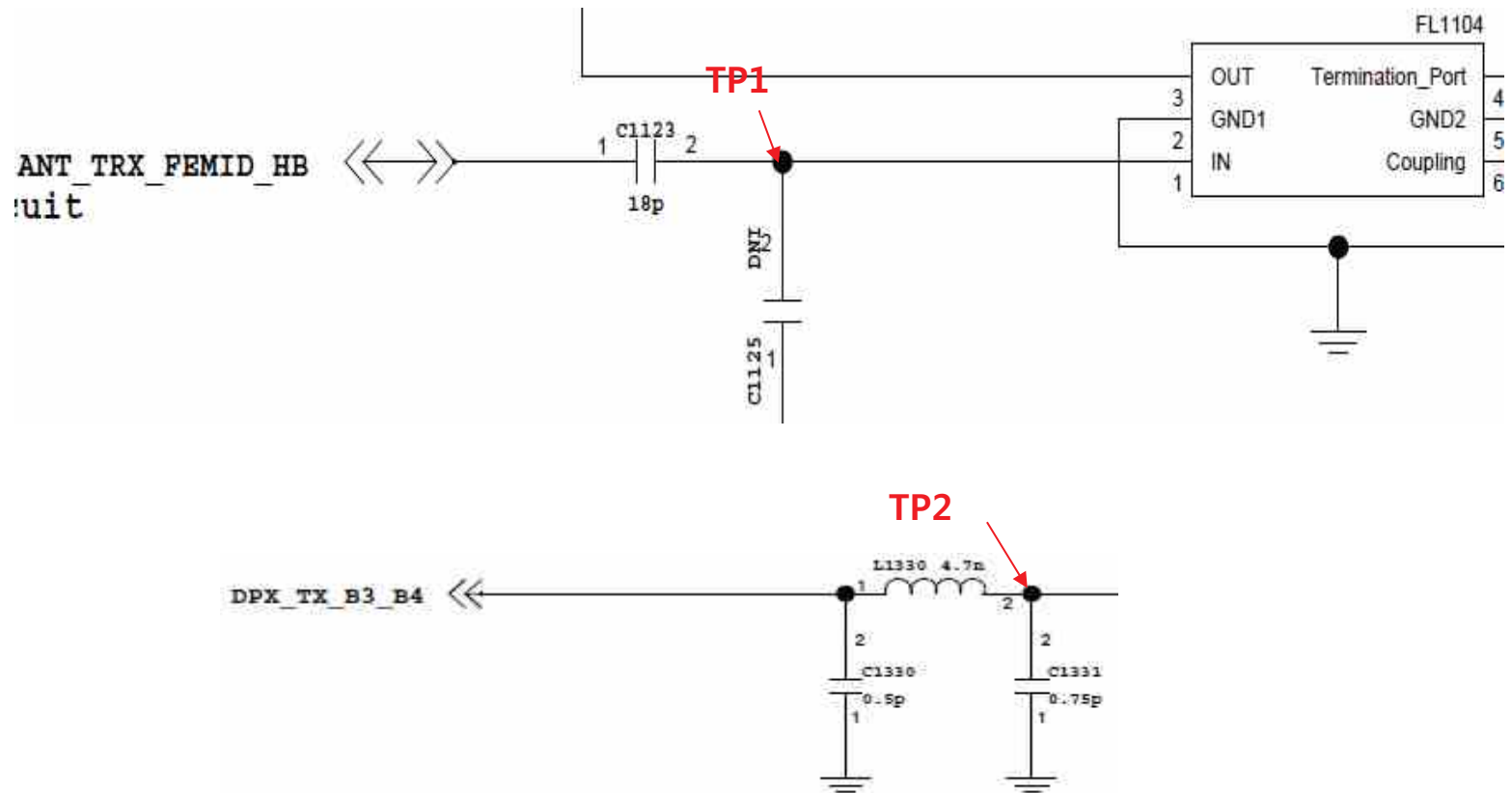
Image



3.7 LTE RF PART

3.7.14 Checking RF Signal TX path(LTE B3)

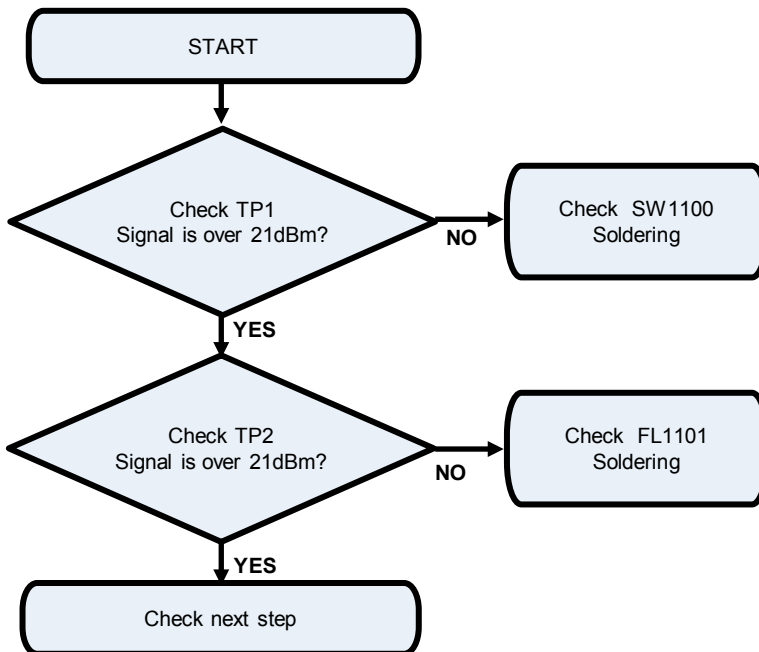
Circuit Diagram



3.7 LTE RF PART

3.7.15 Checking RF Signal TX path(SW, LTE B7/B28)

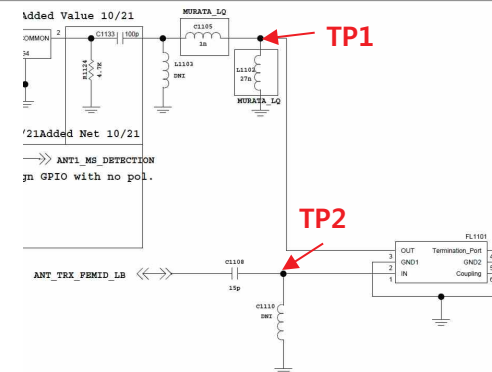
Checking Flow



Image



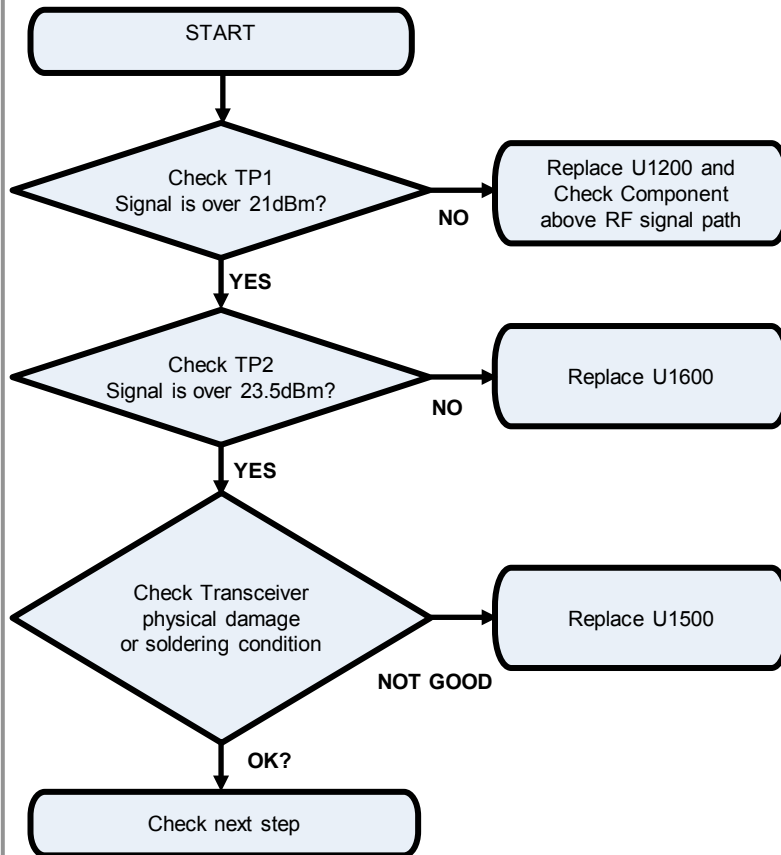
Circuit Diagram



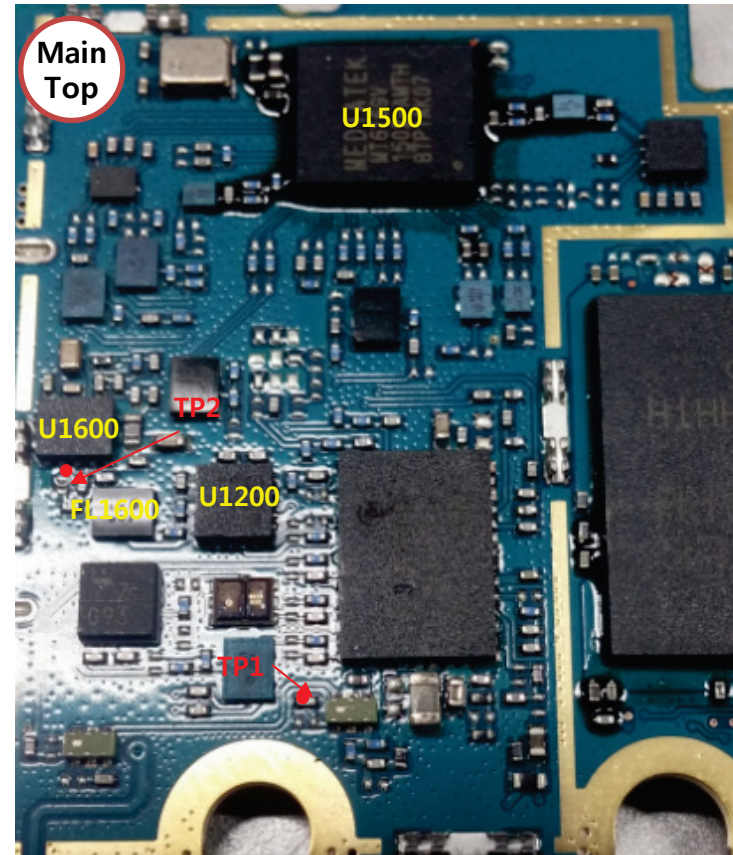
3.7 LTE RF PART

3.7.16 Checking RF Signal TX path(LTE B7)

Checking Flow



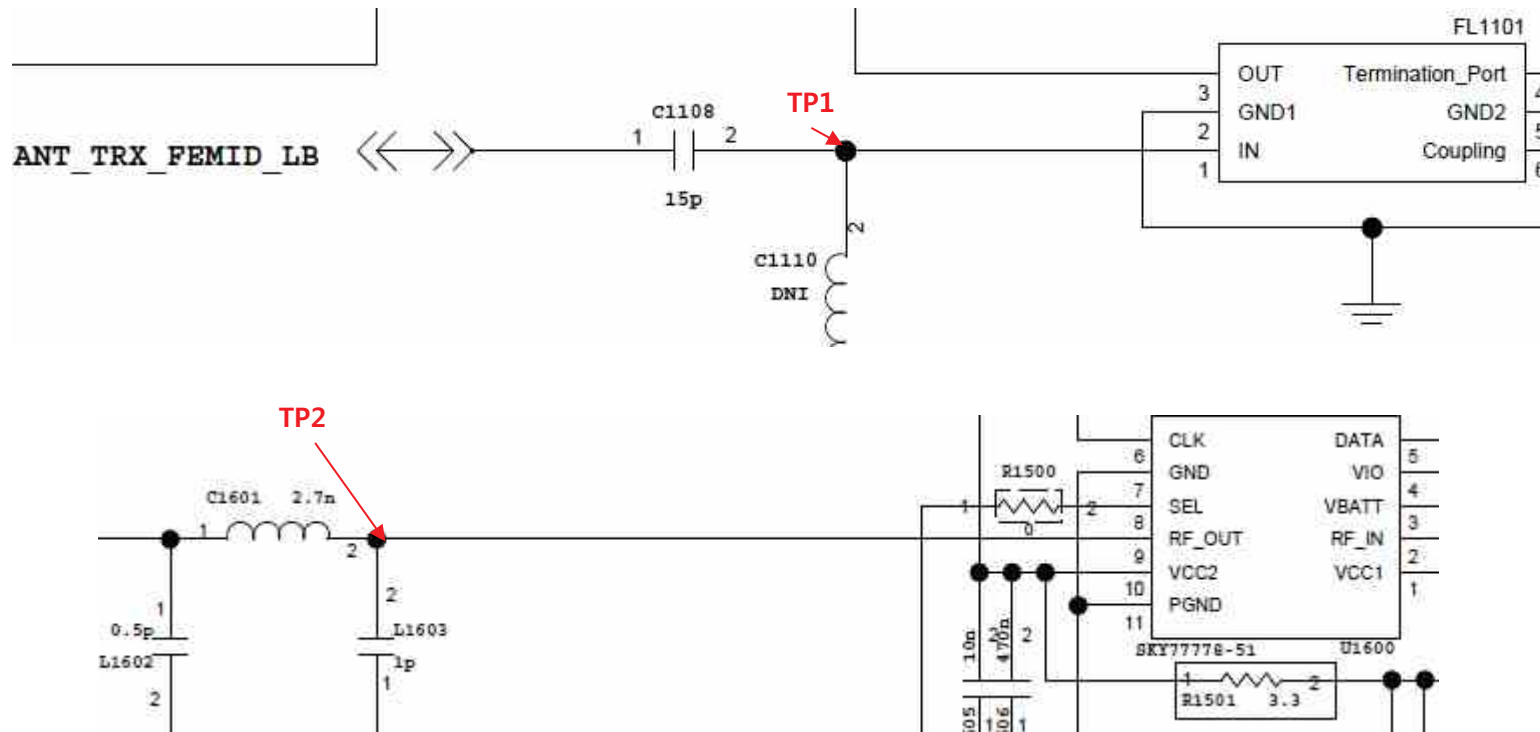
Image



3.7 LTE RF PART

3.7.16 Checking RF Signal TX path(LTE B7)

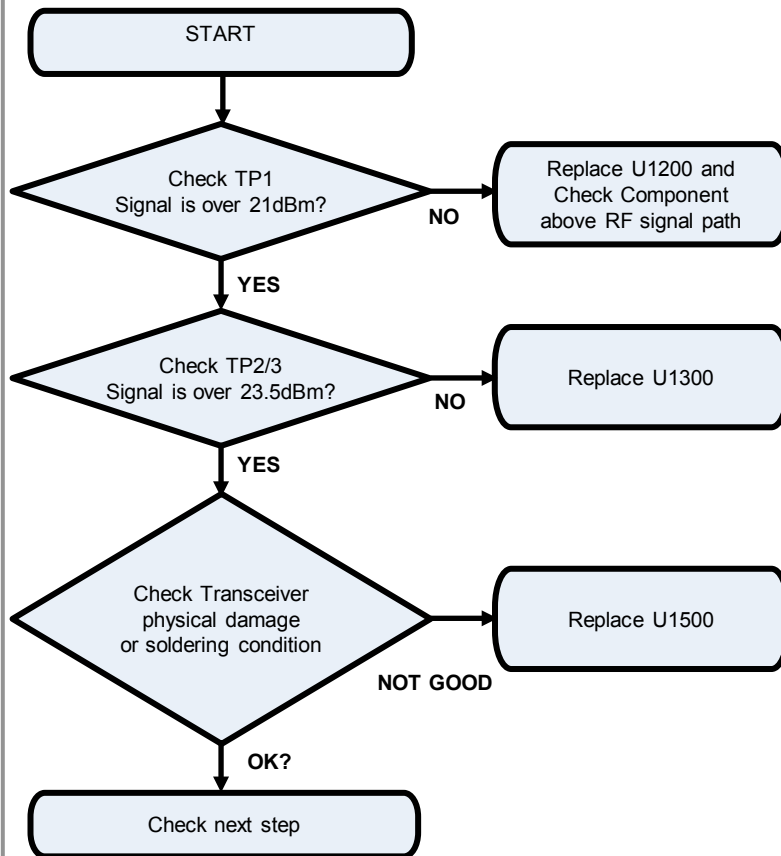
Circuit Diagram



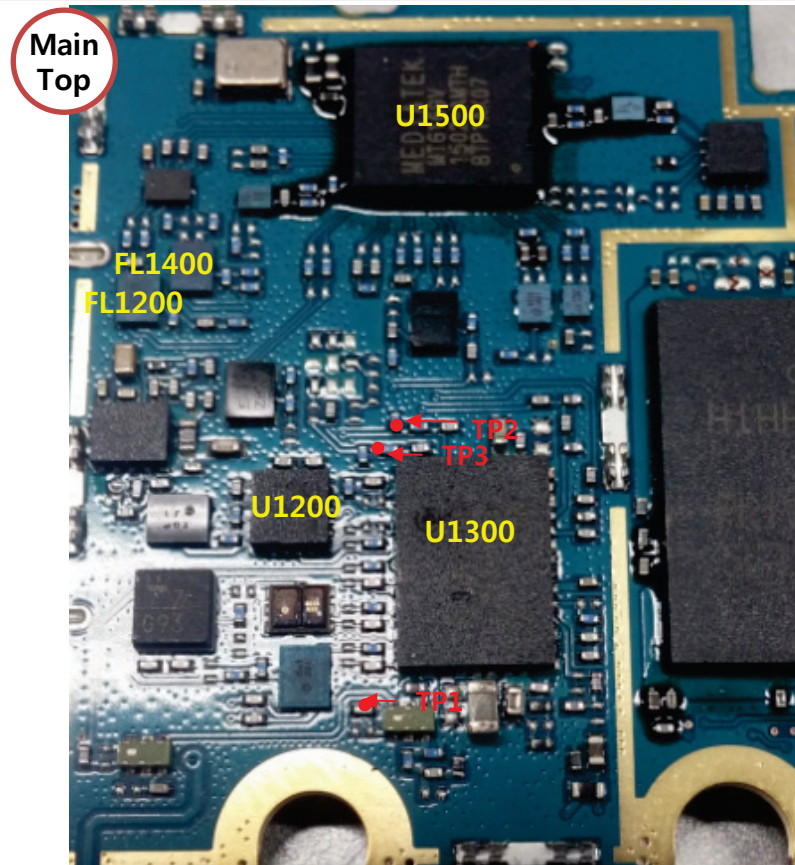
3.7 LTE RF PART

3.7.17 Checking RF Signal TX path(LTE B28)

Checking Flow



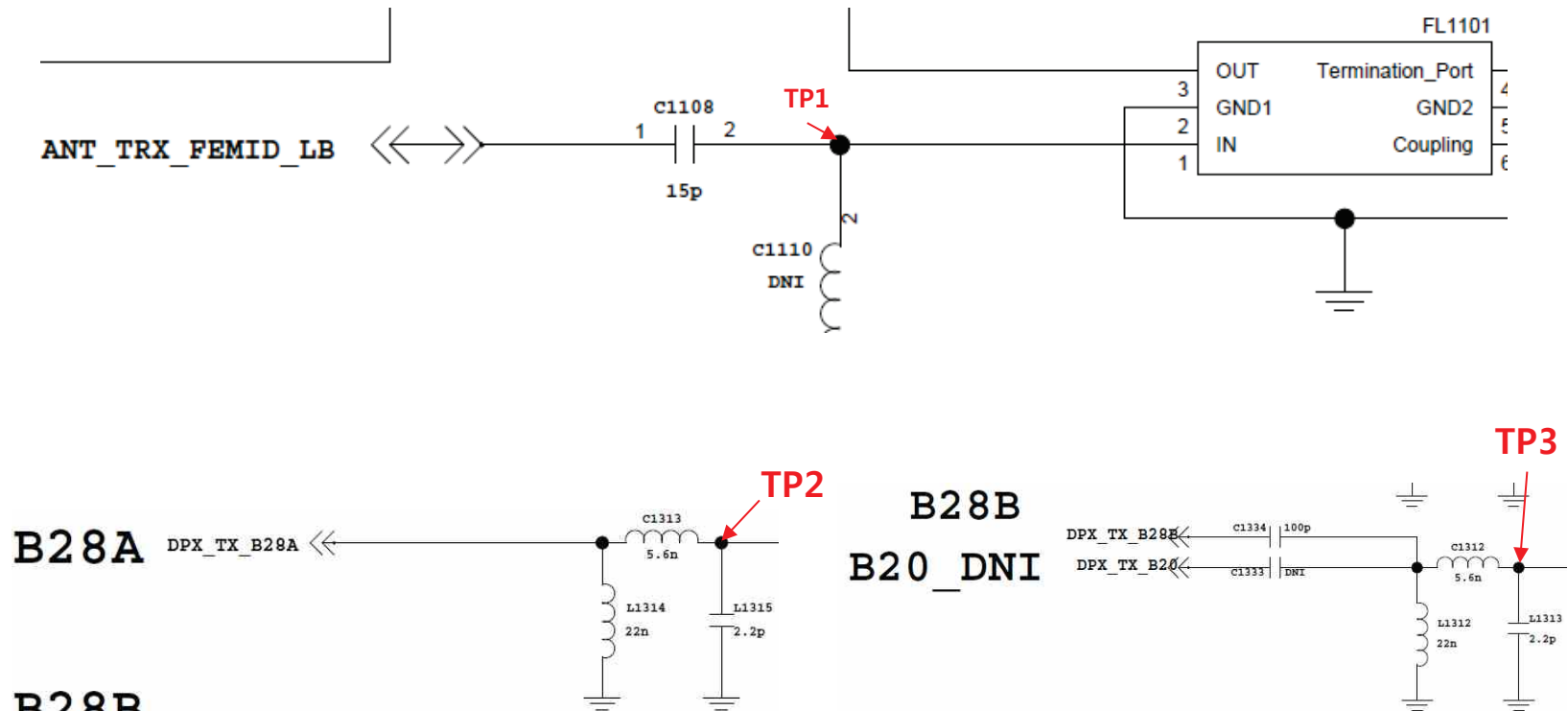
Image



3.7 LTE RF PART

3.7.17 Checking RF Signal TX path(LTE B28)

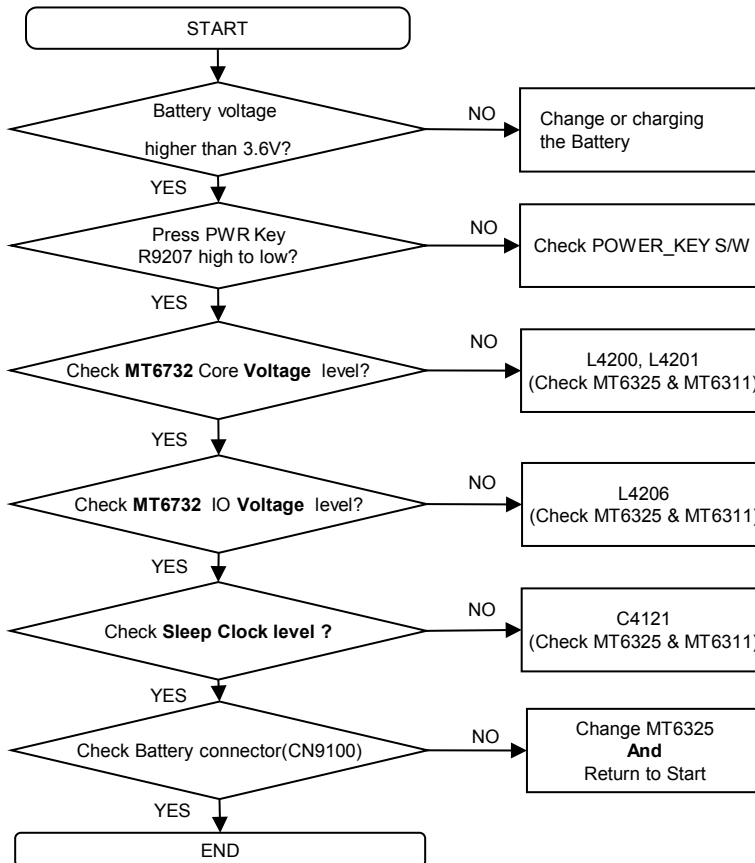
Circuit Diagram



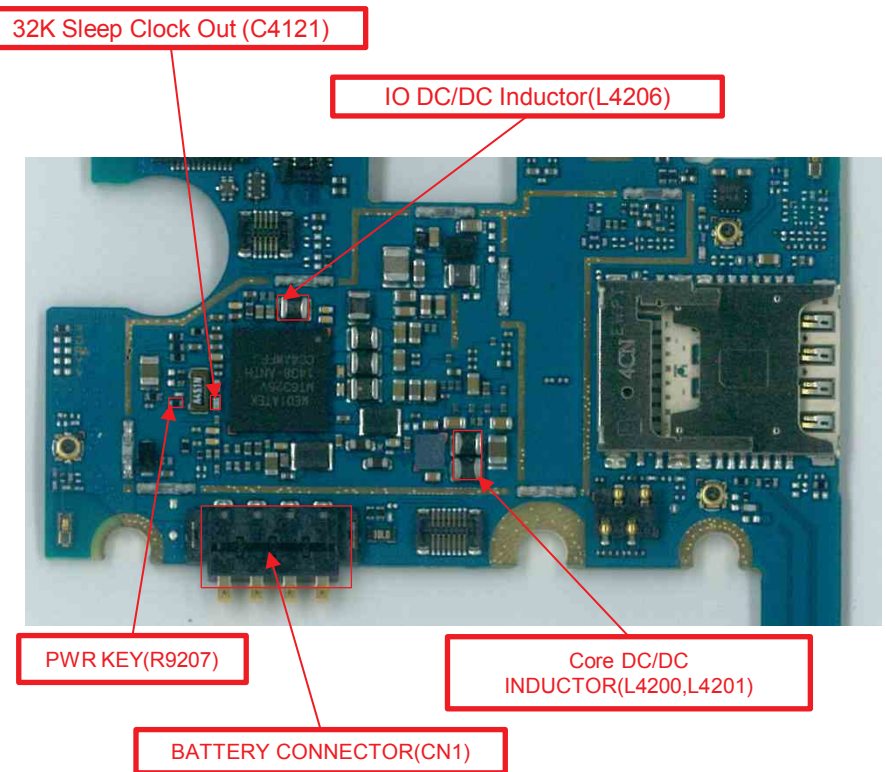
3.8 Power

Checking Power signal (Battery connector, Power Key, PMIC Regulator)

Checking Flow



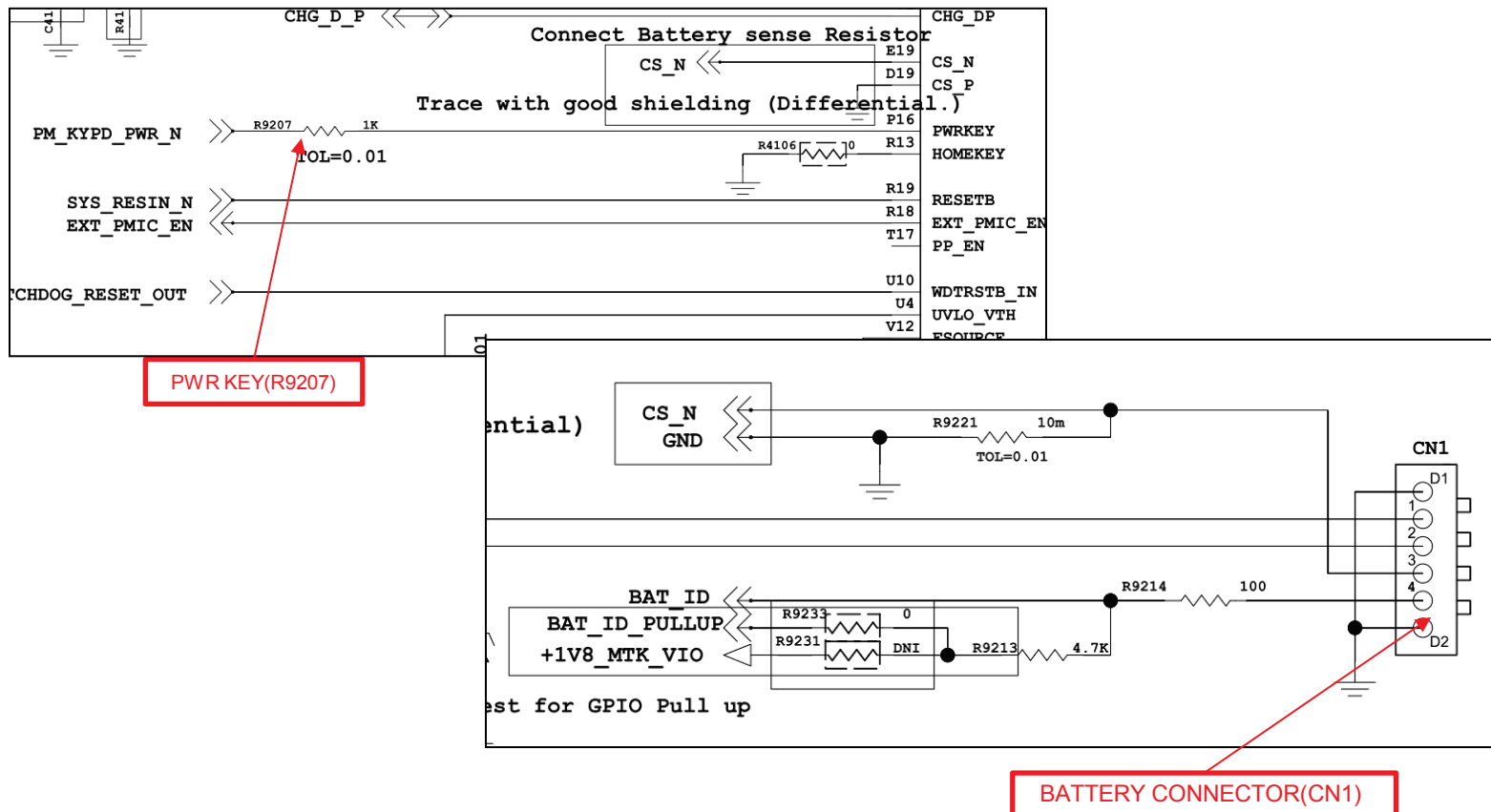
Image



3.8 Power

Checking Power signal (Battery connector, Power Key, PMIC Regulator)

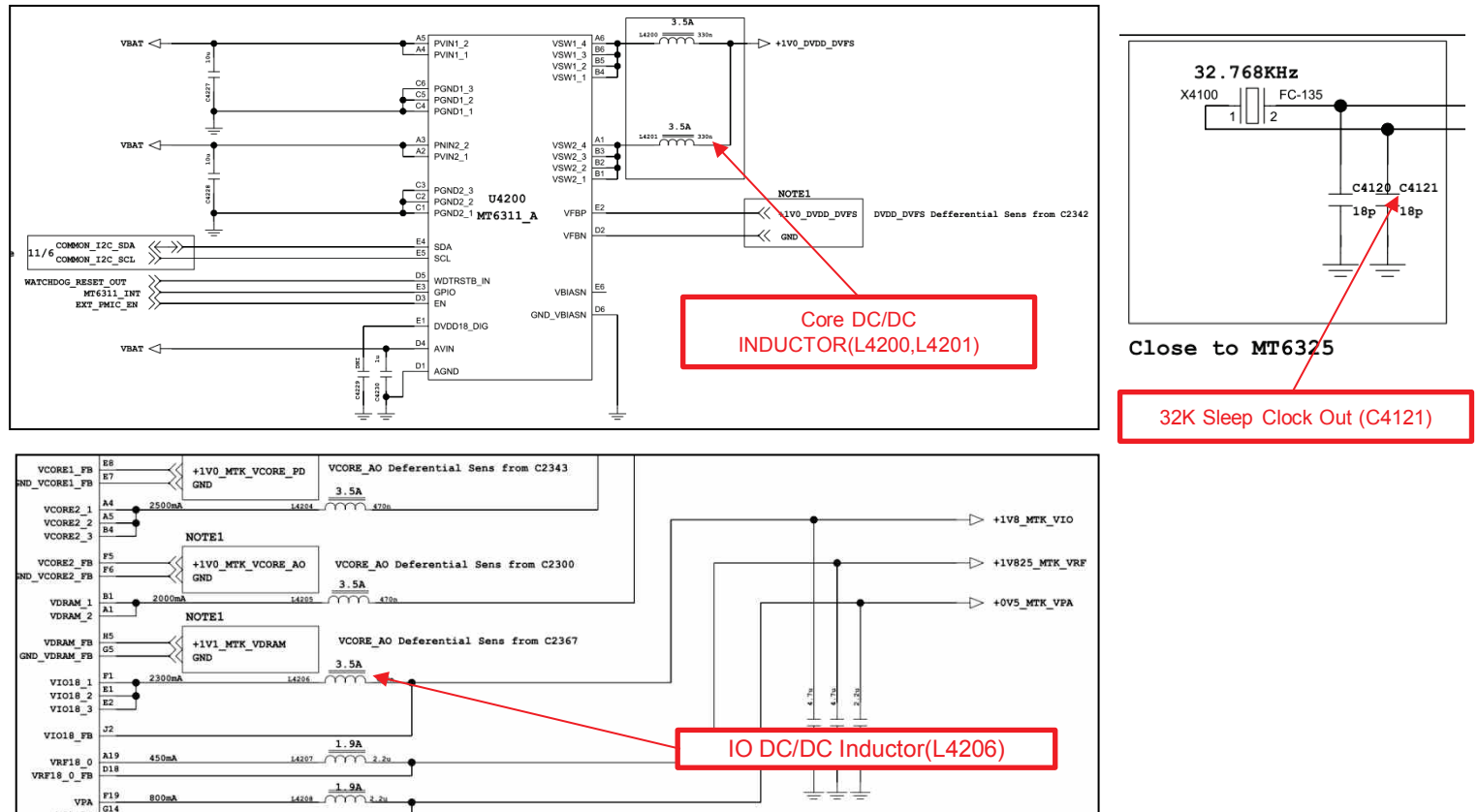
Circuit Diagram



3.8 Power

Checking Power signal (Battery connector, Power Key, PMIC Regulator)

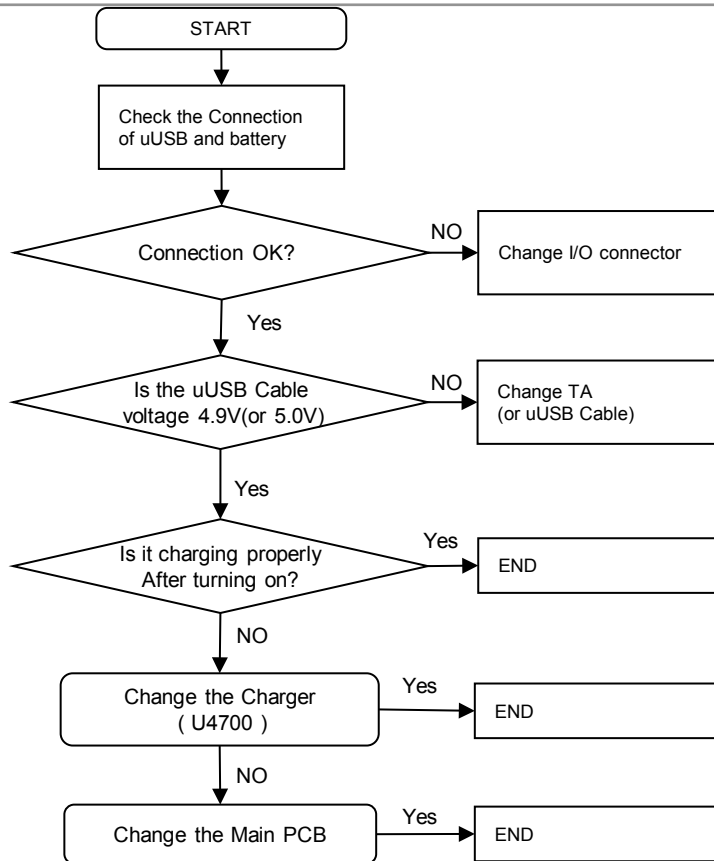
Circuit Diagram



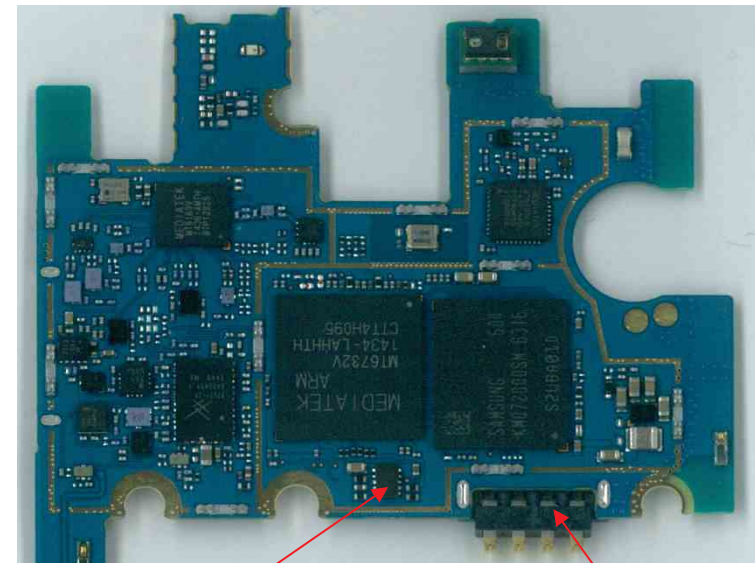
3.9 Charger

The I/O connector and uUSB cable voltage(5.0V) is used as the reference one of PMIC for charging.

Checking Flow



Image

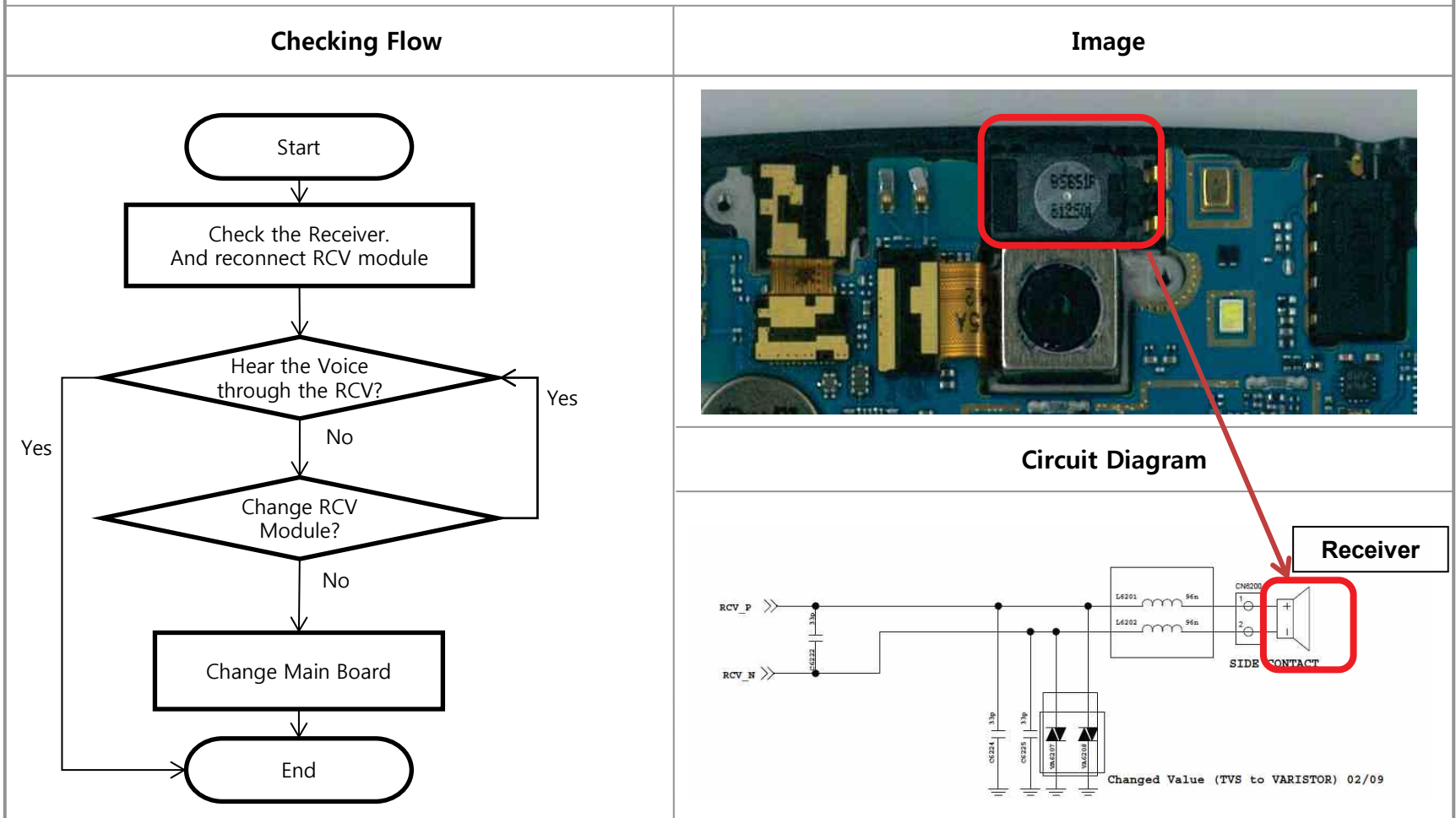


Linear Charger

I/O connector

3.10 Audio Block(3.10.1 Audio receiver)

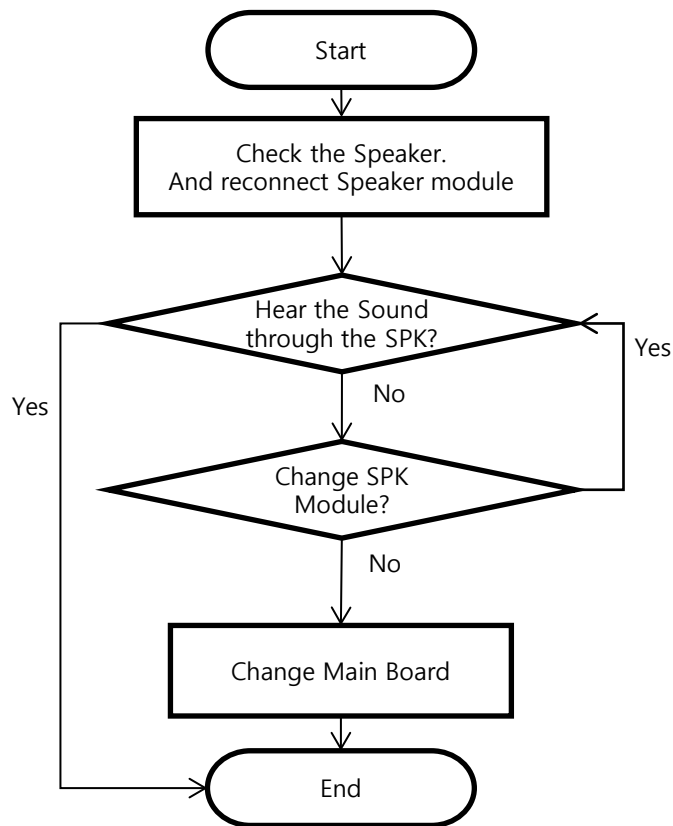
The receiver control signals are generated by MT6325(U4100), the MT6325 chip and the receiver are to be checked out.



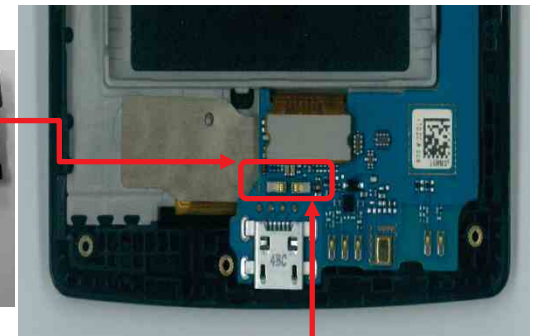
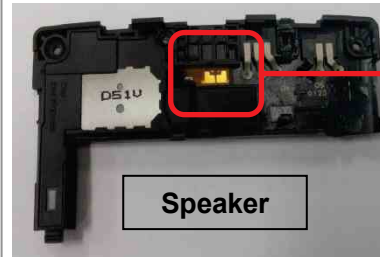
3.10.2 Audio_speaker

The Speaker control signals are generated by MT6325(U4100), the MT6325 chip and the speaker are to be checked out.

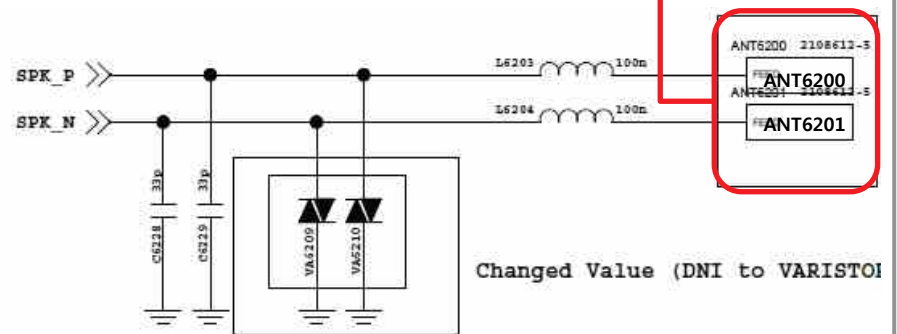
Checking Flow



Image



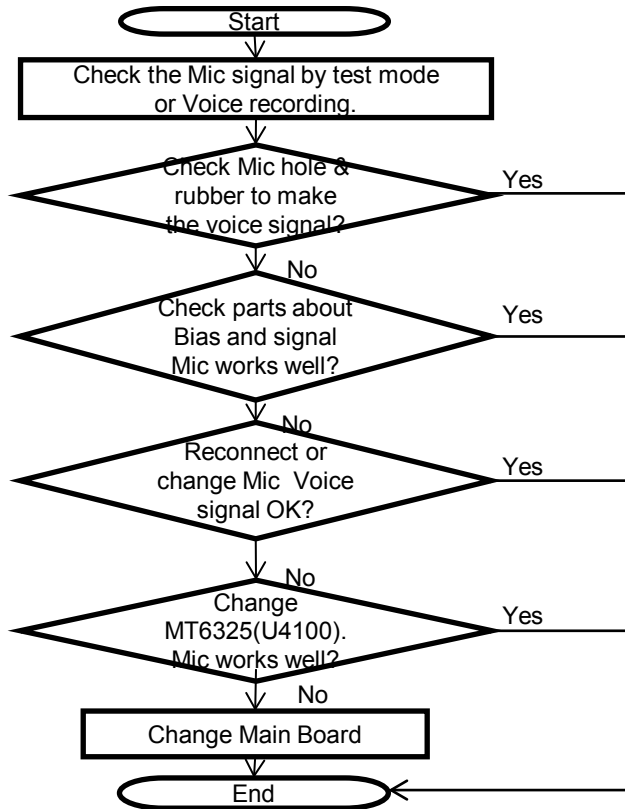
Circuit Diagram



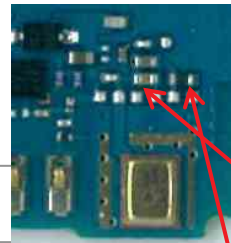
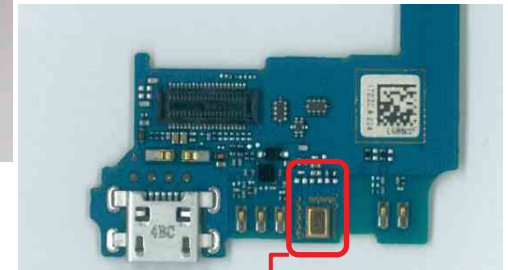
3.10.3 Audio_Main MIC

It's operating voice call(except speakerphone), voice recording, camcorder recording.

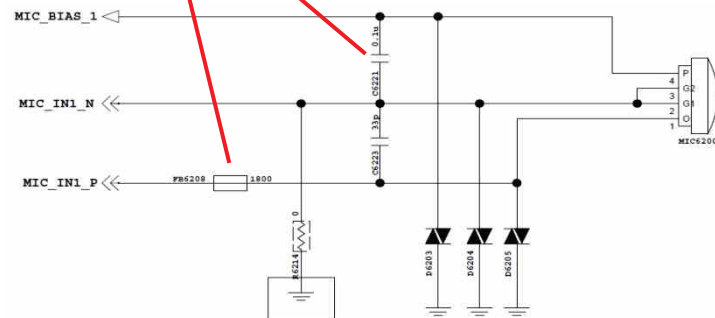
Checking Flow



Image



Circuit Diagram

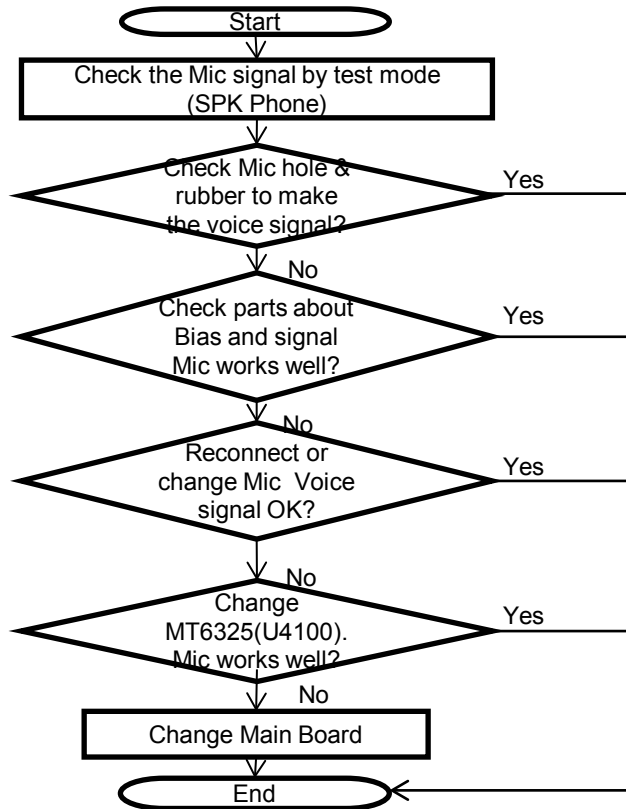


Check FB6208, VMIC voltage level (C6221)

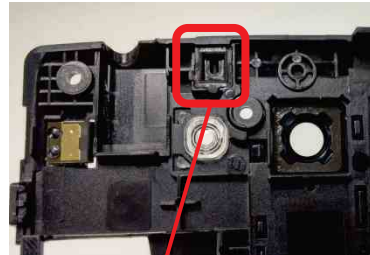
3.10.4 Audio_sub MIC

It's operating Speakerphone call.

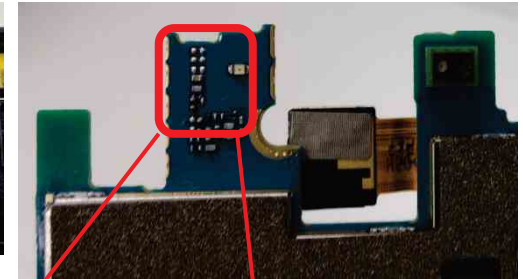
Checking Flow



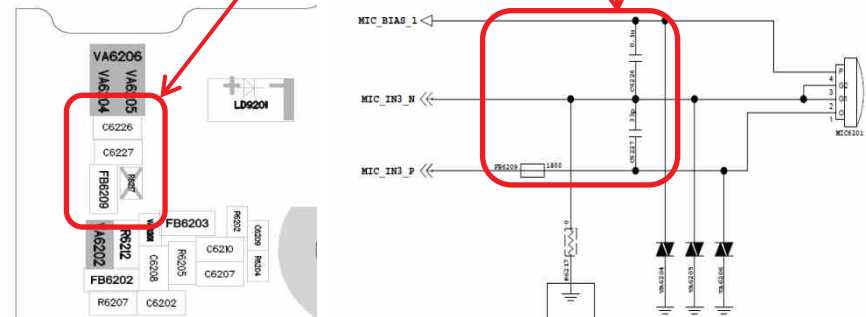
Image



Check SUB Mic rubber



Circuit Diagram

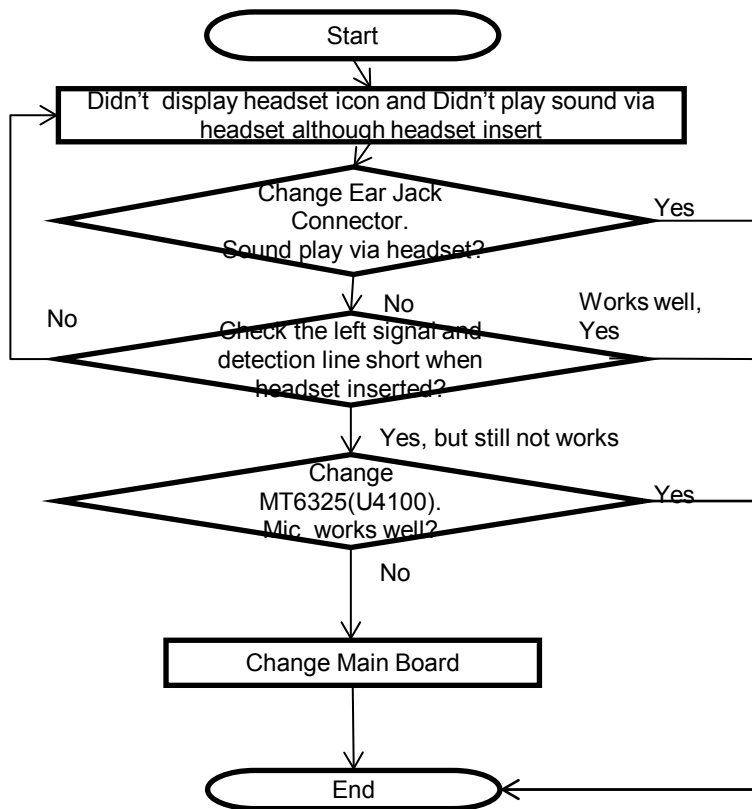


Check FB6209 , VMIC voltage level (C6226)

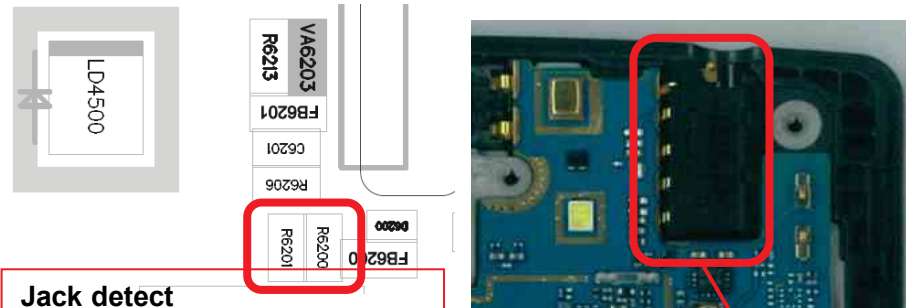
3.10.5 Audio_Ear MIC Jack

Disable detecting headset insert or No sound from Earphone, Check the Ear Mic and MT6325

Checking Flow

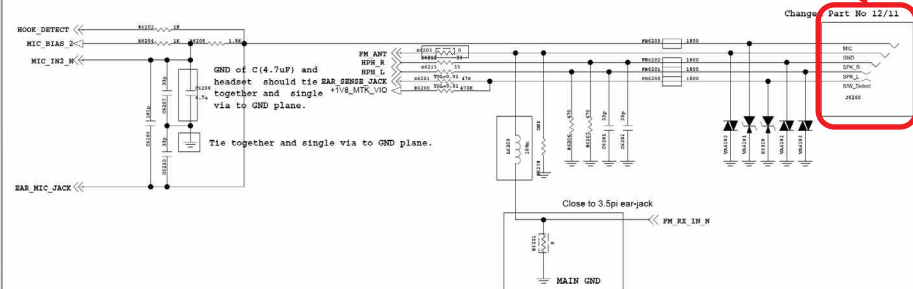


Image



Jack detect Voltage Level Check at R6201
 - 1.8V(before Jack)
 - low(after Jack)

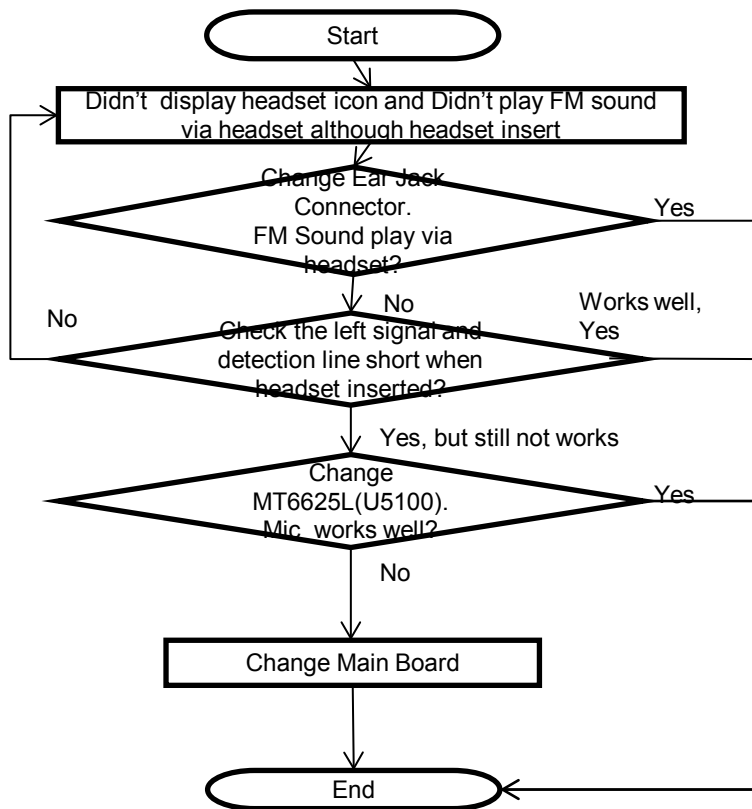
Circuit Diagram



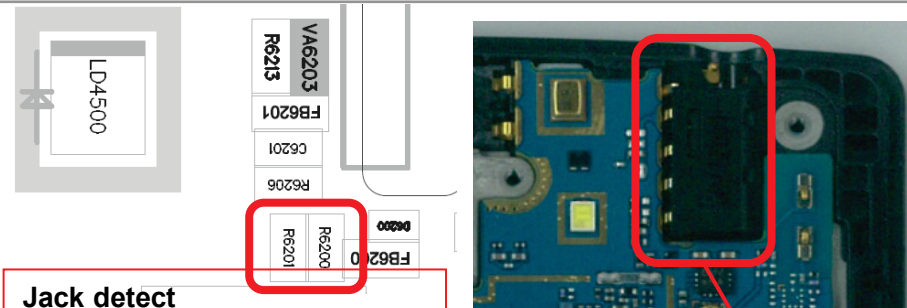
3.10.6 FM_Radio

Disable detecting headset insert or No FM sound from Earphone, Check the Ear Mic and MT6625L

Checking Flow

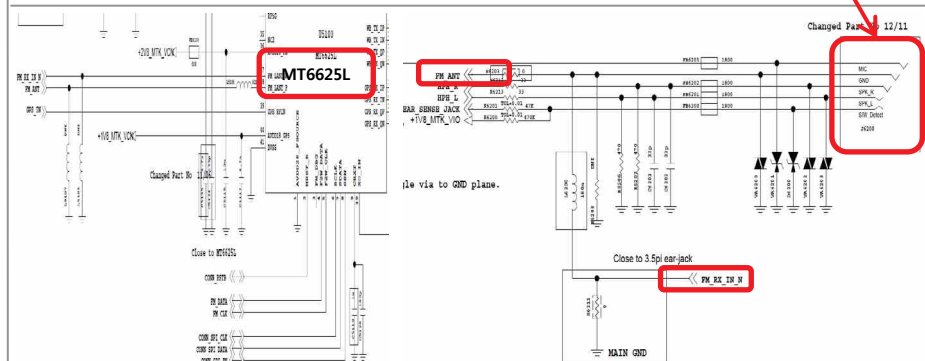


Image



Jack detect Voltage Level Check at R6201
 - 1.8V(before insert)
 - low(after insert)

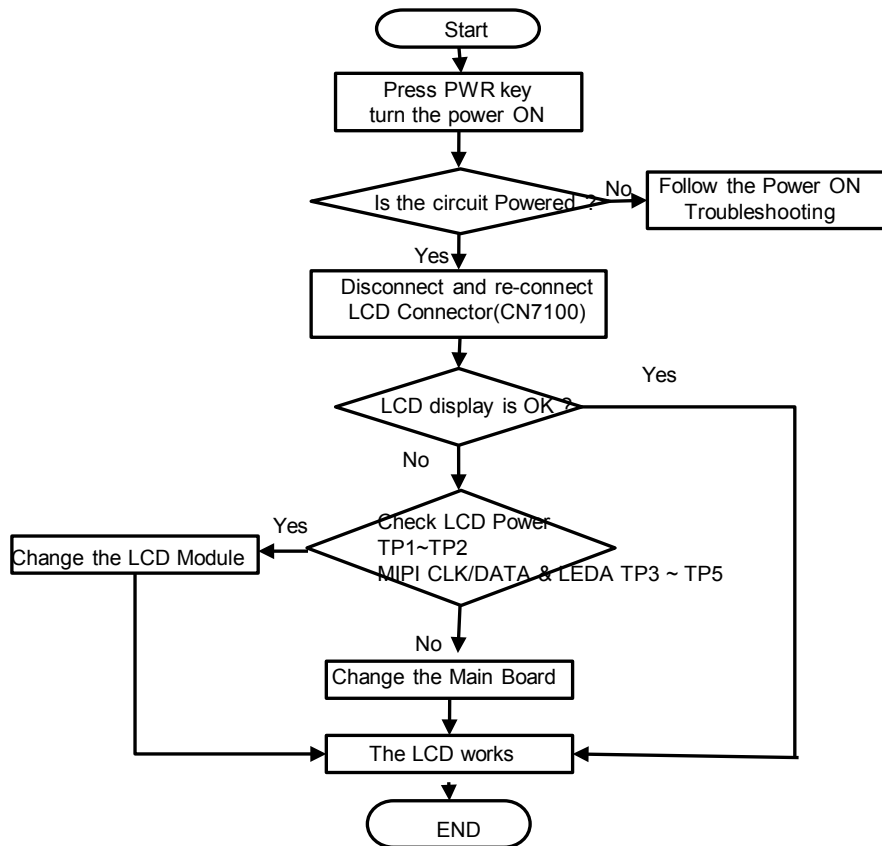
Circuit Diagram



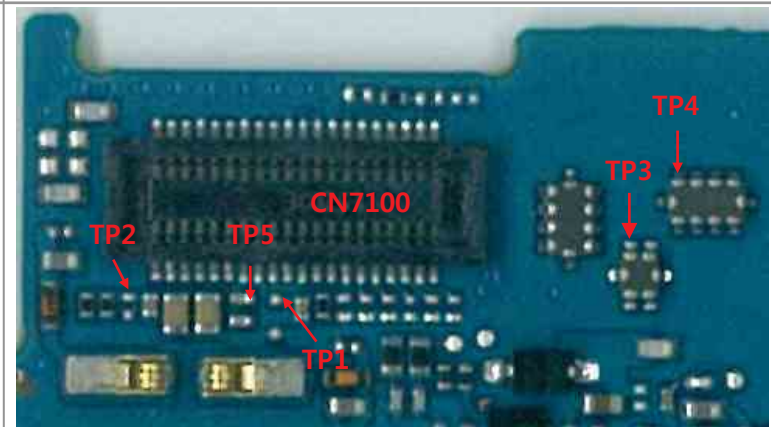
3.11 Checking LCD Block

The LCD control signals are generated by MT6732. Its interface is MIPI having four data lanes and one clock lane.

Checking Flow



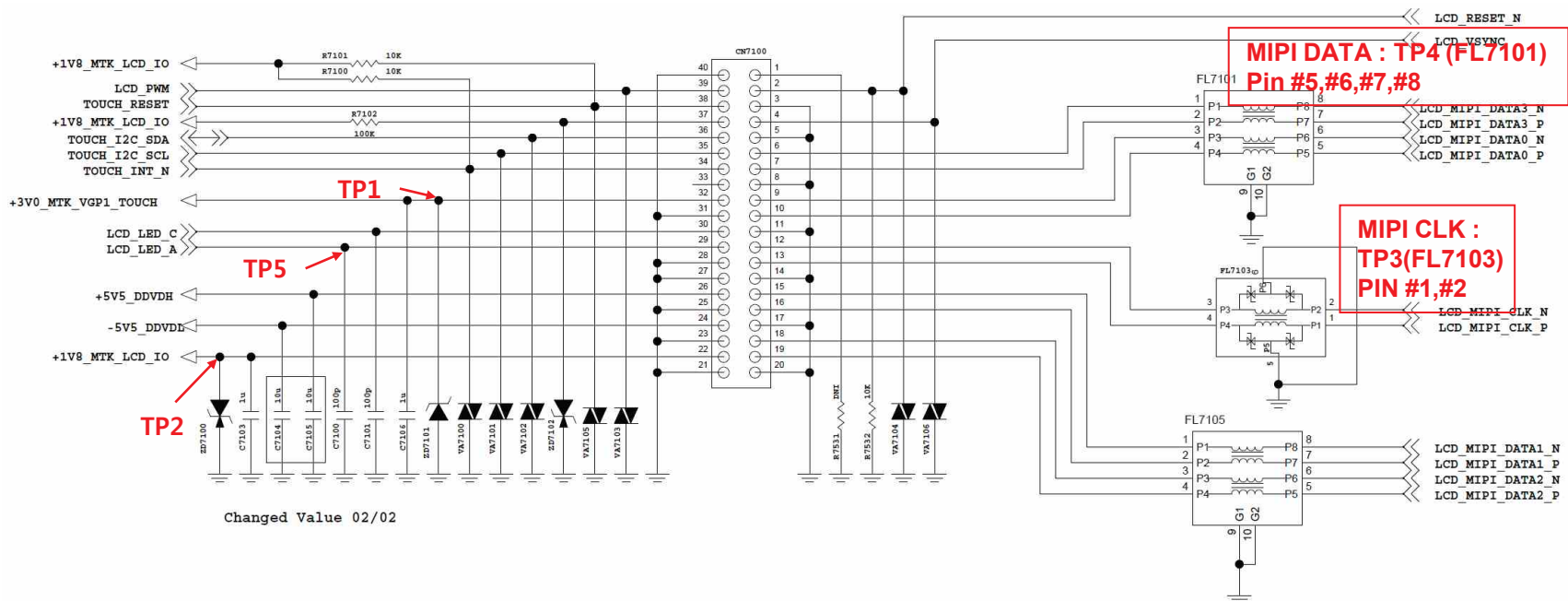
Image



3.11 Checking LCD Block

The LCD control signals are generated by MT6732. Its interface is MIPI having four data lanes and one clock lane.

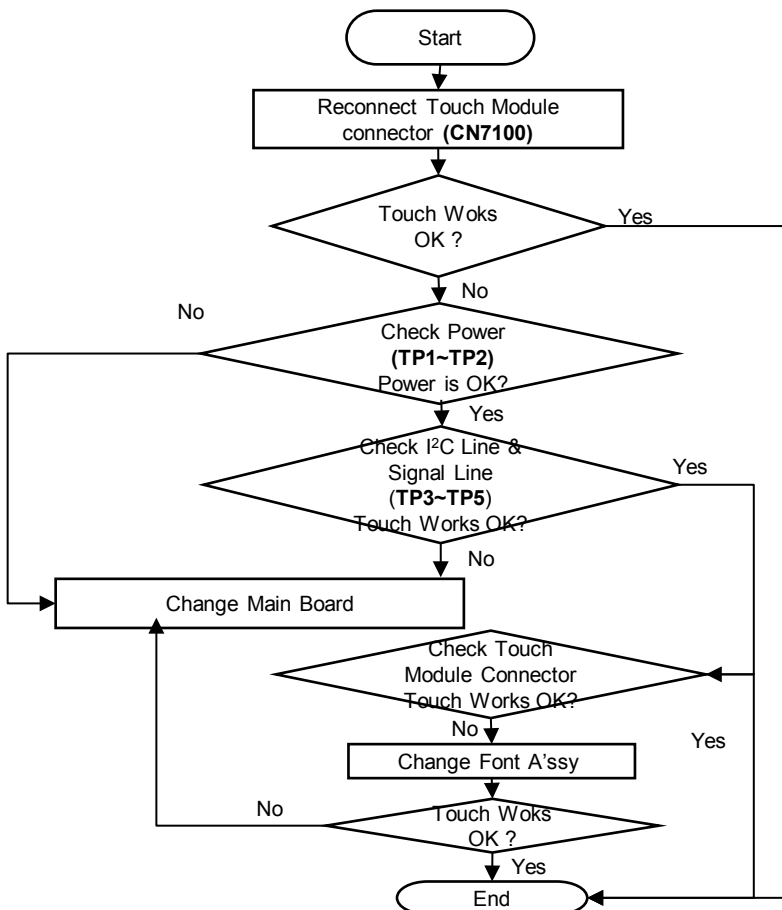
Circuit Diagram



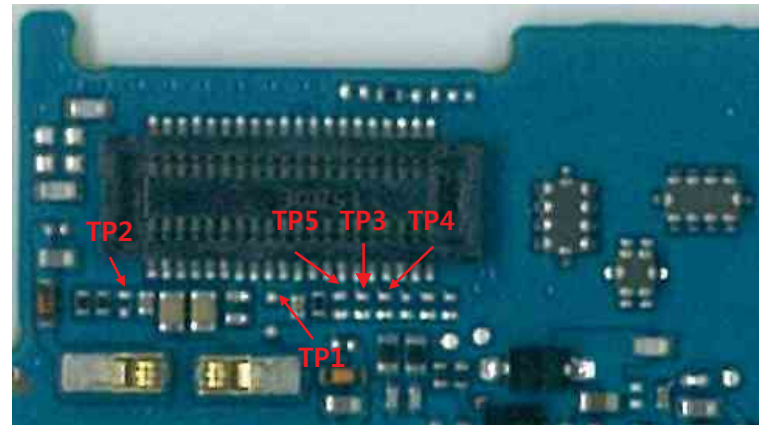
3.12 Checking Touch Block

The Touch control signals are generated by MT6732. It is assembled with LCD.

Checking Flow



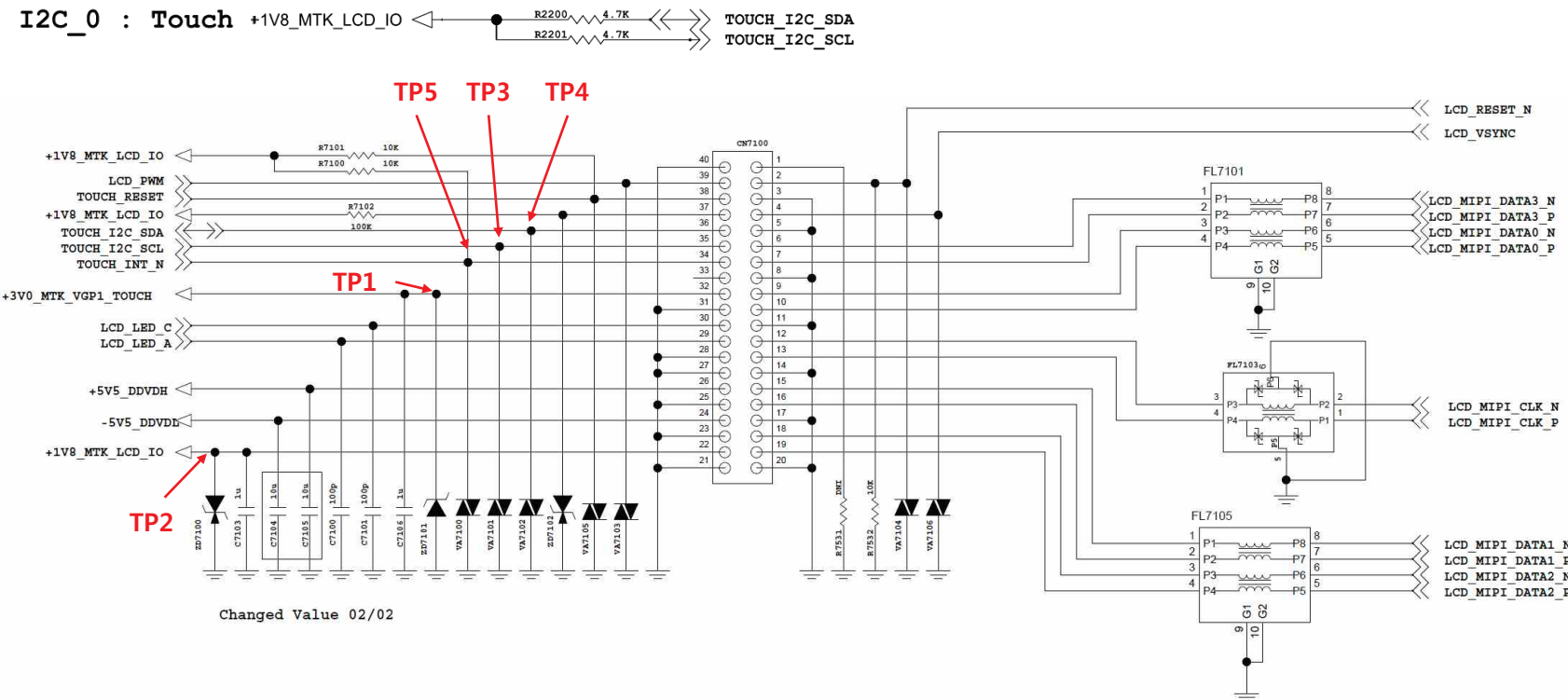
Image



3.12 Checking Touch Block

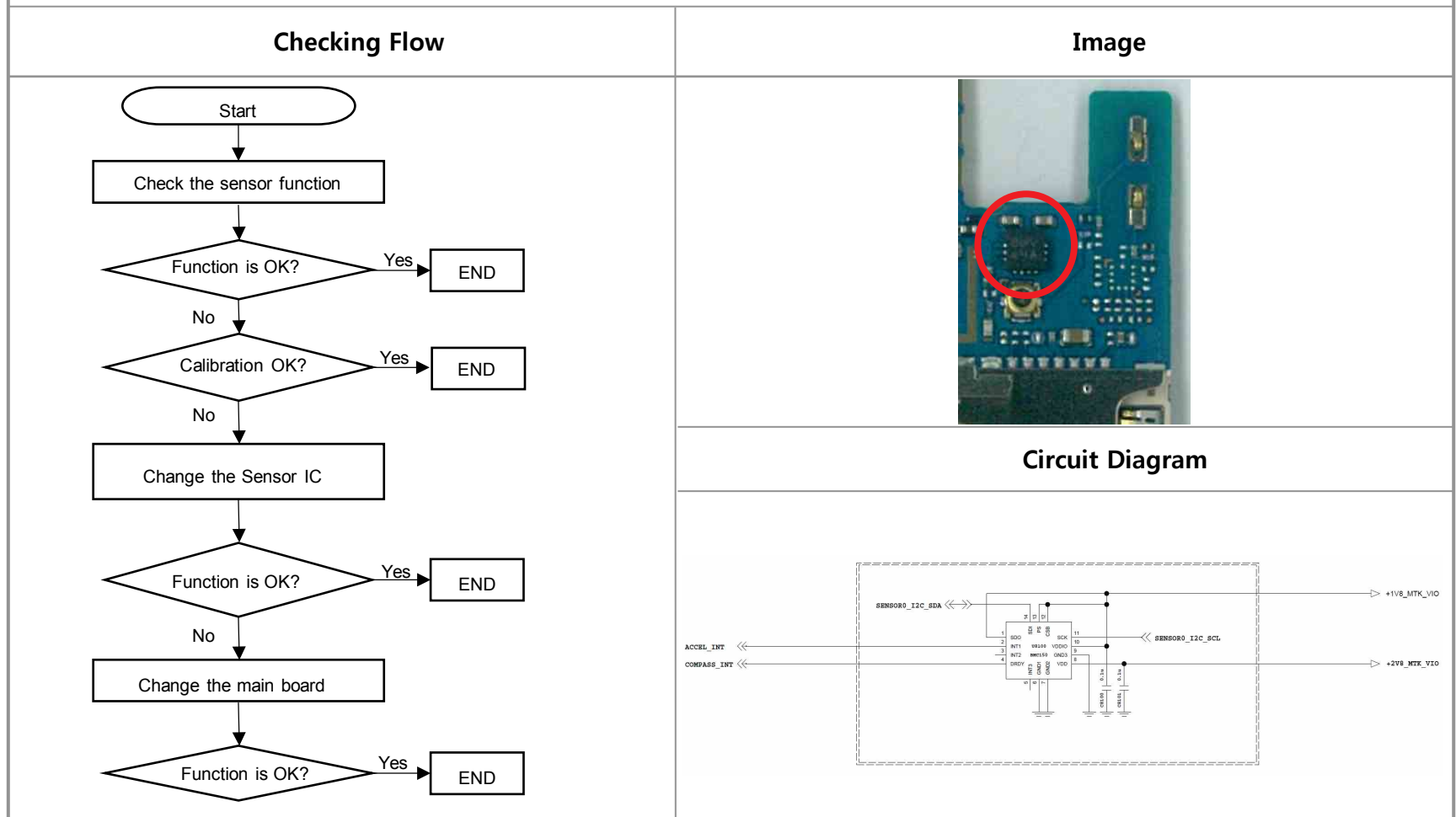
The Touch control signals are generated by MT6732. It is assembled with LCD.

Circuit Diagram



3.13 Checking accelerometer+compass sensor Block

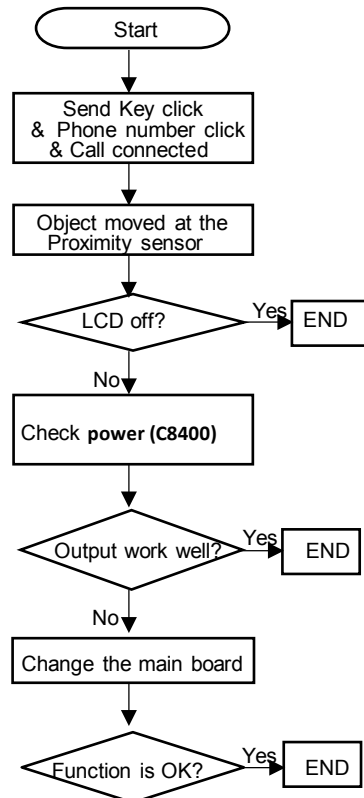
The Accl. & compass sensors are calibrated by using SW algorithm.



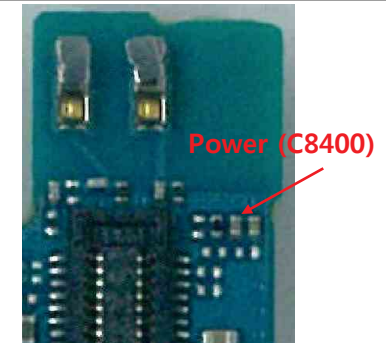
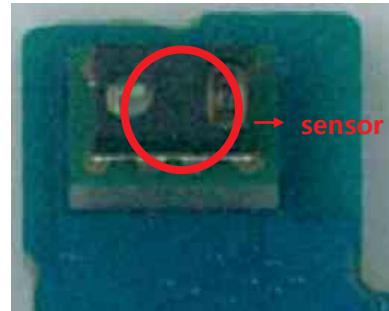
3.13 Checking proximity sensor Block

Proximity Sensor is worked as below: Send Key click → Phone number click → Call connected
 → Object moved at the sensor → Control the screen's on/off operation automatically

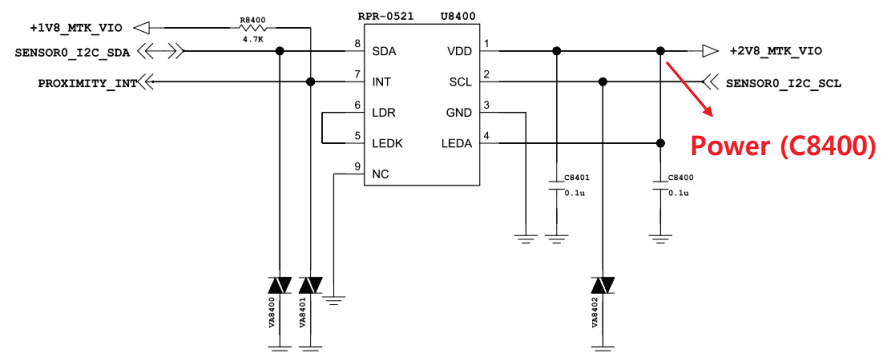
Checking Flow



Image



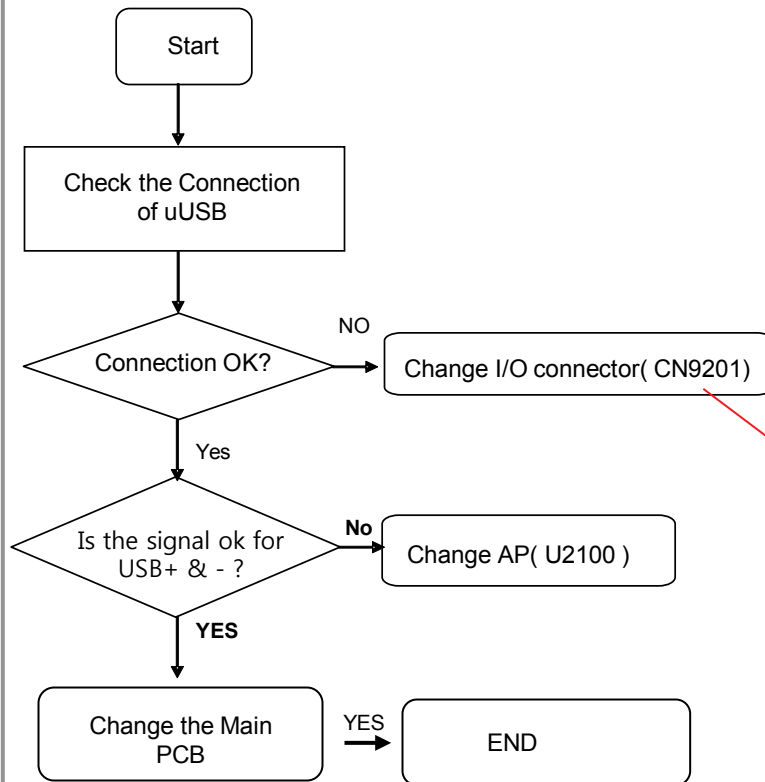
Circuit Diagram



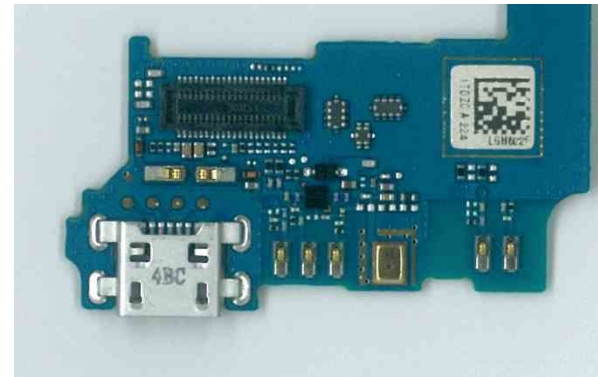
3.14 Checking USB Block

I/O connector is used as the USB port.

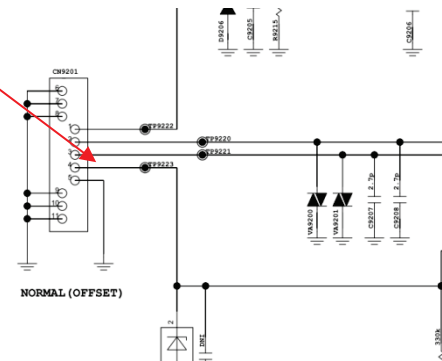
Checking Flow



Image



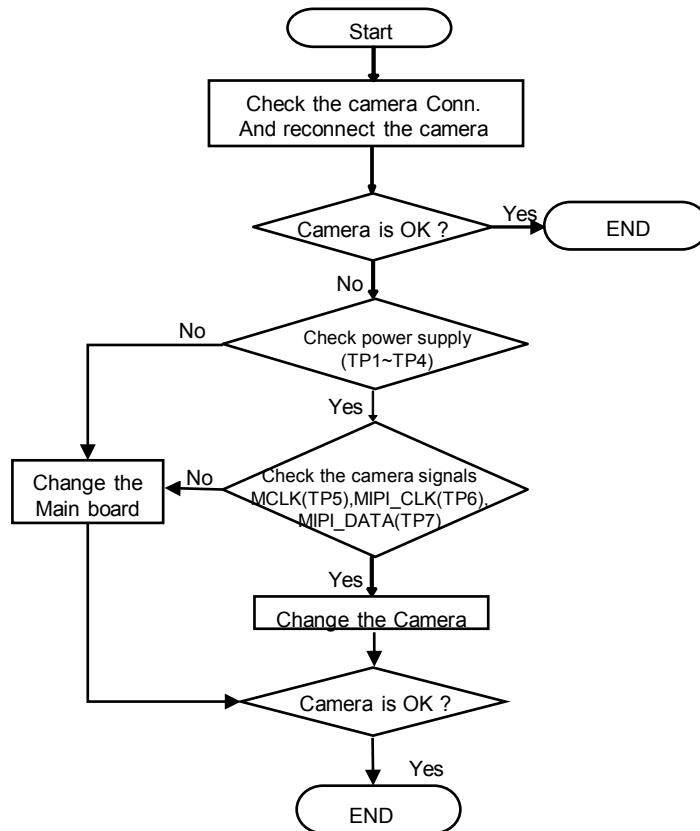
Circuit Diagram



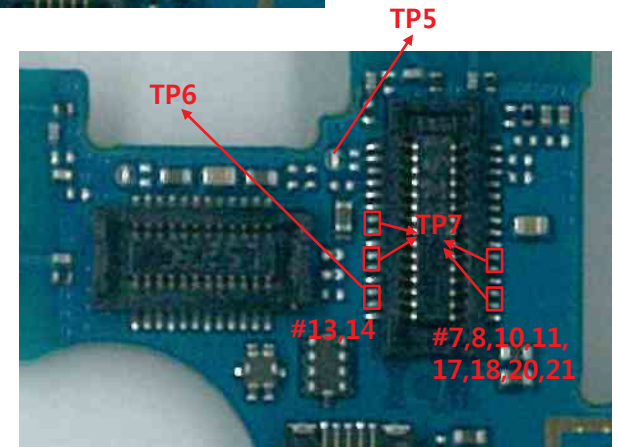
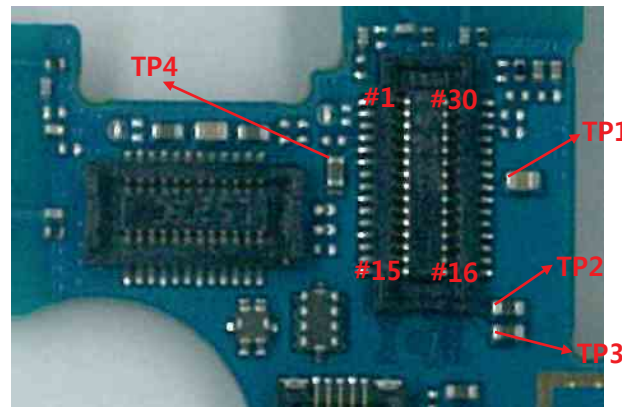
3.15 Checking Main camera Block

8M camera control signals are generated by MT6732 (U2100 : Main Chipset). And powered by MT6325 (U4100 : PMIC).

Checking Flow



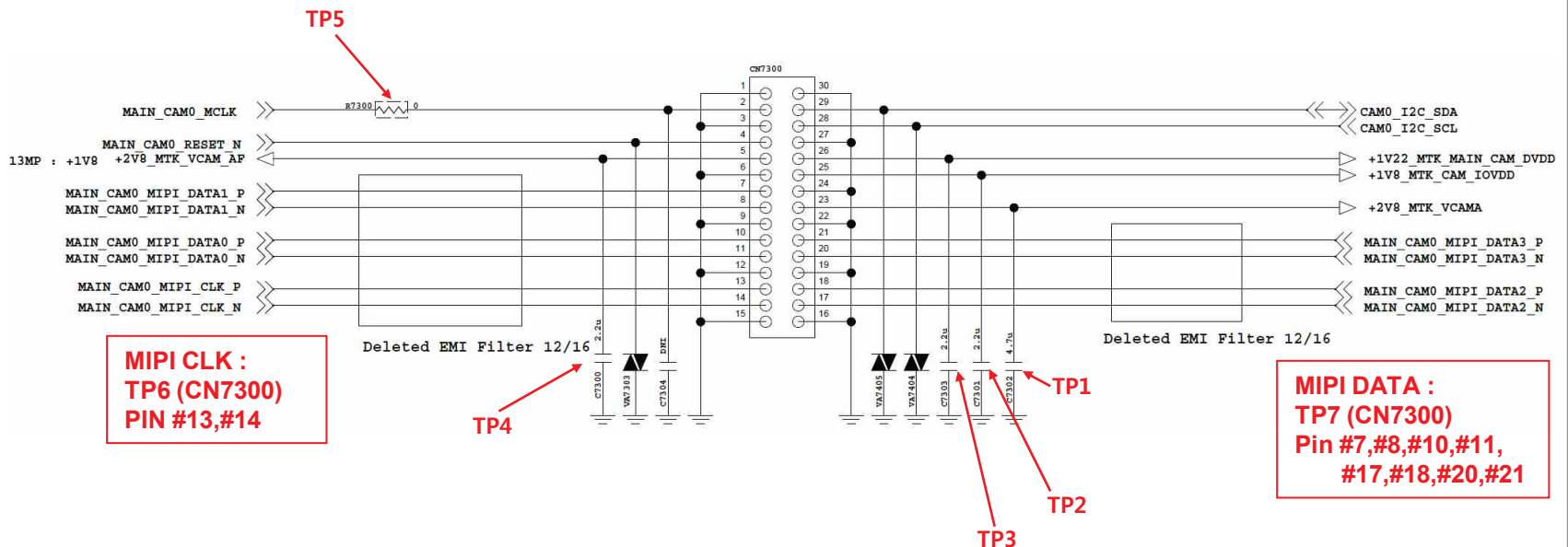
Image



3.15 Checking Main camera Block

8M camera control signals are generated by MT6732 (U2100 : Main Chipset). And powered by MT6325 (U4100 : PMIC).

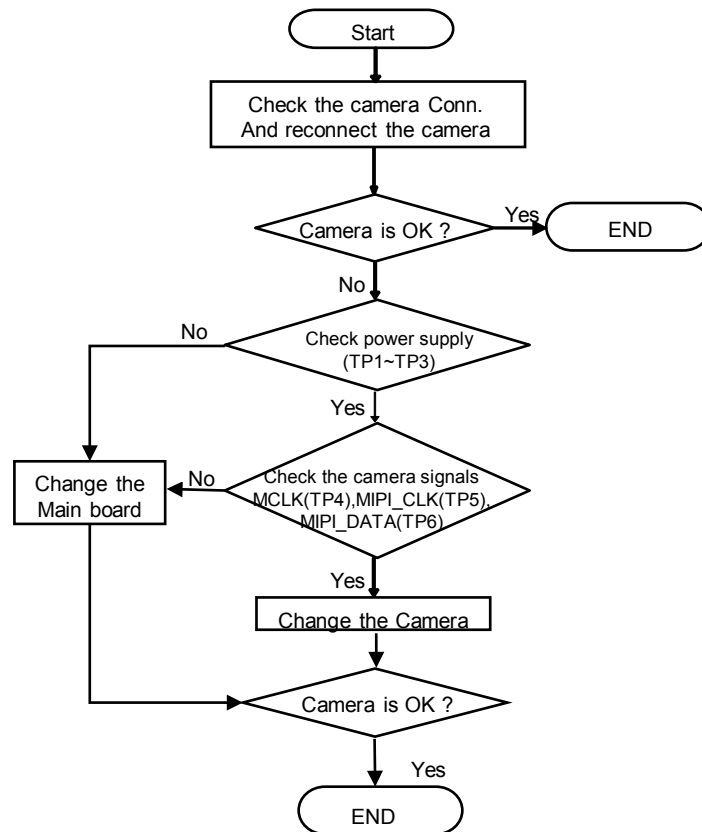
Circuit Diagram



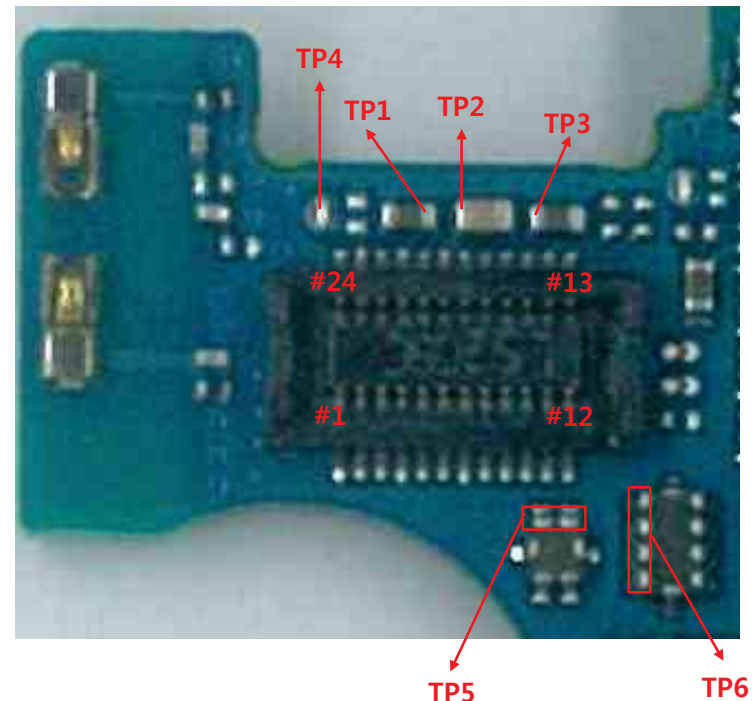
3.15 Checking VT camera Block

5M camera control signals are generated by MT6732 (U2100 : Main Chipset). And powered by MT6325 (U4100 : PMIC).

Checking Flow



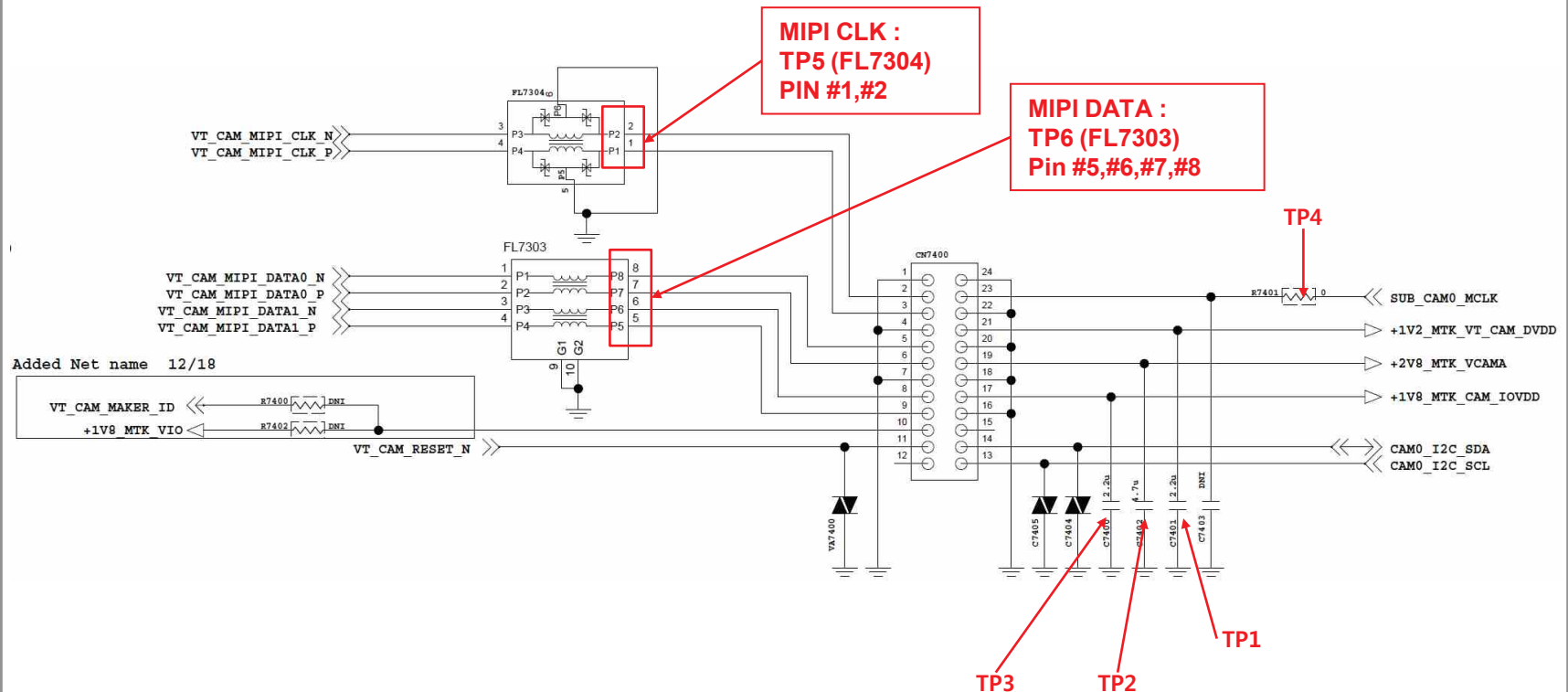
Image



3.15 Checking VT camera Block

5M camera control signals are generated by MT6732 (U2100 : Main Chipset). And powered by MT6325 (U4100 : PMIC).

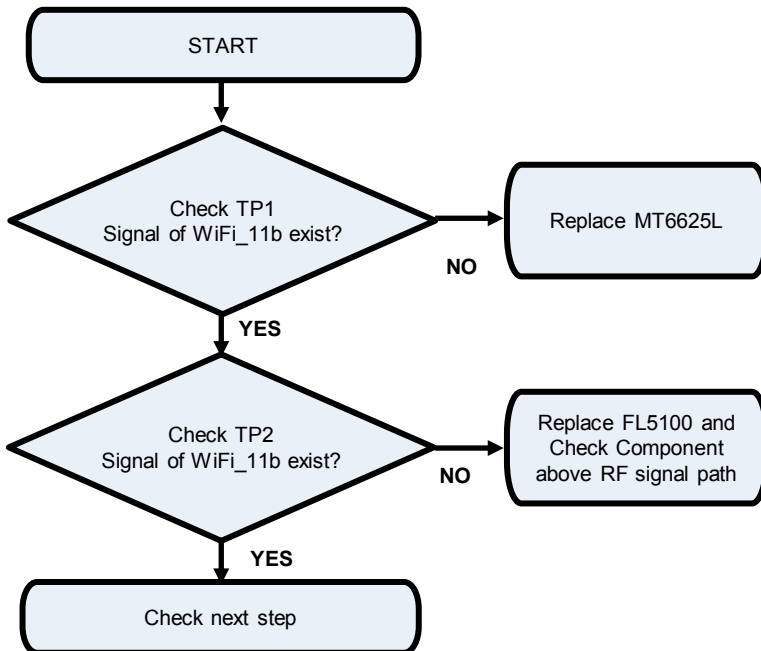
Circuit Diagram



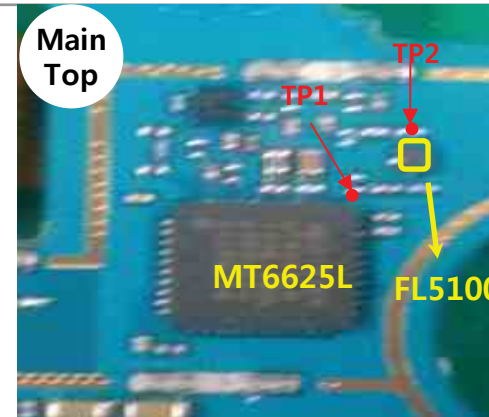
3.16 Connectivity RF PART

3.16.1 Checking RF Signal TRX path(WiFi, BT)

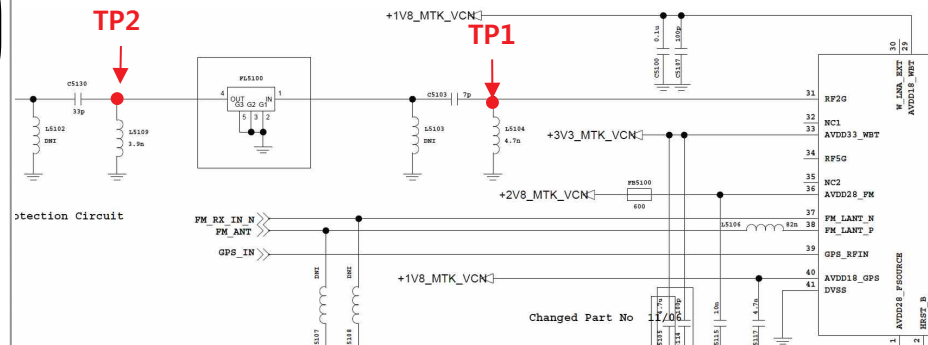
Checking Flow



Image



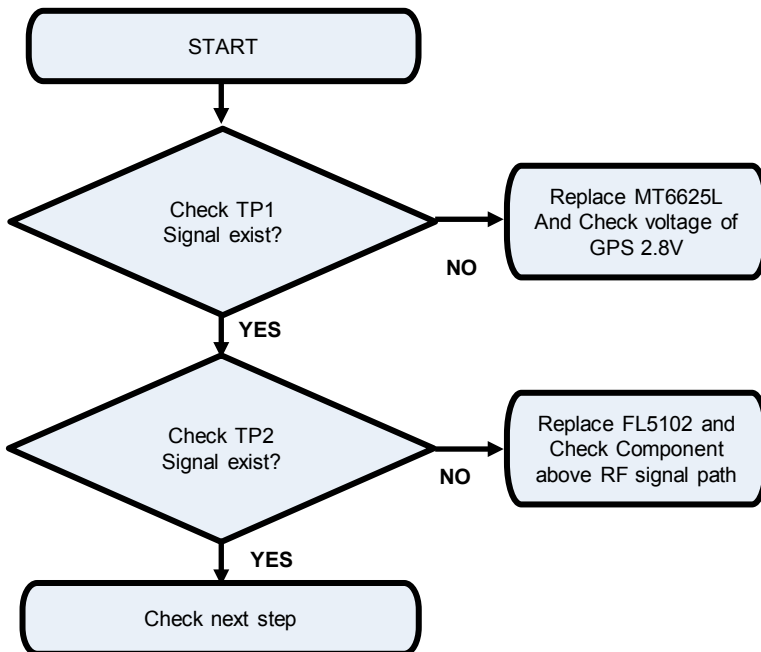
Circuit Diagram



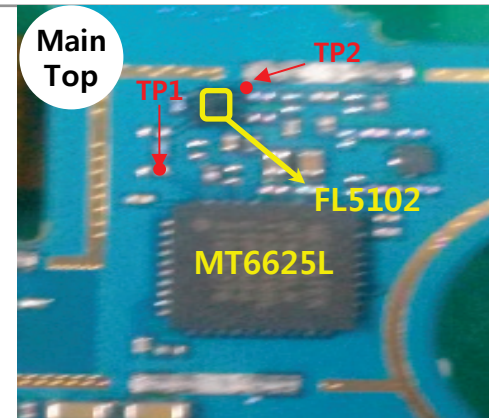
3.16 Connectivity RF PART

3.16.2 Checking RF Signal TRX path(GPS)

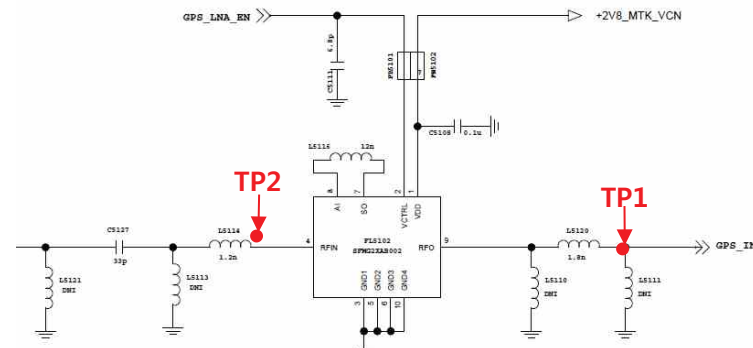
Checking Flow



Image



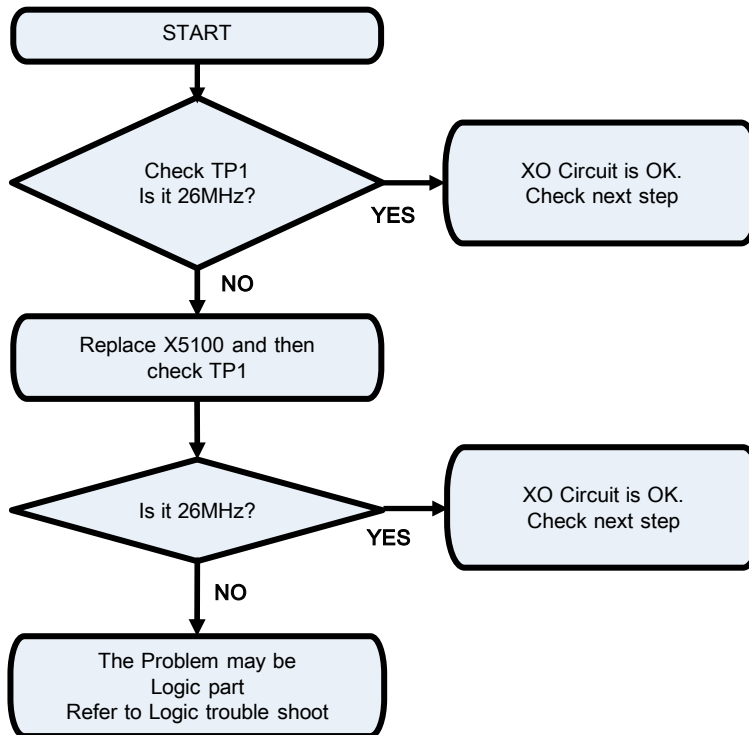
Circuit Diagram



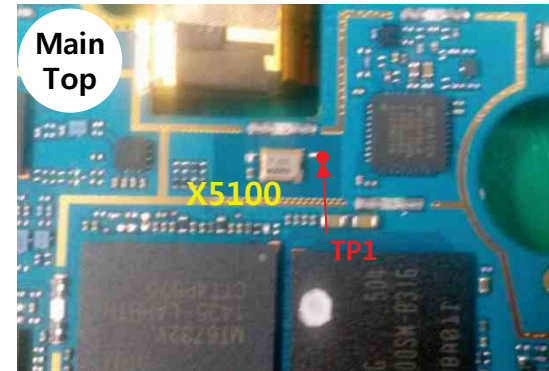
3.16 Connectivity RF PART

3.16.3 The out put frequency(26MHz) of TCXO(X5100) is used as the reference of MT6625L(Connectivity-WLAN, GPS, BT)

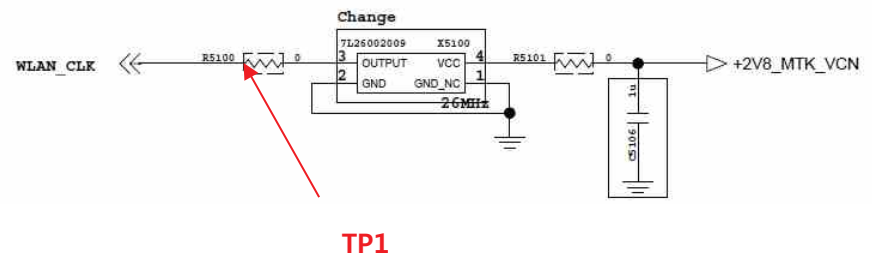
Checking Flow



Image

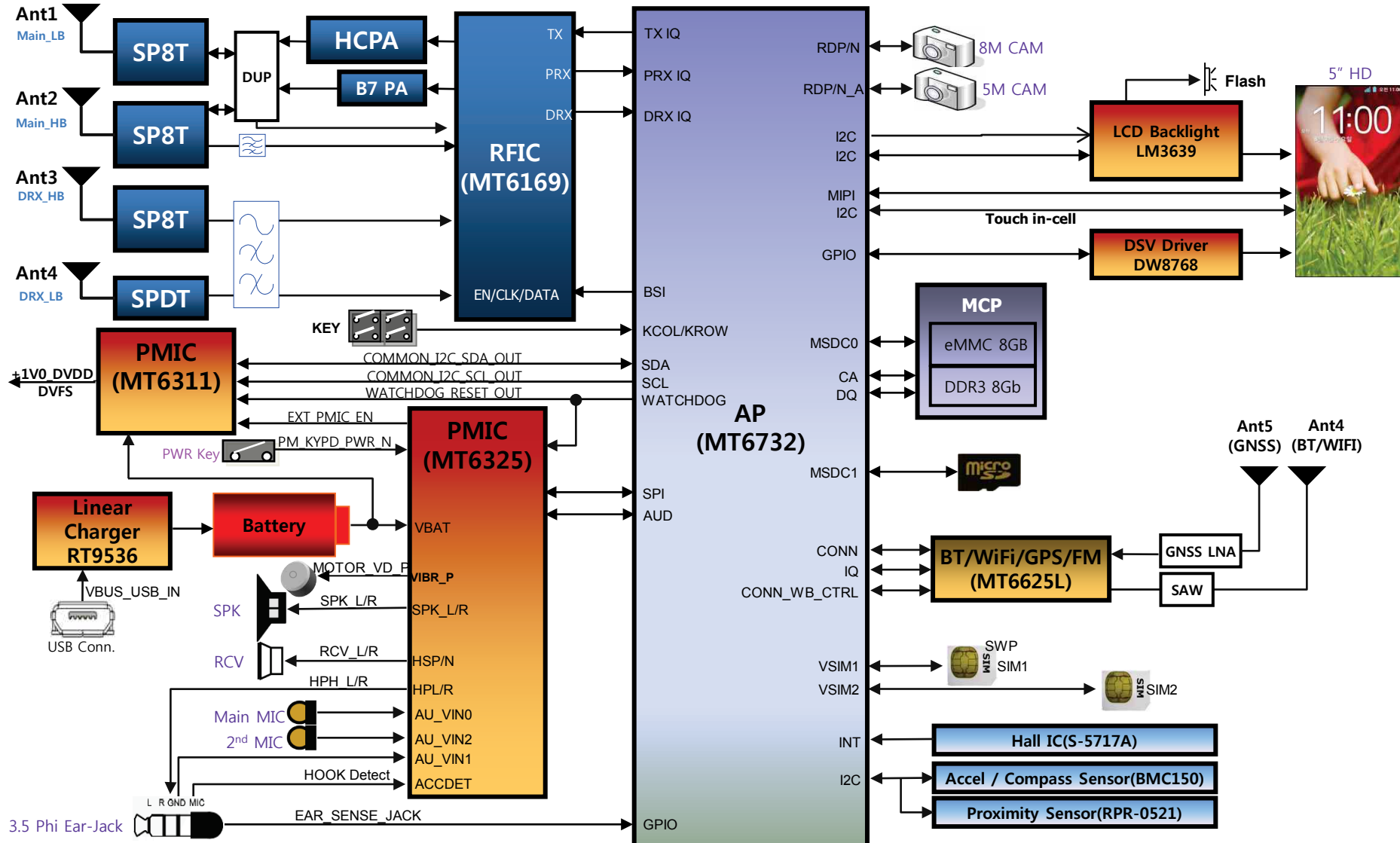


Circuit Diagram



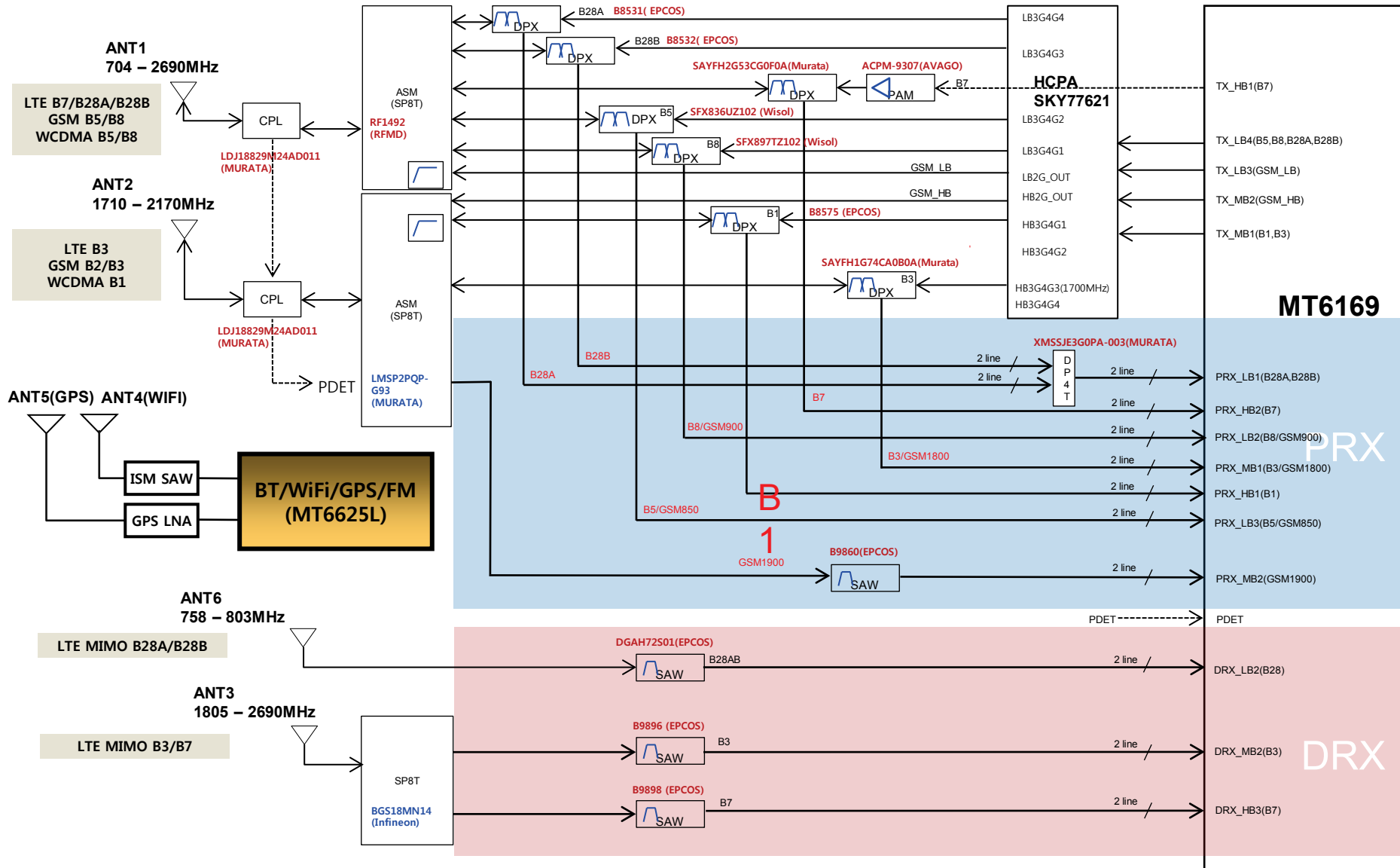
4. BLOCK DIAGRAM

1. Total block diagram

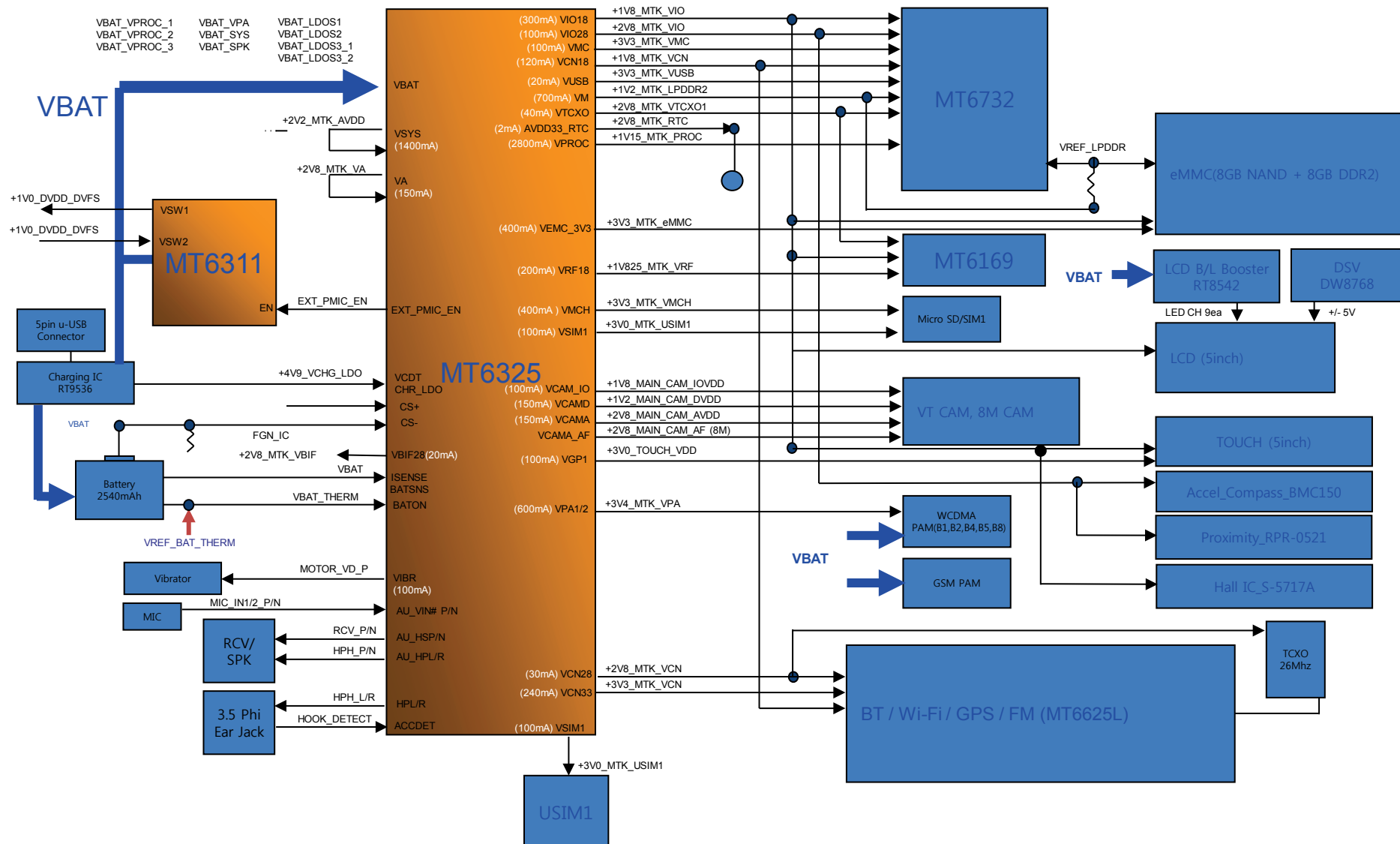


4. BLOCK DIAGRAM

2. RF Main

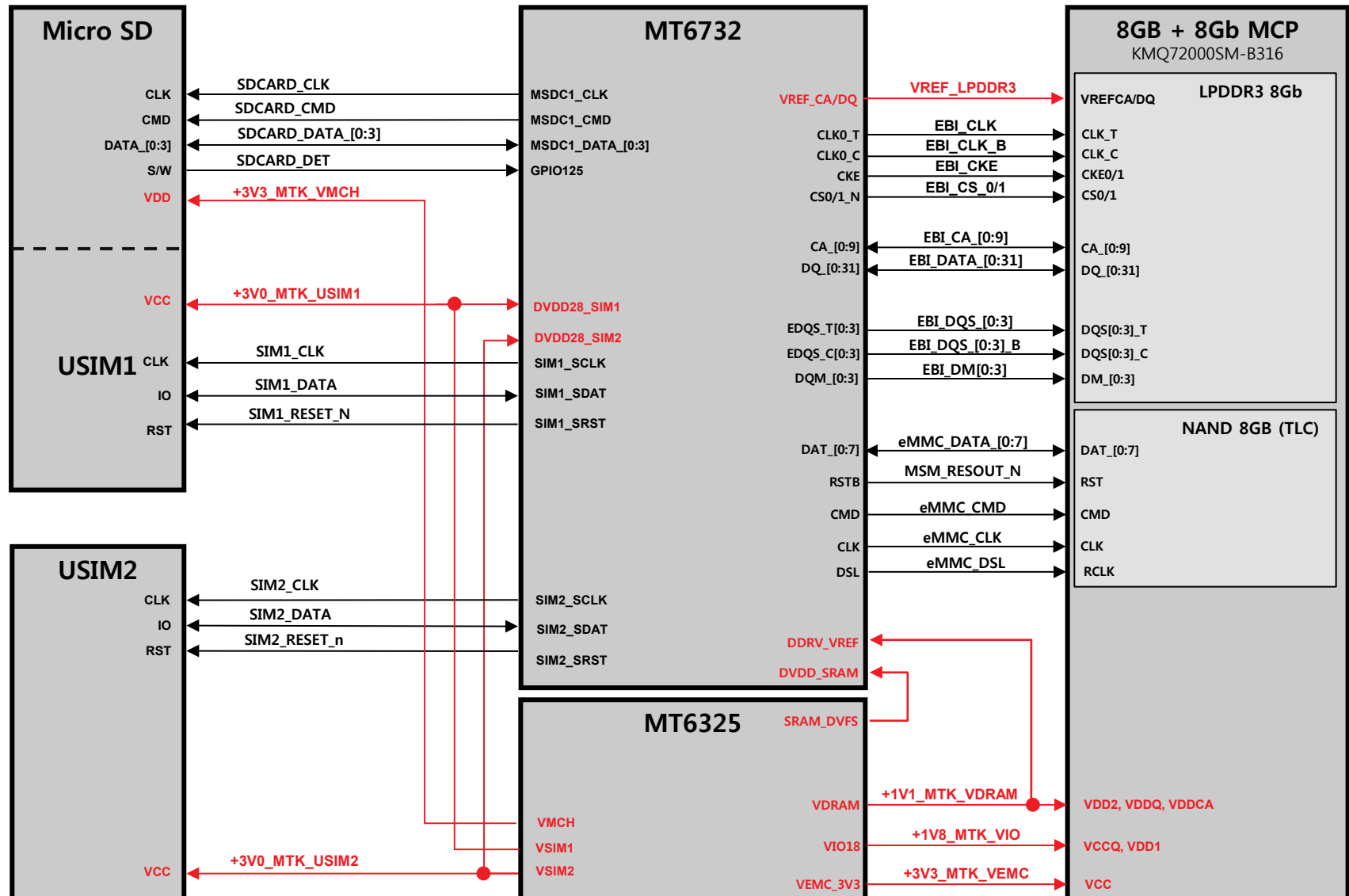


3. Power block diagram

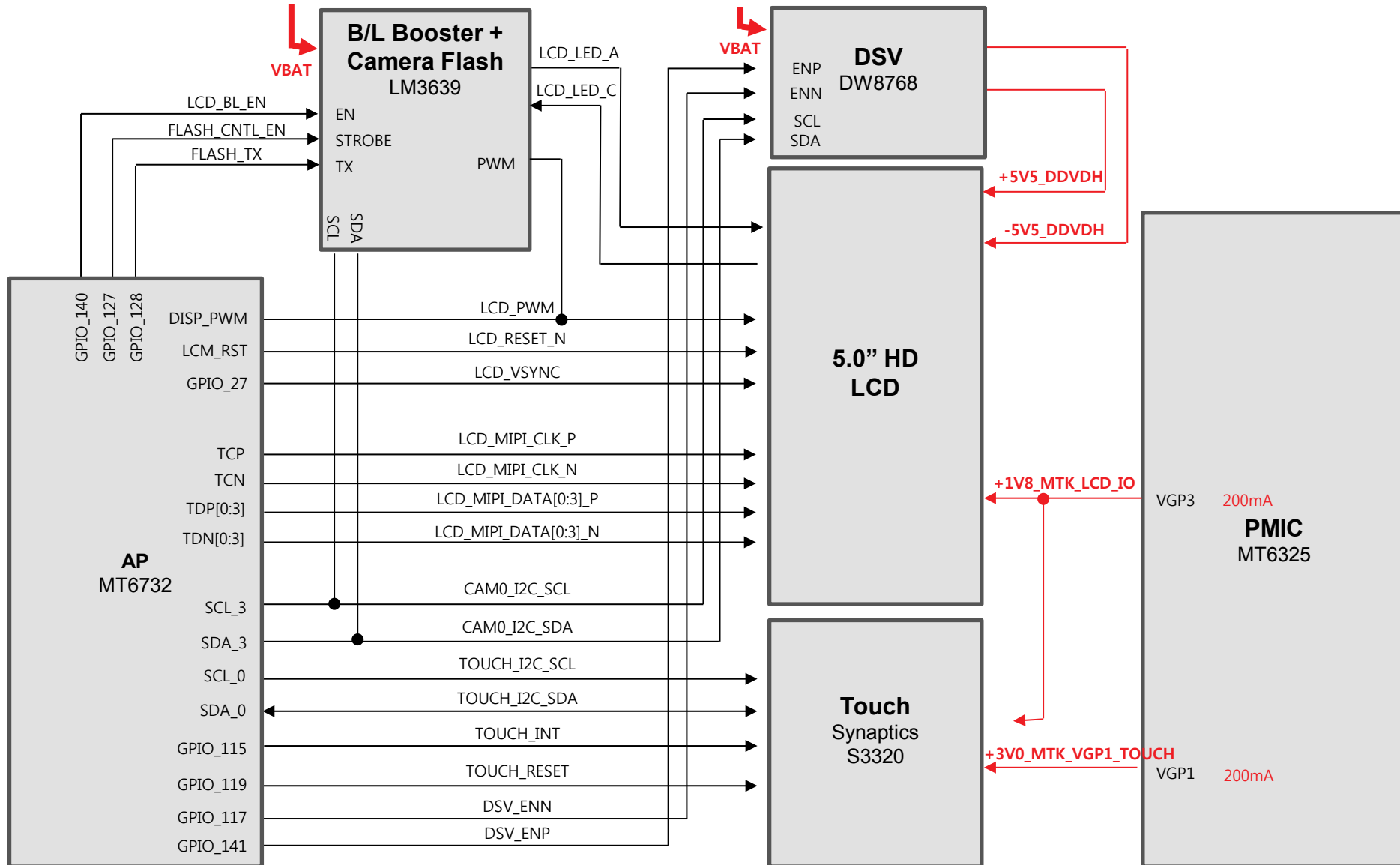


4. BLOCK DIAGRAM

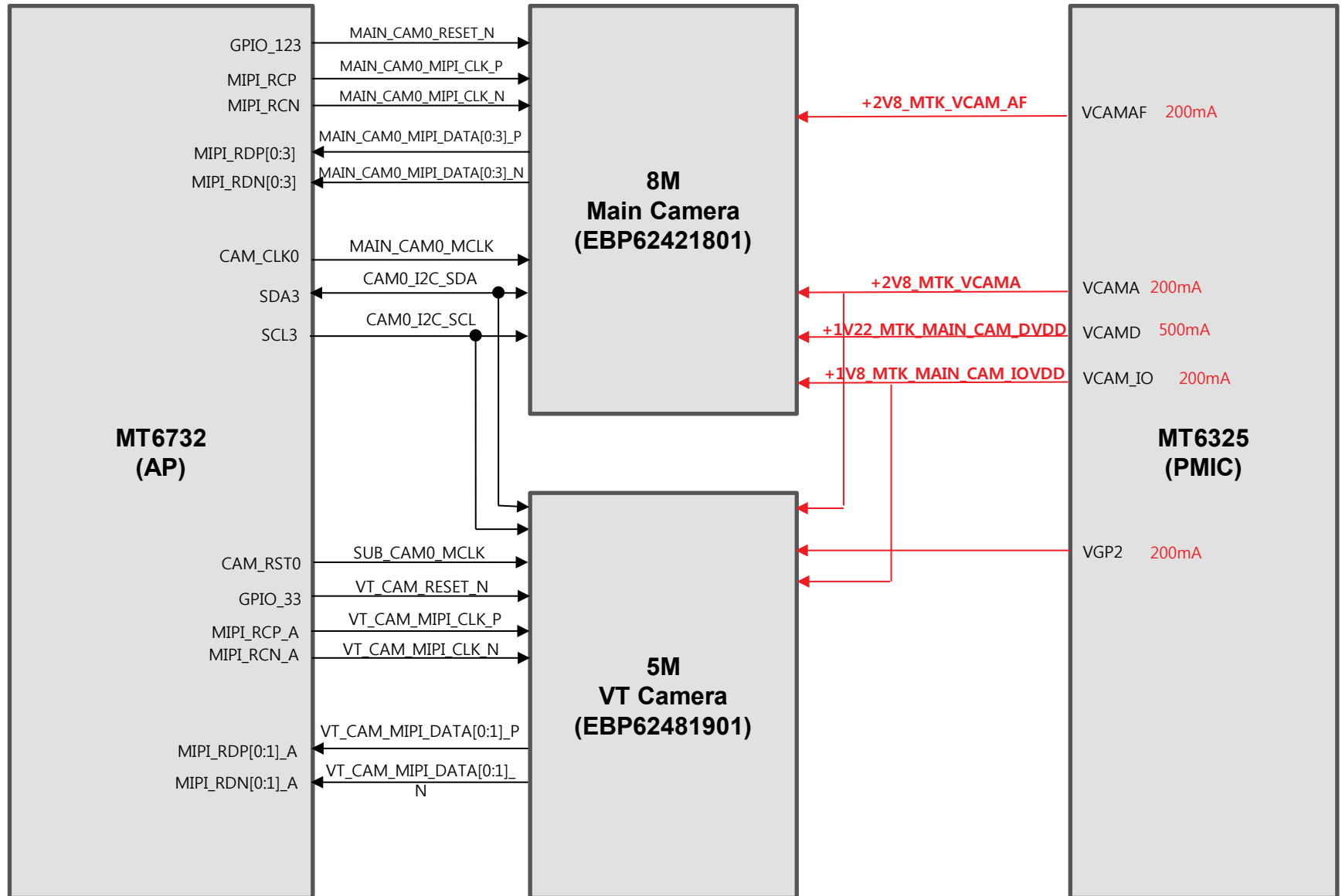
4. Memory / USIM / Micro SD Combo



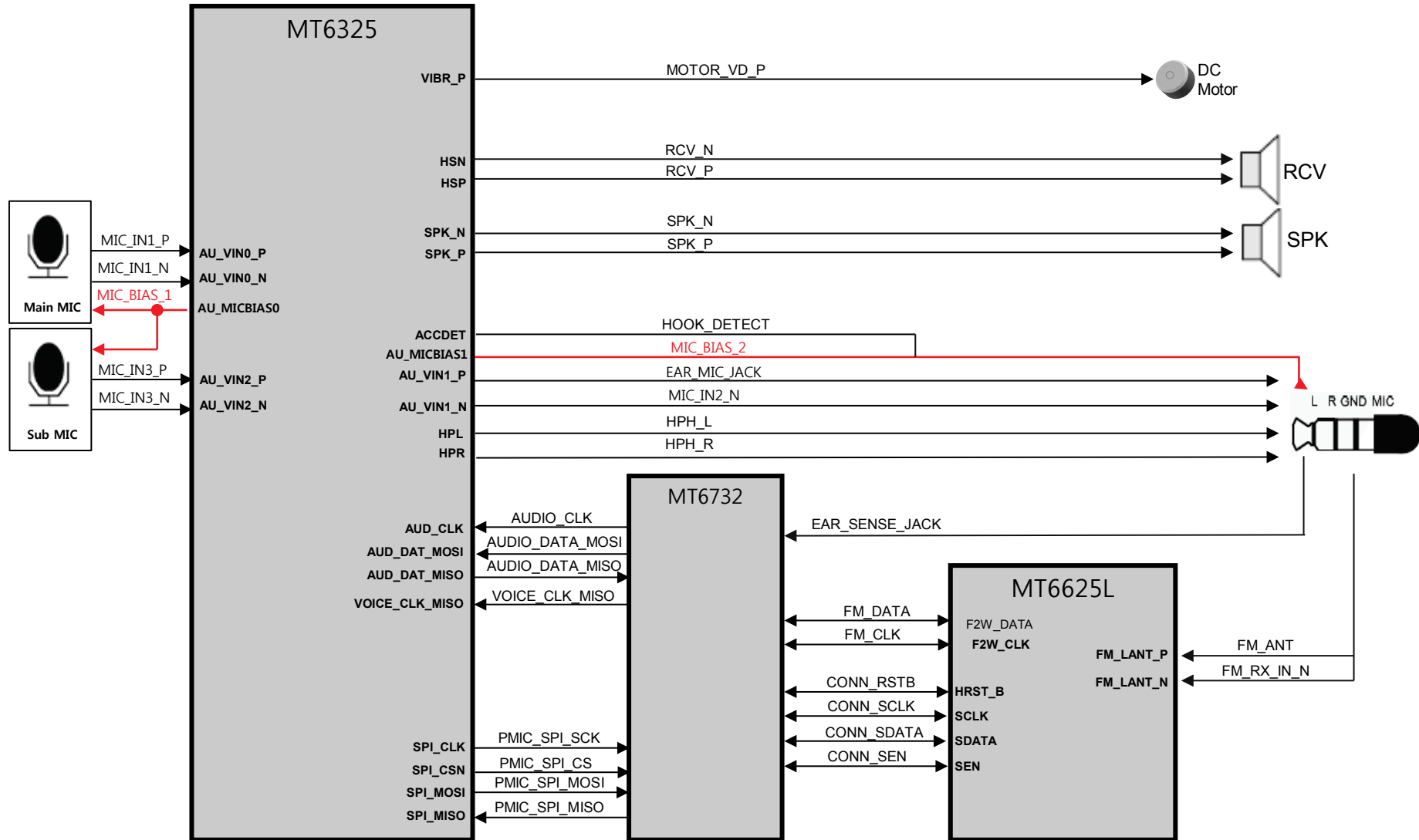
5. LCD(INCELL TOUCH)



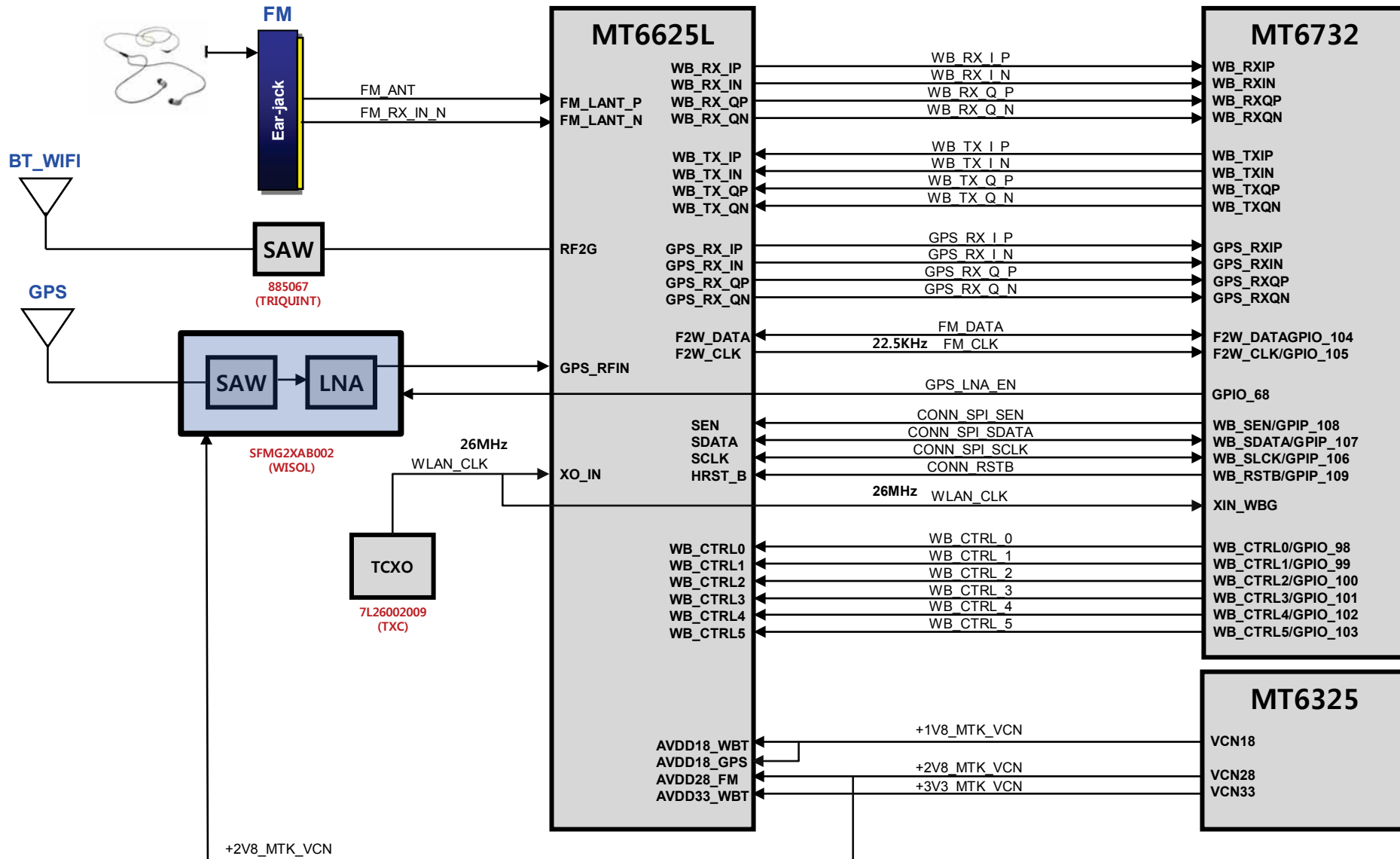
6. Camera



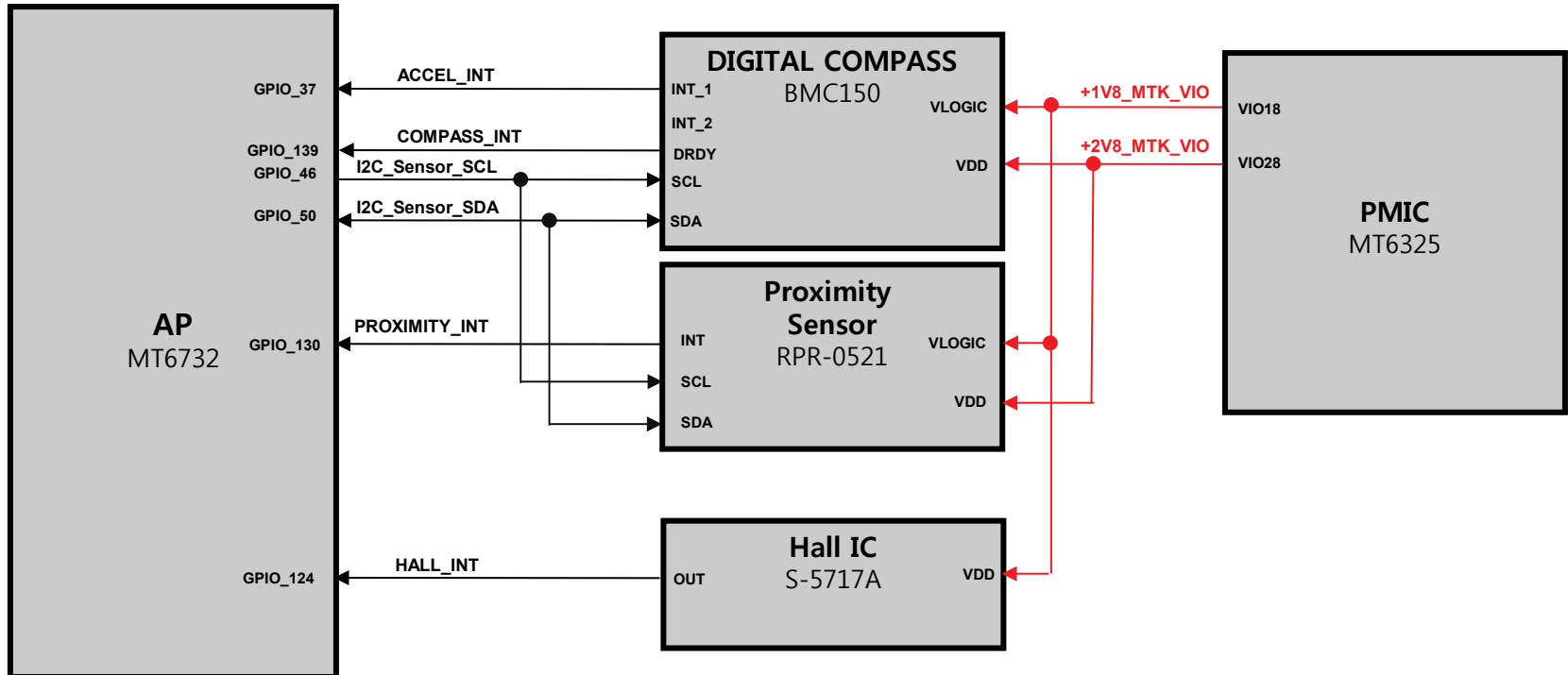
7. AUDIO



8. Connectivity (BT / WIFI / GPS / FM RADIO)

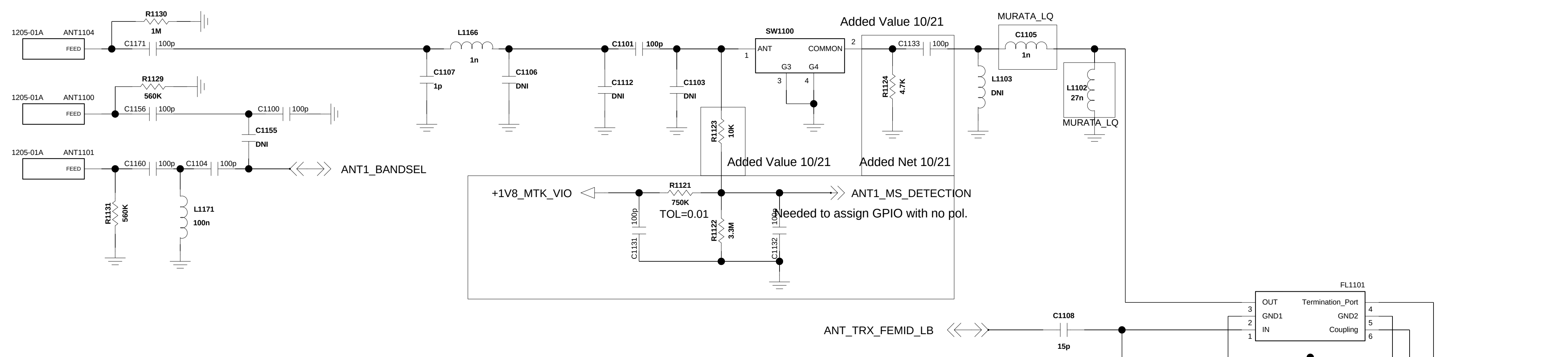


9. Sensors

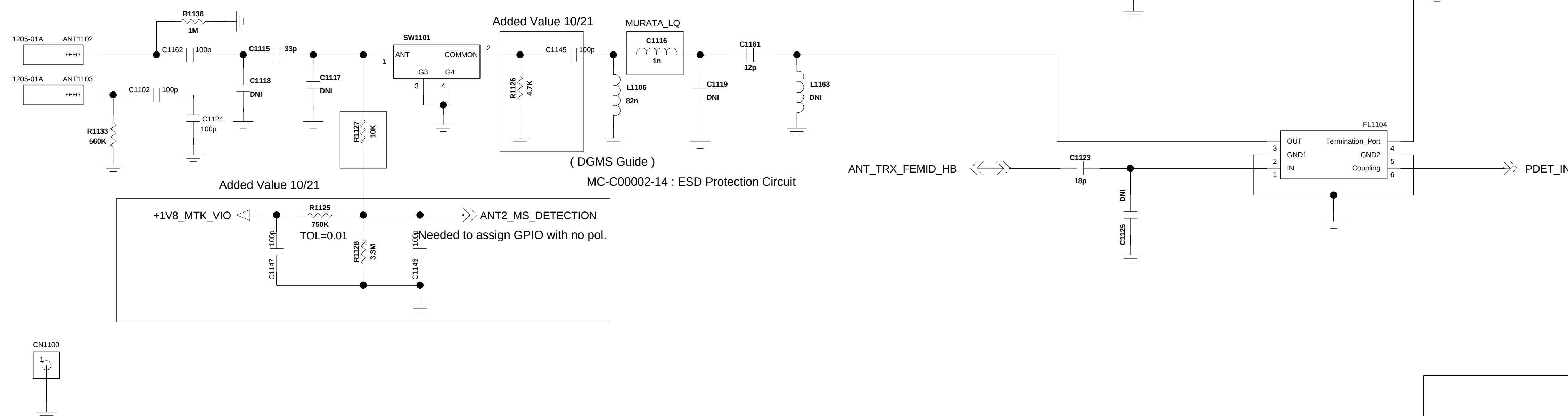


CIRCUIT DIAGRAM

ANT1 (G850/900, W_B5/B8, LTE_B7/B28A/B28B)



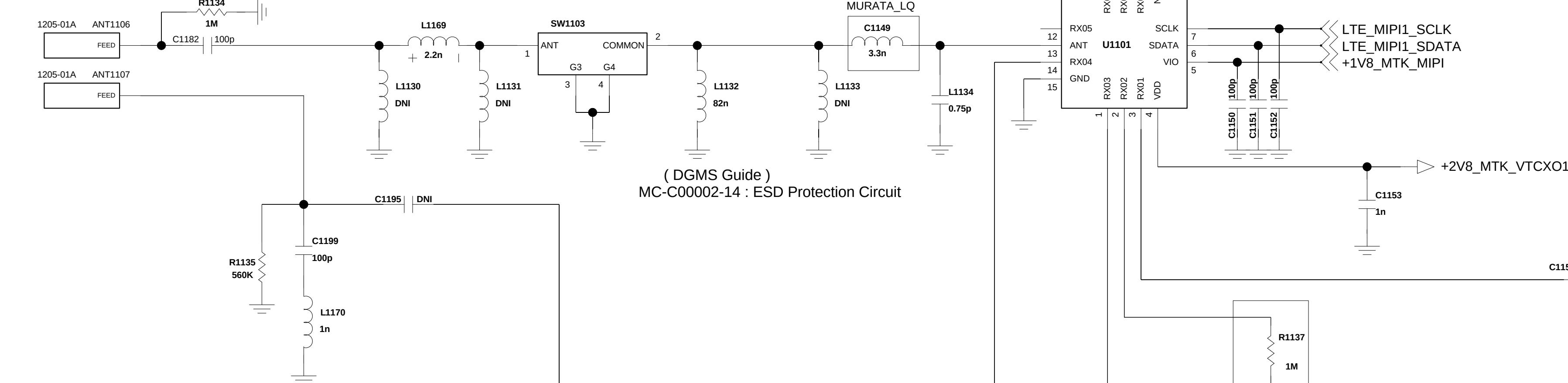
ANT2 (G1800/1900, W_B1, LTE_B3)



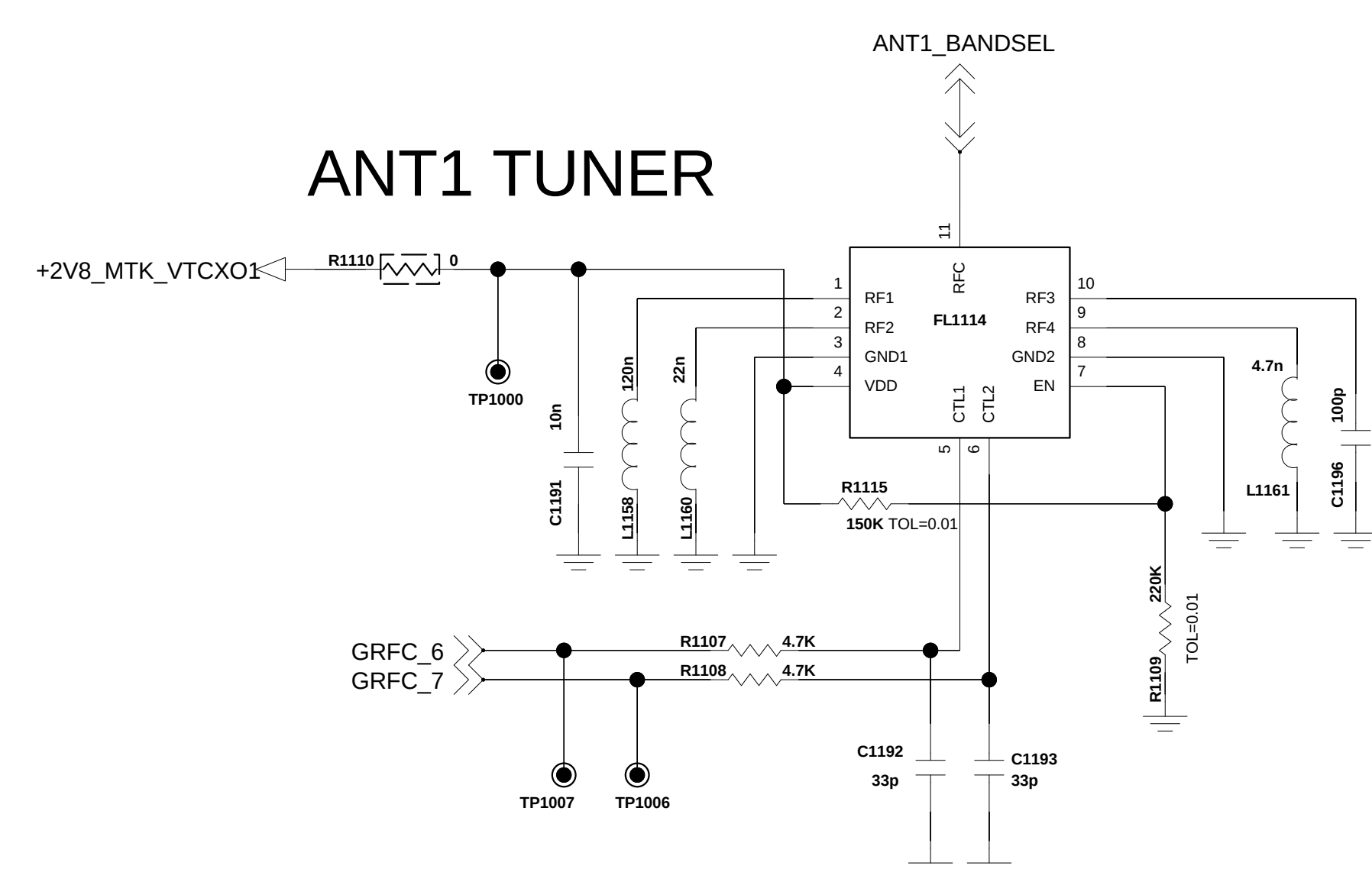
ANT6 (B28 2nd)



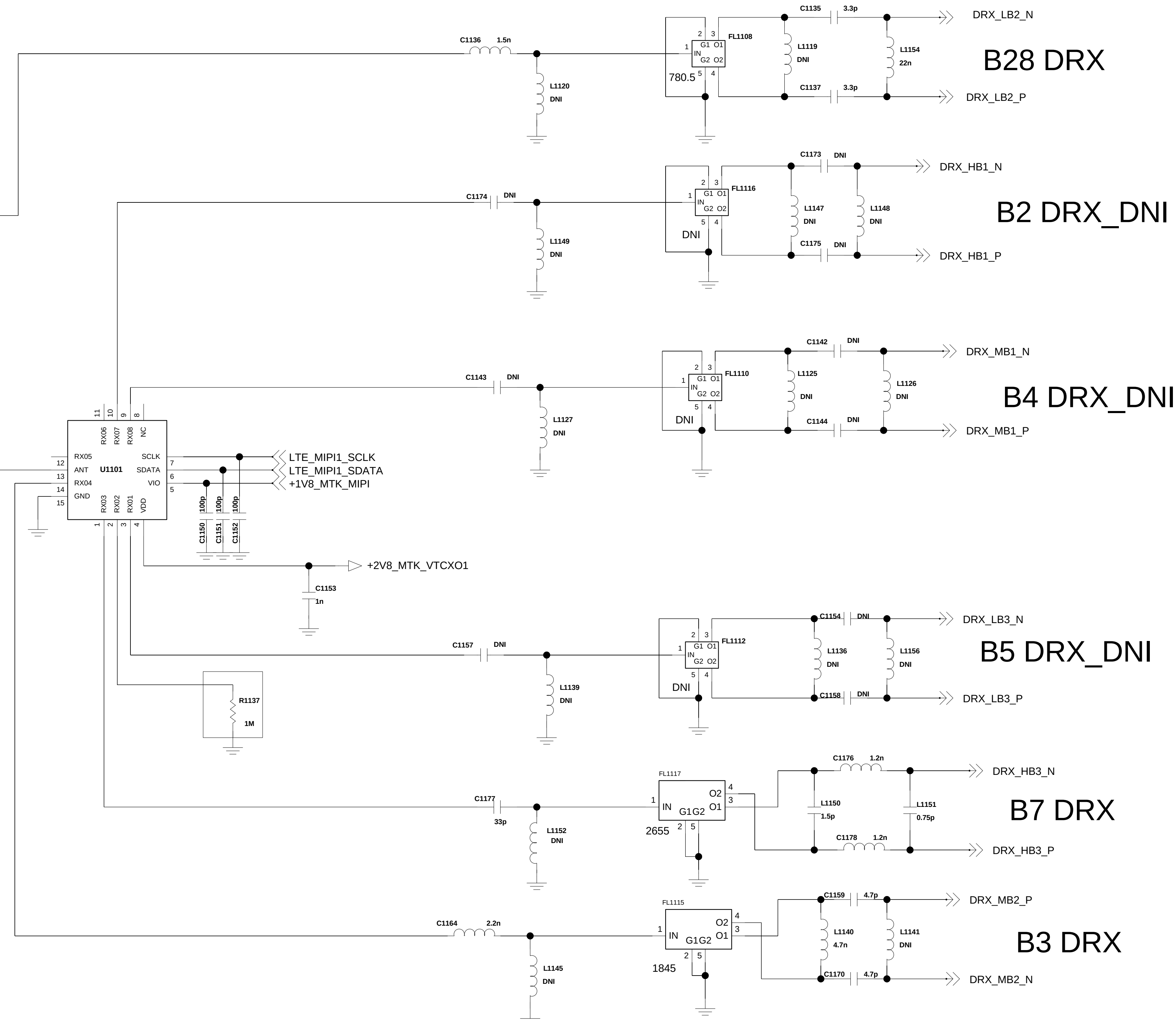
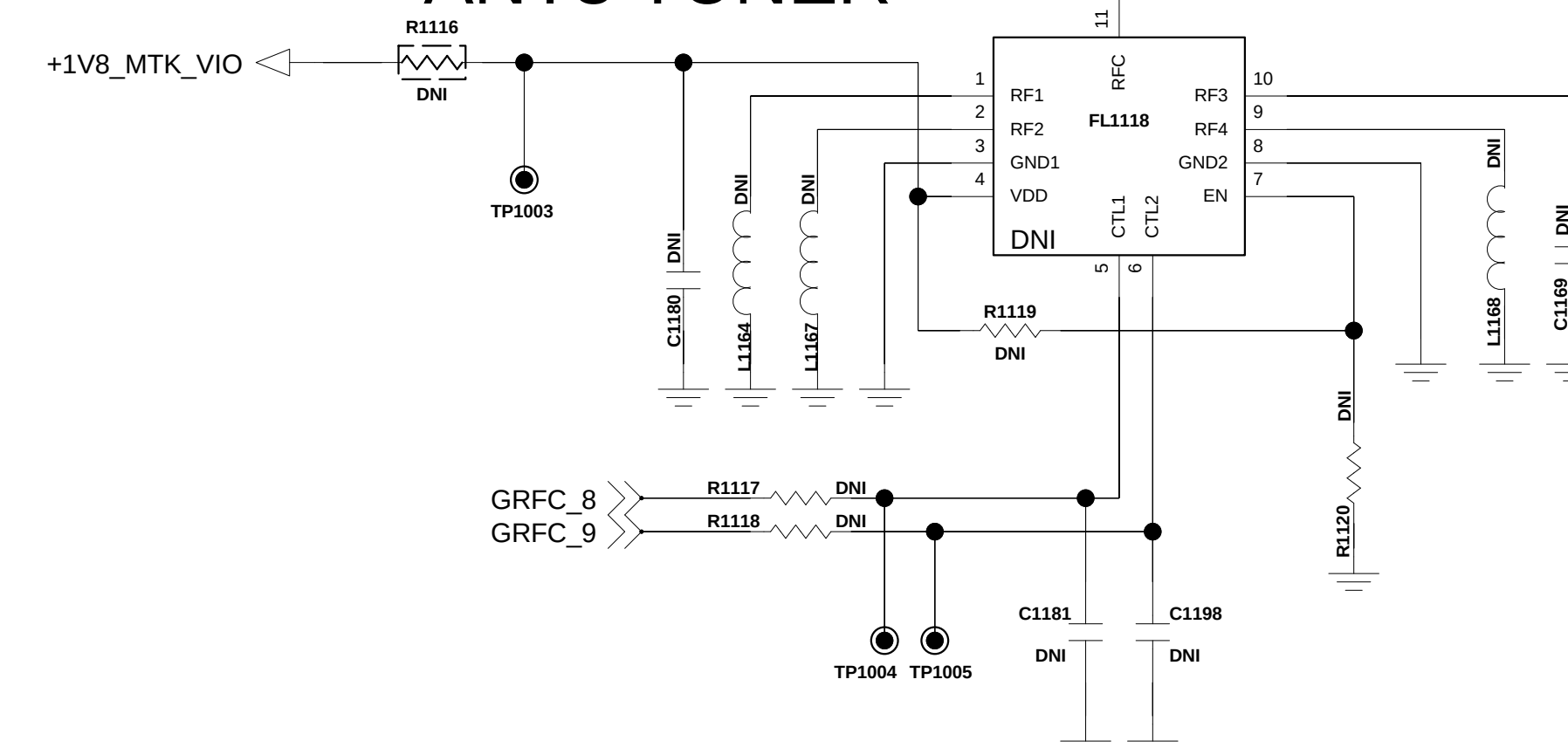
ANT3 (B3,B7 2nd)



ANT1 TUNER

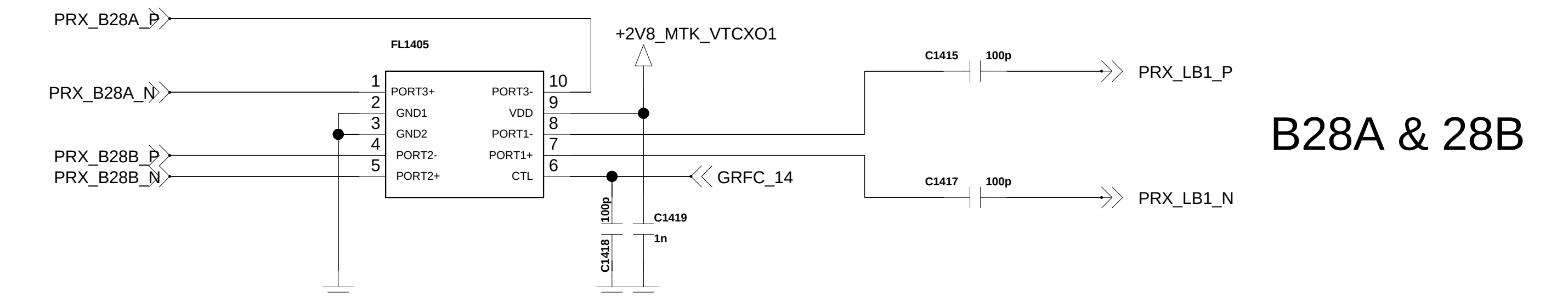
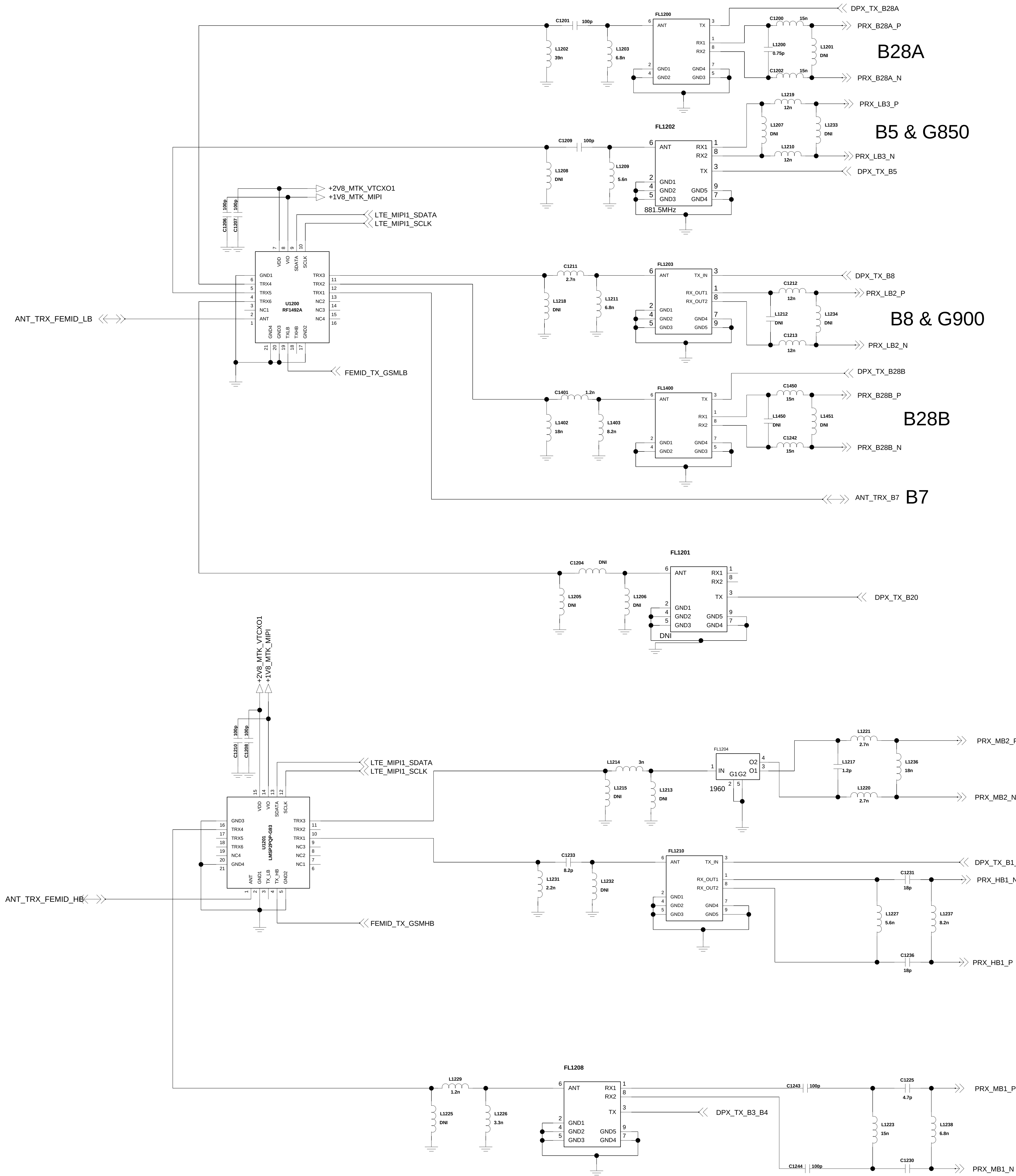


ANT3 TUNER



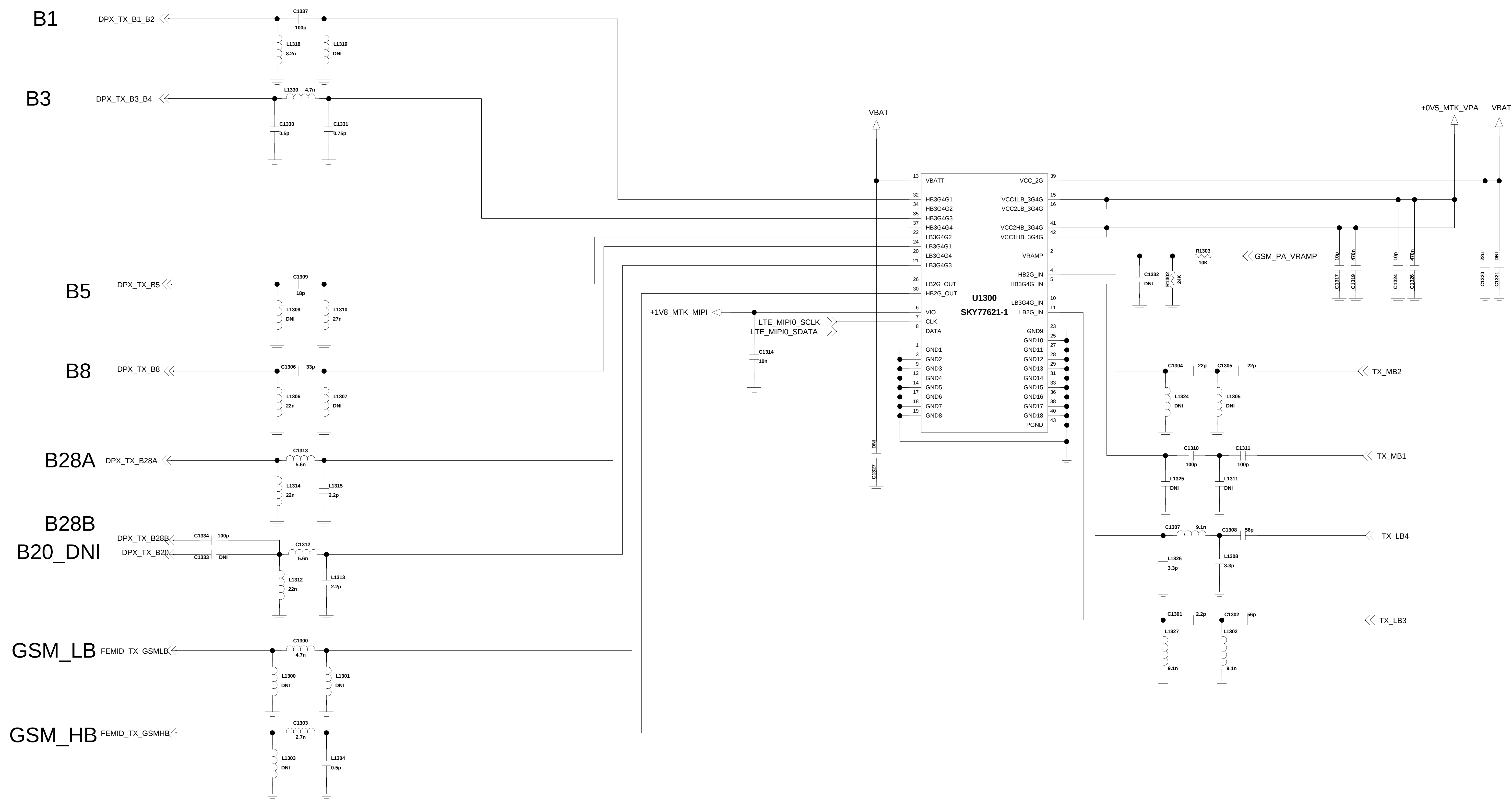
< 1-x-2-1_ ASM >

Rev_1.0



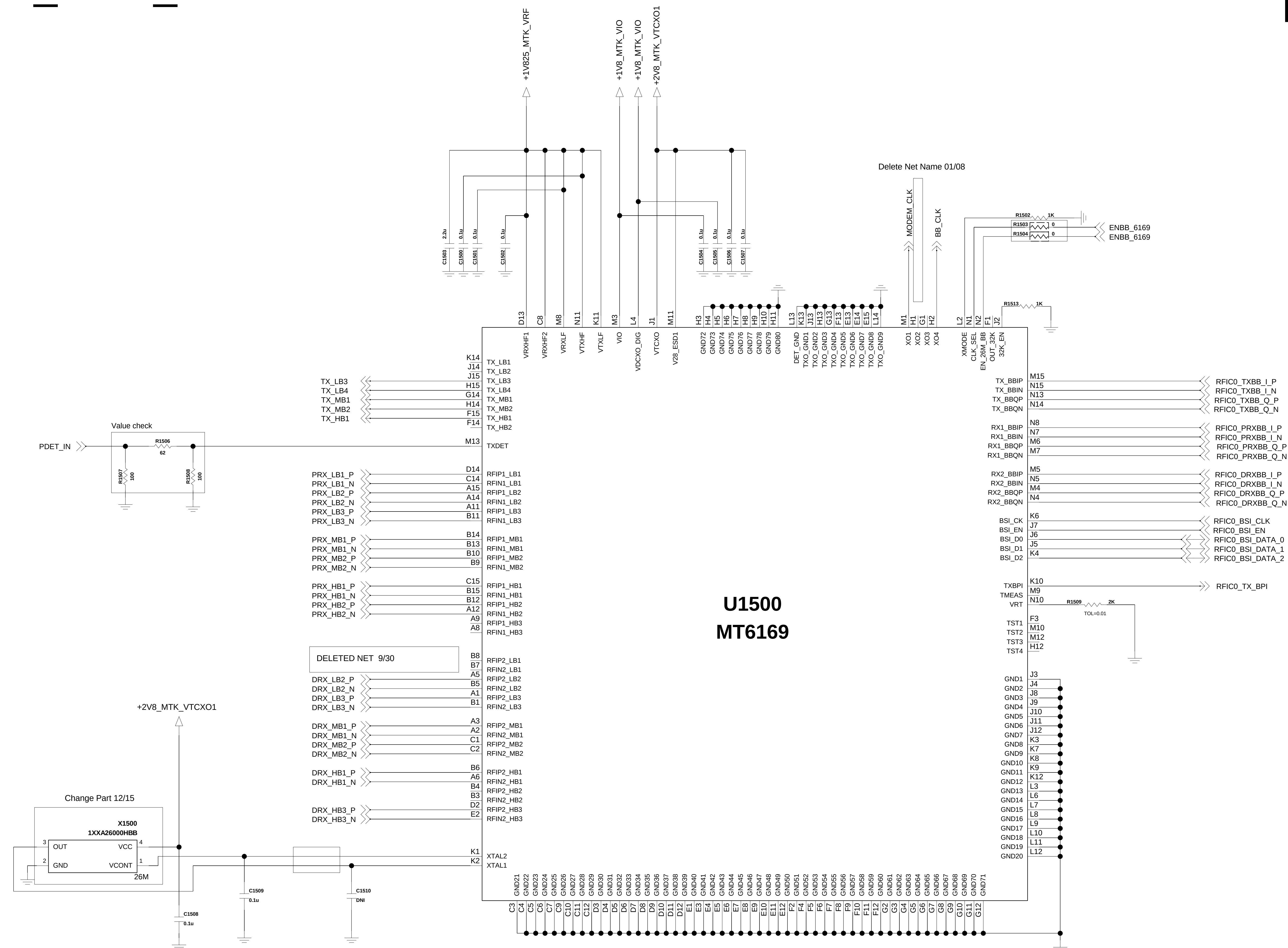
<1-6-3-1_HCPA_SKY77621>

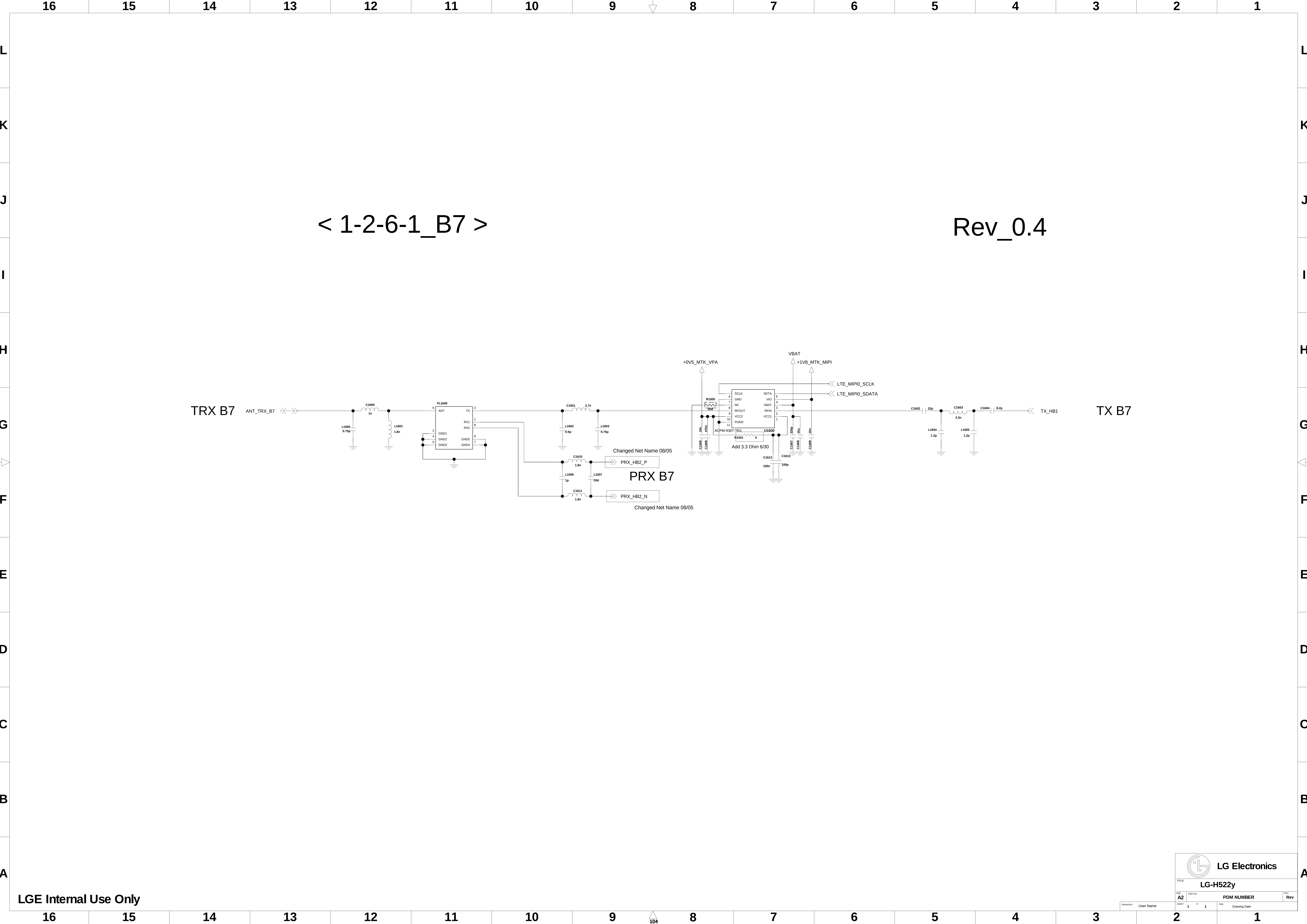
Rev_0.3



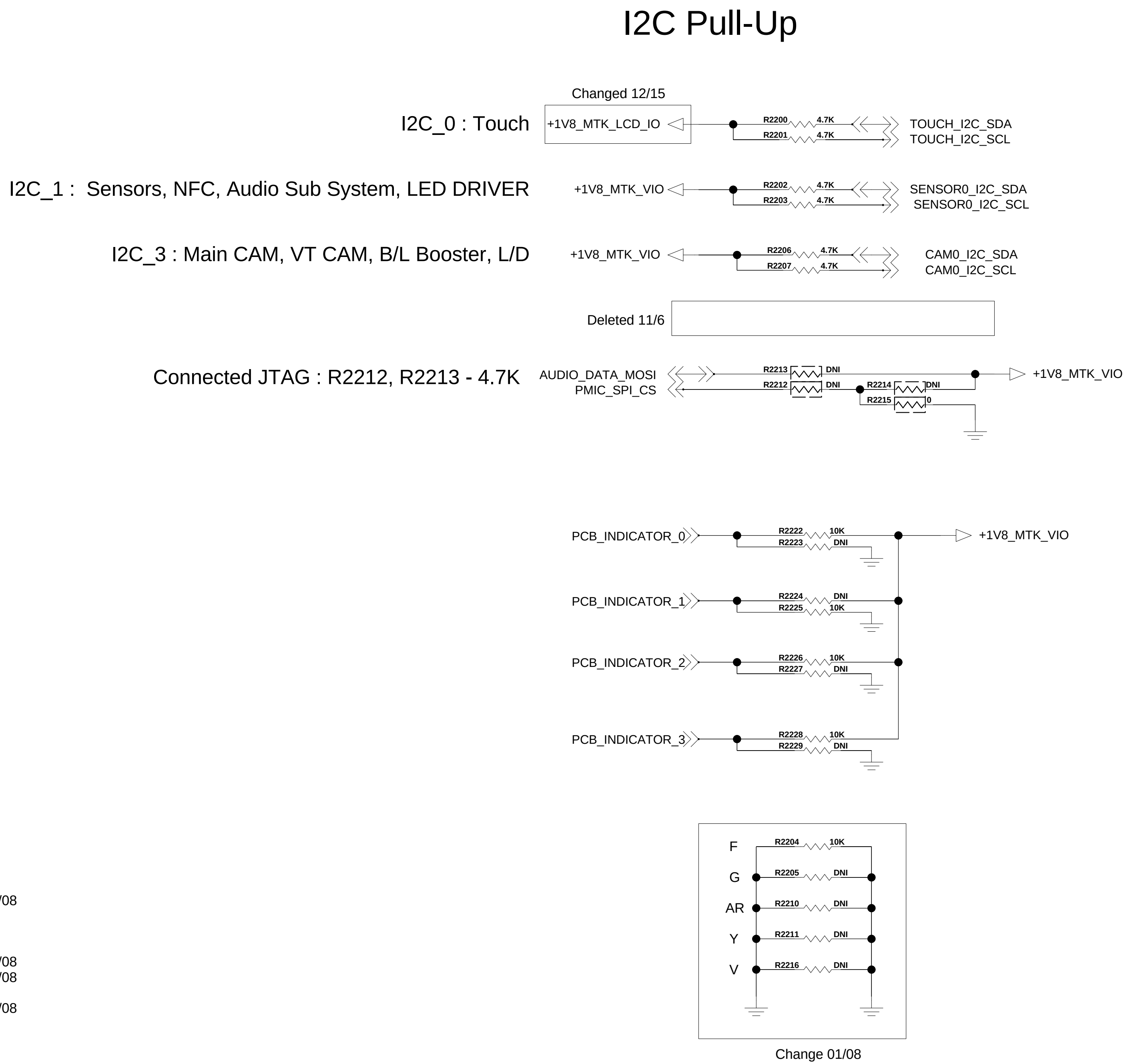
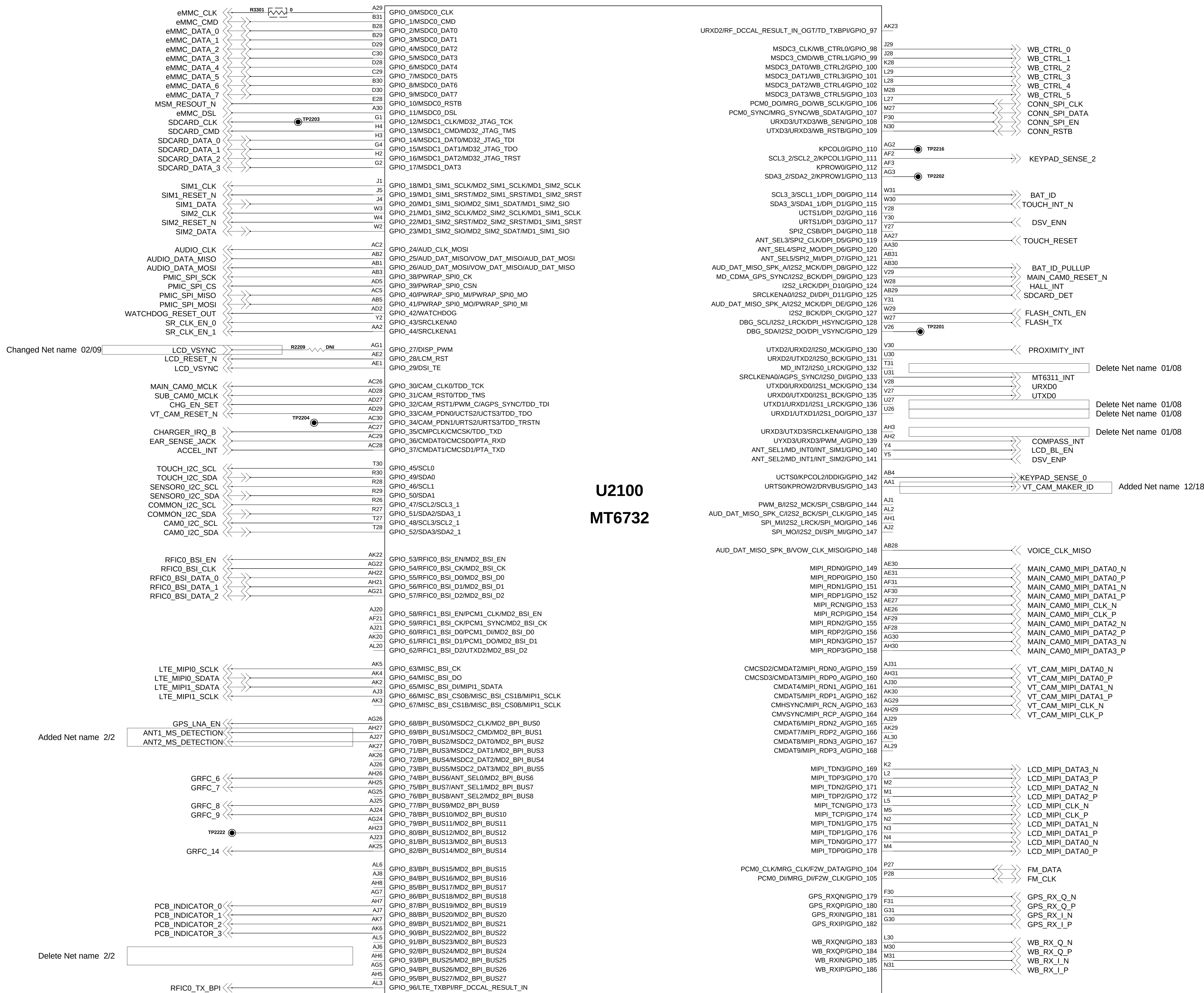
<1-9-5-1_RFIC_MT6169>

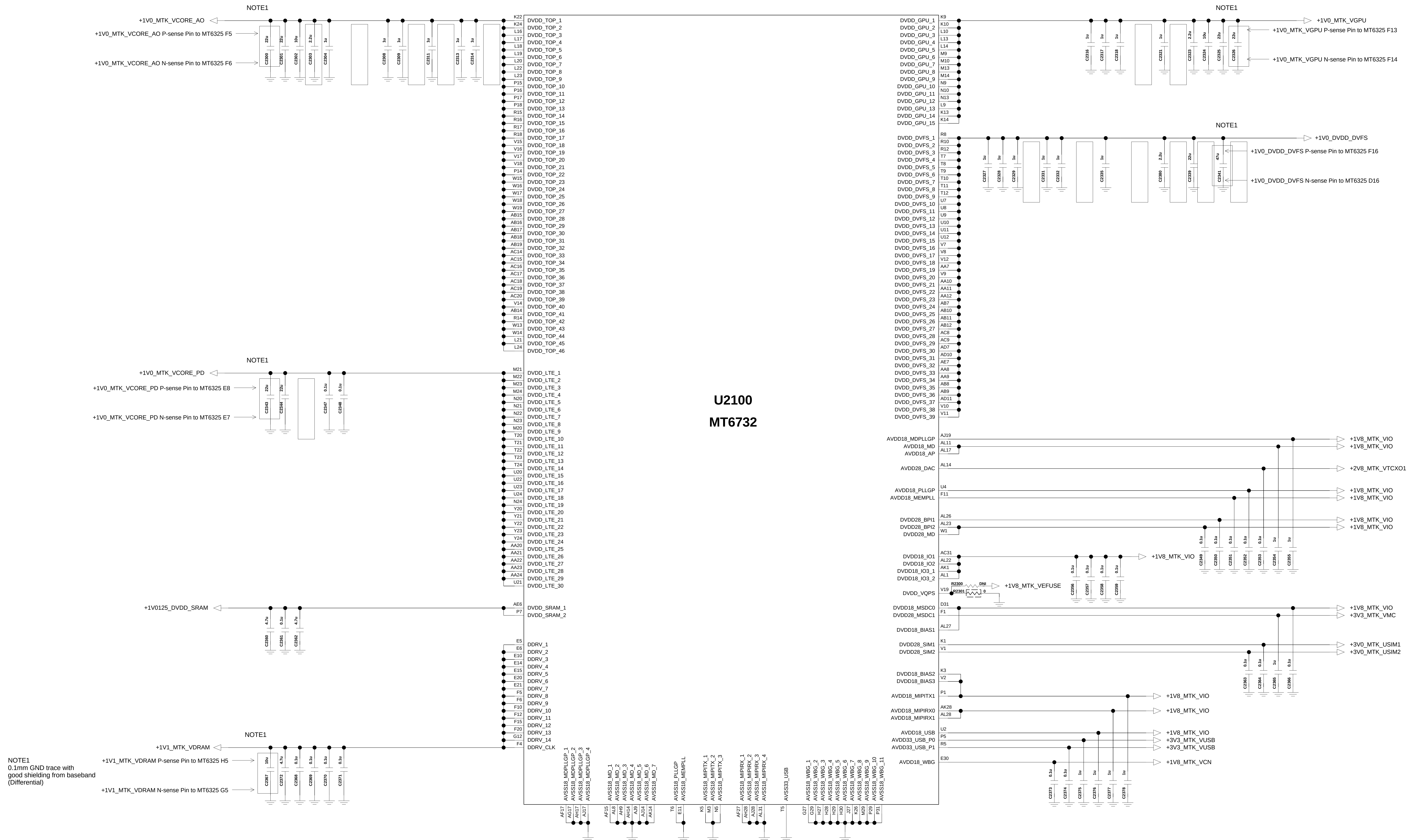
Rev_0.4

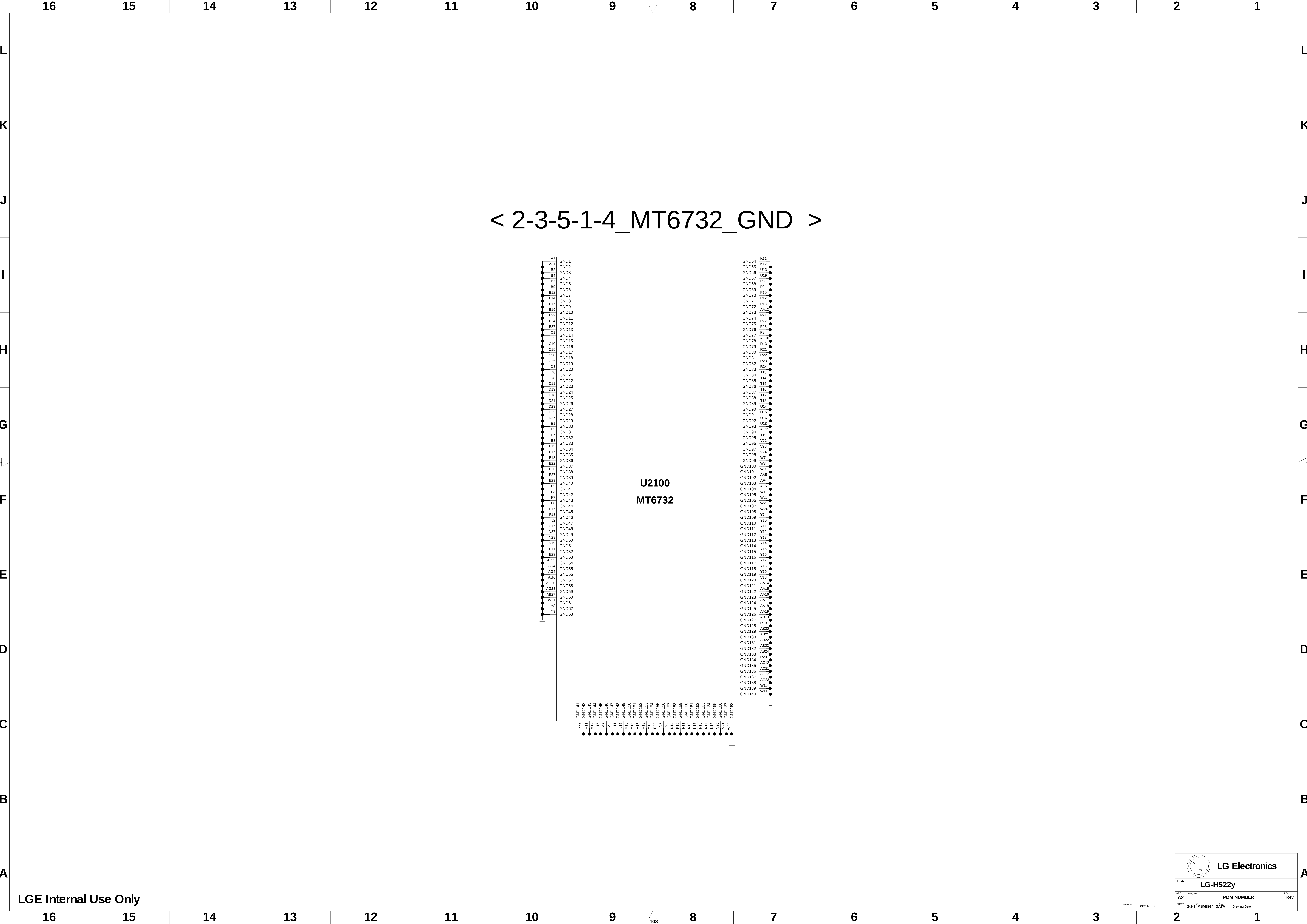




< 2-3-5-1-2_MT6732_GPIO >





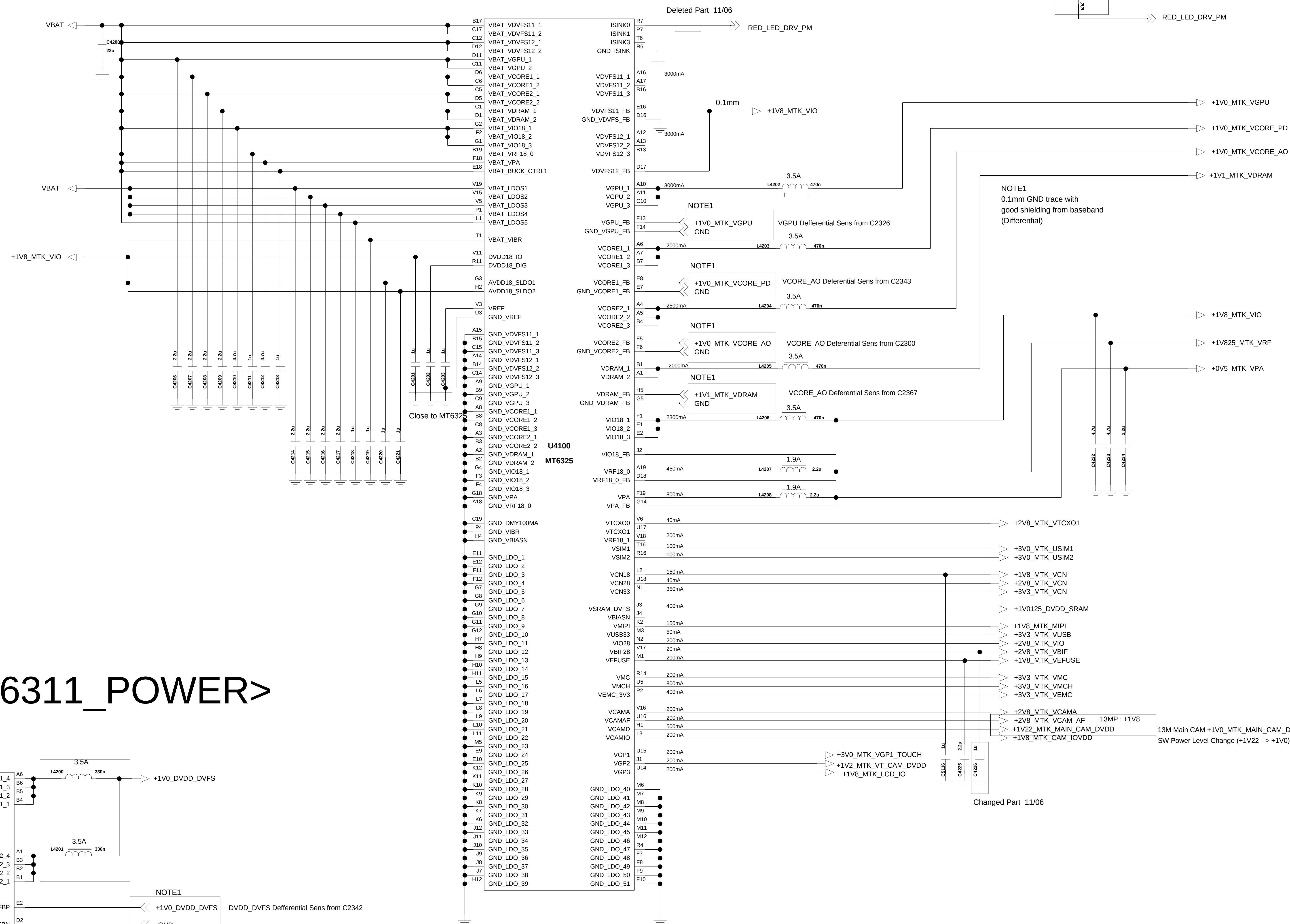


Rev_1.0

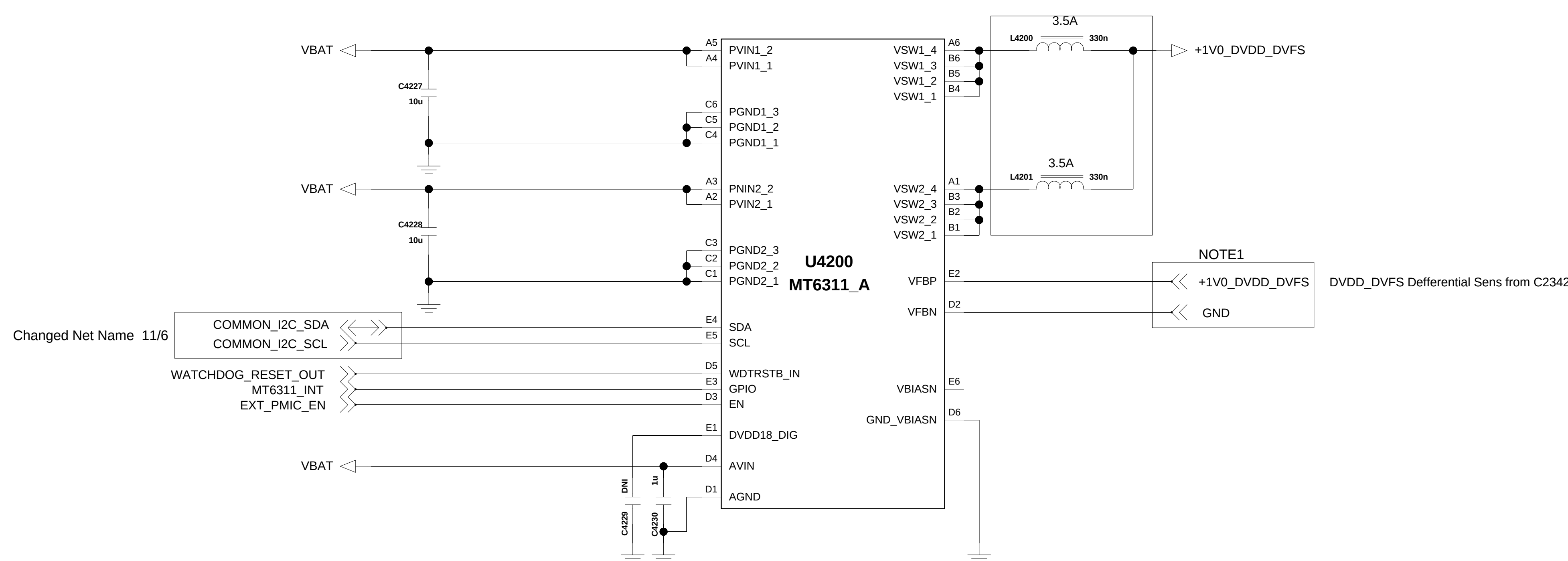


<4-1-12-2 PMIC MT6325 POWER>

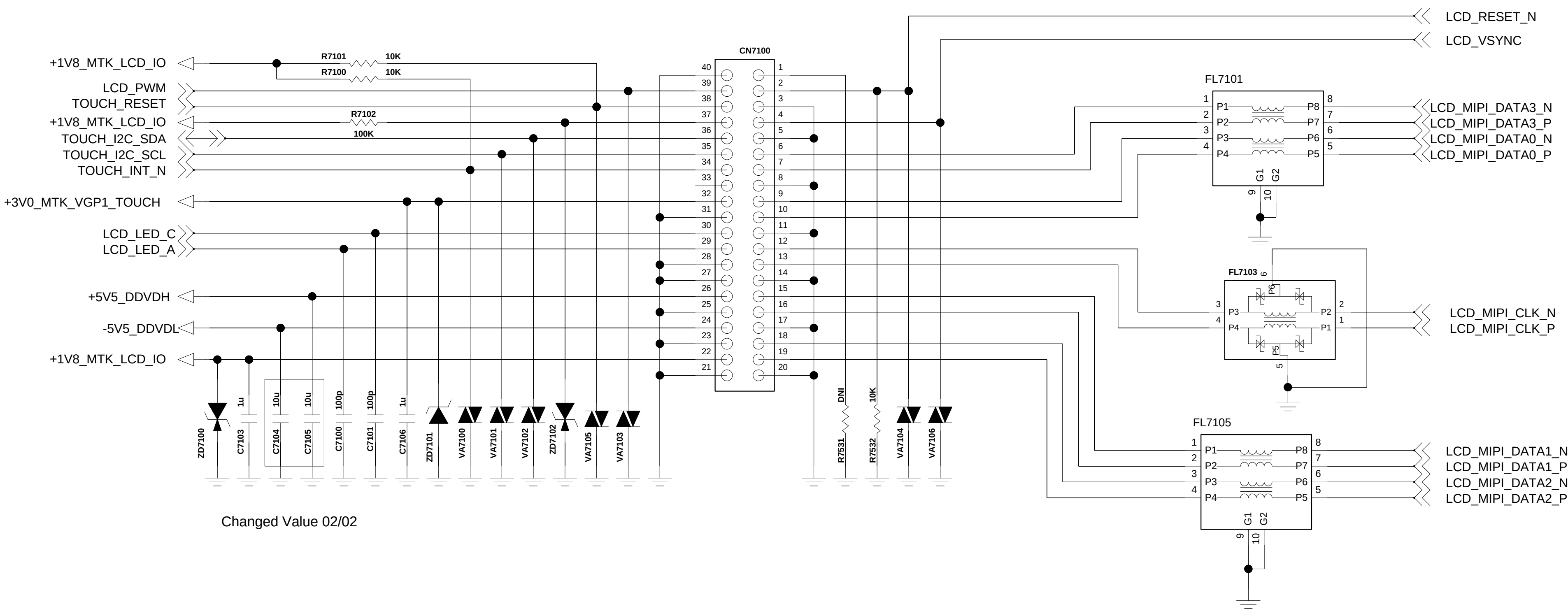
RED LED



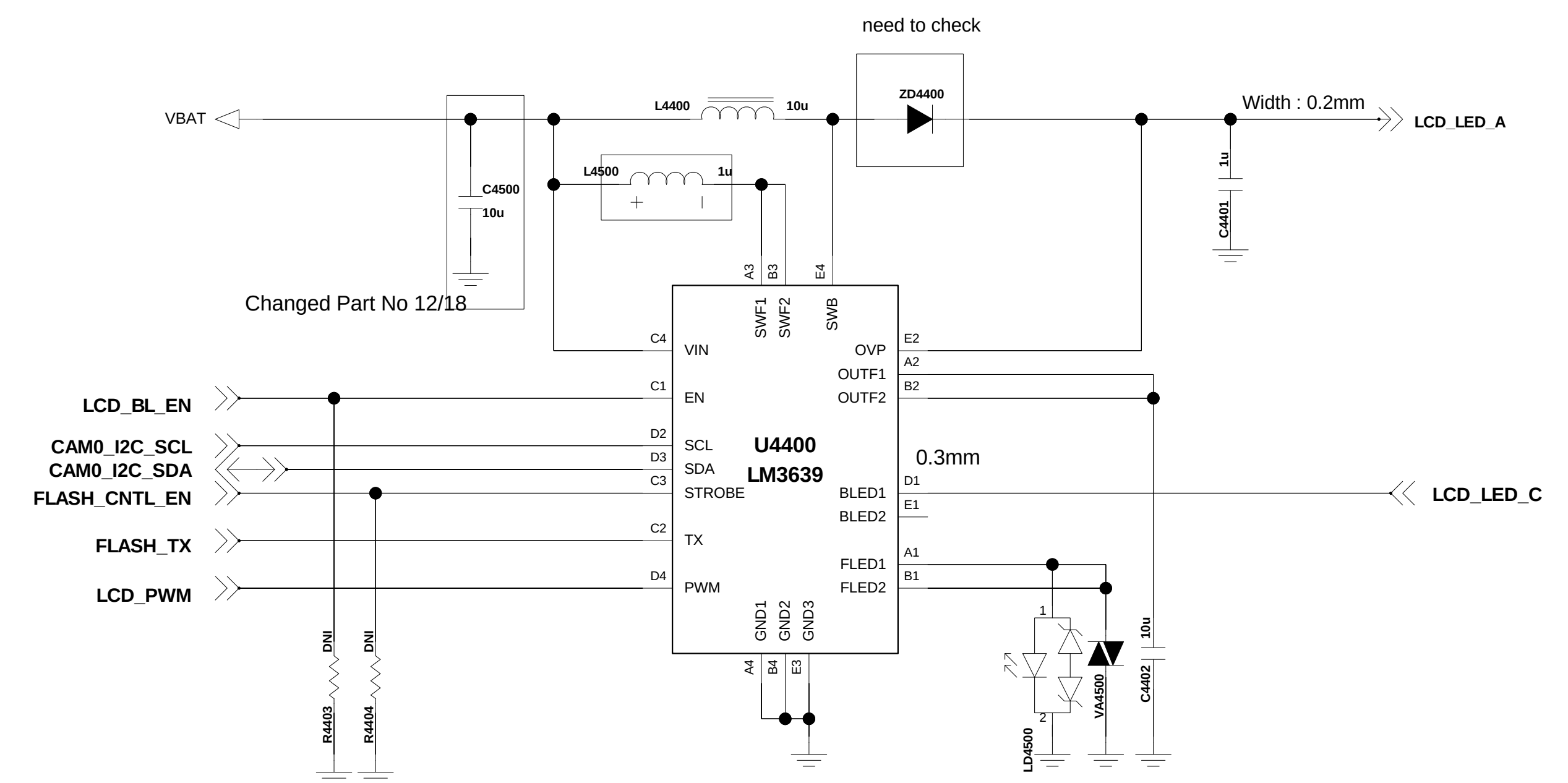
<4-2-7_2'nd PMIC_MT6311_POWER>



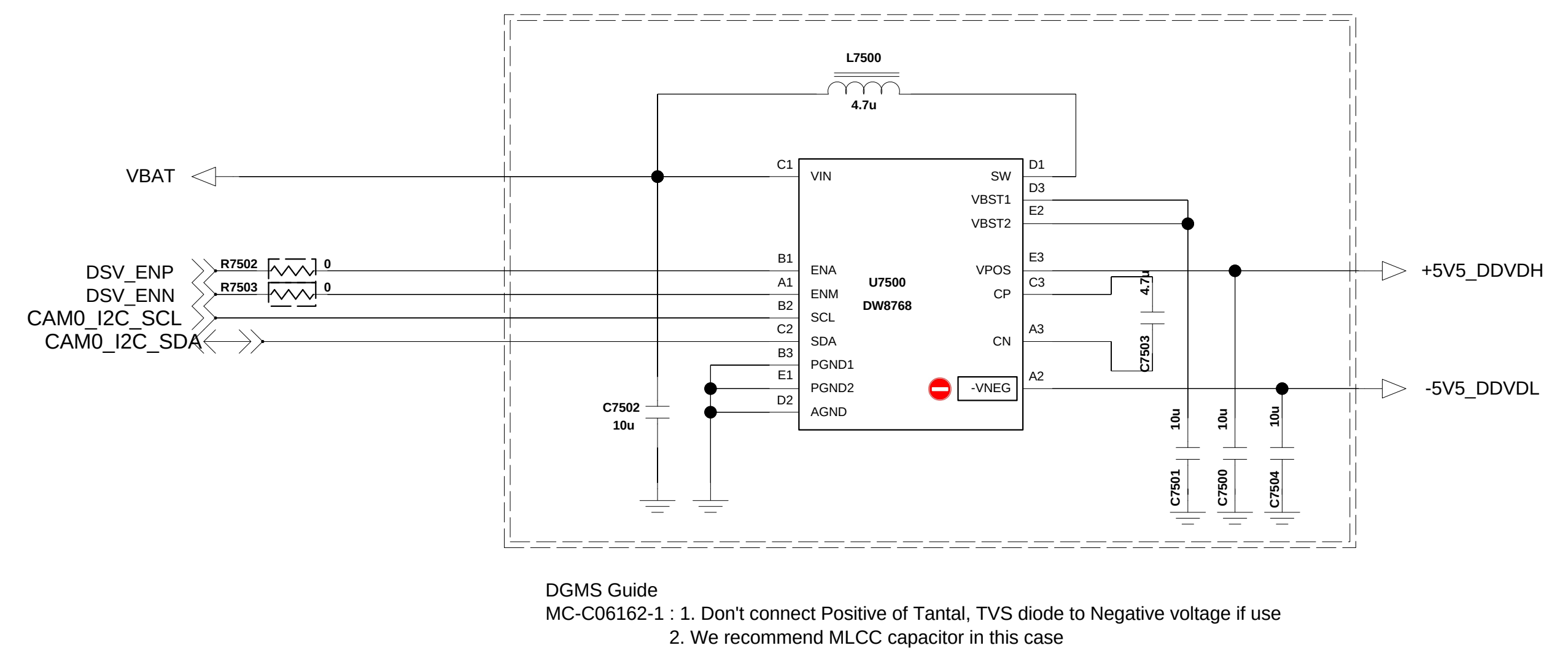
<7-1-5_LCD_HD> Rev_1.1
JDI 5.0 inch HD Hybrid LCD



<4-4-3_LCD_Backlight_IC_RT8542>

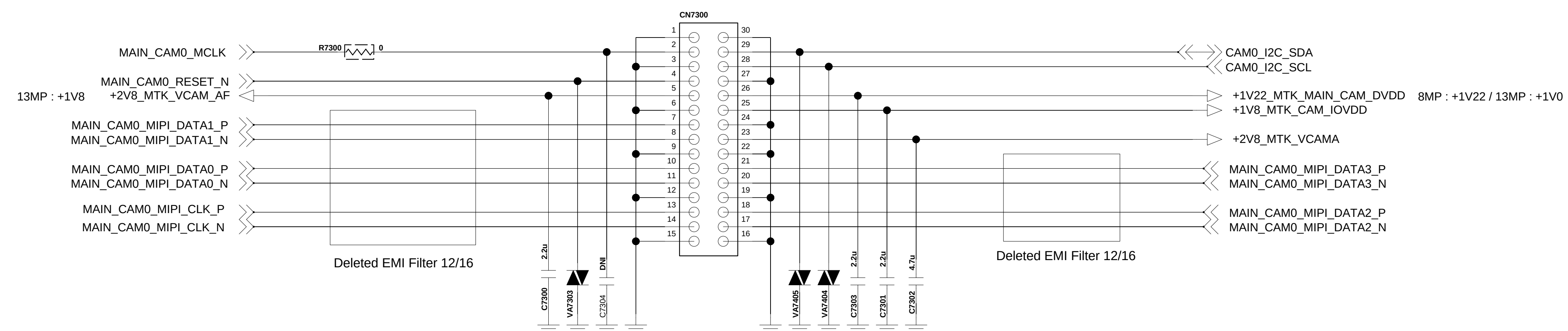


<7-5-8_DSV_DW8768> Rev_0.3

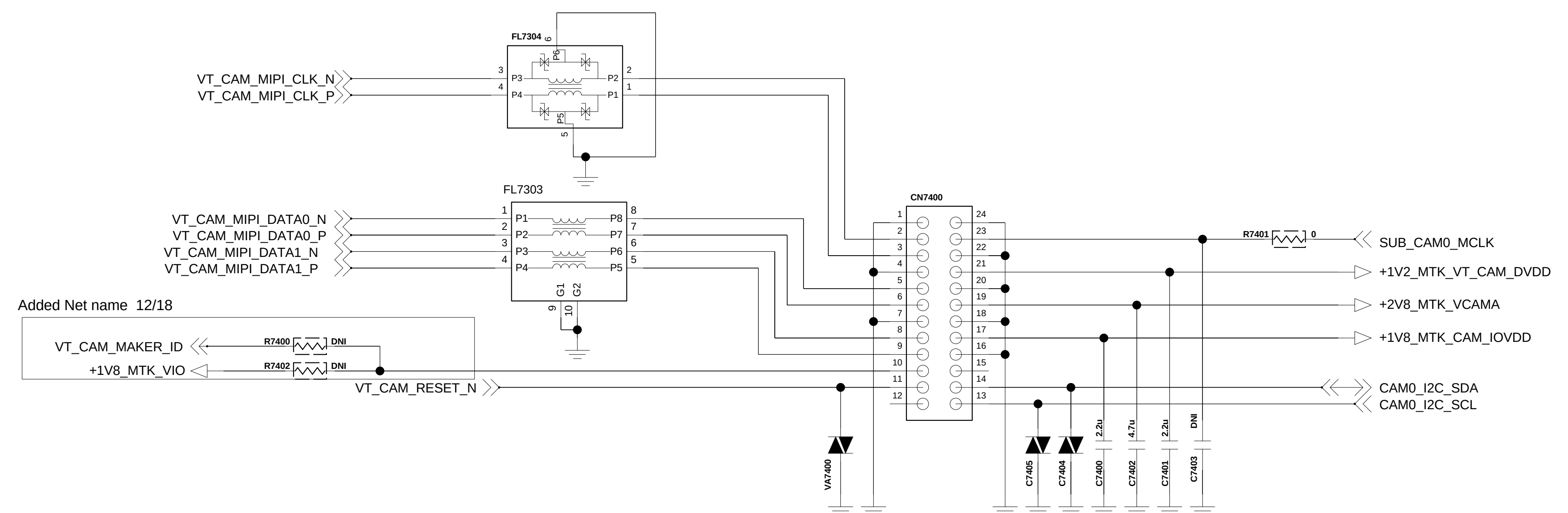


<7-3_Main_Camera> Rev_1.0

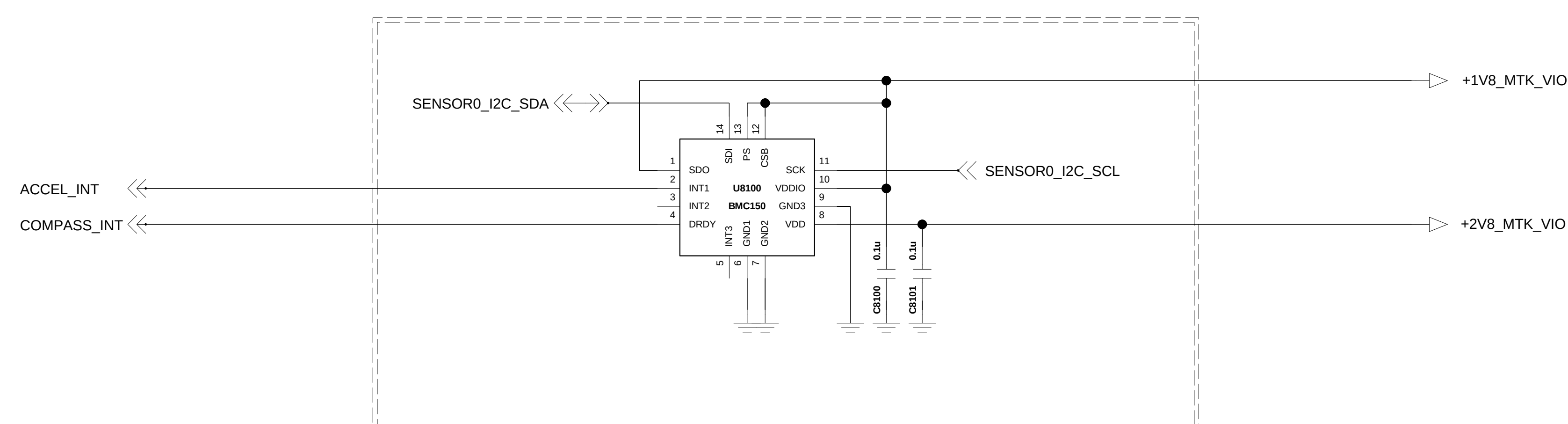
8MP camera Default



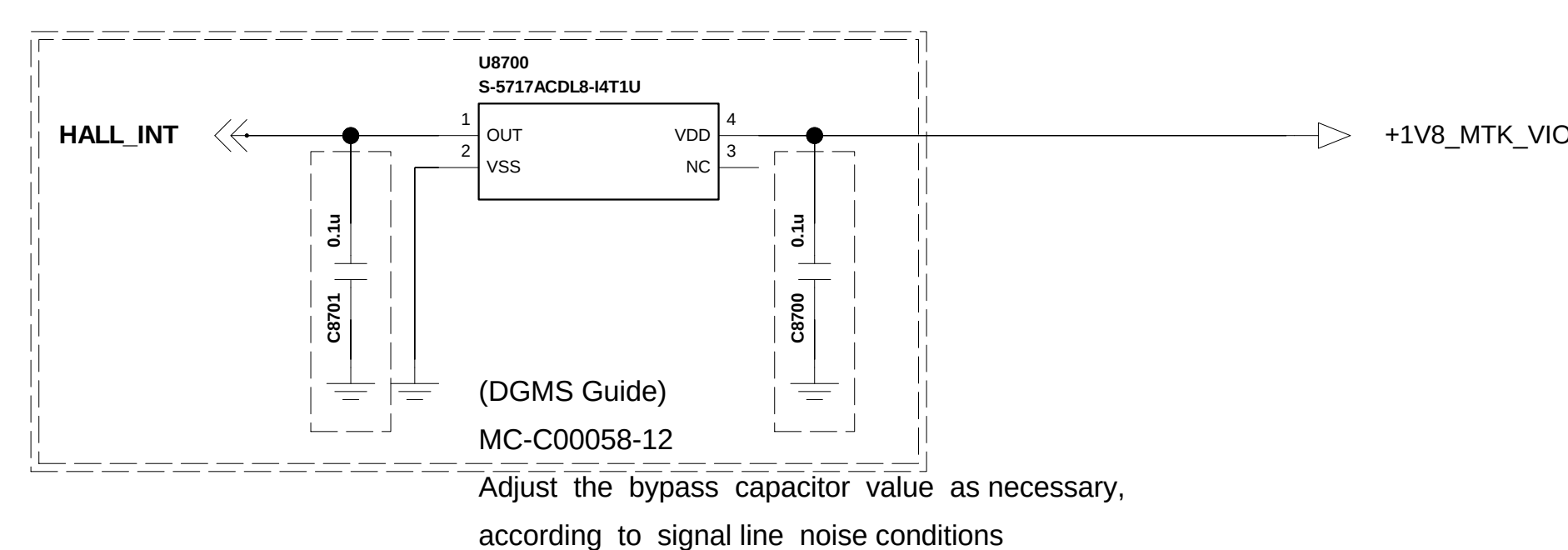
< 7-4-1_VT_Camera_24pin_5M>



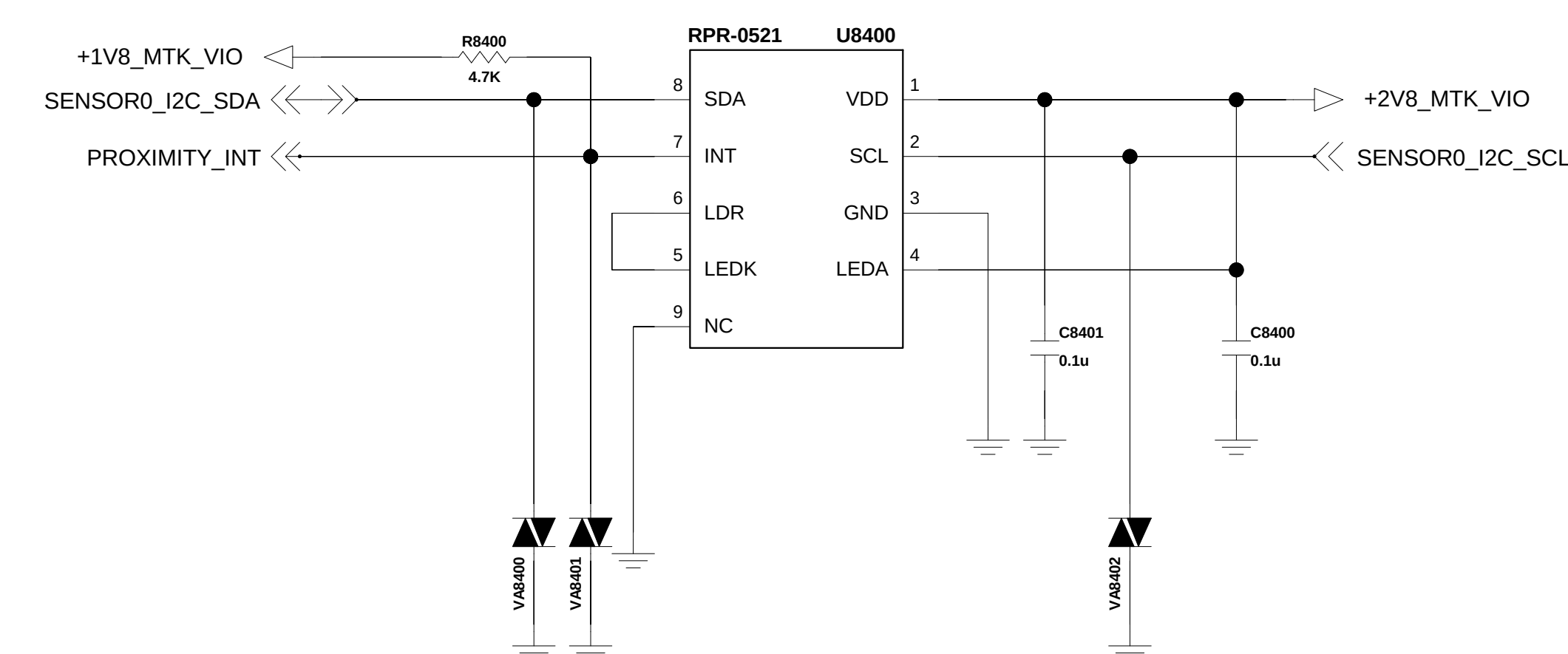
<8-1-2-1_Accel_Compass_BMC150> REV_1.0



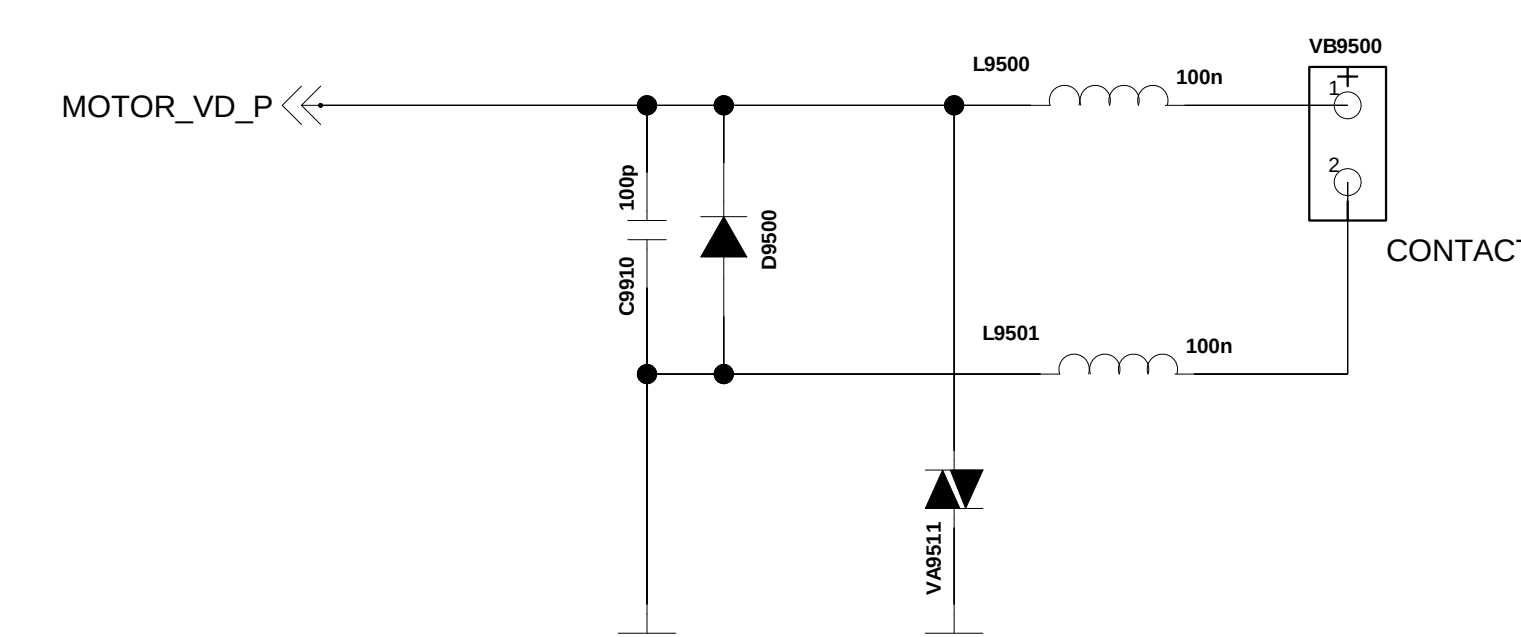
<8-7-1-5 Hall IC S5717> Rev_1.1



<8-4-1-X_Proximity_RPR-0521>

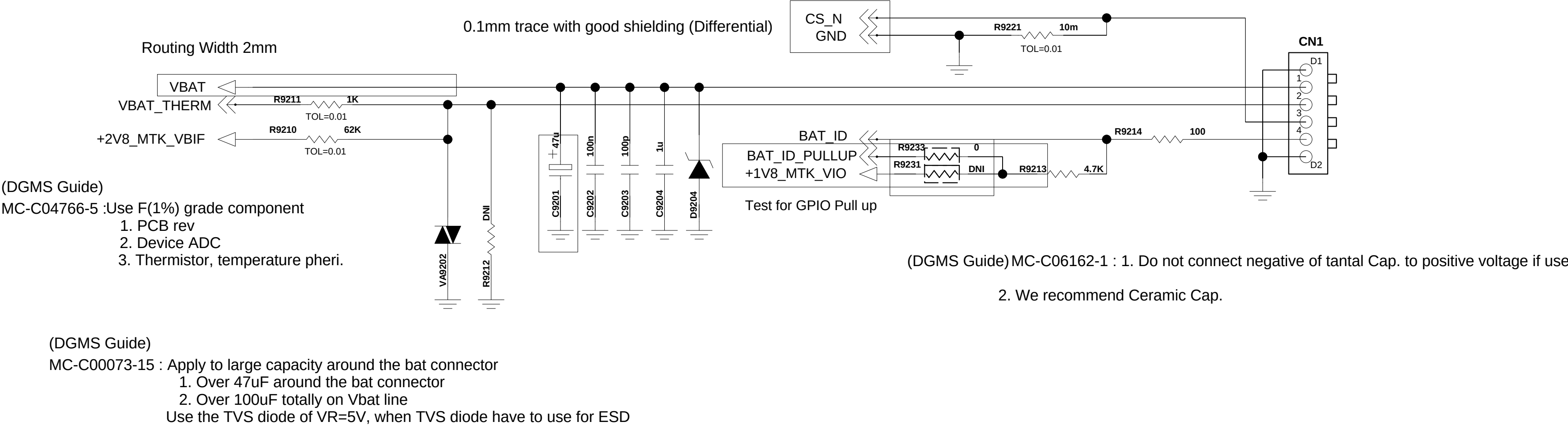


< 9-9-1-1_Motor > Rev 0.3

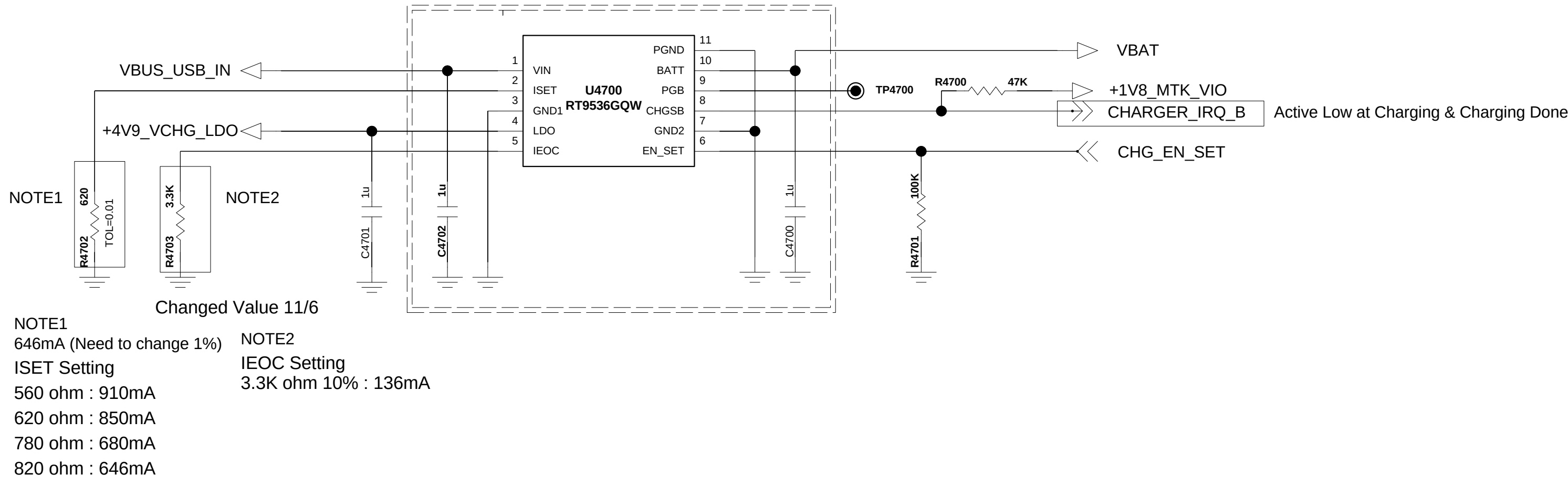


<9-1-2_Battery_CNT_4P>

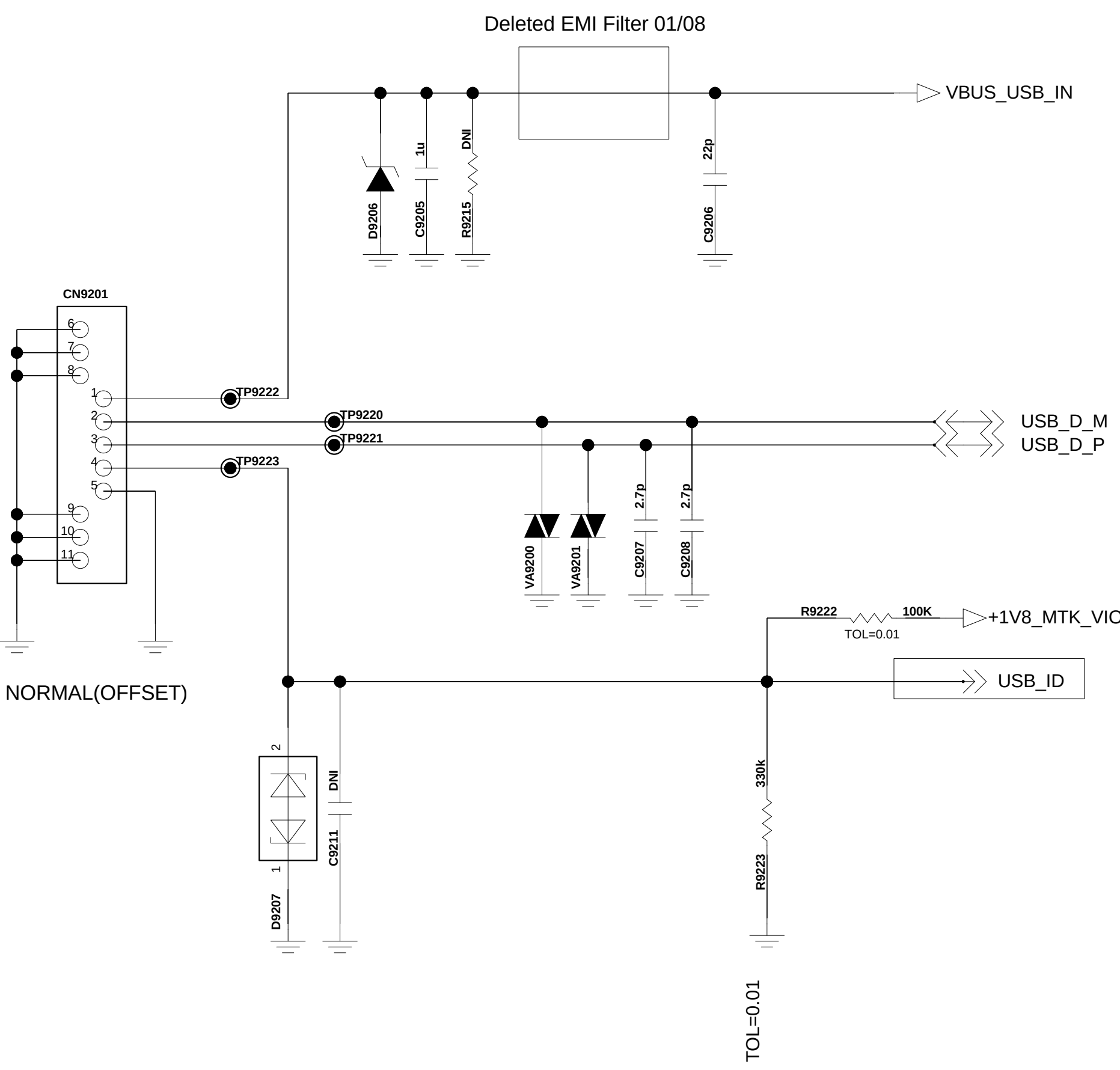
REV.0.6



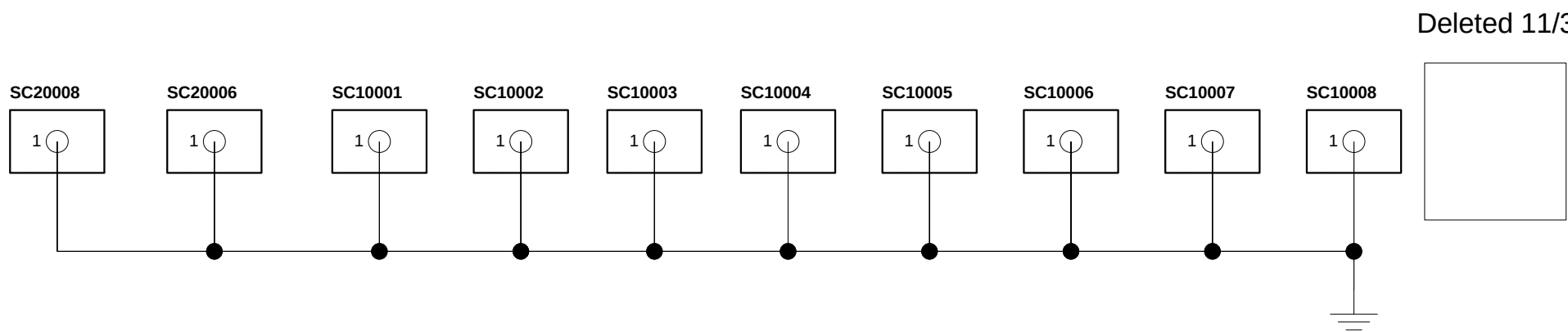
<4-7-1-1_Linear Charger_RT9536> Rev_0.3



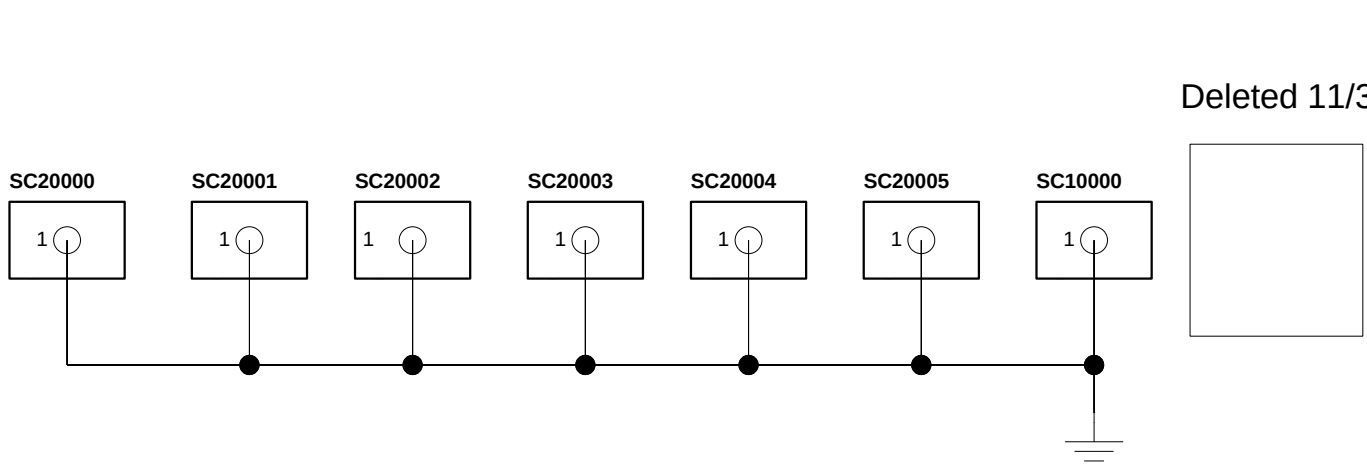
< 9-2-1_IO_USB2.0 > Rev_0.5



<TOP SHIELD CAN CLIP>

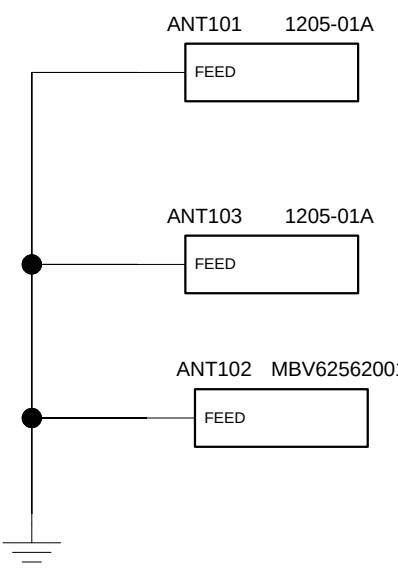


<BOTTOM SHIELD CAN CLIP>

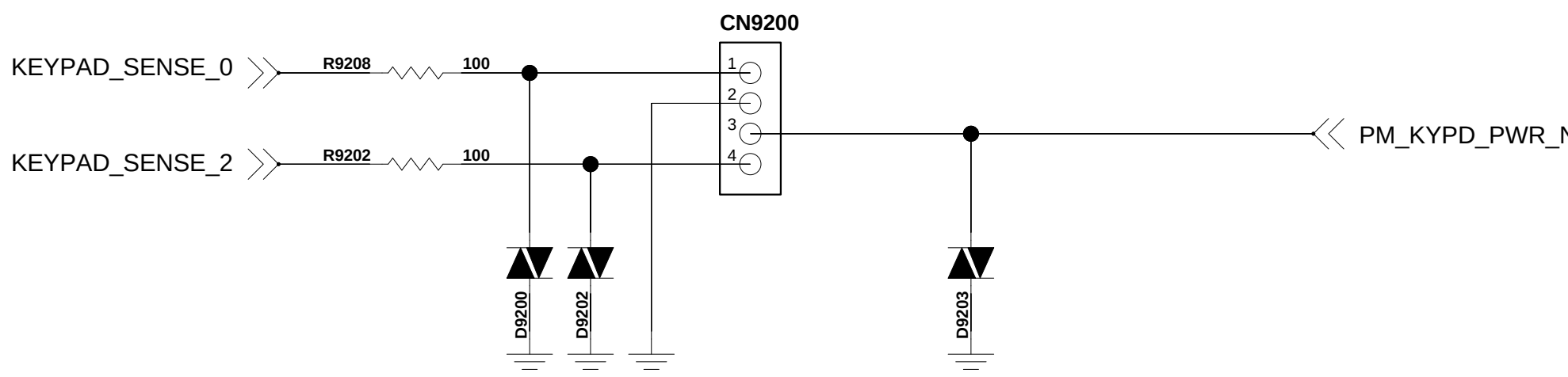


<Coaxial Cable Clip >

<GND Clip >

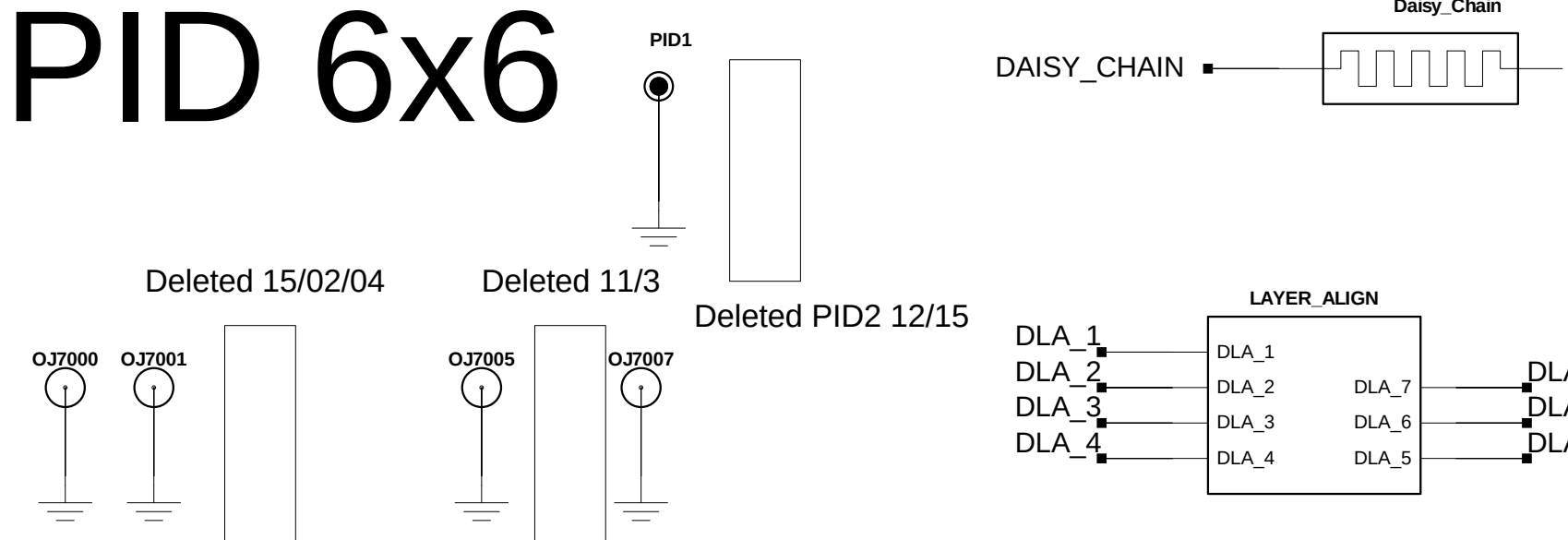


BACK KEY CONNECTOR



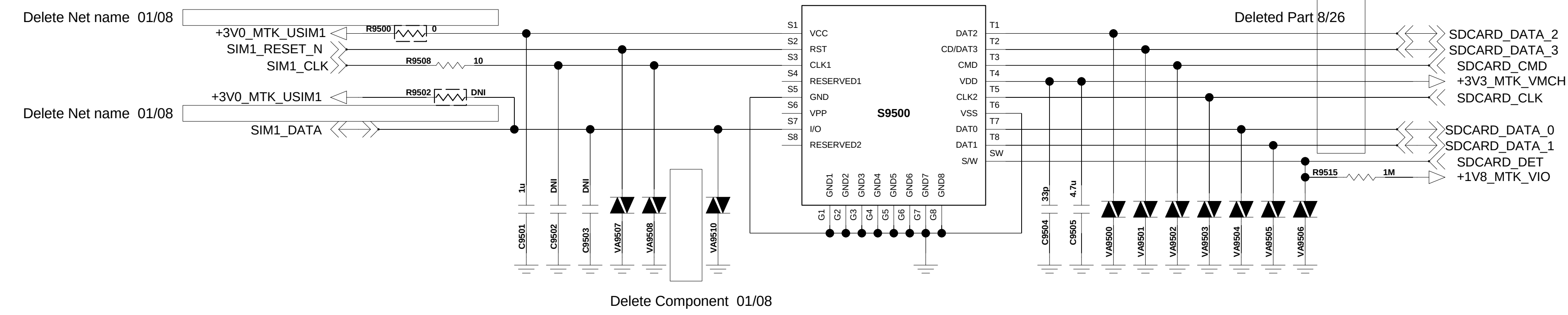
	Sense0-KCOL2	Sense2-KCOL1
GND	UP	DOWN

PID 6x6



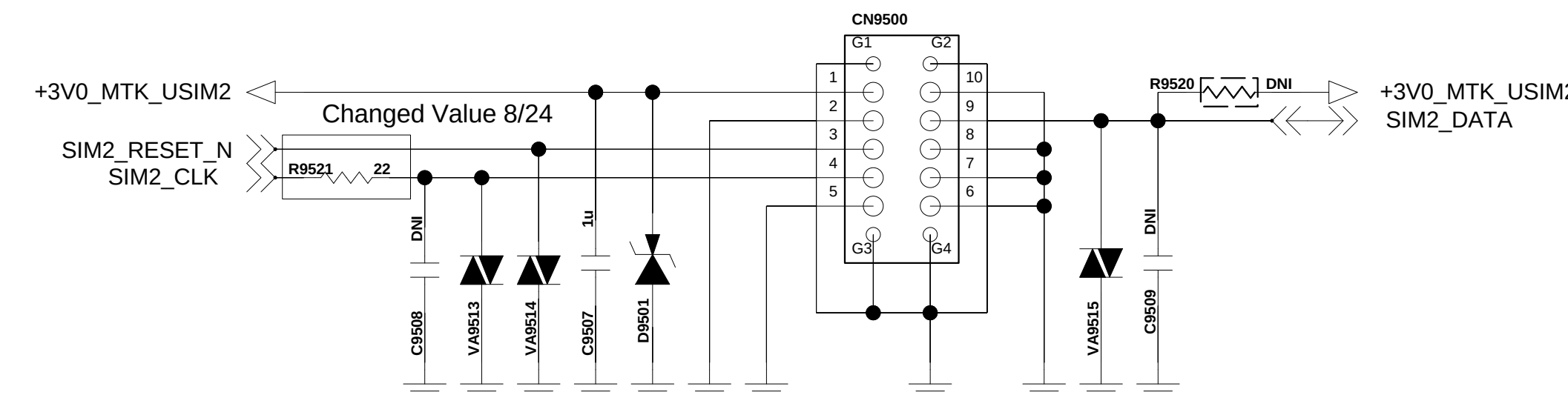
< 9-5-3_Micro_SD_Combo >

Main PCB



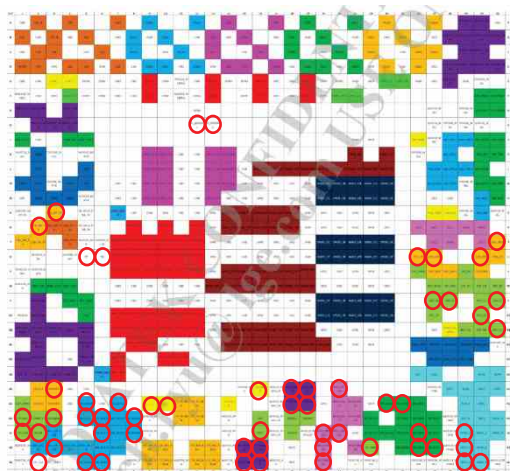
Main PCB

<Dual Sim Connector>



6. BGA PIN MAP

U2100_MT6732_IC,Digital Baseband Processor(Top View)



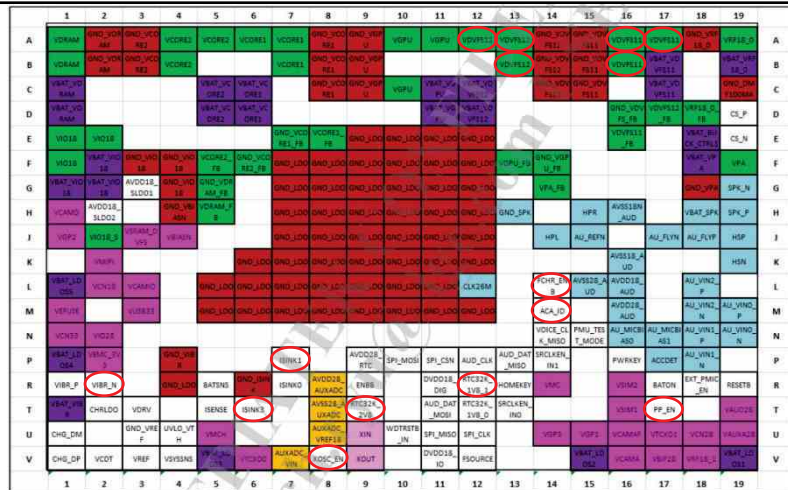
O Not used

U3300_KMQ72000SMB-316,IC,MCP.eMMC4905(Top View)



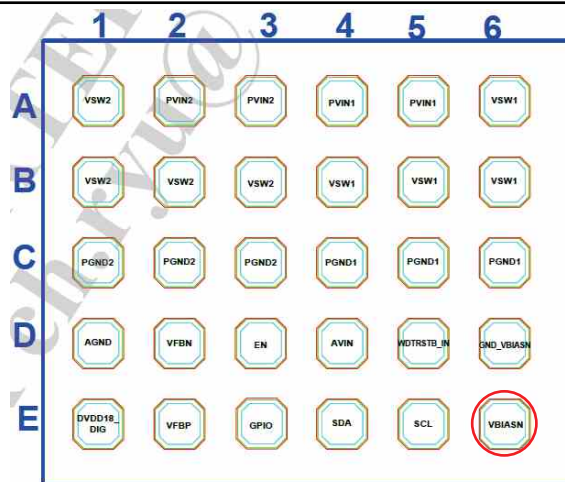
O Not used

U4100_MT6325_IC,PMIC (Top View)



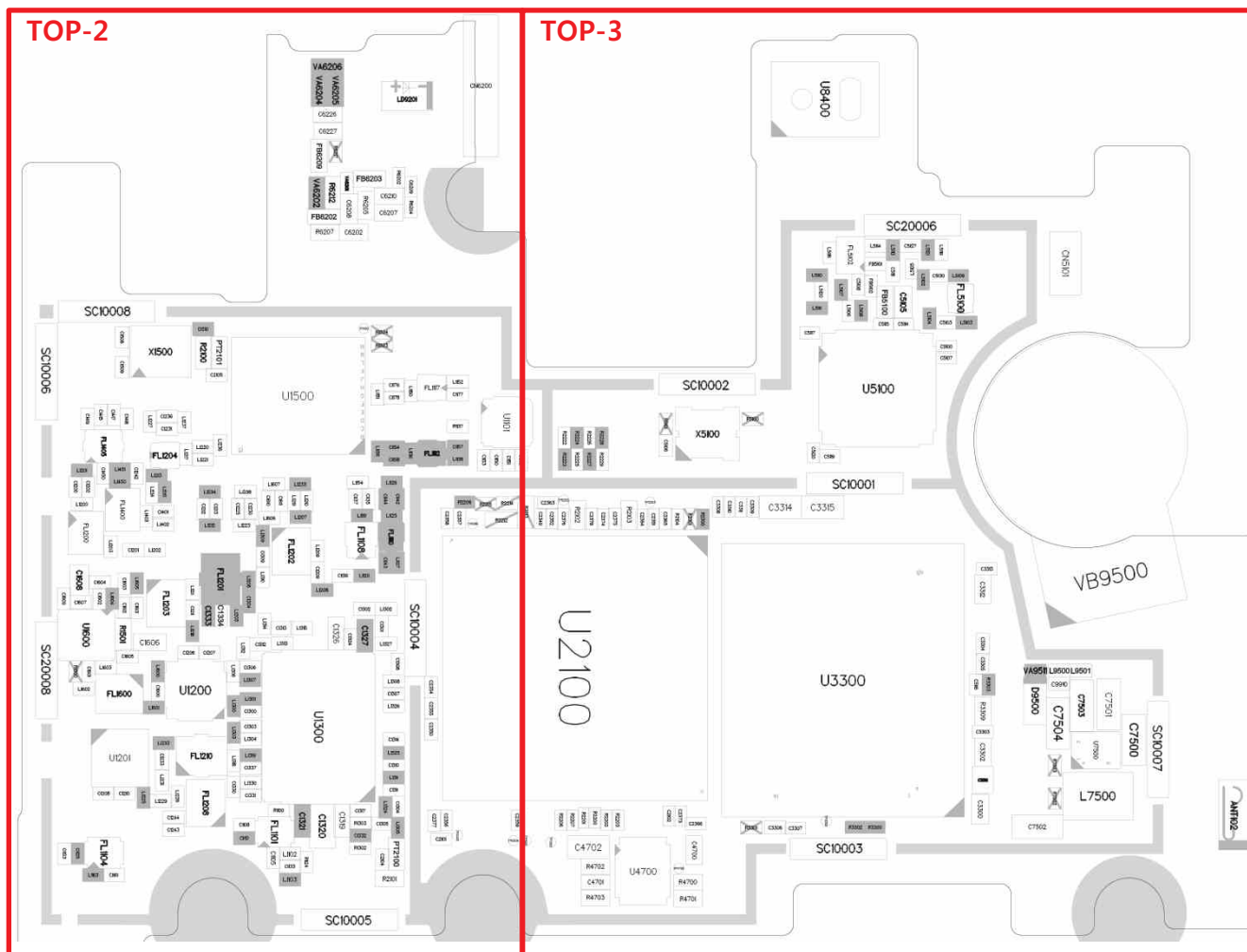
O Not used

U4200_MT6311_IC,2nd PMIC (Top View)



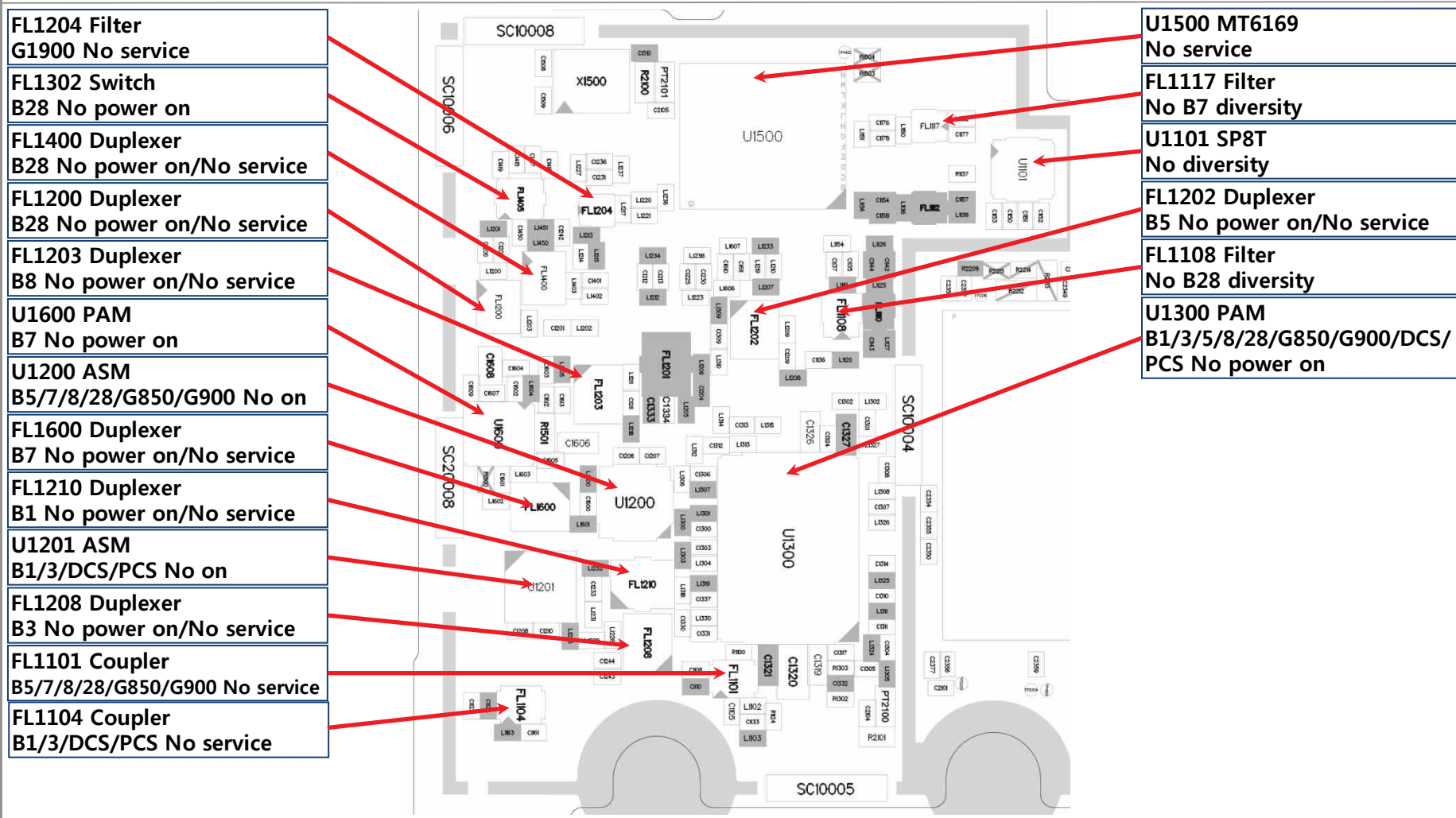
O Not used

LG-H522y-MAIN_TOP-1



7. PCB LAYOUT

LG-H522y-MAIN_TOP-2

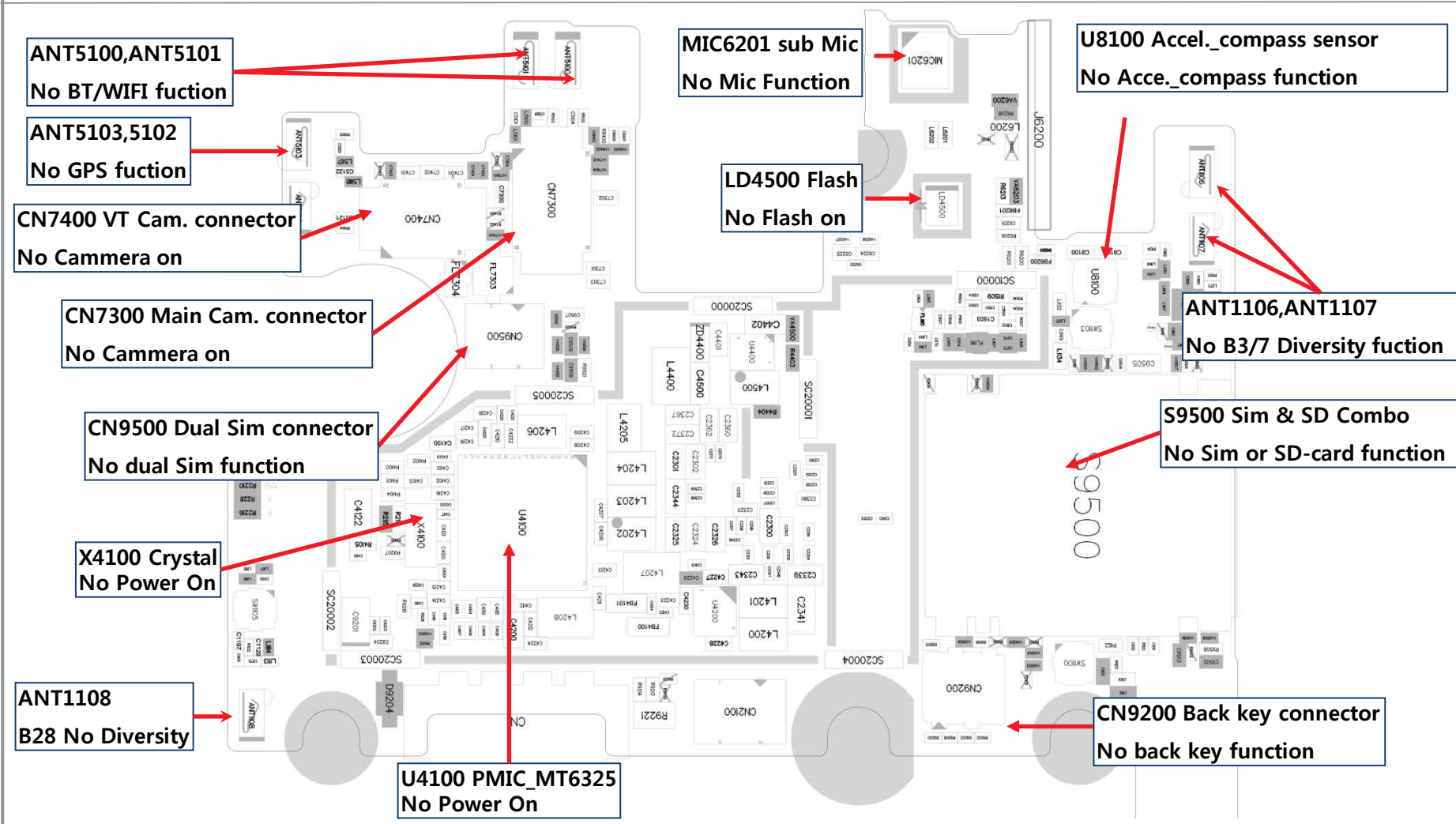


LG-H522y-MAIN_TOP-3

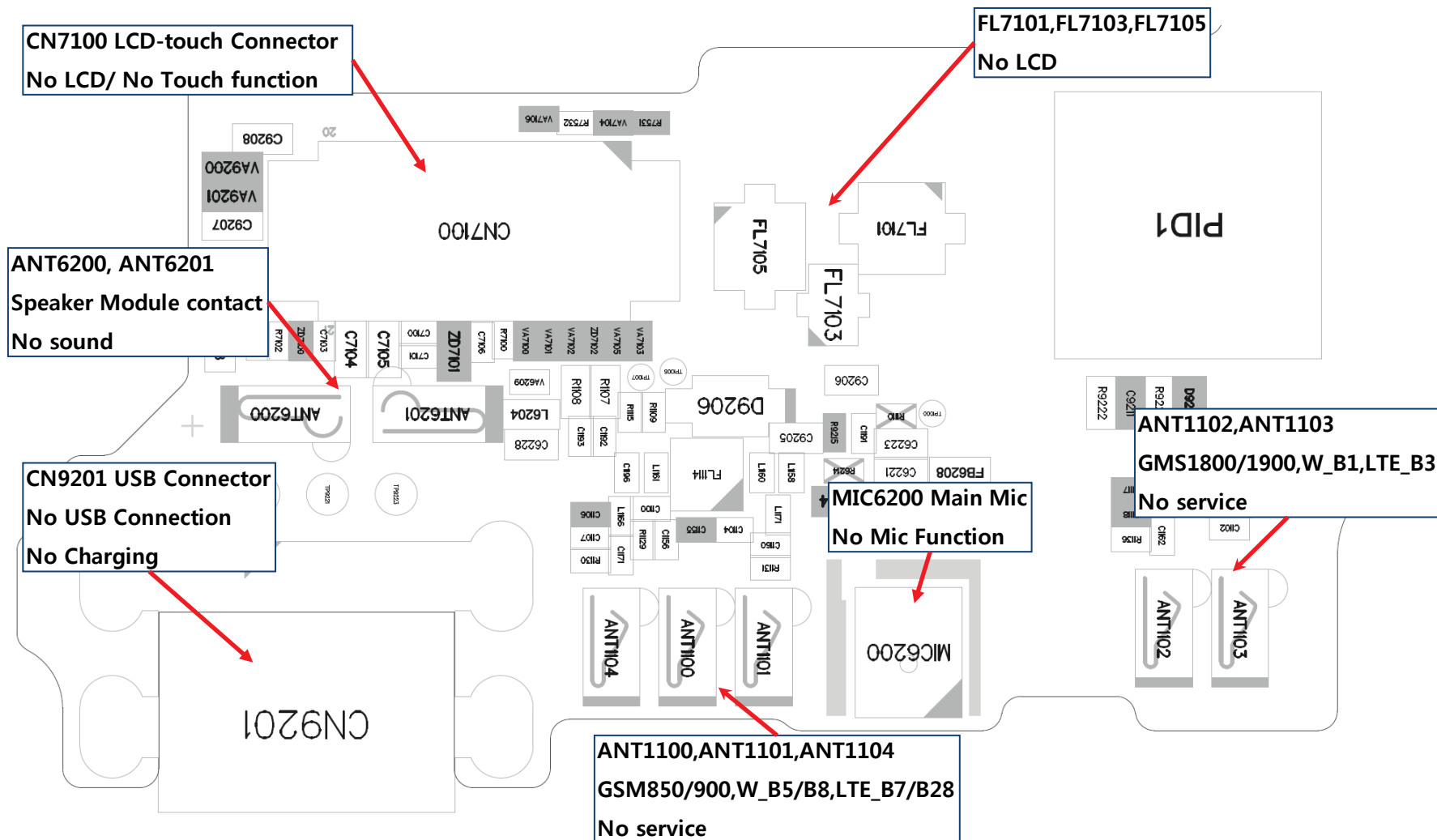


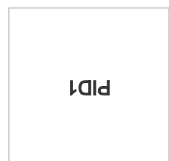
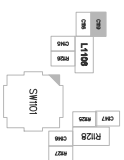
7. PCB LAYOUT

LG-H522y-MAIN_BOT-1

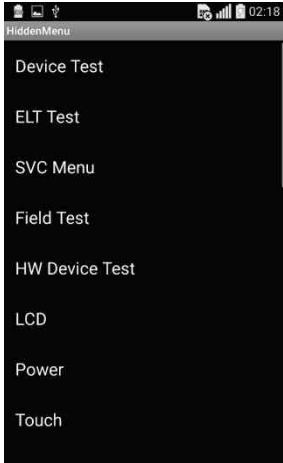


LG-H522y-MAIN_BOT-2



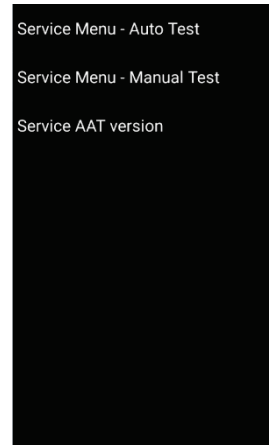


1. Hidden Menu Start



- Start shortcut key:
*#546368#*522#
- Hidden Menu List
: Start the desired menu, click

2. Device Test



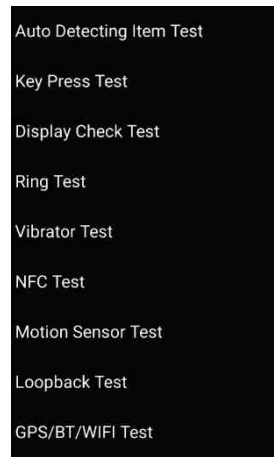
- Service Menu – Auto Test
- Service Menu – Manual Test
- Service AAT version
→ Hidden Menu version Display

2. Device Test



- Service Menu – Auto Test
→ All Test Items are continued one after another.
→ Continuous information on the menu, giving you ability test.

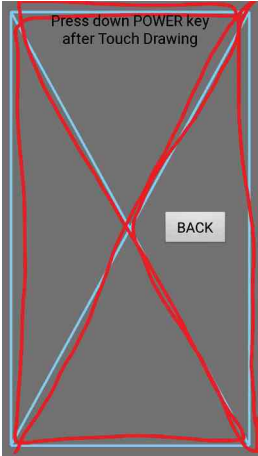
3. Device test List



- Service Menu – Manual Test
→ Each test item can be selected and performed by user.

8. HIDDEN MENU

1. Auto Detecting Item Test Auto Detecting Item Pass <table><tr><th>Item</th><th>Result</th></tr><tr><td>USIM Card</td><td>Pass</td></tr><tr><td>USIM Card2</td><td>Pass</td></tr><tr><td>SD Card</td><td>Pass</td></tr><tr><td>Ear Phone wrong</td><td>Pass</td></tr><tr><td>TA wrong</td><td>Fail</td></tr><tr><td>Battery</td><td>100%</td></tr></table> Don't touch Proximity Sensor Touch 후 근접센서 작동하지 않 PASSFAIL	Item	Result	USIM Card	Pass	USIM Card2	Pass	SD Card	Pass	Ear Phone wrong	Pass	TA wrong	Fail	Battery	100%	2. Key Press Test Proximity Cal Pass:10 Block the sensor with your hand Proximity : FAR Block the light with your hand Light:517.00	3. Display Check Test PASSFAIL	4. Ring Test Ring Test ONSOUND PASSFAIL
Item	Result																
USIM Card	Pass																
USIM Card2	Pass																
SD Card	Pass																
Ear Phone wrong	Pass																
TA wrong	Fail																
Battery	100%																
5. Vibrator Test Vibrator Test ONVIBRATION PASSFAIL	6. Motion Sensor Test Calibration LV : 3 Magnetic Accelerometer (Motion Accel)(N/A) Gyroscope Sensor(N/A) Stand the phone horizontal.	7. Loop Back Test LOOPBACK TEST AUDIOON↔ PASSFAIL	8. GPS/BT/WIFI Test GPS(RF) Test82 Sec GPS TestGLO Test CNo : 23.0 dB CNo : 15.7 dB Bluetooth Scanning Bluetooth Scanning... C4:D9:87:25:85:E5Pass WiFi Scanning WiFi Scan Complete. BSSID: 00:08:9f:dd:04:14Pass PASSRetryFAIL														

9. Touch Draw -Manual	10. Camera(Main) Test	11. Camcorder Test	12. Camera(VT) Test
			

1. Summary

Tool Version	DLL name	USB Driver	
LGFLAS Hv201	LGH522y_2015 0715_LGFLASH v190_Download	LGUnitedMobileDriver _S50MAN311AP22_ML _WHQL_Ver_3.11.3	
Please Check the Version to “B2B”			
H/W			
	Name	Part No.	SW
D/L Cable	Micro 5P (56-open-910K) USB DLC	RAD321 67835	TOT

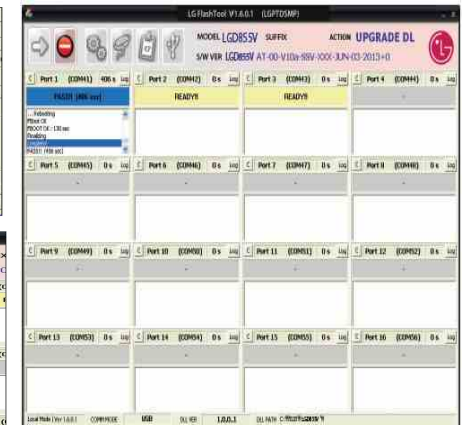
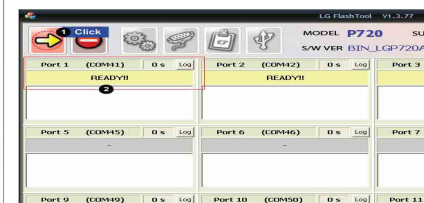
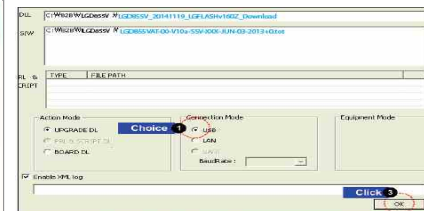
2. . USB COM port Setting



3. USB D/L Cable setting



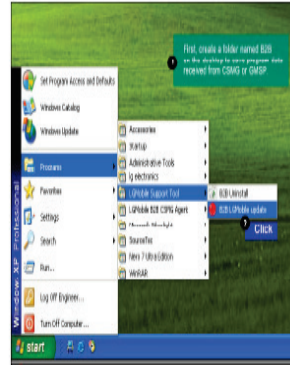
4. Flash tool D/L setting



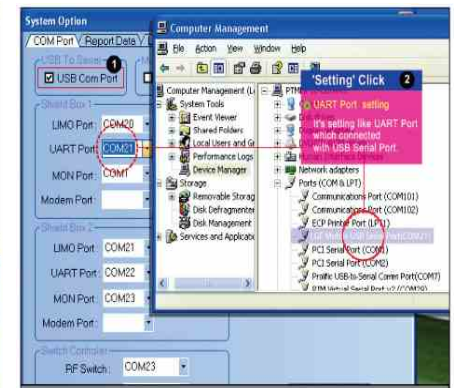
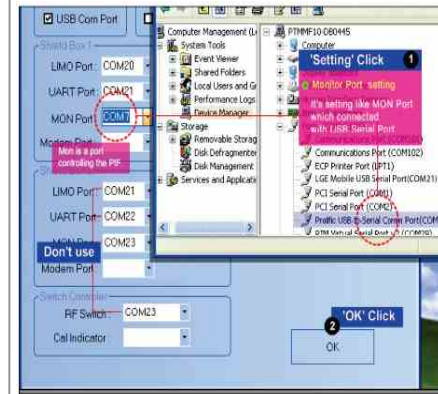
※ If you want more information, please refer B2B's Notification "Download User Guide".

1. Summary

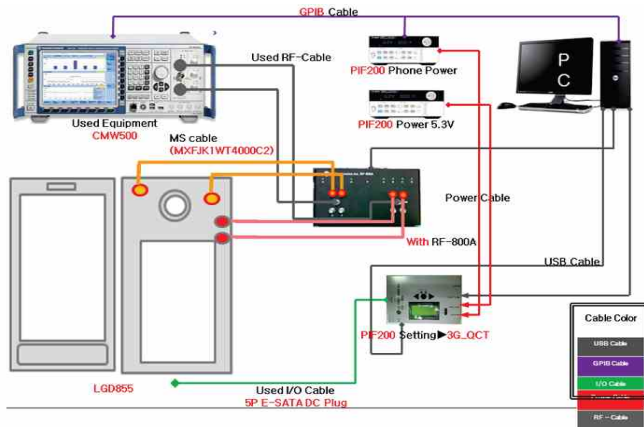
CAL INFORMATION		S/W VERSION
		[TtchyonV20131LGH522Y 20150720ALL File Calibration
Please Check the Version to "B2B"		
H/W		
	Name	Part No.
PIF	PIF200	BJAY0024021
USB Cable	USB Cable 2.0	RAD32247898
Power Cable	DC Power Cable	RAD32247878
I/O Cable	5P E-SATA DC Plug	RAD32167861
RF Cable Main	MXFJM3WX6000	RAD32827895
Power Supply PIF	Power supply 5.2V	
Power Supply Phone	Power supply 5.0V	
PF Test Equipment	CMW500	
Notice	1. Use the Battery (Refer to Attached ppt) 1) Phone states: Power off 2) If do not use the battery, TX fails. 2. Port Setting (Refer to Attached ppt) 1) Uart Port1 : Use the "LGE Mobile USB Serial Port"	
CMW 500 RF Cable connection	Reference to Attached ppt ▷ How to connect between phone RF switch and RF 800A: It has to be connected as below ▷ How to connect between CMW500 and RF800A; It has to connected as below - CMW500 [RF1COM] to RF-800A [left COM] , - CMW500 [RF2COM] to RF-800A [Right COM]	



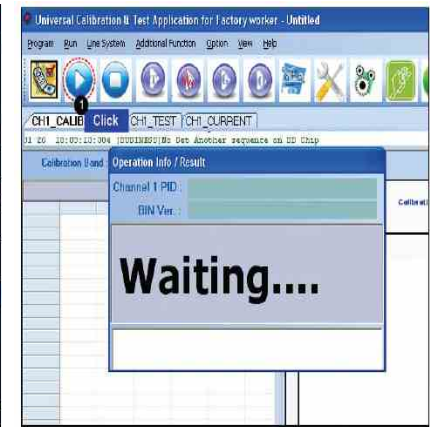
2. . USB COM port Setting



3. Calibration Cable setting

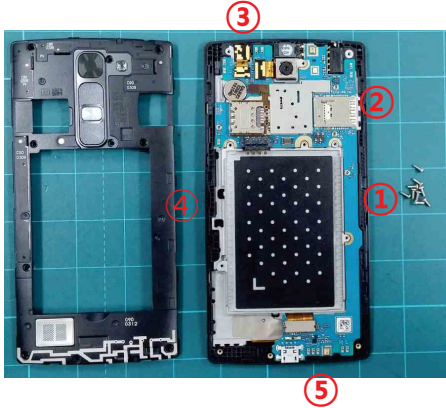
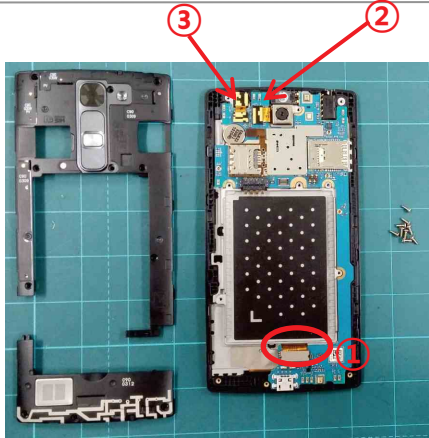
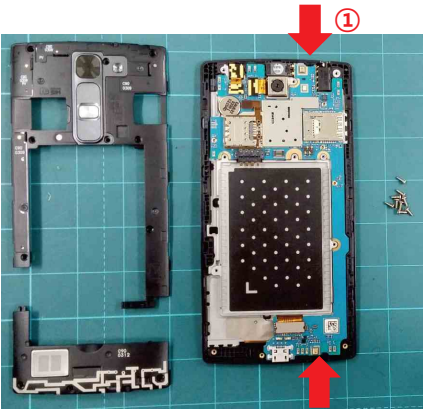


4. Flash tool D/L setting



※ If you want more information, please refer B2B's Notification "RF Calibration User Guide".

11. DISASSEMBLE GUIDE

1. Disassemble Screw (11ea) 	2. Disassemble Rear cover 	3. Disassemble Main Antenna 	4. Disassemble Connector 
5. Disassemble Main PCB  Disassemble Hooks ②	6. Disassemble HW Parts (5ea) 	7. LCD Disassemble (by Heating machine) 	8. Complete Disassembling 

EXPLODED VIEW

13. REPLACEMENT PART LIST

No	P/N	Description	Qty	Location no
1	AGQ88734501	Phone Assembly	1	AGQ000000
2	ACQ88542601	Cover Assembly,EMS	1	ACQ100400
3	ACQ87831971	Cover Assembly,Rear	1	ACQ063300
4	ACQ87852471	Cover Assembly,Rear(SVC)	1	ACQ105800
5	ABH75459905	Button Assembly	1	ABH000000
6	MBG65483302	Button	1	MBG000000
7	MBG65562905	Button	1	MBG000001
8	MCR65887705	Decor	1	MCR000000
9	MJN69388701	Tape,Protect	1	MJN061100
10	EAA63826701	PIFA Antenna,GPS	1	EAA030100
11	EAA63826801	PIFA Antenna,RF	1	EAA030103
12	EAA63884601	PIFA Antenna,WiFi	1	EAA030101
13	EAA63884701	PIFA Antenna,RF	1	EAA030102
14	EAX66264501	PCB,Flexible	1	EAX010700
15	MBL66357701	Cap	1	MBL000000
16	MCK68589401	Cover,Rear	1	MCK063300
17	MCQ68166801	Damper,Camera	1	MCQ009400
18	MCQ68166901	Damper,Motor	1	MCQ049800
19	MCQ68545101	Damper	1	MCQ000000
20	MEZ64319901	Label,After Service	1	MEZ000900
21	MFB63813101	Lens,Flash	1	MFB029600
22	MJN69388601	Tape	1	MJN000000
23	MJN69568101	Tape,Protect	1	MJN061100
24	MKC65379201	Window,Camera	1	MKC009400
25	EAB63728601	Speaker Module	1	EAB010100
26	MBL66318301	Cap	1	EAU010000
27	ACQ87832060	Cover Assembly,Bar	1	ACQ003400
28	ACQ88378051	Cover Assembly	1	ACQ000000
29	ACQ87791651	Cover Assembly,Front	1	ACQ032700
30	ACQ87852305	Cover Assembly,Front(Sub)	1	ACQ033200
31	MCK68588905	Cover,Front	1	MCK032700
32	MDQ64336301	Frame	1	MDQ000000
33	MICE0016905	Insert,Nut	4	MET099501
34	MBL66397101	Cap	1	MBL000000
35	MCQ68166701	Damper	1	MCQ000001
36	MCQ68264701	Damper	1	MCQ000000
37	MCR65966502	Decor	1	MCR000000
38	MDS65013401	Gasket	1	MDS000000
39	MHK64907101	Sheet	1	MHK000000
40	MHK64945601	Sheet	1	MHK000001
41	MJN69408301	Tape	1	MJN000001
42	MJN69428101	Tape	1	MJN000000

13. REPLACEMENT PART LIST

43	MJN69828601	Tape,Protect	2	MJN061100
44	MJN69828602	Tape,Protect	1	MJN061101
45	MJN69828603	Tape,Protect	1	MJN061102
46	EAT62613701	Module,Hybrid Touch LCD	1	EAT130000
47	MCQ68166601	Damper,Camera	1	MCQ009400
48	MDS65190701	Gasket	1	MDS000000
49	EAB63288401	Receiver	1	EAB010400
50	EAG63930601	Jack,Phone	1	EAG140300
51	EAU62283801	Motor,DC	1	EAU010000
52	EBP62481901	Camera Module	1	EBP000000
53	MJN69527608	Tape,USP Film	1	MJN107400
54	MJN69527701	Tape,Protect	1	MJN061100
55	MLAZ0038301	Label	1	MEZ000000
56	EBR81711401	PCB Assembly,Main	1	EBR071900
57	EBR80675101	PCB Assembly,Main,Insert	1	EBR071500
58	ABM74756101	Can Assembly,Shield	1	ABM070302
59	MBK64233001	Can,Shield	1	MBK070300
60	MDS65330701	Gasket	2	MDS000000
61	MEV65654601	Insulator	1	MEV000000
62	ABM75116201	Can Assembly,Shield	1	MBK070300
63	MBK64213501	Can,Shield	1	MBK070300
64	MES63658301	Indicator,LED	1	MES043800
65	BRAH0001305	Resin,PC	0	RAA050100
66	EBP62421801	Camera Module	1	EBP000000
67	EBR76354001	PCB Assembly,Flexible	1	EBR070500
68	EBR76354101	PCB Assembly,Flexible,SMT	1	EBR070400
69	EBR76328601	PCB Assembly,Flexible,SMT Top	1	EBR070300
70	EAG63393301	Connector,BtoB	1	CN9501
71	EAX66307501	PCB,Flexible	1	EAX010700
72	EBR76354201	PCB Assembly,Flexible,SMT Bottom	1	EBR070200
73	EAG64229801	Socket,DIMM/SIMM	1	J9500
74	EBR81021701	PCB Assembly,Flexible,Insert	1	EBR070100
75	MDS65330901	Gasket	1	MDS000000
76	MCQ68365101	Damper	1	MCQ000000
77	EBR81711601	PCB Assembly,Main,SMT	1	EBR071800
78	BRAH0001304	Resin,PC	0	RAA050100
79	EBR81730801	PCB Assembly,Main,SMT Top	1	EBR071700
80	EAE62286801	Capacitor,Ceramic,Chip	1	C1613
81	EAE62487001	Capacitor,Low ESL	1	C3301
82	EAE62502901	Capacitor,Ceramic,Chip	1	C1608
83	EAE62505701	Capacitor,Ceramic,Chip	1	C4702
84	EAE62522101	Capacitor,Ceramic,Chip	1	C1320
85	EAE62762301	Capacitor,Ceramic,Chip	2	C2365,C2375

13. REPLACEMENT PART LIST

86	EAE62843001	Capacitor,Ceramic,Chip	4	C7500,C7501,C7502,C7504
87	EAE62924001	Capacitor(High Frequency),Ceramic,Chip	1	C5103
88	EAE62945801	Capacitor(High Frequency),Ceramic,Chip	1	L1606
89	EAE62945901	Capacitor(High Frequency),Ceramic,Chip	3	L1217,L1604,L1605
90	EAE62946201	Capacitor(High Frequency),Ceramic,Chip	3	C1301,L1313,L1315
91	EAE62946401	Capacitor(High Frequency),Ceramic,Chip	2	L1308,L1326
92	EAE62946601	Capacitor(High Frequency),Ceramic,Chip	2	C1225,C1230
93	EAE62946801	Capacitor(High Frequency),Ceramic,Chip	1	C5111
94	EAE62946901	Capacitor(High Frequency),Ceramic,Chip	2	C1233,C1604
95	EAE62947001	Capacitor(High Frequency),Ceramic,Chip	2	C1317,C1324
96	EAE62962201	Capacitor(High Frequency),Ceramic,Chip	5	C1331,L1151,L1200,L1600,L1603
97	EAE63067401	Capacitor,Ceramic,Chip	1	C7503
98	EAE63222701	Capacitor,Ceramic,Chip	3	C1319,C1326,C1606
99	EAG63652301	C-Clip	1	ANT102
100	EAG64389801	C-Clip	2	ANT101,ANT103
101	EAG64451301	Connector,Terminal	1	CN5101
102	EAG64451501	Connector,Terminal	1	CN1100
103	EAH61532901	Diode,Switching	1	D9500
104	EAH62033201	Diode,TVS	1	VA6201
105	EAM62252001	Filter,Bead	2	FB5101,FB5102
106	EAM62673701	Filter,Duplexer	1	FL1208
107	EAM62792601	Filter,Separator,Switch	1	FL1405
108	EAM62792701	Filter,Duplexer	1	FL1202
109	EAM62930901	Filter,Duplexer	1	FL1210
110	EAM62955301	Filter,Duplexer	1	FL1203
111	EAM63210001	Filter,Separator,Switch	1	U1101
112	EAM63290701	Filter,Duplexer	1	FL1200
113	EAM63431001	Filter,Duplexer	1	FL1400
114	EAM63450301	Filter,Saw	1	FL1108
115	EAM63730001	Filter,Saw	1	FL5100
116	EAM63830201	Filter,Duplexer	1	FL1600
117	EAM63910601	Filter,Saw	1	FL1117
118	EAM63910701	Filter,Saw	1	FL1204
119	EAN62677301	IC,Charger	1	U4700
120	EAN62702001	IC,Power Amplifier	1	U1300
121	EAN62867801	IC,Power Amplifier	1	U1600
122	EAN62969801	IC,Hall Effect Switch	1	U8700
123	EAN63106301	IC,RF Amplifier	1	FL5102
124	EAN63366602	IC,MCP,eMMC	1	U3300
125	EAN63389001	IC,Digital Baseband Processor,4G	1	U2100
126	EAN63406501	IC,RF Transceiver,4G	1	U1500
127	EAN63486501	IC,WiFi	1	U5100

13. REPLACEMENT PART LIST

128	EAN63569001	IC, Proximity	1	U8400
129	EAN63688501	IC, DC, DC Converter	1	U7500
130	EAP61767701	Inductor, Multilayer, Chip	1	L1214
131	EAP61866401	Inductor, Multilayer, Chip	3	C1307, L1302, L1327
132	EAP61866601	Inductor, Multilayer, Chip	2	L9500, L9501
133	EAP61866701	Inductor, Multilayer, Chip	3	L5105, L5106, L5119
134	EAP62107801	Inductor, Wire Wound, Chip	1	L7500
135	EAP62107901	Inductor, Multilayer, Chip	1	C1136
136	EAP62108001	Inductor, Multilayer, Chip	1	L1231
137	EAP62108101	Inductor, Multilayer, Chip	5	C1211, C1303, C1601, L1220, L1221
138	EAP62108201	Inductor, Multilayer, Chip	2	C1603, L1226
139	EAP62108301	Inductor, Multilayer, Chip	3	L1237, L1318, L1403
140	EAP62108401	Inductor, Multilayer, Chip	5	C1212, C1213, L1210, L1219, L5116
141	EAP62108501	Inductor, Multilayer, Chip	2	L1236, L1402
142	EAP62108601	Inductor, Multilayer, Chip	4	L1154, L1306, L1312, L1314
143	EAP62108701	Inductor, Multilayer, Chip	2	L1102, L1310
144	EAP62186301	Inductor, Multilayer, Chip	3	C1610, C1611, L1601
145	EAP62225901	Inductor, Multilayer, Chip	2	C1105, C1600
146	EAP62226001	Inductor, Multilayer, Chip	5	C1176, C1178, C1401, L1229, L5114
147	EAP62226101	Inductor, Multilayer, Chip	1	L5109
148	EAP62226201	Inductor, Multilayer, Chip	1	C1300
149	EAP62226301	Inductor, Multilayer, Chip	4	C1312, C1313, L1209, L1227
150	EAP62226401	Inductor, Multilayer, Chip	3	L1203, L1211, L1238
151	EAP62226601	Inductor, Multilayer, Chip	5	C1200, C1202, C1242, C1450, L1223
152	EAP62226701	Inductor, Multilayer, Chip	1	L1202
153	EAP62546201	Inductor, Multilayer, Chip	1	L5120
154	EAP62886101	Inductor, Multilayer, Chip	2	L1330, L5104
155	EAT61914101	Module, FEM(Front End Module)	1	U1200
156	EAT62613601	Module, FEM(Front End Module)	1	U1201
157	EAU61983901	Oscillator, TCXO	1	X5100
158	EAU62743401	Oscillator, VCTCXO	1	X1500
159	EAX66407101	PCB, Main	1	EAX010000
160	EBC61856201	Resistor, Chip	1	R2104
161	EBC62615501	Resistor, Chip	1	R4702
162	ECA30240001	Coupler, RF Directional	2	FL1101, FL1104
163	ECCH0000147	Capacitor, Ceramic, Chip	2	C6207, C6210
164	ECCH0000198	Capacitor, Ceramic, Chip	2	C3302, C3312
165	ECCH0002001	Capacitor, Ceramic, Chip	1	C6226
166	ECCH0005604	Capacitor, Ceramic, Chip	2	C3314, C3315
167	ECCH0009101	Capacitor, Ceramic, Chip	28	C1508, C1509, C2102, C2349, C2350, C2351, C2352, C2356, C2357, C2358, C2359, C2363, C2364, C2366, C2373, C2374, C3303, C3305, C3307, C3308, C3309, C3310, C3311, C3313, C5100, C5108, C8700, C8701

13. REPLACEMENT PART LIST

168	ECCH0009103	Capacitor, Ceramic, Chip	26	C1133, C1150, C1151, C1152, C1201, C1206, C1207, C1208, C1209, C1210, C1243, C1244, C1310, C1311, C1334, C1337, C1415, C1417, C1418, C1612, C2101, C5107, C5114, C5120, C6209, C9910
169	ECCH0009104	Capacitor, Ceramic, Chip	1	C1177
170	ECCH0009106	Capacitor, Ceramic, Chip	4	C1314, C1605, C1609, C5115
171	ECCH0009109	Capacitor, Ceramic, Chip	1	C1607
172	ECCH0009208	Capacitor, Ceramic, Chip	3	C1330, L1304, L1602
173	ECCH0009211	Capacitor, Ceramic, Chip	2	C1135, C1137
174	ECCH0009216	Capacitor, Ceramic, Chip	3	C1304, C1305, C1602
175	ECCH0009217	Capacitor, Ceramic, Chip	2	C1302, C1308
176	ECCH0009228	Capacitor, Ceramic, Chip	1	C5117
177	ECCH0009502	Capacitor, Ceramic, Chip	1	L1150
178	ECCH0009504	Capacitor, Ceramic, Chip	4	C1123, C1231, C1236, C1309
179	ECCH0009520	Capacitor, Ceramic, Chip	1	C1108
180	ECCH0017301	Capacitor, Ceramic, Chip	12	C2104, C2105, C2354, C2355, C2376, C2377, C2378, C3304, C3306, C5106, C5116, C5119
181	ECCH0017601	Capacitor, Ceramic, Chip	3	C3300, C5105, C6208
182	ECZH0000830	Capacitor, Ceramic, Chip	2	C6202, C6227
183	ECZH0001215	Capacitor, Ceramic, Chip	2	C4700, C4701
184	ECZH0025911	Capacitor, Ceramic, Chip	1	C1161
185	ECZH0025916	Capacitor, Ceramic, Chip	3	C1306, C5127, C5130
186	ECZH0025920	Capacitor, Ceramic, Chip	2	C1153, C1419
187	EDLH0015104	LED, Chip	1	LD9201
188	ERHY0000166	Resistor, Chip	2	R2100, R2101
189	ERHY0009311	Resistor, Chip	1	R1100
190	ERHY0009504	Resistor, Chip	2	R6202, R6204
191	ERHY0009505	Resistor, Chip	5	R1303, R2222, R2224, R2227, R2228
192	ERHY0009507	Resistor, Chip	1	R1137
193	ERHY0009519	Resistor, Chip	1	R1302
194	ERHY0009526	Resistor, Chip	7	R1124, R2200, R2201, R2202, R2203, R2206, R2207
195	ERHZ0000220	Resistor, Chip	1	R2103
196	ERHZ0000250	Resistor, Chip	1	R3309
197	ERHZ0000267	Resistor, Chip	1	R4703
198	ERHZ0000294	Resistor, Chip	1	R2102
199	ERHZ0000401	Resistor, Chip	1	R1501
200	ERHZ0000406	Resistor, Chip	1	R4701
201	ERHZ0000463	Resistor, Chip	1	R6212
202	ERHZ0000484	Resistor, Chip	1	R6207
203	ERHZ0000486	Resistor, Chip	1	R4700
204	ERHZ0000529	Resistor, Chip	1	R6205
205	MBV62321701	Clip	10	SC10001, SC10002, SC10003, SC10004, SC10005, SC10006, SC10007, SC10008, SC20006, SC20008
206	SAFO0000401	Wire Pad, Open	2	R1500, R2214
207	SAFO0000501	Wire Pad, Open	2	R2212, R2213
208	SAFP0000401	Wire Pad, Short	10	R1503, R1504, R2215, R2301, R3301, R5100, R5101, R6217, R7502, R7503
209	SETY0007301	Thermistor, NTC	2	PT2100, PT2101

13. REPLACEMENT PART LIST

210	SFBH0008101	Filter,Bead	1	FB5100
211	SFBH0008102	Filter,Bead	1	FB6209
212	SFBH0008105	Filter,Bead	2	FB6202,FB6203
213	EBR81748701	PCB Assembly,Main,SMT Bottom	1	EBR071600
214	EAB62950801	Microphone,Condenser	2	MIC6200,MIC6201
215	EAE62286801	Capacitor,Ceramic,Chip	1	C9202
216	EAE62502901	Capacitor,Ceramic,Chip	4	C4227,C4228,C7104,C7105
217	EAE62505701	Capacitor,Ceramic,Chip	1	C4401
218	EAE62506501	Capacitor,Ceramic,Chip	2	C4210,C4212
219	EAE62522101	Capacitor,Ceramic,Chip	1	C4200
220	EAE62685301	Capacitor,Ceramic,Chip	1	C9205
221	EAE62762301	Capacitor,Ceramic,Chip	3	C4114,C7103,C9501
222	EAE62767801	Capacitor,Ceramic,Chip	2	C2341,C4122
223	EAE62843001	Capacitor,Ceramic,Chip	1	C4402
224	EAE62883101	Capacitor(High Frequency),Ceramic,Chip	2	C9207,C9208
225	EAE62945801	Capacitor(High Frequency),Ceramic,Chip	1	C1107
226	EAE62946601	Capacitor(High Frequency),Ceramic,Chip	2	C1159,C1170
227	EAE62962301	Capacitor,Ceramic,Chip	8	C4206,C4207,C4208,C4209,C4214,C4215,C4216,C4217
228	EAE62969601	Capacitor,Ceramic,Chip	1	C4500
229	EAE63941501	Capacitor,TA,Polymer	1	C9201
230	EAG63393201	Connector,BtoB	1	CN9500
231	EAG63530401	Connector,Terminal Block	1	CN1
232	EAG63613001	C-Clip	2	ANT6200,ANT6201
233	EAG63652201	C-Clip	2	ANT5100,ANT5101
234	EAG63772101	Connector,RF	4	SW1100,SW1101,SW1103,SW1105
235	EAG64091101	Socket,DIMM/SIMM	1	S9500
236	EAG64149801	Connector,I/O	1	CN9201
237	EAG64370001	Connector,Terminal Block	1	CN9200
238	EAG64389801	C-Clip	10	ANT1100,ANT1101,ANT1102,ANT1103,ANT1104,ANT1106,ANT1107,ANT1108,ANT5102,ANT5103
239	EAH61872601	Diode,TVS	1	D9206
240	EAH61992701	Diode,Schottky	1	ZD4400
241	EAH62033201	Diode,TVS	1	D6200
242	EAM62071001	Filter,Bead	2	FB4100,FB4101
243	EAM62291101	Filter,EMI/Power	2	FL7103,FL7304
244	EAM62530001	Filter,EMI/Power	3	FL7101,FL7105,FL7303
245	EAM62710701	Filter,Bead	1	FB6208
246	EAM63030601	Filter,Separator,Switch	1	FL1114
247	EAM63930901	Filter,Saw	1	FL1115
248	EAN62572401	IC,DC,DC Converter	1	U4400
249	EAN62637201	IC,Acceleration Sensor	1	U8100
250	EAN63546901	IC,PMIC	1	U4100
251	EAN63706801	IC,DC,DC Converter	1	U4200

13. REPLACEMENT PART LIST

252	EAP61866601	Inductor,Multilayer,Chip	1	L1171
253	EAP61866701	Inductor,Multilayer,Chip	1	L1115
254	EAP62106301	Inductor,Wire Wound,Chip	1	L4400
255	EAP62108001	Inductor,Multilayer,Chip	2	C1164,C1197
256	EAP62108201	Inductor,Multilayer,Chip	1	C1149
257	EAP62108601	Inductor,Multilayer,Chip	1	L1160
258	EAP62225901	Inductor,Multilayer,Chip	3	C1116,L1166,L1170
259	EAP62226201	Inductor,Multilayer,Chip	1	L1140
260	EAP62226401	Inductor,Multilayer,Chip	1	L1113
261	EAP62546001	Inductor,Multilayer,Chip	4	L4203,L4204,L4205,L4206
262	EAP62686901	Inductor,Multilayer,Chip	1	L1158
263	EAP62886101	Inductor,Multilayer,Chip	1	L1161
264	EAP62906001	Inductor,Wire Wound,Chip	2	L6201,L6202
265	EAP63067001	Inductor,Multilayer,Chip	1	L1169
266	EAP63107201	Inductor,Multilayer,Chip	2	L4200,L4201
267	EAP63146501	Inductor,Multilayer,Chip	1	L4202
268	EAP63166001	Inductor,Wire Wound,Chip	2	L4207,L4208
269	EAP63185901	Inductor,Multilayer,Chip	1	L4500
270	EAV63012101	LED,Flash	1	LD4500
271	EBC62316201	Resistor,Chip	1	R1109
272	EBC62858401	Resistor,Chip	2	R1121,R1125
273	ECCH0000115	Capacitor,Ceramic,Chip	2	C4121,C9206
274	ECCH0000117	Capacitor,Ceramic,Chip	1	C4120
275	ECCH0000196	Capacitor,Ceramic,Chip	1	L1134
276	ECCH0000198	Capacitor,Ceramic,Chip	13	C1503,C2303,C2323,C2380,C4118,C4119,C4224,C4225,C7300,C7301,C7303,C7400,C7401
277	ECCH0002001	Capacitor,Ceramic,Chip	3	C6221,C8100,C8101
278	ECCH0004904	Capacitor,Ceramic,Chip	1	C9507
279	ECCH0005604	Capacitor,Ceramic,Chip	3	C2302,C2324,C2367
280	ECCH0006201	Capacitor,Ceramic,Chip	4	C2360,C2362,C2372,C9505
281	ECCH0007804	Capacitor,Ceramic,Chip	1	C4112
282	ECCH0009101	Capacitor,Ceramic,Chip	19	C1500,C1501,C1502,C1504,C1505,C1506,C1507,C2347,C2348,C2353,C2361,C2368,C2369,C2370,C2371,C4113,C4117,C8400,C8401
283	ECCH0009103	Capacitor,Ceramic,Chip	29	C1100,C1102,C1104,C1124,C1129,C1130,C1131,C1132,C1145,C1146,C1147,C1156,C1160,C1162,C1165,C1171,C1179,C1182,C1196,C1199,C4123,C4124,C5104,C5121,C5128,C5129,C7100,C7101,C9203
284	ECCH0009104	Capacitor,Ceramic,Chip	3	C1115,C1192,C1193
285	ECCH0009106	Capacitor,Ceramic,Chip	1	C1191
286	ECCH0017301	Capacitor,Ceramic,Chip	35	C2100,C2103,C2304,C2308,C2309,C2311,C2313,C2314,C2316,C2317,C2318,C2321,C2327,C2328,C2329,C2331,C2332,C2335,C4104,C4105,C4106,C4107,C4108,C4109,C4110,C4111,C4115,C4116,C4201,C4202,C4203,C4220,C4221,C4226,C7106
287	ECCH0017501	Capacitor,Ceramic,Chip	7	C2300,C2301,C2325,C2326,C2339,C2343,C2344
288	ECCH0017601	Capacitor,Ceramic,Chip	4	C4222,C4223,C7302,C7402
289	ECZH0000813	Capacitor,Ceramic,Chip	1	C1101

13. REPLACEMENT PART LIST

290	ECZH0000830	Capacitor,Ceramic,Chip	6	C6201,C6223,C6224,C6225,C6228,C6229
291	ECZH0001215	Capacitor,Ceramic,Chip	10	C4100,C4101,C4102,C4103,C4211,C4213,C4218,C4219,C4230,C9204
292	ECZH0025916	Capacitor,Ceramic,Chip	4	C5101,C5122,C6222,C9504
293	ELCH0001444	Inductor,Multilayer,Chip	3	L6200,L6203,L6204
294	ELCH0003841	Inductor,Multilayer,Chip	1	L1132
295	ELCH0004717	Inductor,Multilayer,Chip	1	L1106
296	ENBY0034201	Connector,BtoB	1	CN7400
297	ENBY0036001	Connector,BtoB	1	CN7100
298	ENBY0040701	Connector,BtoB	1	CN7300
299	ERHY0000254	Resistor,Chip	3	R1107,R1108,R9213
300	ERHY0000298	Resistor,Chip	2	R1122,R1128
301	ERHY0003201	Resistor,Chip	1	R9207
302	ERHY0003301	Resistor,Chip	1	R9214
303	ERHY0009302	Resistor,Chip	1	R9211
304	ERHY0009316	Resistor,Chip	6	R1129,R1131,R1133,R1135,R5102,R5104
305	ERHY0009503	Resistor,Chip	4	R1507,R1508,R9202,R9208
306	ERHY0009504	Resistor,Chip	2	R1502,R1513
307	ERHY0009505	Resistor,Chip	5	R1123,R1127,R7100,R7101,R7532
308	ERHY0009506	Resistor,Chip	1	R7102
309	ERHY0009507	Resistor,Chip	5	R1130,R1132,R1134,R1136,R9515
310	ERHY0009526	Resistor,Chip	2	R1126,R8400
311	ERHY0009531	Resistor,Chip	1	R1506
312	ERHY0009537	Resistor,Chip	1	R1115
313	ERHY0009553	Resistor,Chip	2	R5103,R5105
314	ERHY0035601	Resistor,Chip	1	R9221
315	ERHZ0000204	Resistor,Chip	2	R2114,R9222
316	ERHZ0000205	Resistor,Chip	1	R4104
317	ERHZ0000236	Resistor,Chip	1	R1509
318	ERHZ0000287	Resistor,Chip	1	R6201
319	ERHZ0000288	Resistor,Chip	1	R6200
320	ERHZ0000295	Resistor,Chip	1	R4102
321	ERHZ0000308	Resistor,Chip	1	R9210
322	ERHZ0000402	Resistor,Chip	1	R9508
323	ERHZ0000405	Resistor,Chip	1	R2216
324	ERHZ0000441	Resistor,Chip	1	R9521
325	ERHZ0000463	Resistor,Chip	1	R6213
326	ERHZ0000484	Resistor,Chip	1	R6206
327	ERHZ0000529	Resistor,Chip	2	R4101,R4105
328	ERHZ0000538	Resistor,Chip	2	R4100,R9223
329	EXXY0018701	Crystal	1	X4100
330	MBV62321701	Clip	7	SC10000,SC20000,SC20001,SC20002,SC20003,SC20004,SC20005
331	MEZ65049701	Label	1	PID1

13. REPLACEMENT PART LIST

332	FAB31918701	Screw,Machine	11	FAB010000
333	FAB32218901	Screw,Tapping	0	FAB020000
334	MEZ64615406	Label,Approval	1	MEZ002100
335	MJN69527603	Tape,USP Film	1	MJN107400
336	SAF30821601	Software Assembly,Common	1	SAF010100
337	SAD35268201	Software,Application	1	SAD010100
338	SAD35451501	Software,USB Driver	1	SAD010500
339	SAD35491701	Software,OSP	1	SAD010600
340	AAD87375201	Addition Assembly	1	AAD000000
341	ACQ88371001	Cover Assembly,Battery	1	ACQ004100
342	MCK68549401	Cover,Battery	1	MCK004100
343	EAB62808211	Earphone,Stereo	1	EAB010200
344	EAC62018207	Rechargeable Battery,Lithium Ion	1	EAC020100
345	EAD62377902	Cable,Assembly	1	EAD010000
346	EAY62751801	Adapters	1	EAY060000
347	MBM65178201	Card,Quick Reference	1	MBM062600
348	MFL69182801	Manual,Operation	0	MFL053800
349	AGF78179301	Package Assembly	1	AGF000000
350	MAF63149903	Bag,Vinyl	1	MAF086500
351	MAY66007601	Box,Master	0	MAY047100
352	MAY66232701	Box,Carton	0	MAY010800
353	MAY66232702	Box,Carton	0	MAY010801
354	MAY66232703	Box,Carton	0	MAY010802
355	MBEC0002901	Box,Carton	0	MAY010803
356	MEZ63874703	Label,Unit Box	2	MEZ084100
357	MEZ64781701	Label,Master Box	0	MEZ047200
358	MEZ65015204	Label,Unit Box	1	MEZ084101
359	MJN68155201	Tape	0	MJN000000
360	MPCY0019330	Pallet	0	MGA000000
361	MAY66471607	Box,Carton	0	MAY010800
362	MAY66471721	Box,Pallet Sleeve	0	MAY115200
363	MBL65957705	Cap,Box	0	MBL007001
364	MLAJ0004402	Label,Master Box	0	MEZ047200
365	MLAZ0050901	Label	0	MEZ000000
366	MPCY0012403	Pallet	0	MGA000000
367	MAF63267402	Bag,Vinyl	1	MAF086500
368	MAY66470404	Box,Master	0	MAY047100
369	MAY67287719	Box,Unit	1	MAY084000
370	MEZ64571201	Label,Unit Box	1	MEZ084000
371	MEZ64686921	Label,Unit Box	1	MEZ084100
372	MLAC0004541	Label,Barcode	1	MEZ084101
373	MLAJ0004402	Label,Master Box	0	MEZ047200

